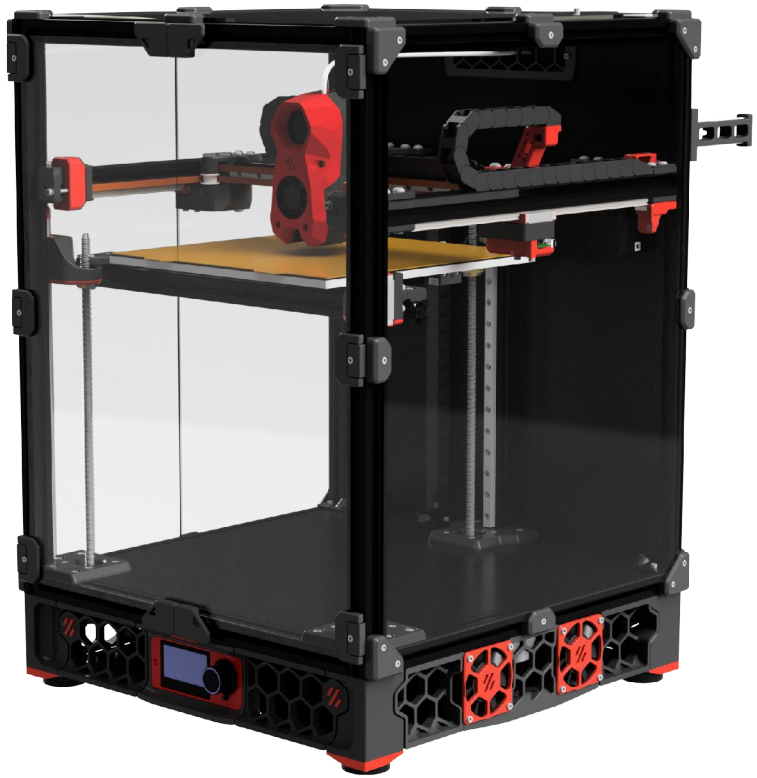
VORON



VORON TRIDENT ASSEMBLY

We build space shuttles with gardening tools
so anyone can have a space shuttle of their own.

VERSION 2025-11-28

Before you begin on your journey, a word of caution:

You are about to assemble a robot in the comfort of your own home. This machine can maim, burn, and electrocute you if you are not careful. Please do not become the first VORON fatality. There is no special Reddit flair for that.

Please read the entire manual before you start assembly. As you begin wrenching, please check our Discord channels for any tips and questions that may halt your progress.

Most of all, good luck!

THE VORON TEAM

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PART PRINTING GUIDELINES

The VORON Team has provided the following print guidelines for you to follow in order to have the best chance at success with your parts. There are often questions about substituting materials or changing printing standards, but we recommend that you follow these:

3D PRINTING PROCESS

Fused Deposition Modeling (FDM)

MATERIAL

ABS or ASA

LAYER HEIGHT

Recommended: 0.2mm

EXTRUSION WIDTH

Recommended: Forced 0.4mm

INFILL TYPE

Grid, Gyroid, Honeycomb, Triangle or Cubic

INFILL PERCENTAGE

Recommended: 40%

WALL COUNT

Recommended: 4

SOLID TOP/BOTTOM LAYERS

Recommended: 5

PRINT IT FORWARD (PIF)

Often times community members that have issues printing ABS will bootstrap themselves into a VORON using our Print It Forward program. This is a service where approved members with VORON printers can make you a functional set of parts to get your own machine up and running. Check Discord if you have any interest in having someone help you out.

FILE NAMING

By this time you should have already downloaded our STL files from the VORON GitHub. You might have noticed that we have used a unique naming convention for the files. This is how to use them.

PRIMARY COLOR

Example z_joint_lower_x4.stl

These files will have nothing at the start of the filename.

ACCENT COLOR

Example [a]_tensioner_left.stl

We have added “[a]” to the front of any STL file that is intended to be printed with accent color.

QUANTITY REQUIRED

Example: [a]_z_belt_clip_lower_x4.stl

If any file ends with “_x#”, that is telling you the quantity of that part required to build the machine.

HOW TO GET HELP

If you need assistance with your build, we’re here to help. Head on over to our Discord group and post your questions. This is our primary medium to help VORON Users and we have a great community that can help you out if you get stuck.

https://discord.gg/voron

https://forum.vorondesign.com

REPORTING ISSUES

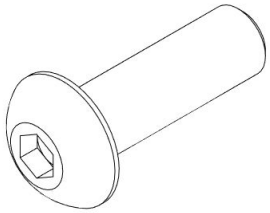
Should you find an issue in the documentation or have a suggestion for an improvement please consider opening an issue on GitHub (https://github.com/VoronDesign/Voron-Trident/issues).
When raising an issue please include the relevant page numbers and a short description; annotated screenshots are also very welcome. We periodically update the manual based on the feedback we get.

THIS IS JUST A REFERENCE

This manual is designed to be a simple reference manual. Building a VORON can be a complex endeavour and for that reason we recommend downloading the CAD files off our Github repository if there are sections you need clarification on. It can sometimes be easier to follow along when you have the whole assembly in front of you.

GitHub https://github.com/vorondesign

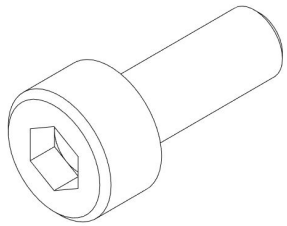
DOCS https://docs.vorondesign.com



BUTTON HEAD CAP SCREW (BHCS)

Metric fastener with a domed shape head and hex drive. Most commonly found in locations where M5 fasteners are used.

ISO 7380-1



SOCKET HEAD CAP SCREW (SHCS)

Metric fastener with a cylindrical head and hex drive. The most common fastener used on the VORON.

ISO 4762

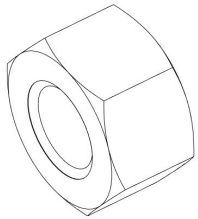
FLAT HEAD COUNTERSUNK SCREW (FHCS)

Metric fastener with a cone shaped head and a flat top.

ISO 10642

SELF TAPPING SCREW

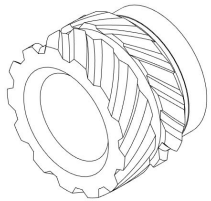
Fastener with a pronounced thread profile that is driven directly into plastic.



HEX NUT

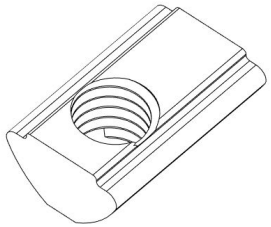
Hex nuts couple with screws to create a tight, secure joint. You'll see these used in both M3 and M5 variants throughout this guide.

ISO 4032



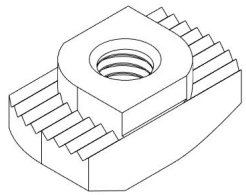
HEAT SET INSERT

Heat inserts with a soldering tip so that they melt the plastic when installed. As the plastic cools, it solidifies around the knurls and ridges on the insert for excellent resistance to both torque and pull-out.



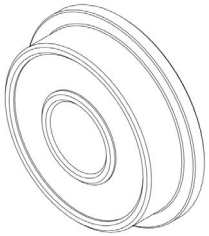
POST INSTALL T-SLOT NUT (T-NUT)

Nut that can be inserted into the slot of an aluminium profile. Used in both M3 and M5 variants throughout this guide. Often also called "roll-in t-nut".



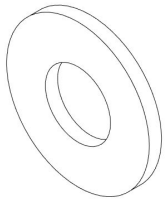
HAMMERHEAD NUT

Nut that can be inserted into the slot of an aluminum profile. Once in place, the nut is secured by rotating the screw, which locks it in place within the slot and prevents it from sliding out.



F695 BEARING

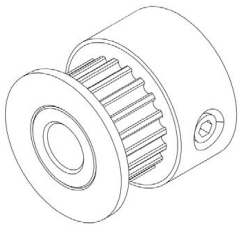
A ball bearing with a flange used in various gantry locations.



SHIM

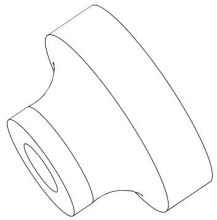
Not to be confused with stamped washers. These are used in all M5 call-out locations in this manual.

DIN 988



PULLEY

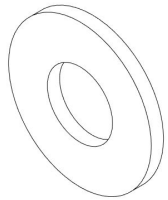
GT2 pulley used on the motion system of the VORON.



THUMB NUT

Used in the print bed as a spacer.

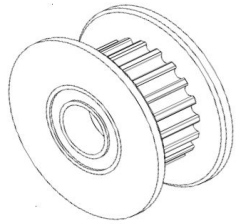
DIN 466-B



WASHER

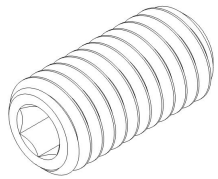
Usually stamped from sheet metal this type of spacer is not as consistent in thickness as the shims are. Only used in M3 size.

DIN 125



IDLER

GT2 idler used in the motion system of the VORON.



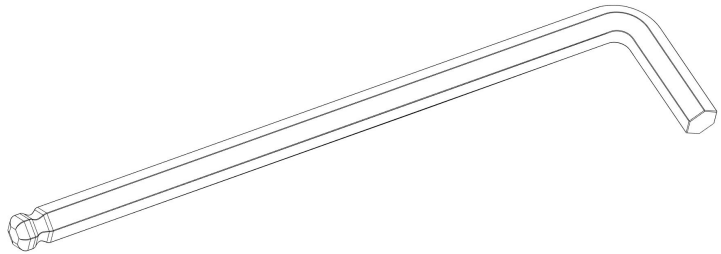
SET SCREW

Small headless fastener with an internal drive. Used in pulleys and other gears. Also called a grub screw.

ISO 4026

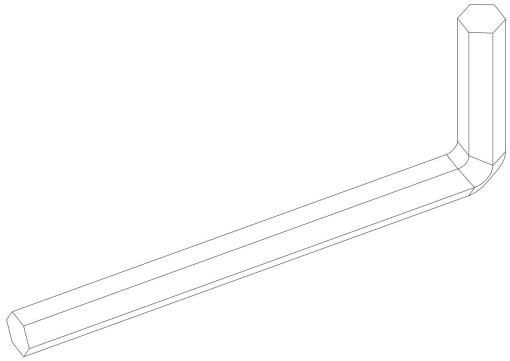
BALL-END DRIVER

Some parts of this design require the use of a ball-end hex driver for assembly. We recommend you get 2.0 mm, 2.5 mm, 3.0 mm and 4.0 mm ones.



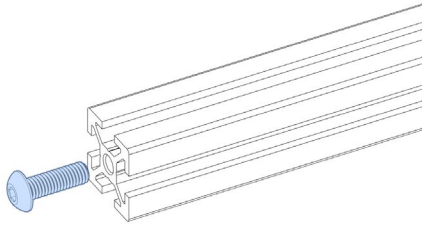
2.5MM HEX DRIVER

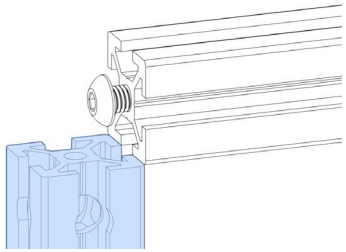
The 2.5mm hex driver will see a lot of use in this build. A quality driver is strongly recommended. Refer to the sourcing guide for suggestions.



ADDITIONAL TOOLS

We provide additional tool recommendations in our sourcing guide. Visit https://vorondesign.com/sourcing_guide and switch to the “VORON Tools” tab at the bottom of the page.





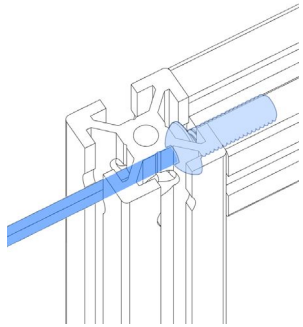
BLIND JOINT BASICS

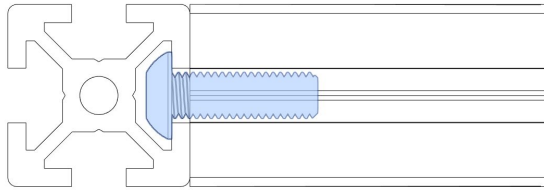
Blind Joints provide a cost effective and rigid assembly method.

The head of the BHCS is slid into the channel of another extrusion and securely fastened through a small access hole in the extrusion.

If you've never assembled one before we recommend you watch the linked guide.

https://voron.link/onjwmcd





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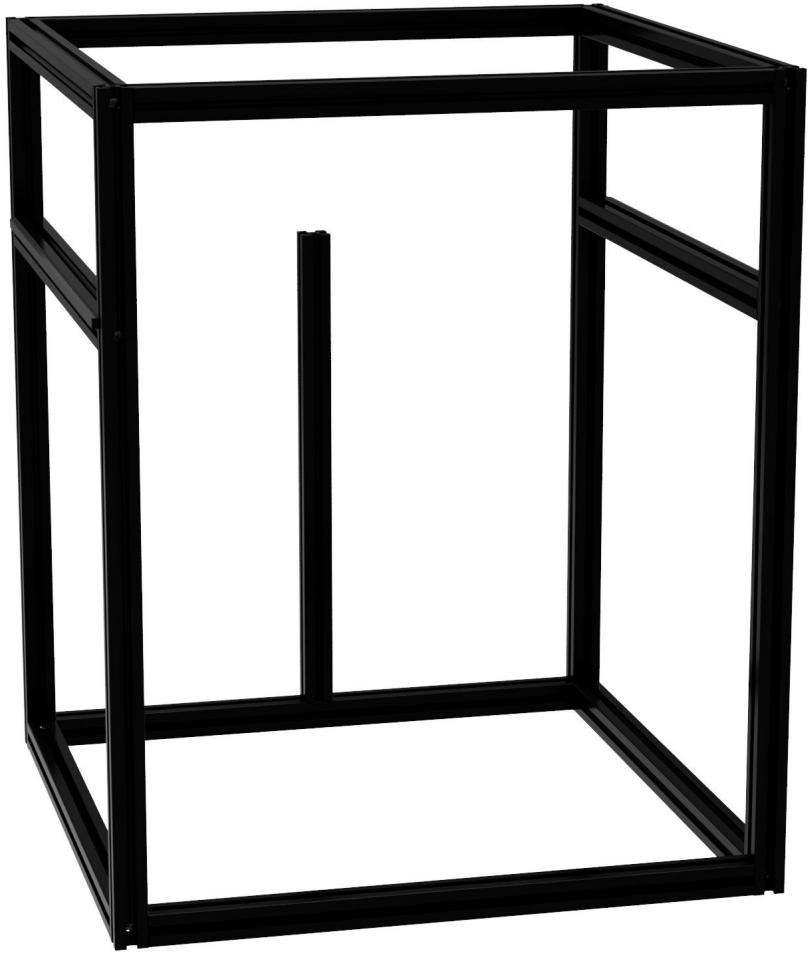
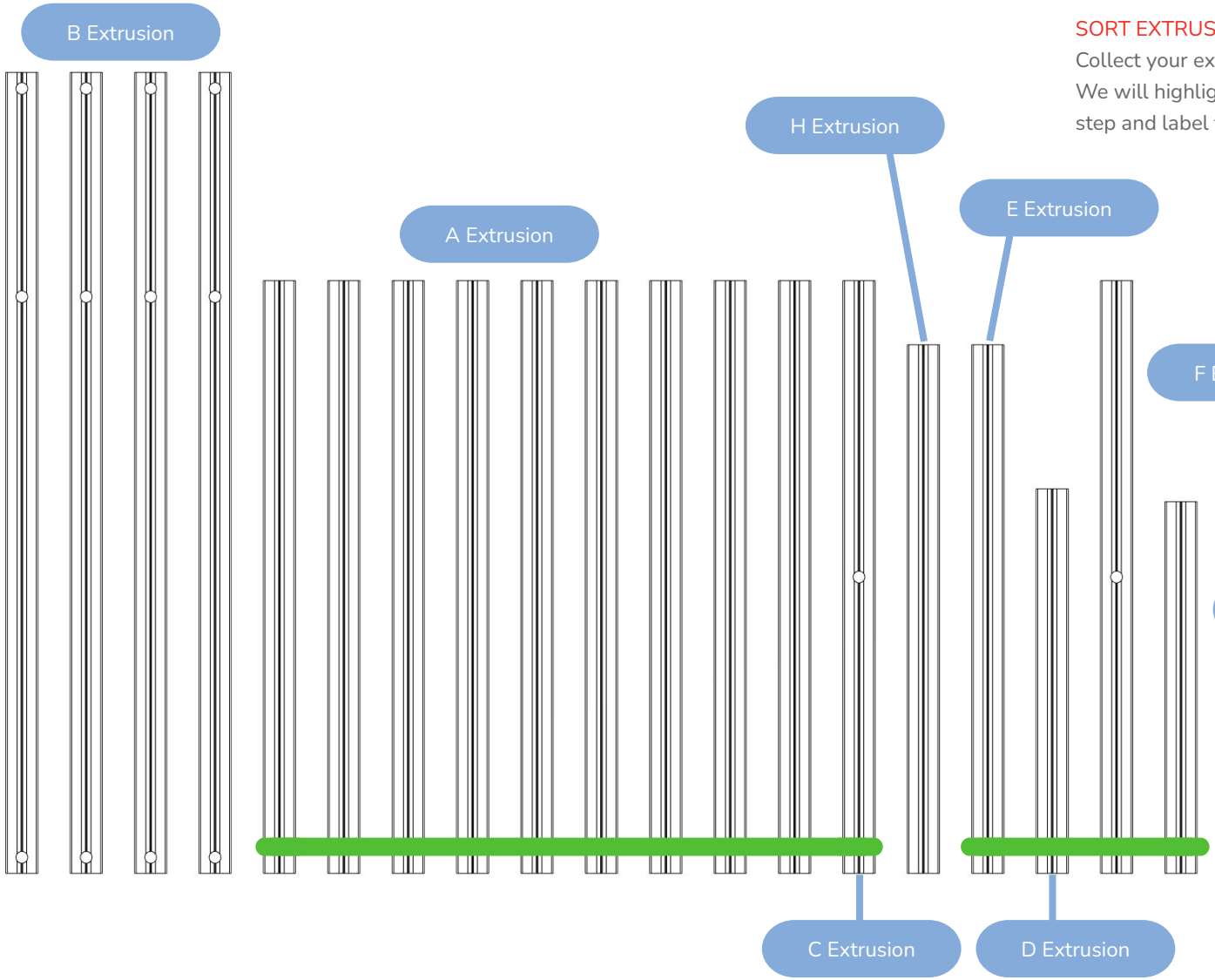


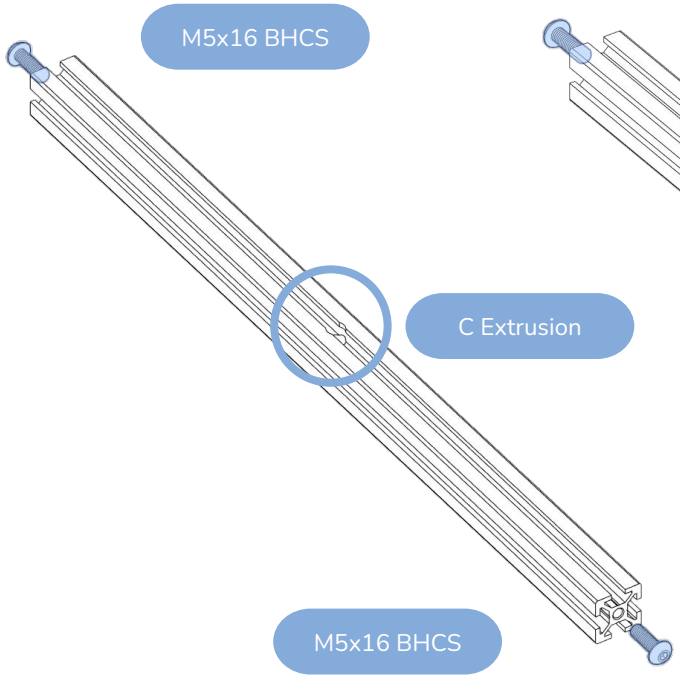


SORT EXTRUSIONS

Collect your extrusions and sort them by length. We will highlight the extrusions used in each step and label them as shown on this page.

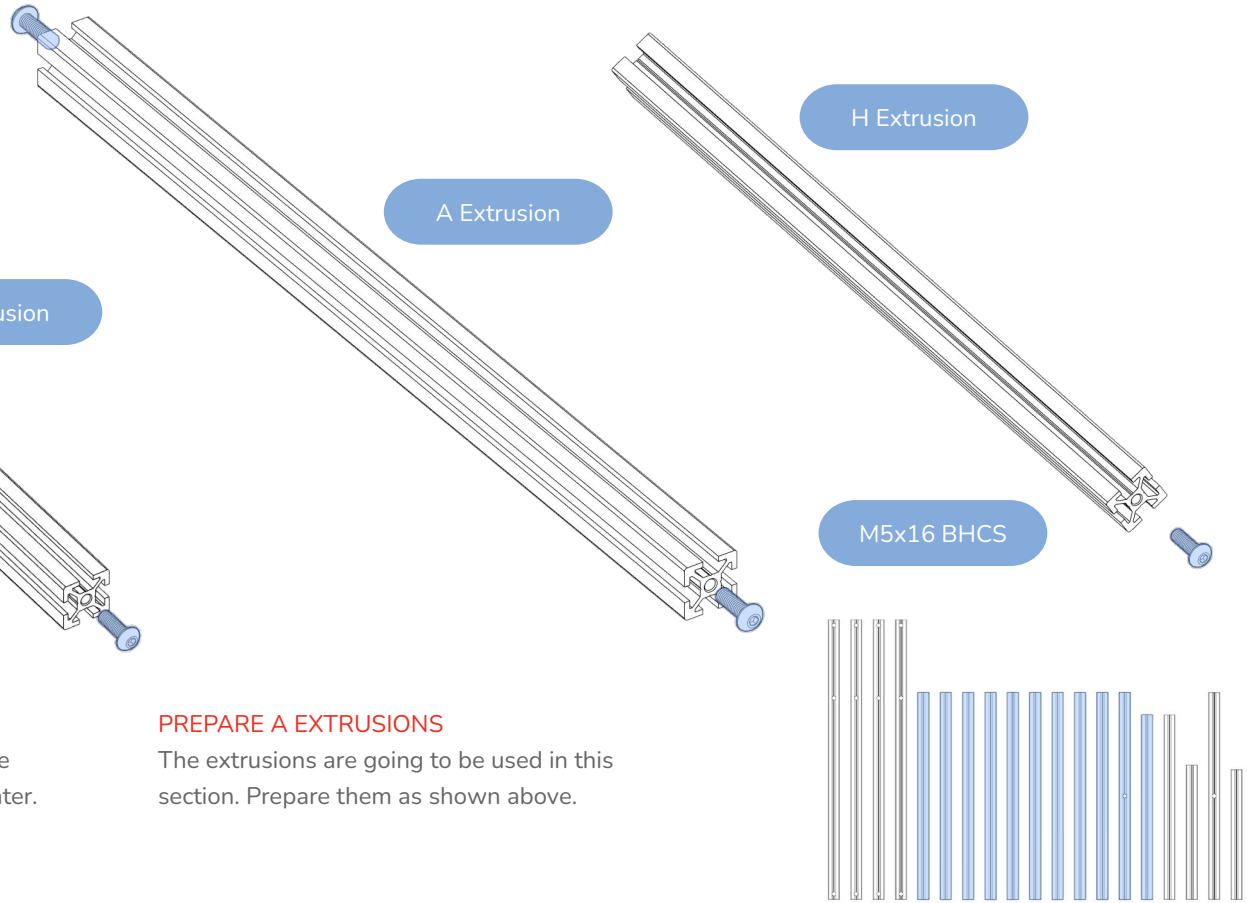
BUILD SIZE DEPENDENT

The length of marked extrusions depends on the build size. The size shown is for a 250 size Trident. Add 50/100mm for a 300/350 sized Trident.



LOWER BACK EXTRUSION

This extrusion will be used throughout the manual to mark the lower back of the printer.

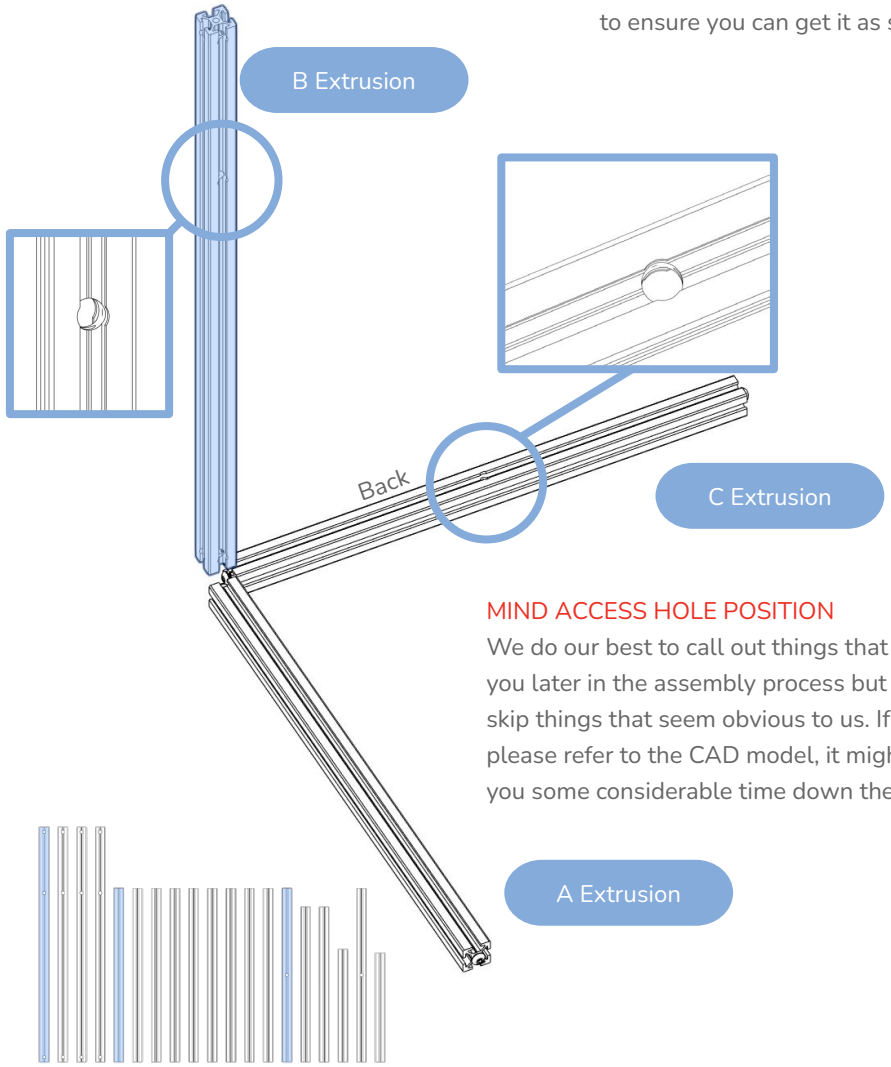


PREPARE A EXTRUSIONS

The extrusions are going to be used in this section. Prepare them as shown above.

BUILD ON A FLAT SURFACE

Build the frame on a glass or granite surface to ensure you can get it as square as possible.



MIND ACCESS HOLE POSITION

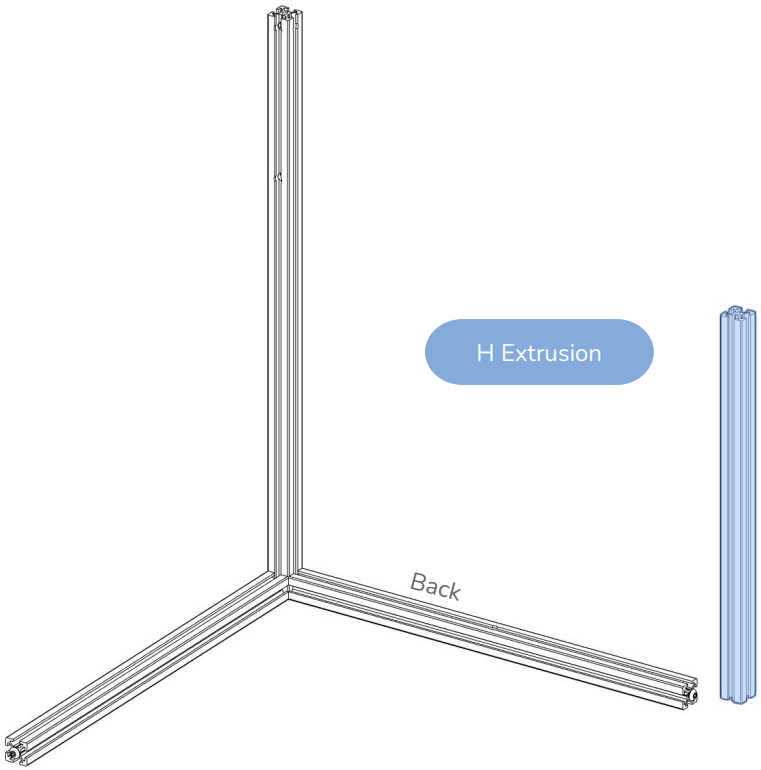
We do our best to call out things that may bite you later in the assembly process but may skip things that seem obvious to us. If in doubt please refer to the CAD model, it might save you some considerable time down the road.

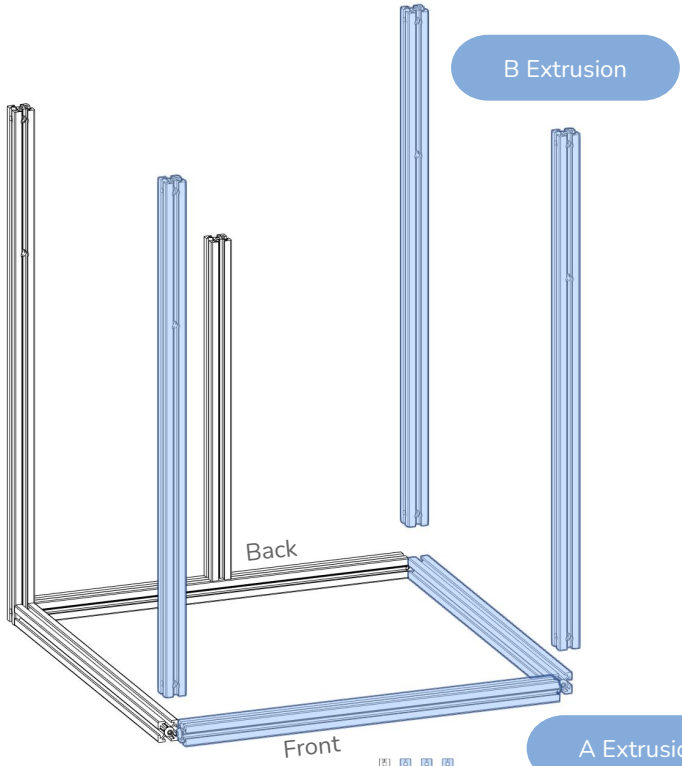
FIRST BLIND JOINT

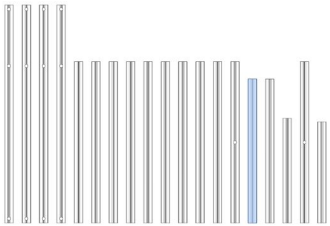
This design relies on blind joints to assemble the frame. We outlined the basics of blind joints on page 10.

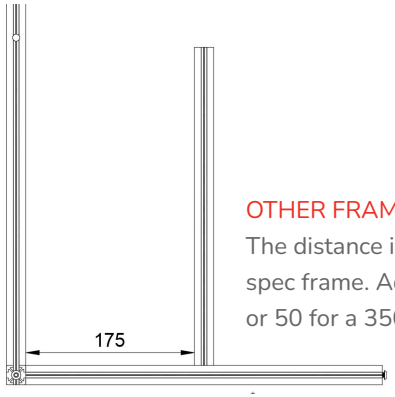
If you've never assembled one before we recommend you watch the linked guide.

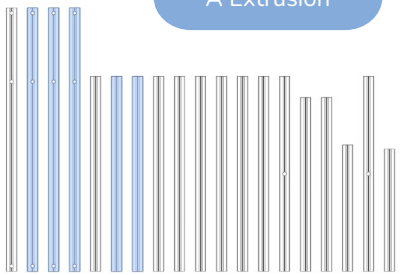
https://voron.link/onjwmcd





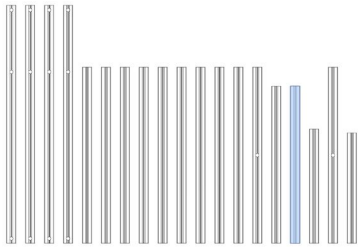


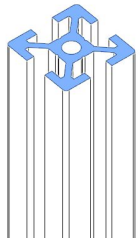




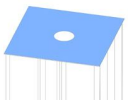
OTHER FRAME SIZES

The distance is shown for a 250 spec frame. Add 25mm for a 300 or 50 for a 350 sized frame.





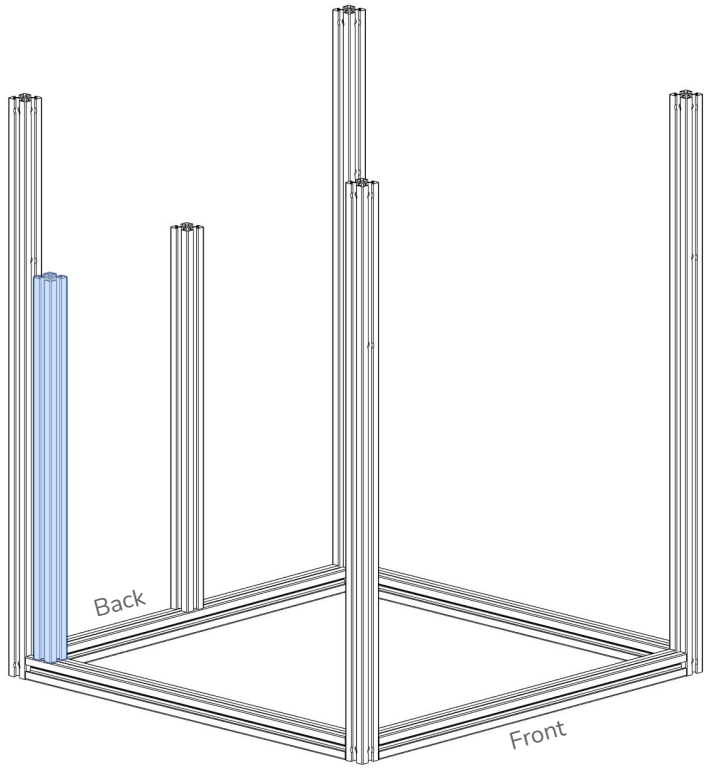
E Extrusion



COVER IN TAPE

We'll be using an extrusion as a spacer.
Apply a single layer of tape to the ends to prevent scratches.

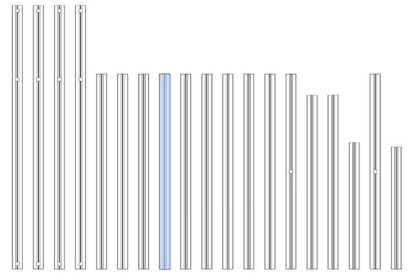
If you're not confident in the length and squareness of your extrusions print the included spacer instead.

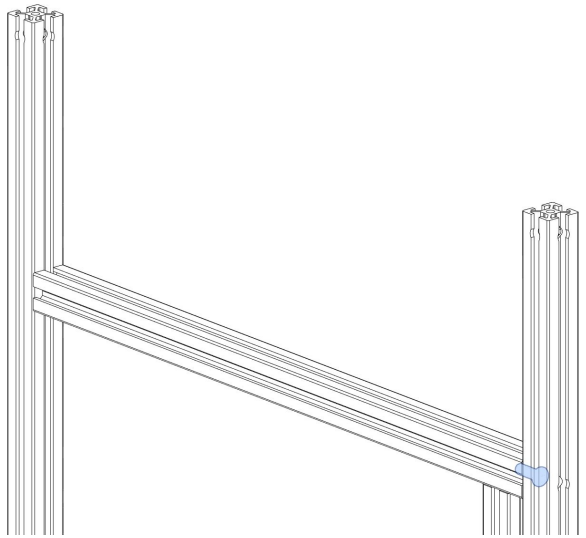


FINDING THE RIGHT POSITION

If you are building a 250 size Trident use the 330mm E extrusion as a spacer to locate the Y extrusions.

Alternatively you can print the spacer that is included in the released files.

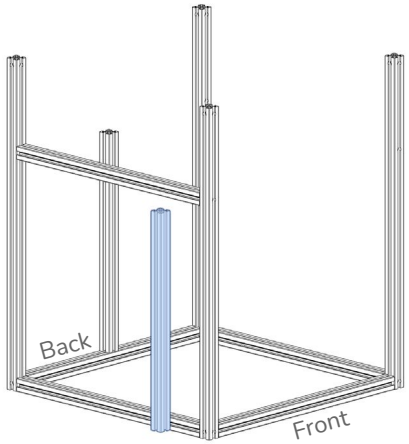


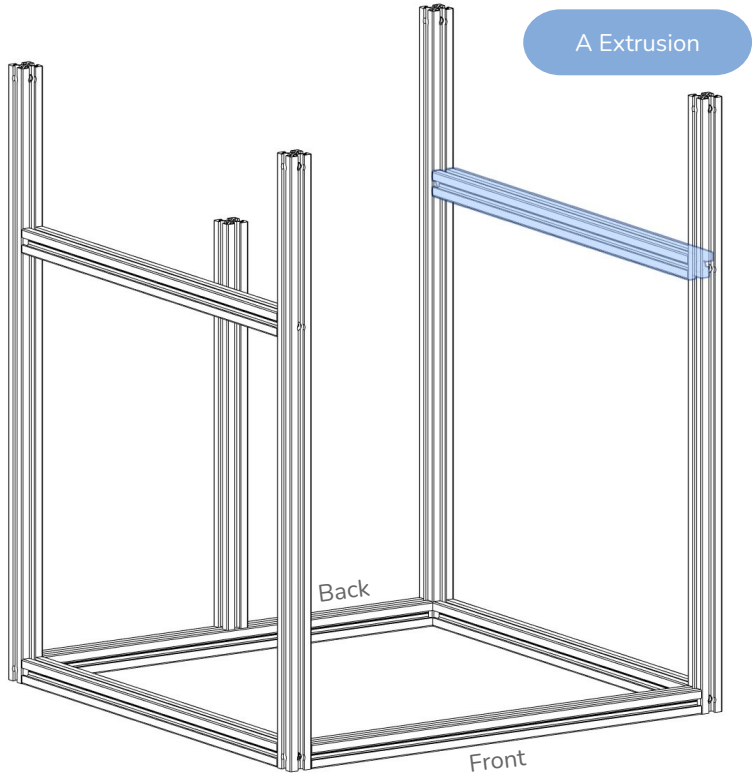


FINDING THE RIGHT POSITION

Using the 330mm extrusion as a spacer insert the Y rail and secure it using the access holes in the upright extrusions.

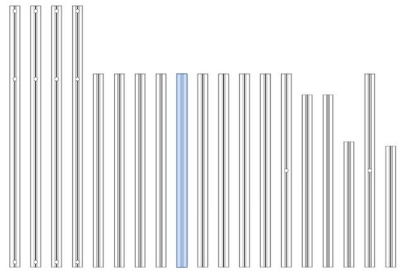
Make sure that the extrusions sit at an 90° angle and are free of any rotation/twist. The outsides of the extrusions should be flush.



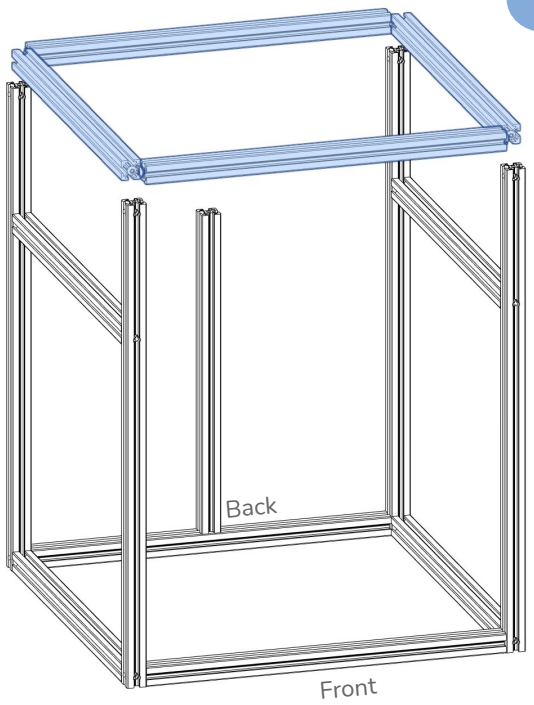


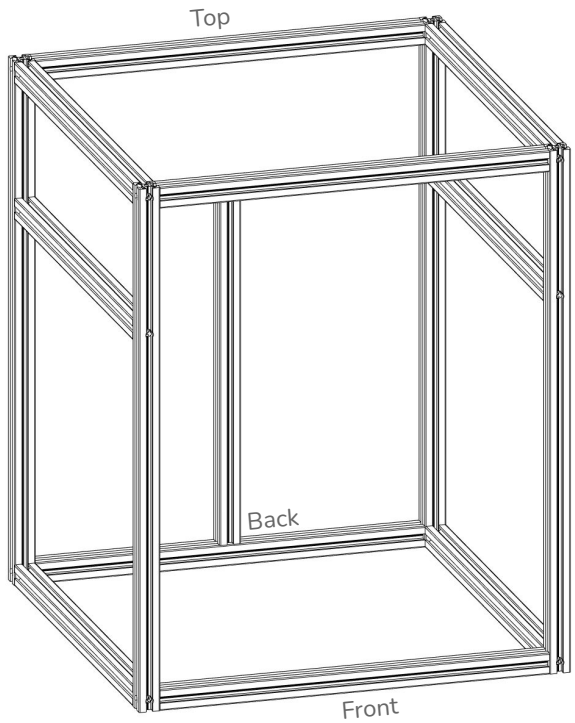
RINSE AND REPEAT

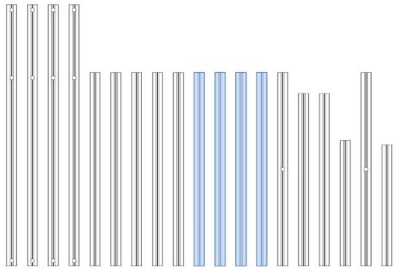
Remove the spacer and repeat the steps for the other Y extrusion.

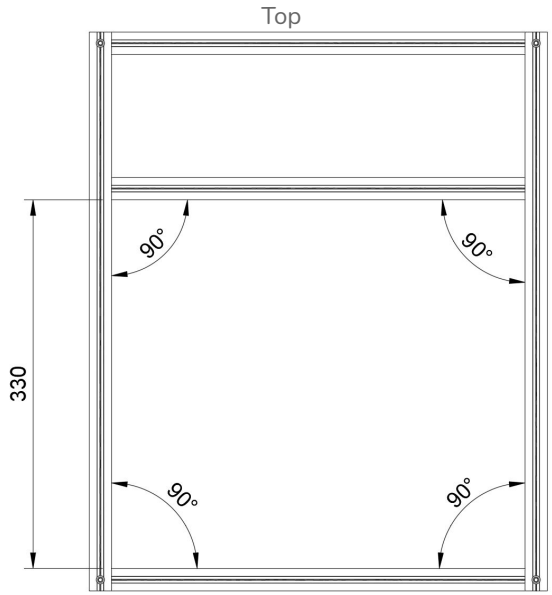


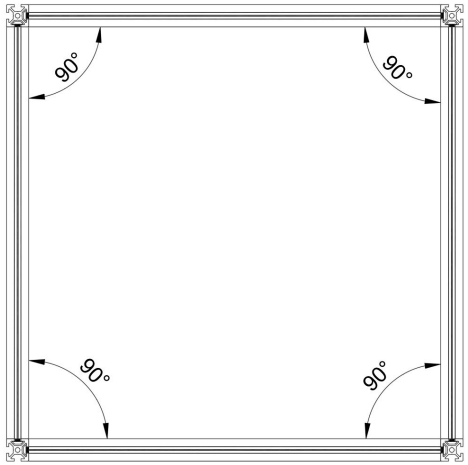
A Extrusion







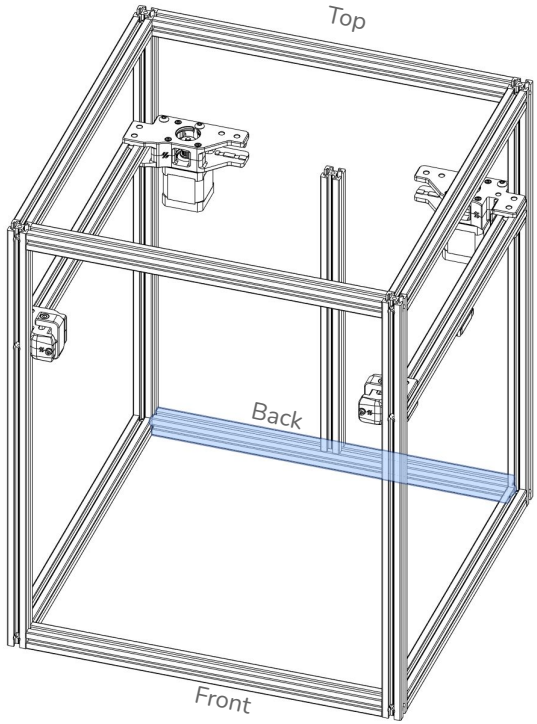


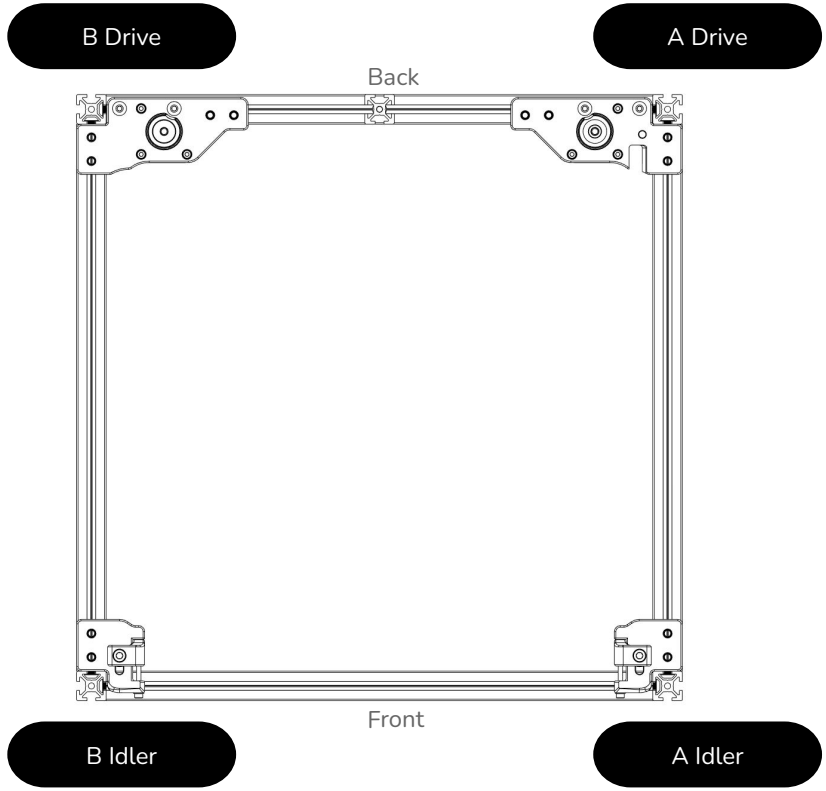


CHECK FOR SQUARENESS

Verify the angle of all corners and the overall squareness by measuring the diagonals. Refer to the second half of the linked video for additional information.

https://voron.link/kdtpzam



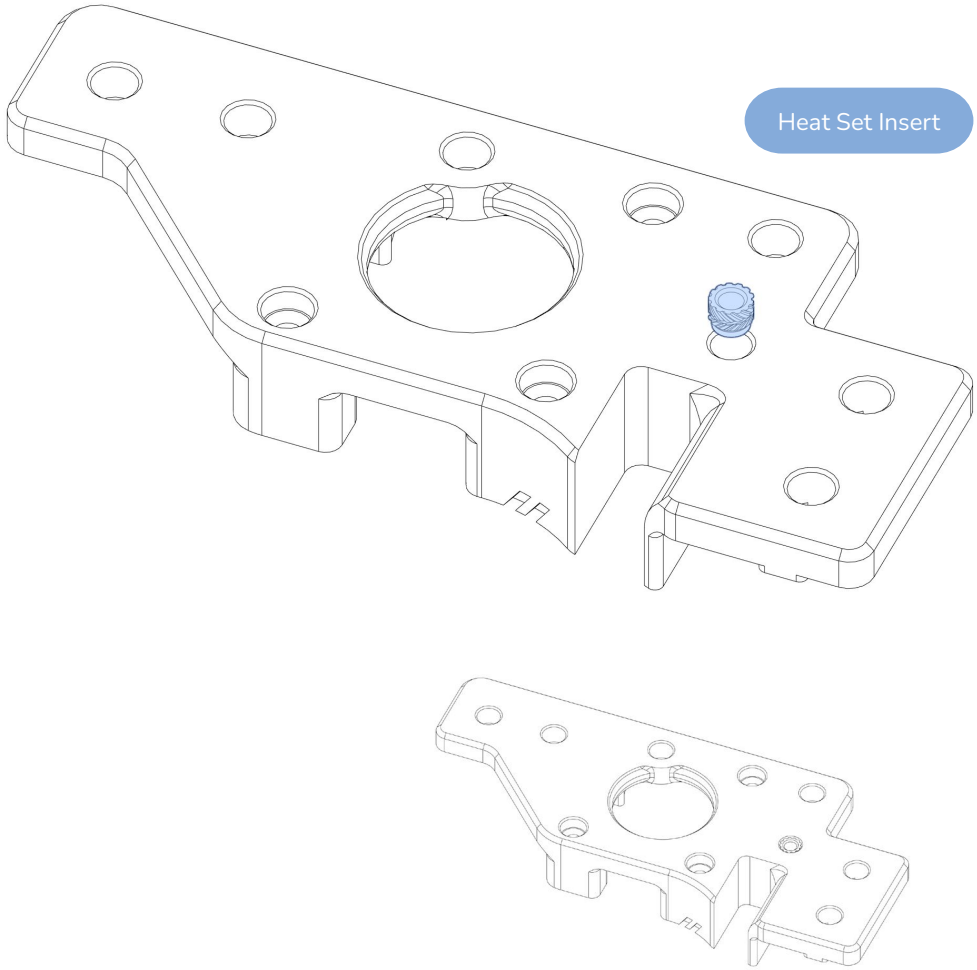


WHY IS THIS HERE?

As you likely skipped over the advice to flip through the entire manual we added graphics like these to assist you with the orientation of the part before you actually put them on the printer.

OVERVIEW

Individual chapters start with an overview of the components that will be built/added to the printer in the chapter.

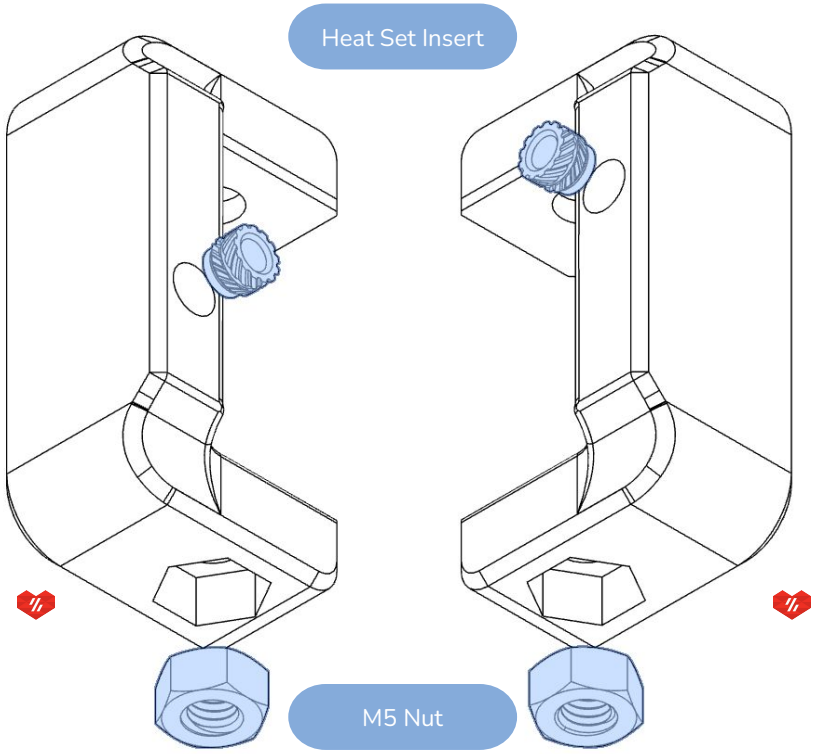


HEAT SET INSERTS

This design relies heavily on heat set inserts. Make sure you have the proper inserts (check the hardware reference for a close up picture and the BOM for dimensions).

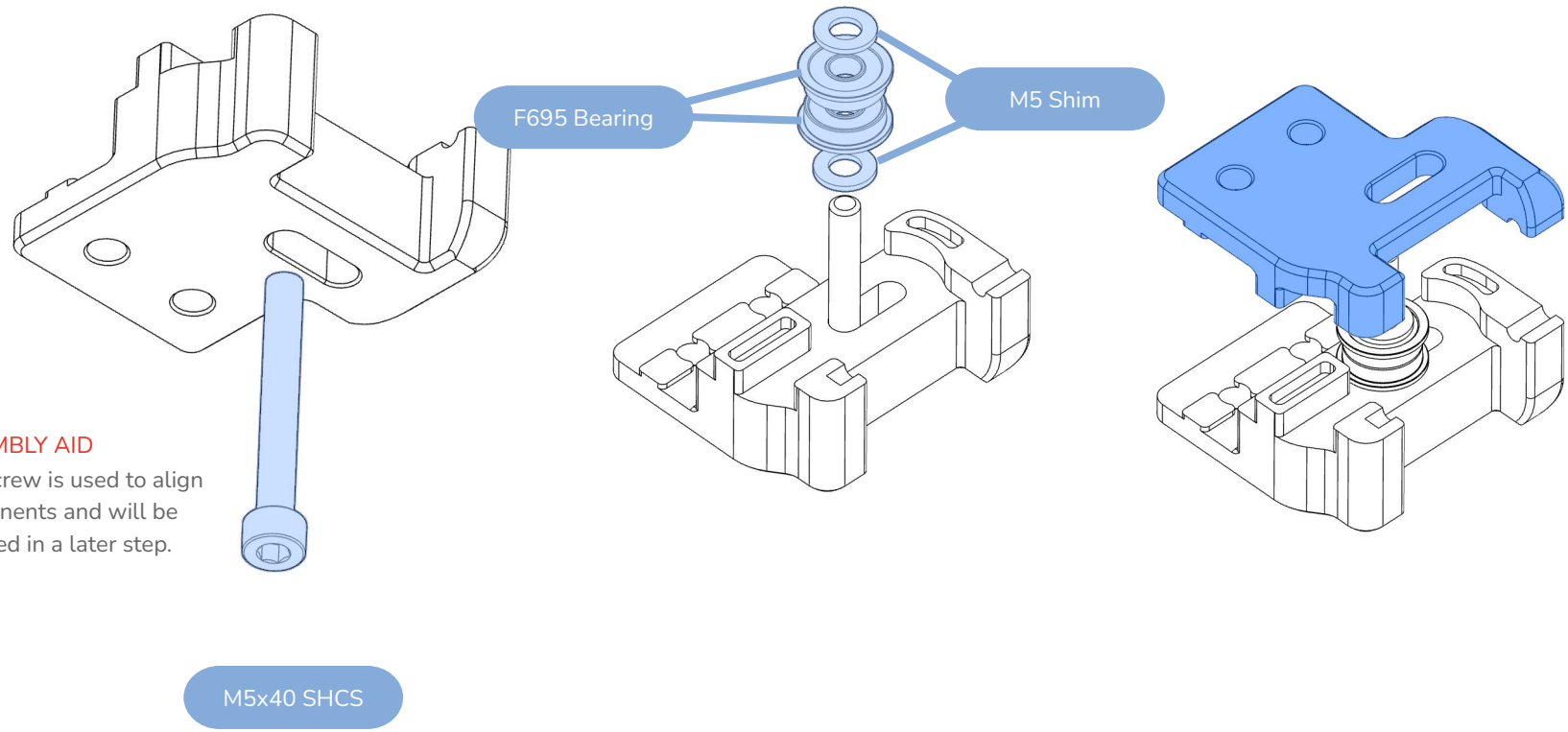
If you've never worked with heat set inserts before we recommend you watch the linked guide.

https://voron.link/m5ybt4d



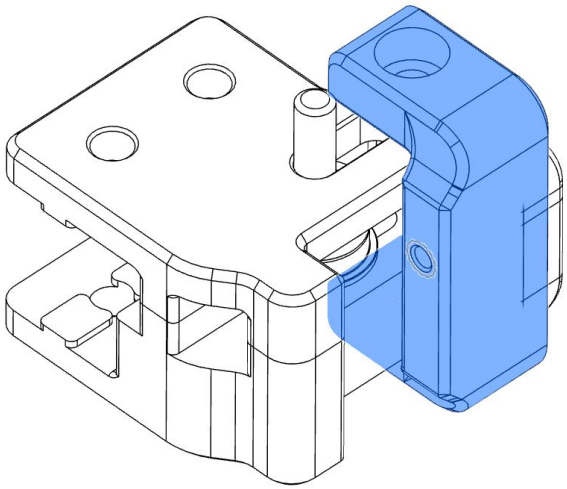
ACCENT PART?

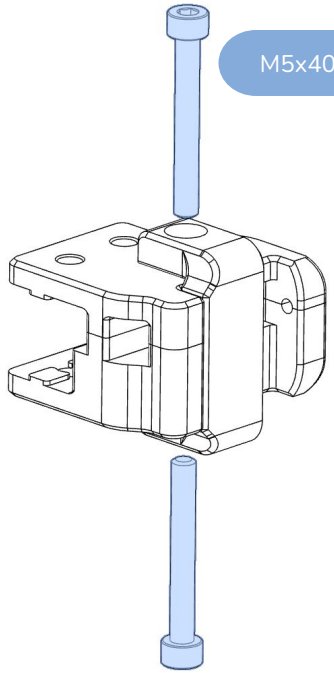
Look for VORON heart next to the part.
It indicates that this is an accent part.

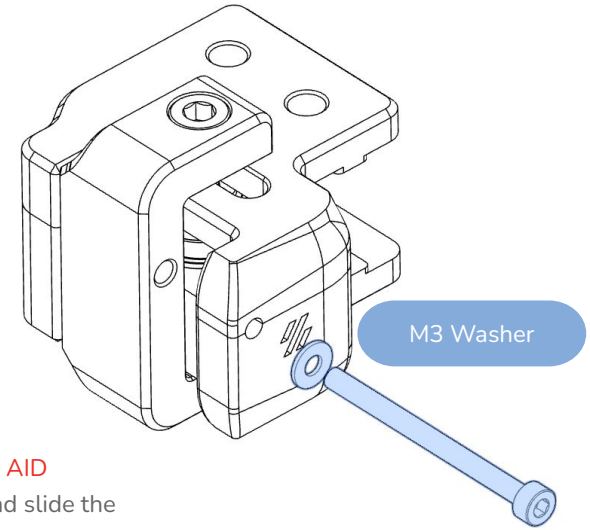


ASSEMBLY AID

This screw is used to align components and will be removed in a later step.

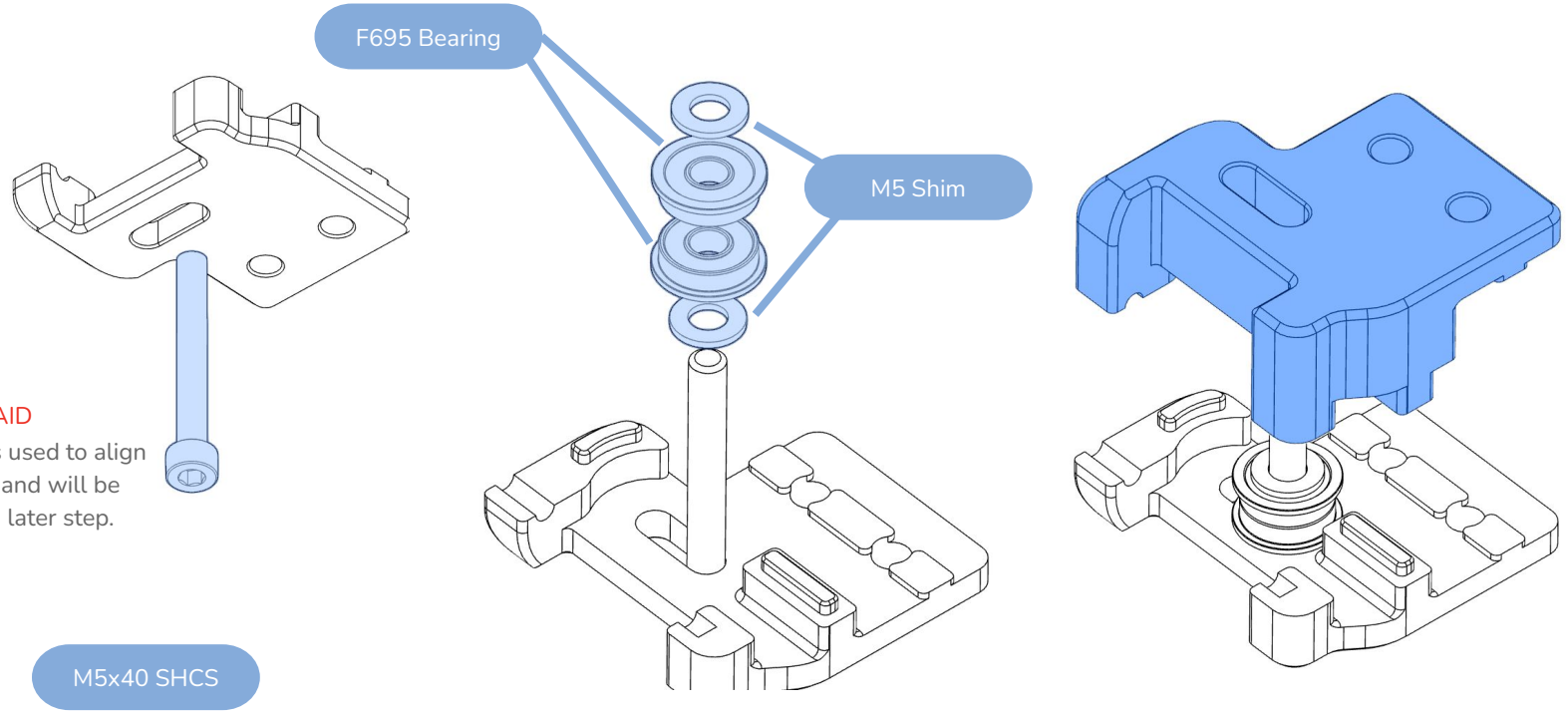






REMOVE ASSEMBLY AID

Remove the screw and slide the tension arm into place.



ASSEMBLY AID

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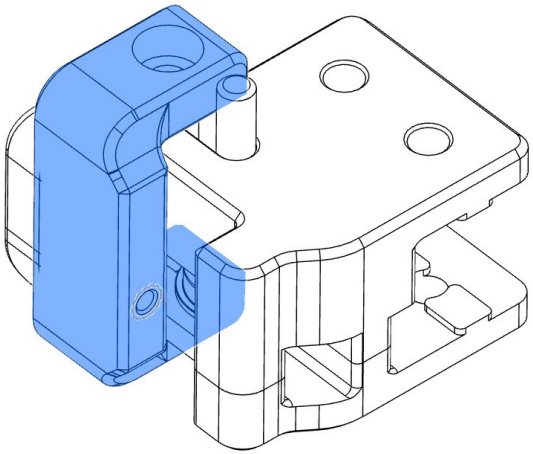
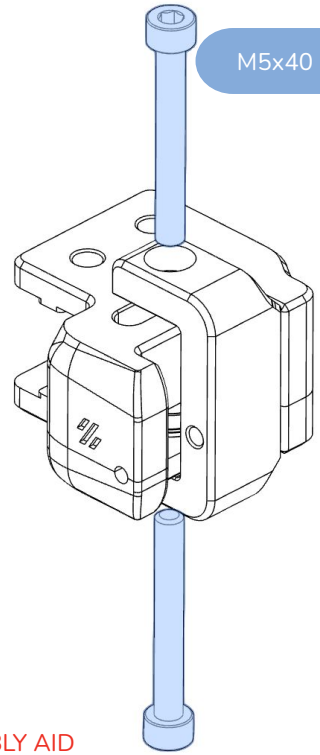
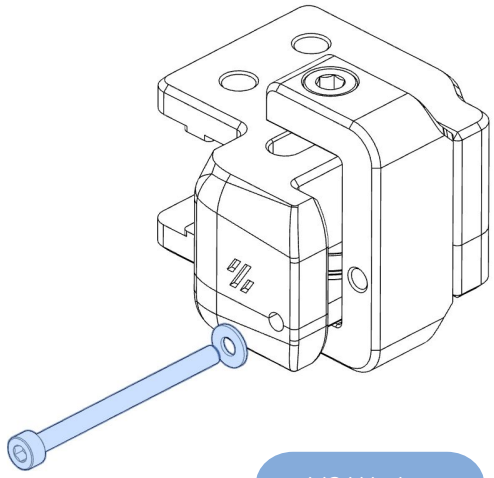




M5x40 SHCS

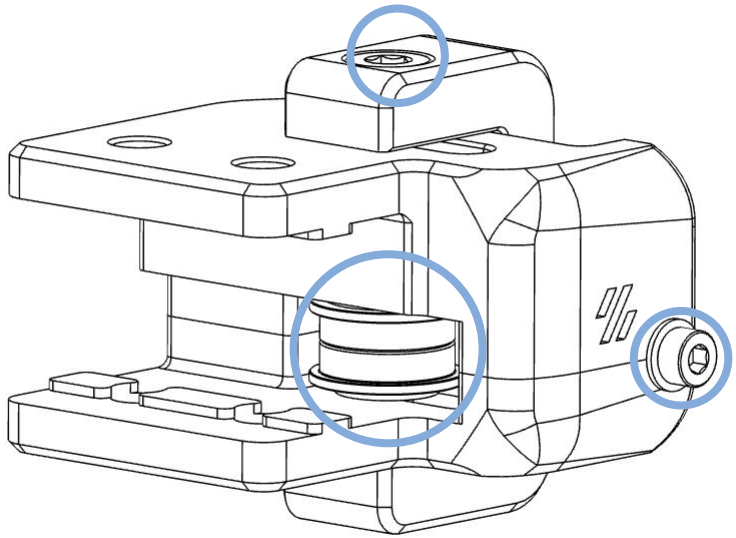
REMOVE ASSEMBLY AID

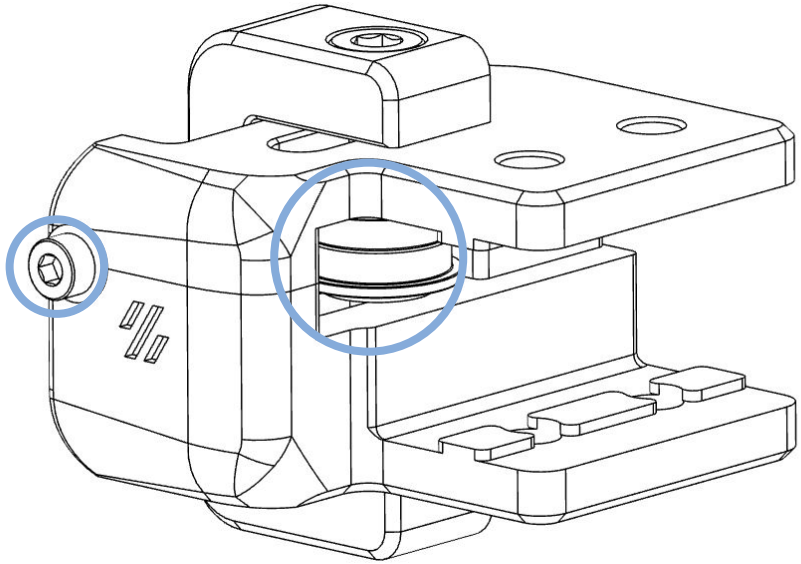
Remove the screw and slide the tension arm into place.



M3 Washer

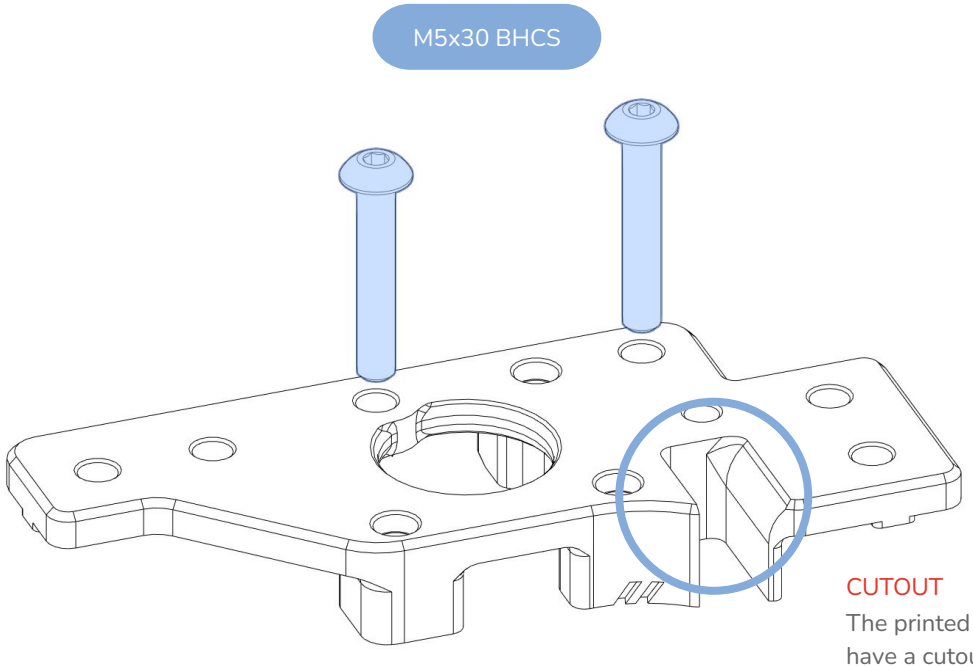
M3x40 SHCS





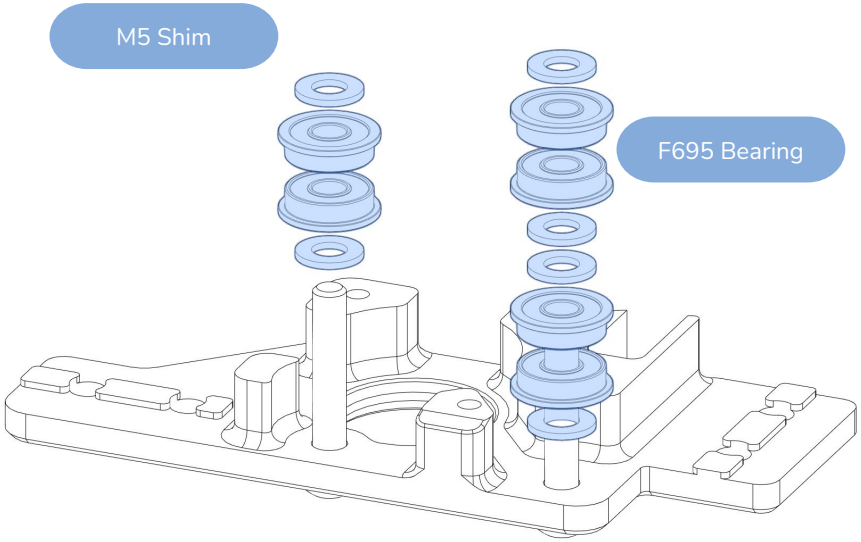
CHECK YOUR WORK

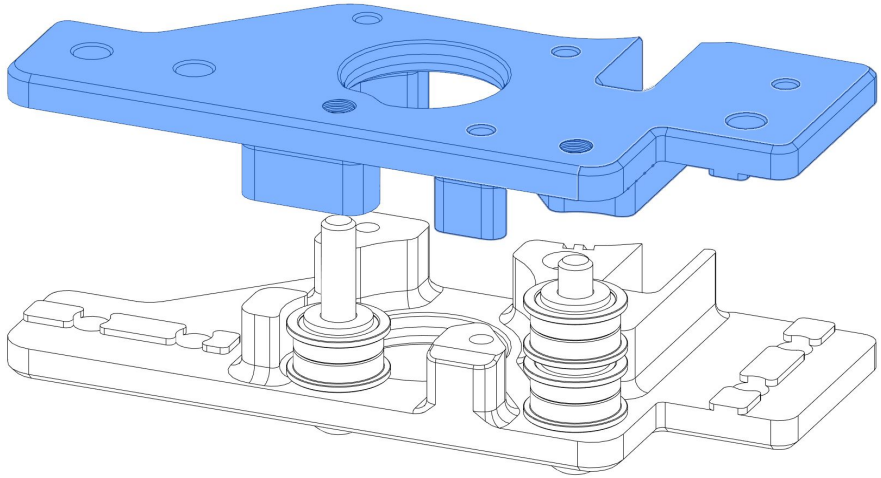
Compare your assembled parts to the graphics shown here. Pay attention to the features highlighted by the circles.



CUTOUT

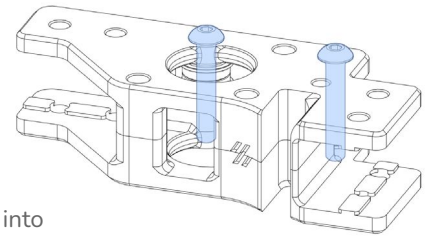
The printed parts for the A drive have a cutout.





UPSIDE DOWN ASSEMBLY

For ease of assembly we recommend to assemble the A and B drives upside down.



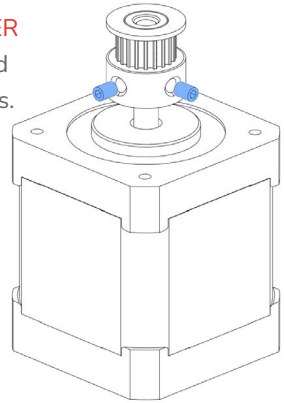
DON'T OVERTIGHTEN

The M5 screws are threaded directly into plastic.

GT2 20 Tooth Pulley

APPLY THREADLOCKER

Make sure to use thread locker on the set screws.



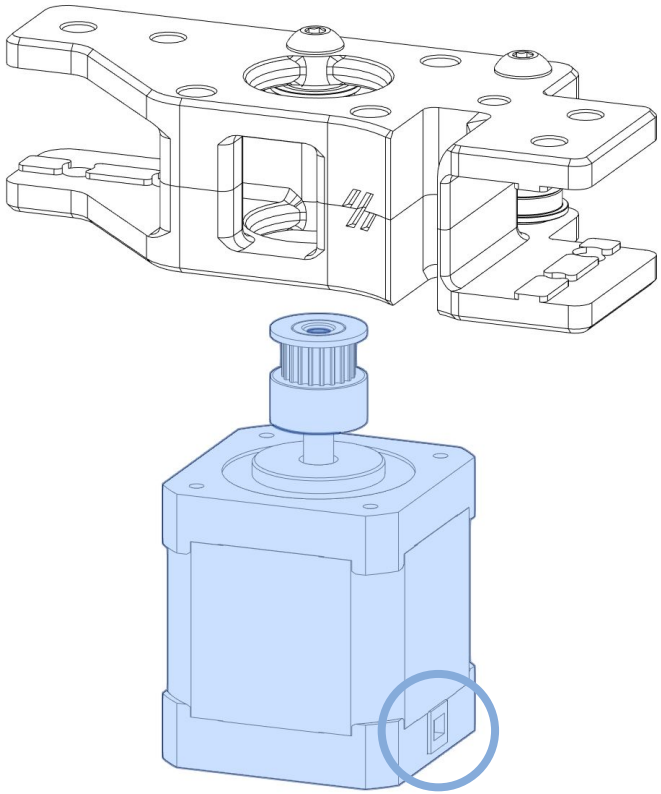
SET SCREW

aka the root of all issues

Insert both set screws and use thread locker on all set screws. Use a high quality hex driver to prevent the hex profile from stripping. Ball-end drivers are not recommended.

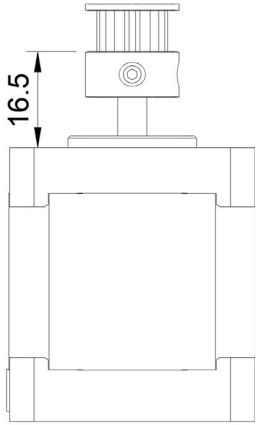
Loose set screws account for the majority of issues that our users report. Save yourself hours of troubleshooting and apply thread locker to all set screws during the build.

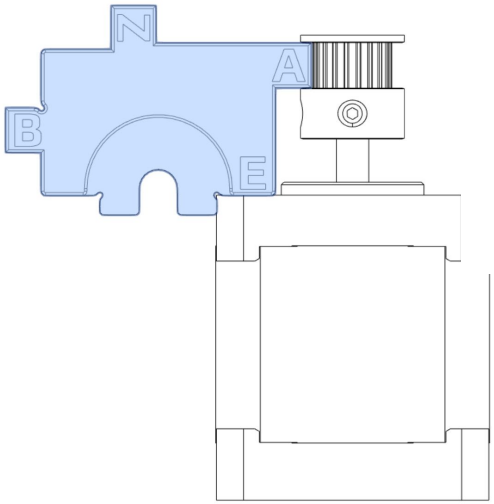
See the product's application notes for instructions - keep away from printed parts.



NEMA 17 Stepper

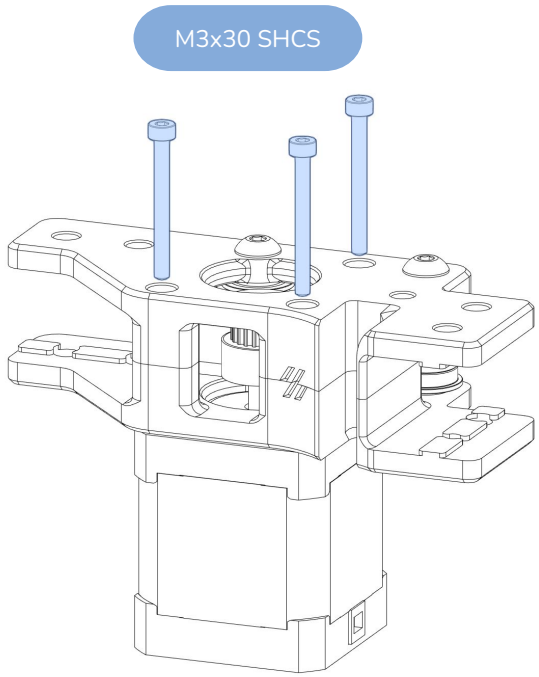
https://voron.link/fx10m8e

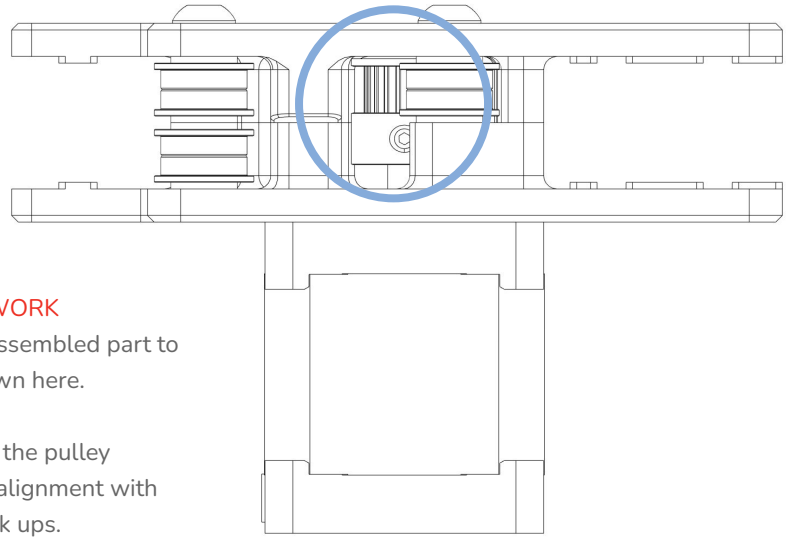




MOTOR ORIENTATION

Pay attention to the orientation of the cable exit. The wires from the motors will be pointing away from each other once fully assembled.

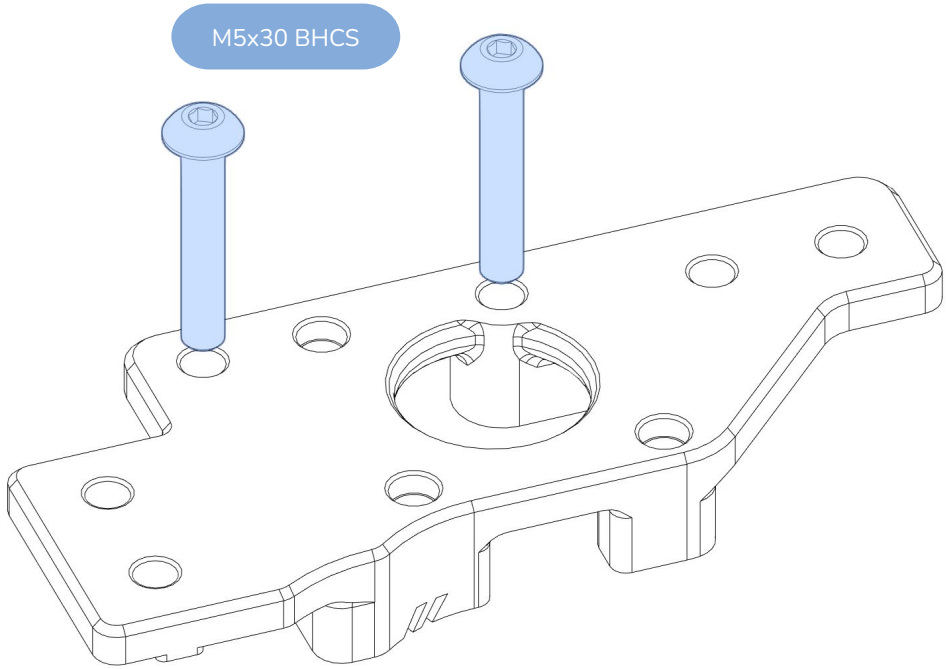


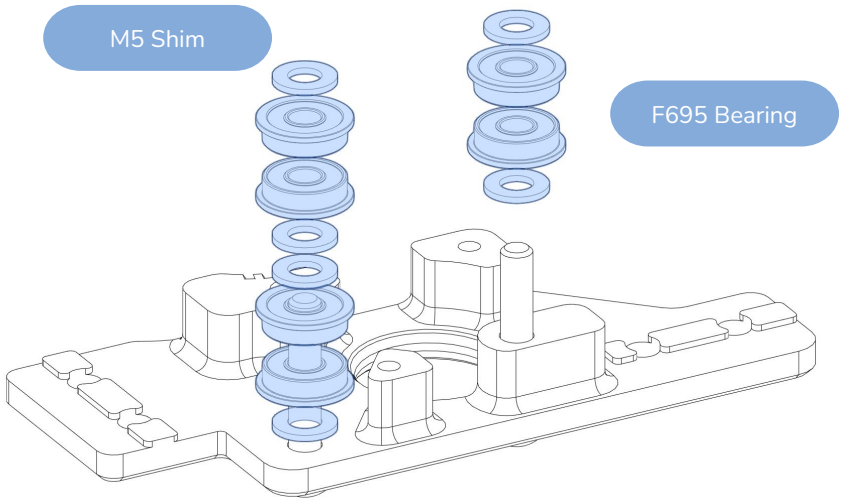


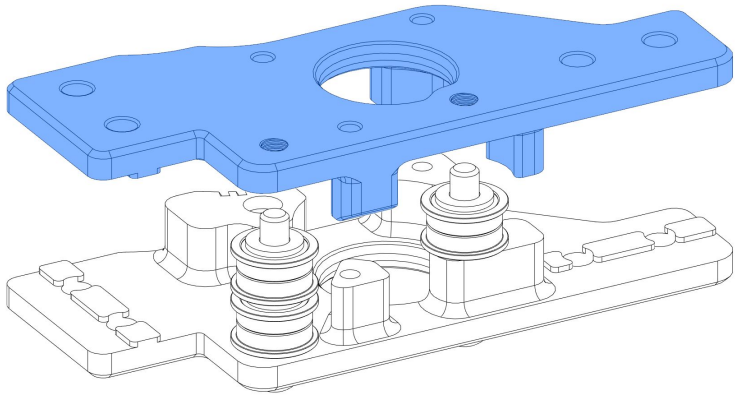
CHECK YOUR WORK

Compare your assembled part to the graphic shown here.

Pay attention to the pulley orientation and alignment with the bearing stack ups.

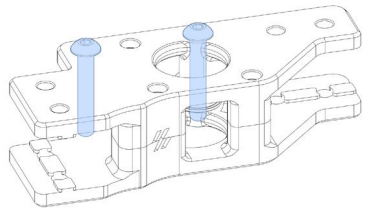

M5x30 BHCS





UPSIDE DOWN ASSEMBLY

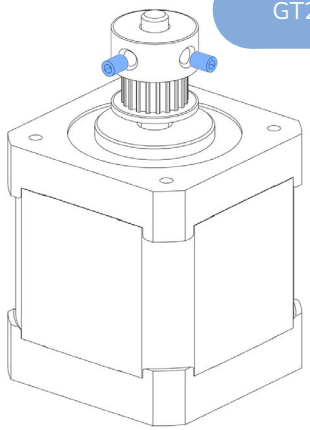
For ease of assembly we recommend to assemble the A and B drives upside down.



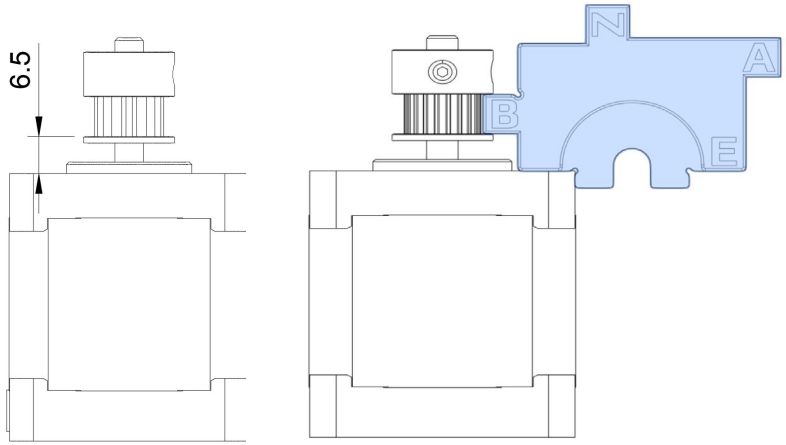
DON'T OVERTIGHTEN

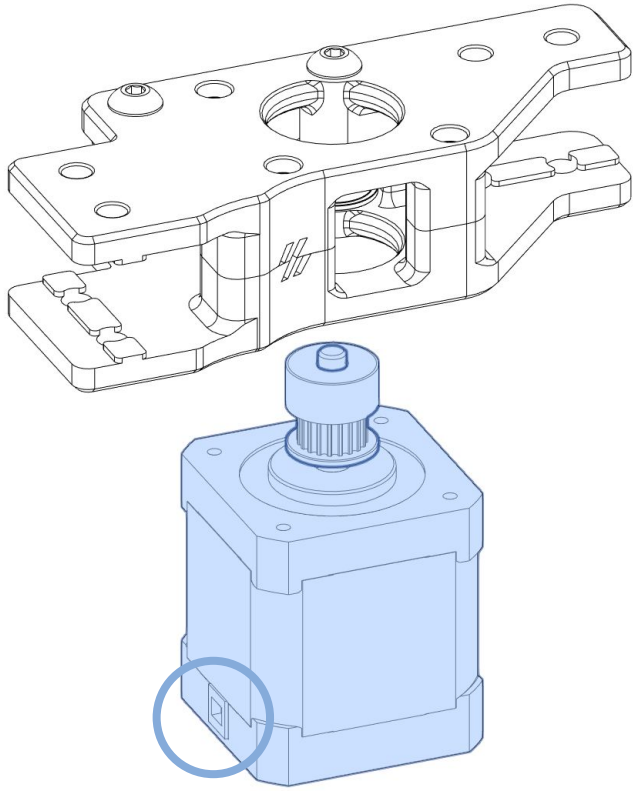
The M5 screws are threaded directly into plastic.

GT2 20 Tooth Pulley



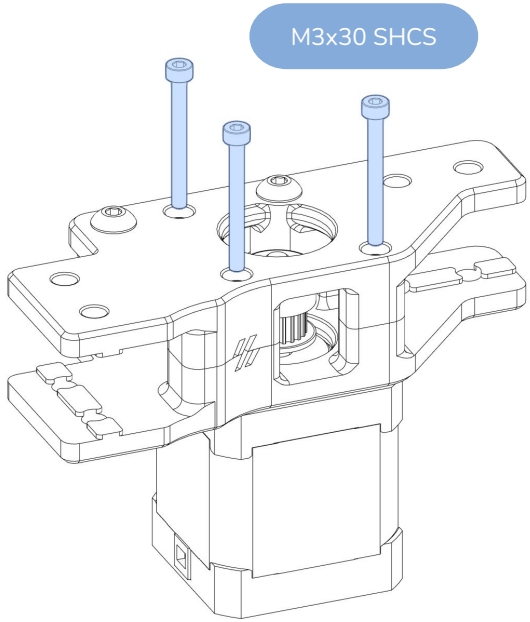
NEMA 17 Stepper

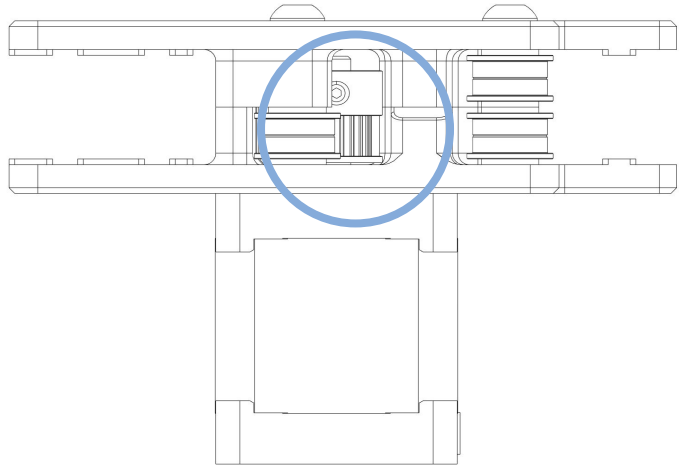




MOTOR ORIENTATION

Pay attention to the orientation of the cable exit.





CHECK YOUR WORK

Compare your assembled part to the graphic shown here.

Pay attention to the pulley orientation and alignment with the bearing stacks.

The first VORON printer was released to the public on March 10 2016.

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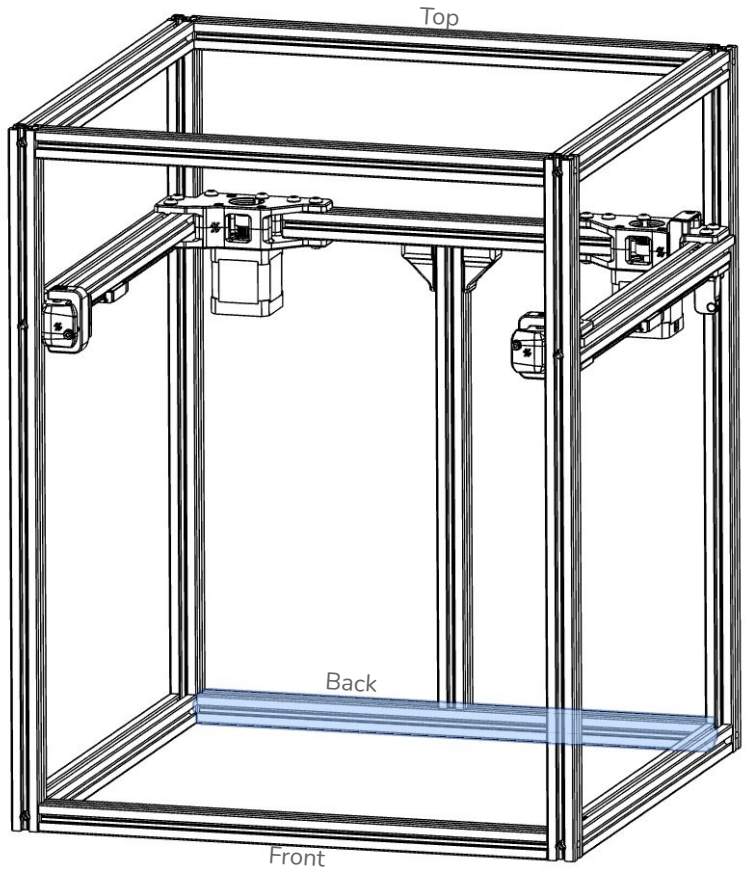
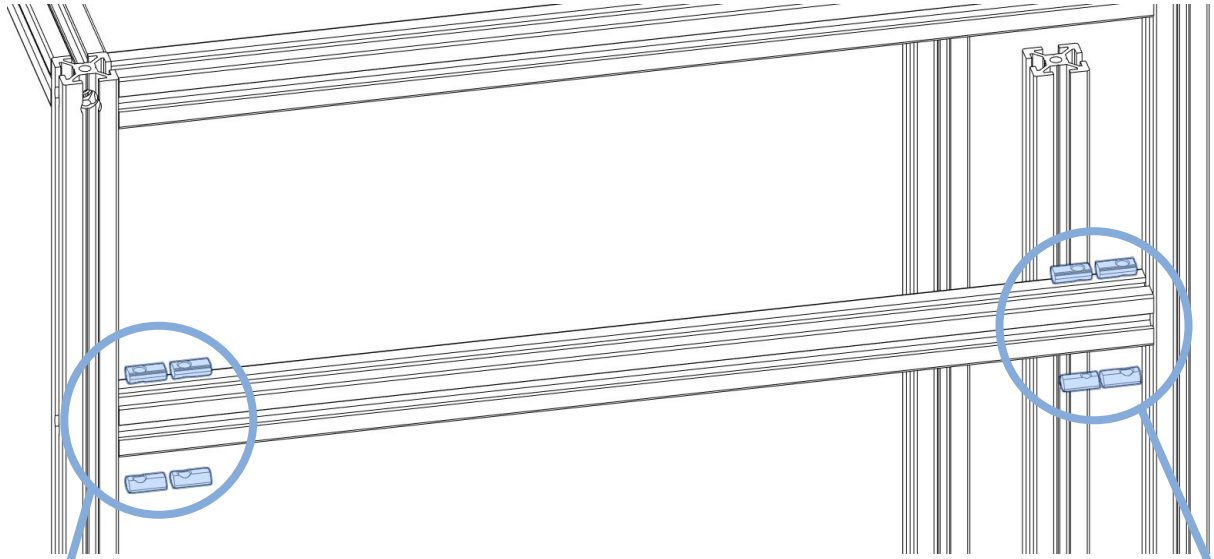
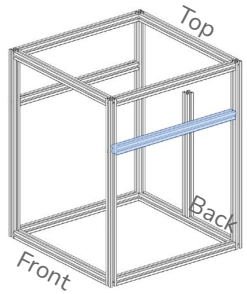
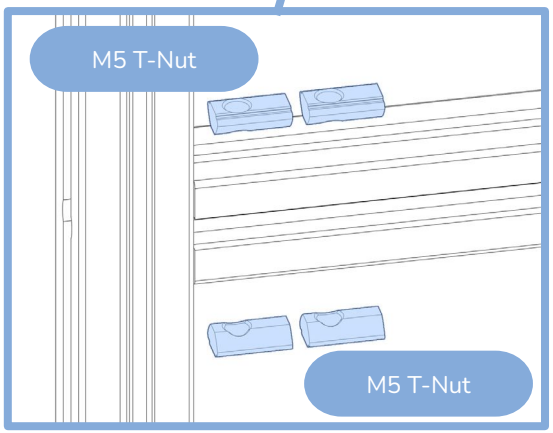

Top
Back
Front

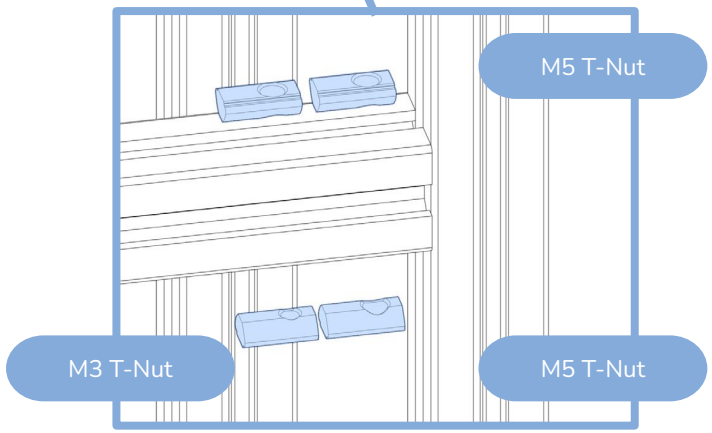


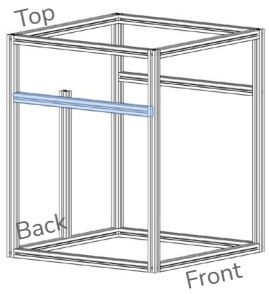


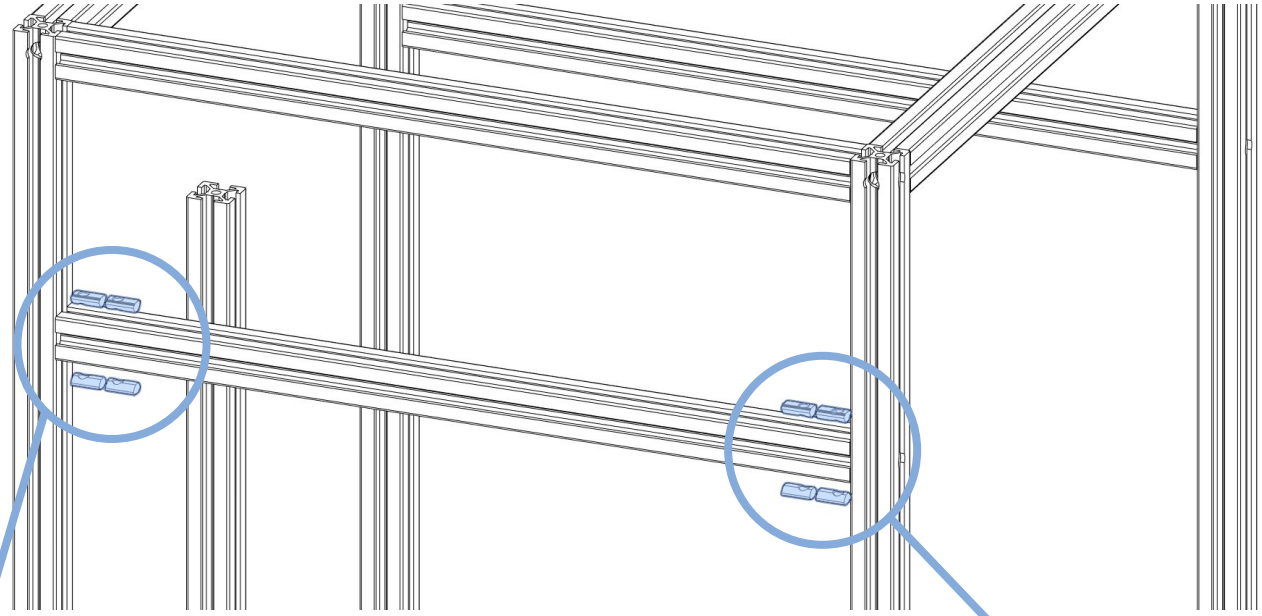


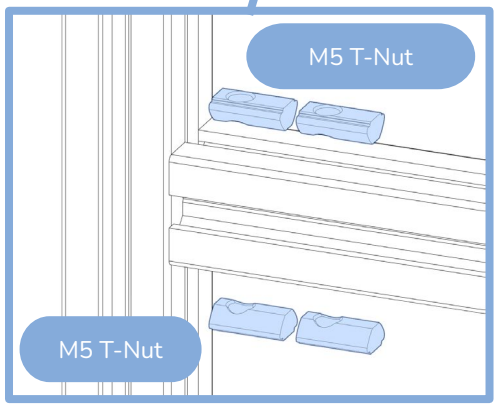
T-NUT ORIENTATION

Insert the t-nuts as shown in highlight.









T-NUT ORIENTATION

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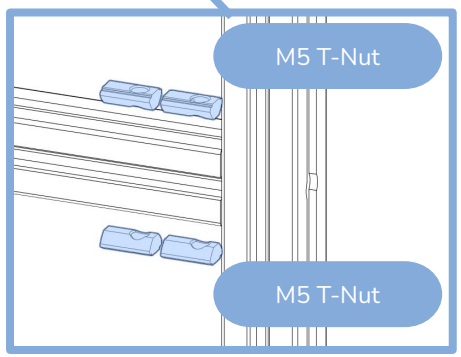
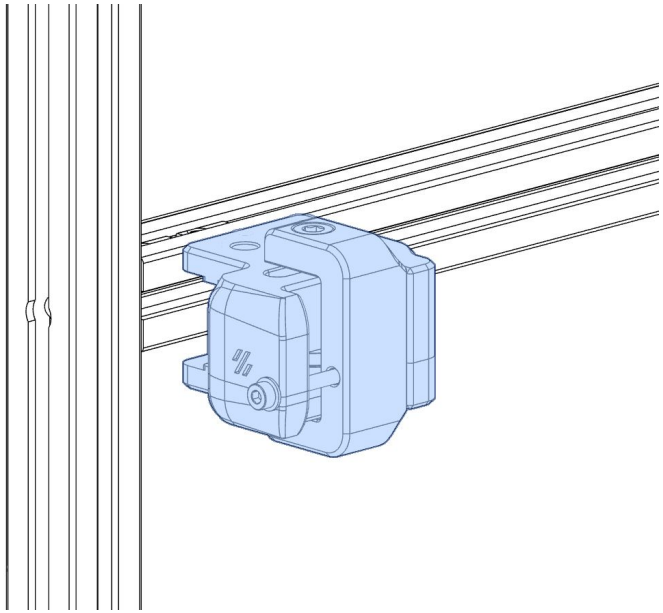
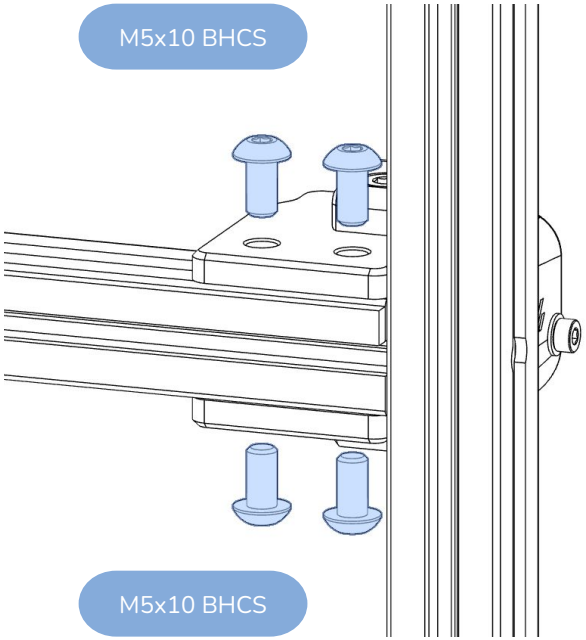
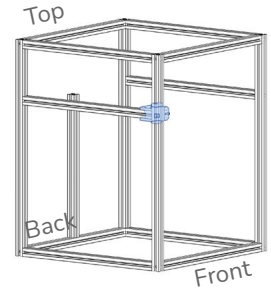


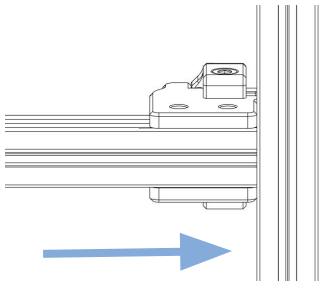


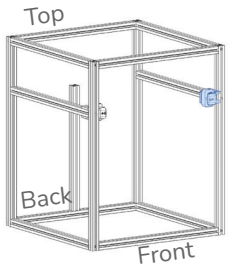


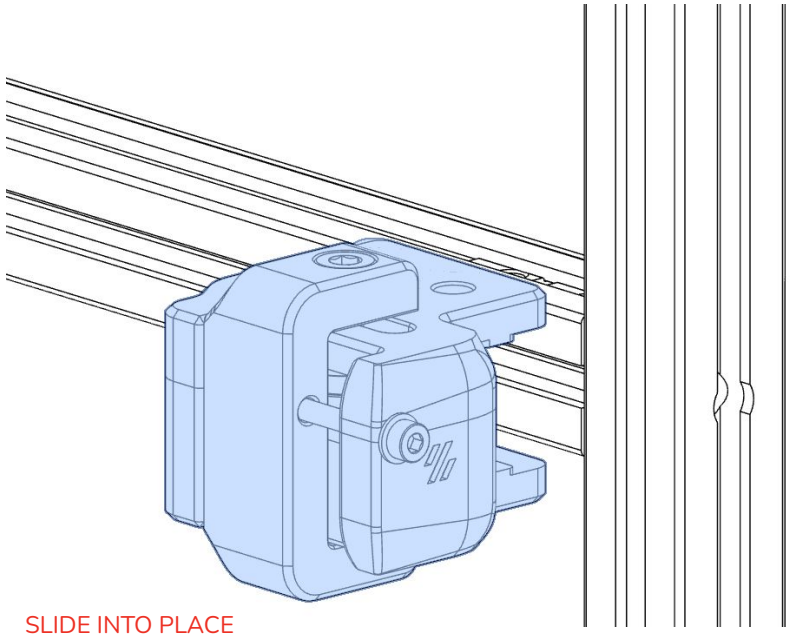


SLIDE INTO PLACE

Slightly loosen the M5 screw and slide into place.







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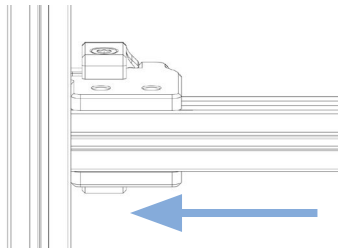
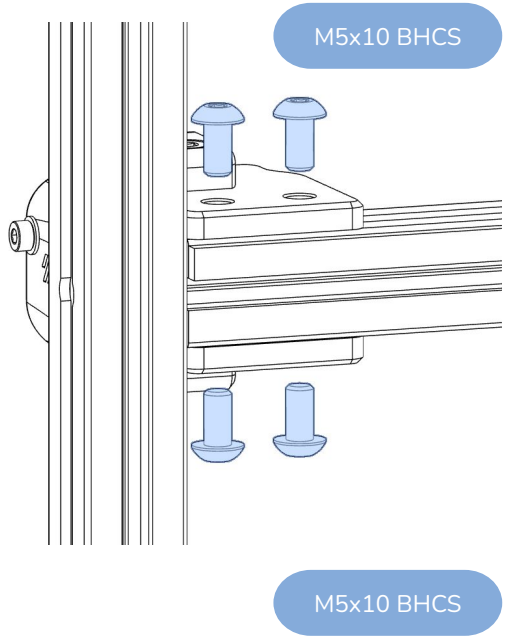
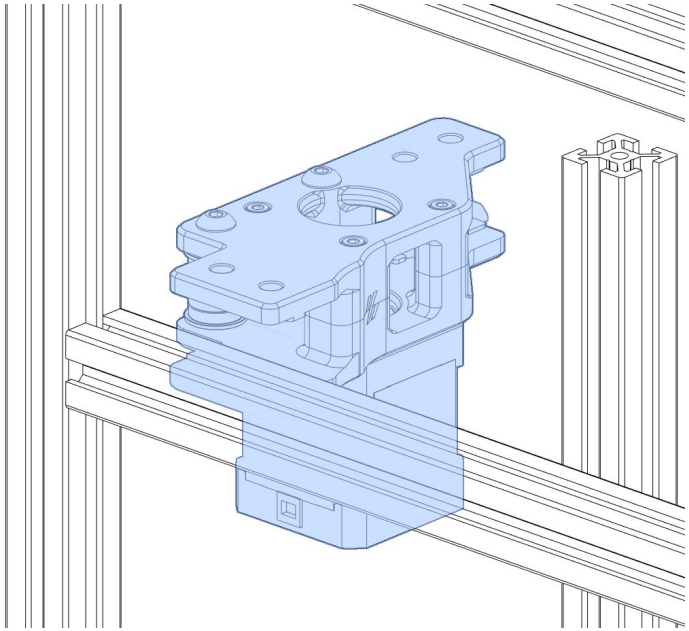
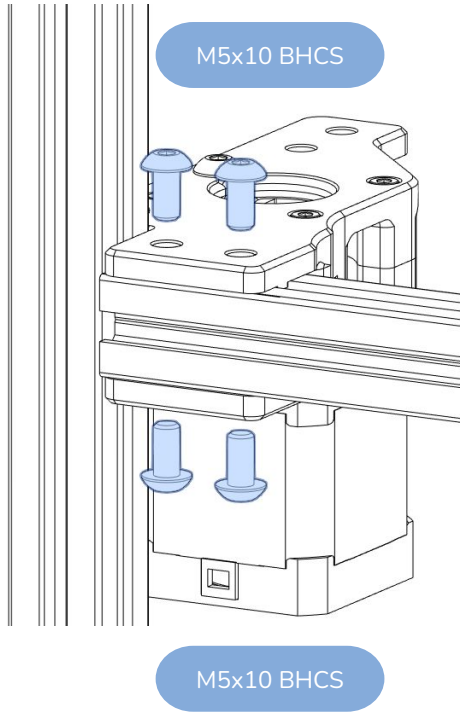
Slightly loosen the M5 screw and slide into place.



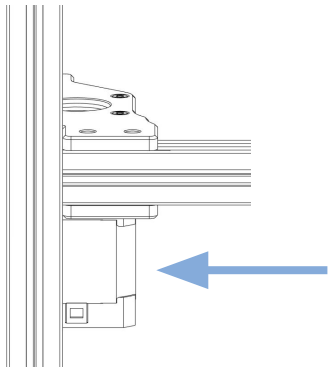


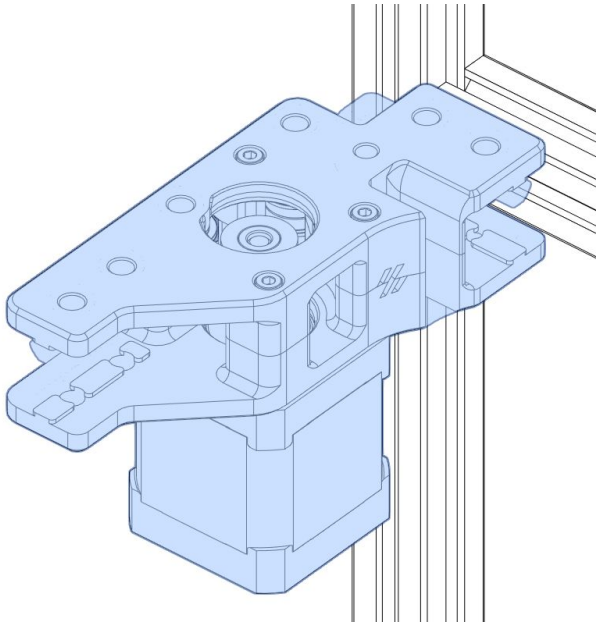


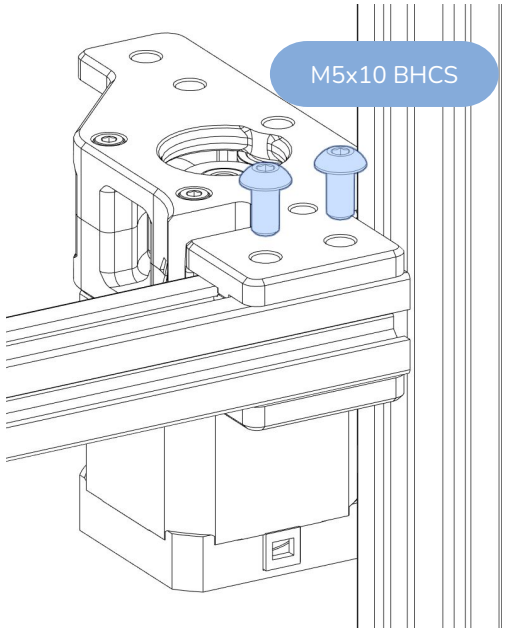


SLIDE INTO PLACE

Loosen the screws and slide into place.

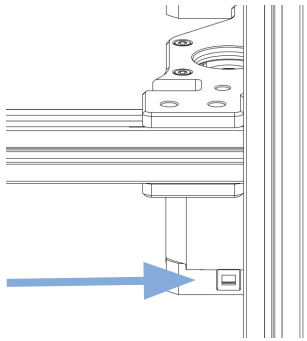




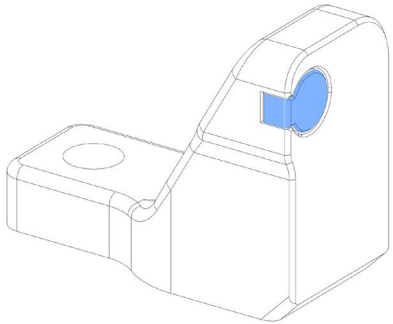


SLIDE INTO PLACE

Loosen the screws and slide into place.

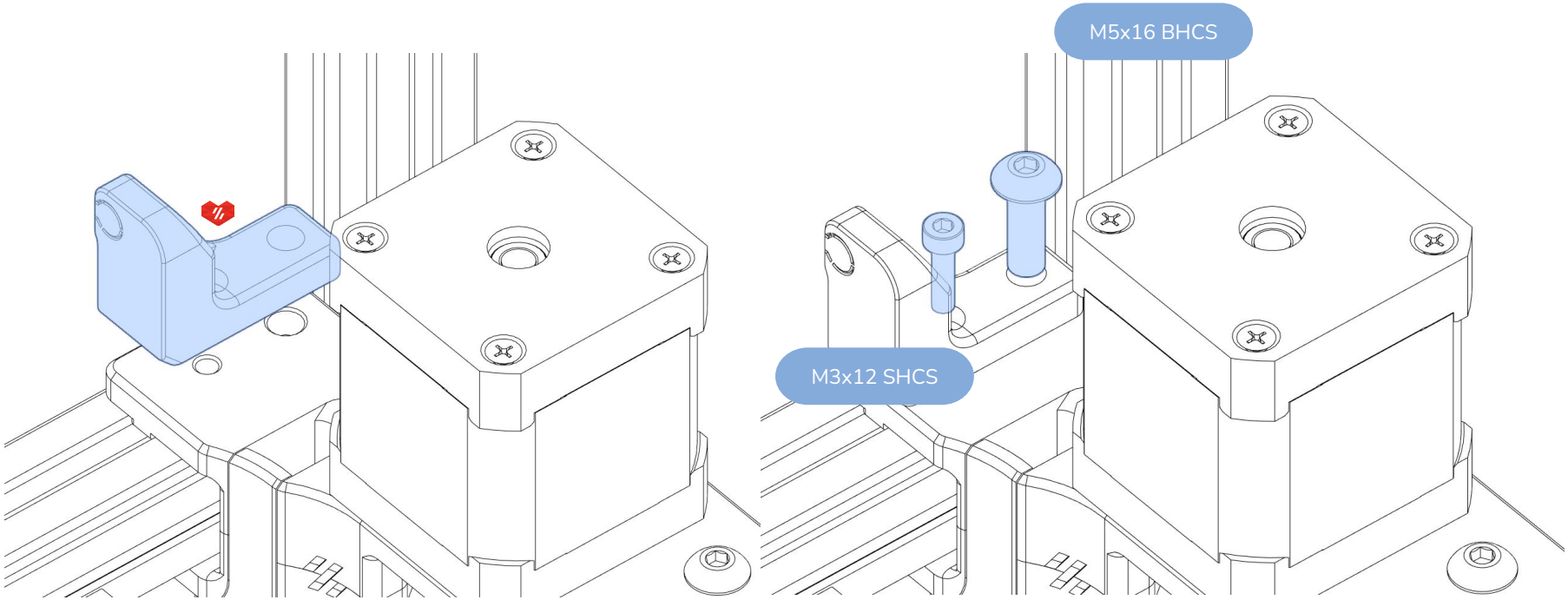


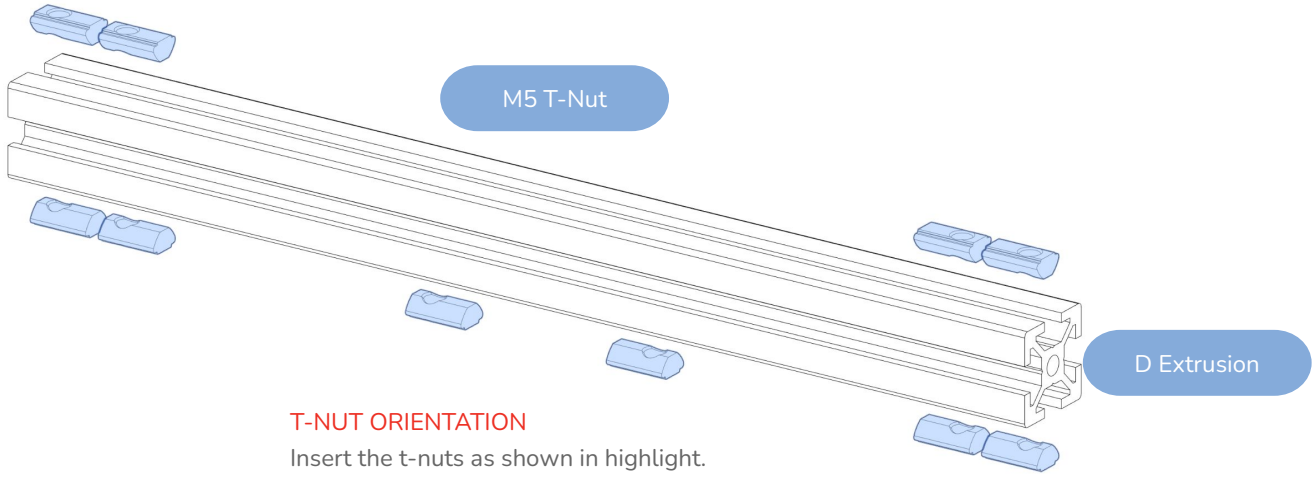
Skip this step, we use physical endstops and no hall effect ones.



OPTION: HALL EFFECT ENDSTOP

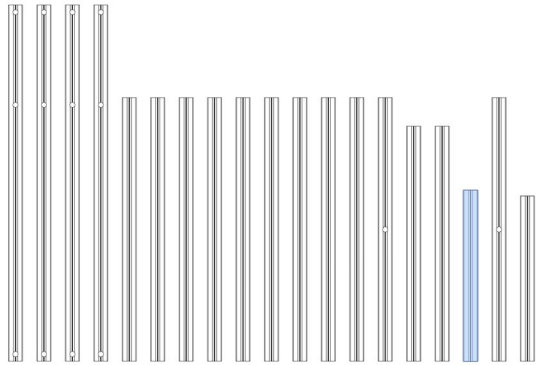
If you are using a Hall Effect Endstop board remove the highlighted part. Replace it with a magnet during initial calibration. See: https://voron.link/hxd3cv0.

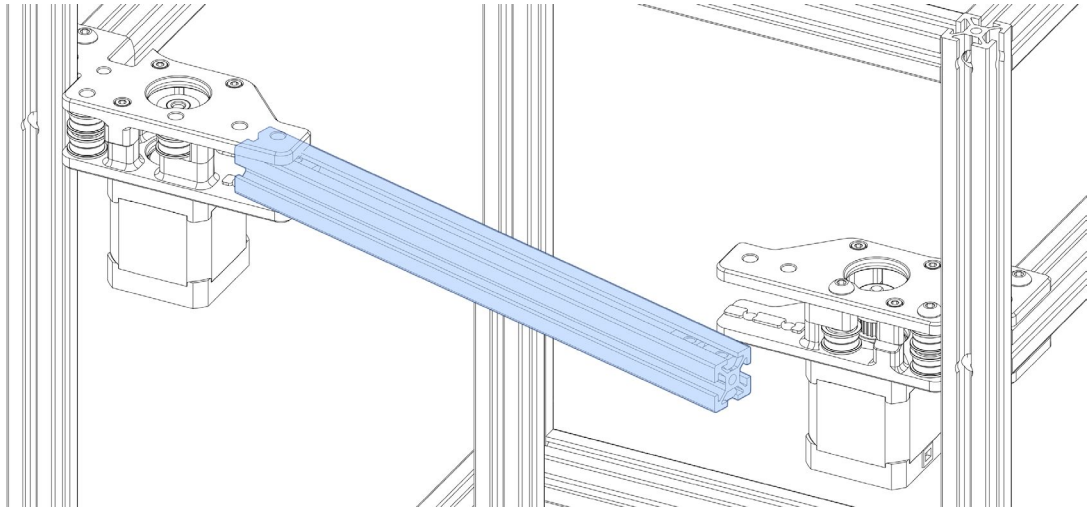




T-NUT ORIENTATION

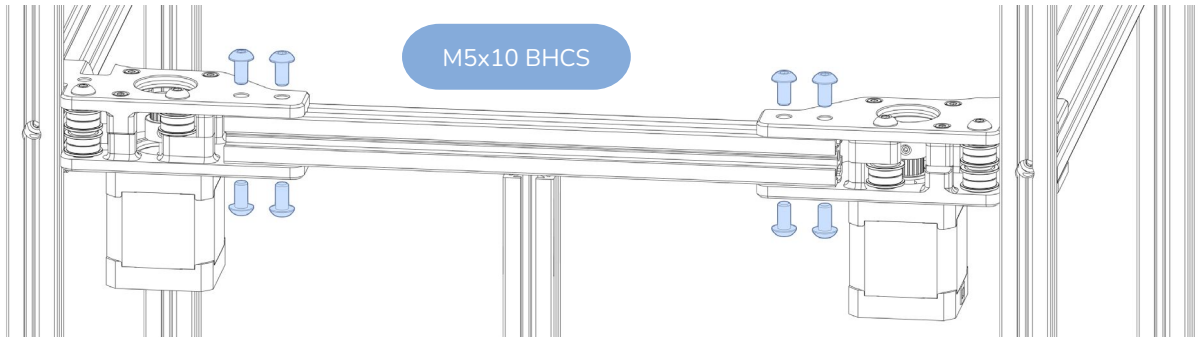
Insert the t-nuts as shown in highlight.

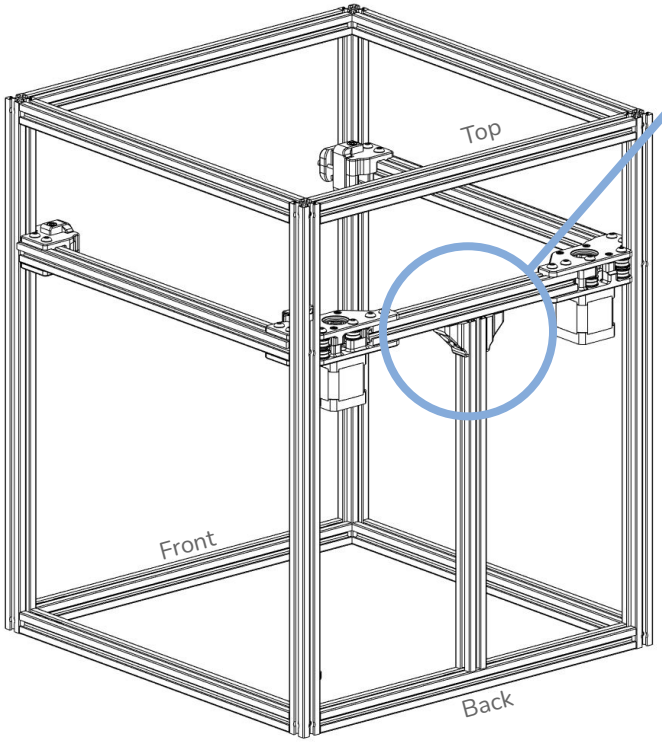


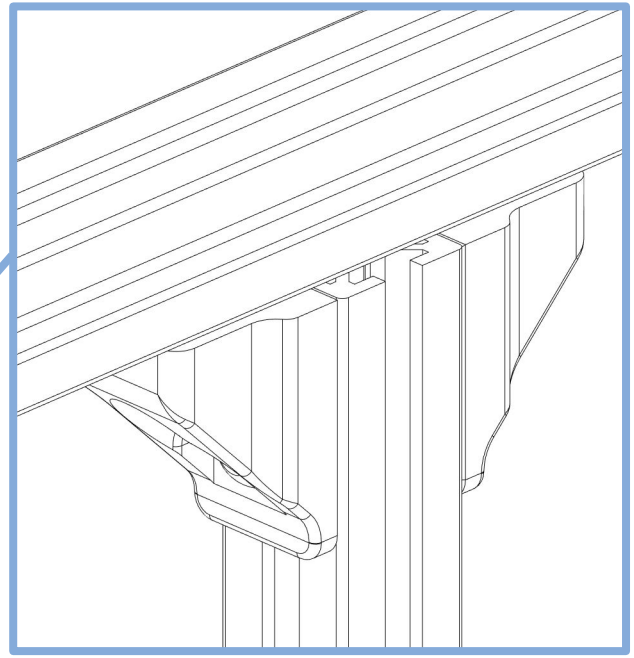


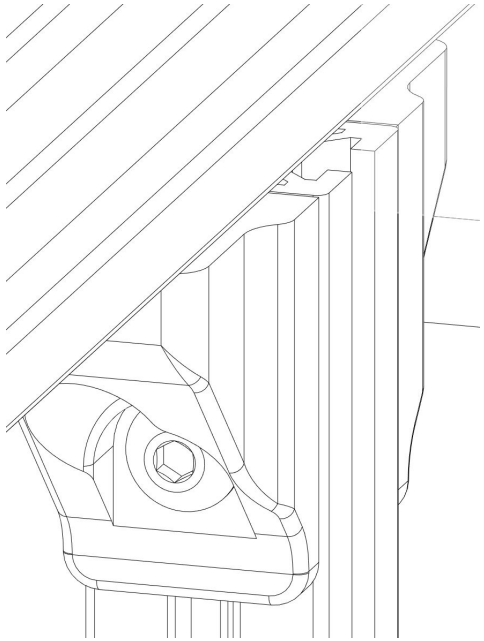
SLIDE INTO PLACE

The rear crossbar can be slid into place. You may need to loosen the M5 screws.



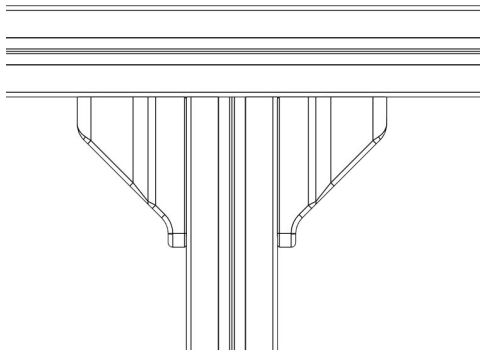


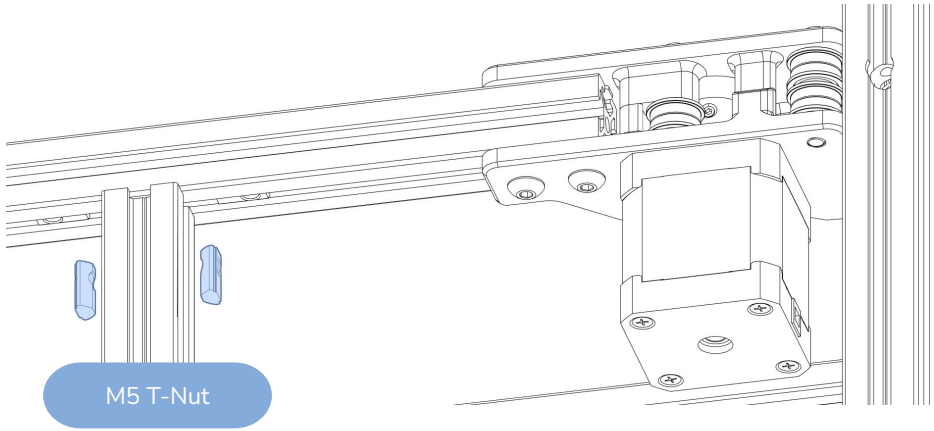


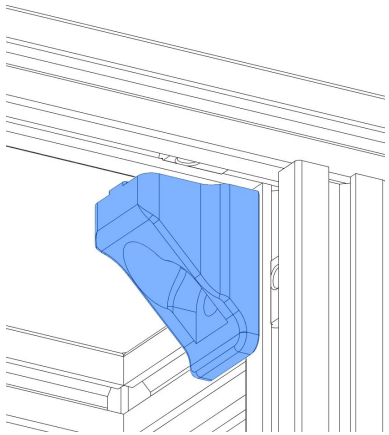


OFFSET EXTRUSION

The extrusion between the A and B drives is offset inwards. The upright extrusion extends past it towards the back of the printer.

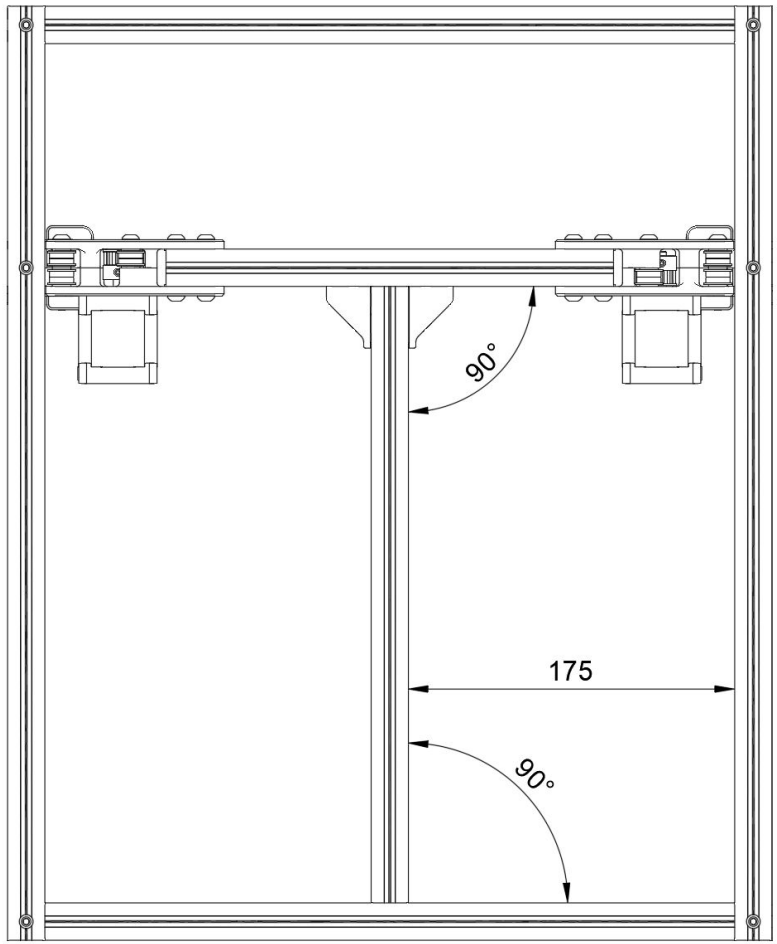



M5 T-Nut



M5x10 BHCS

Top



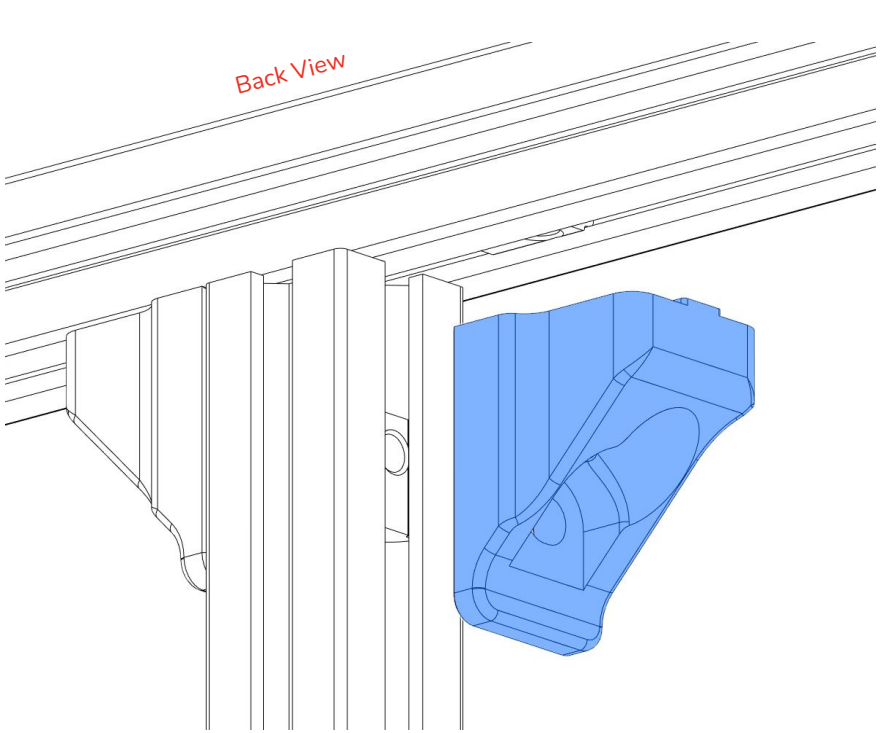
ADJUST POSITIONING

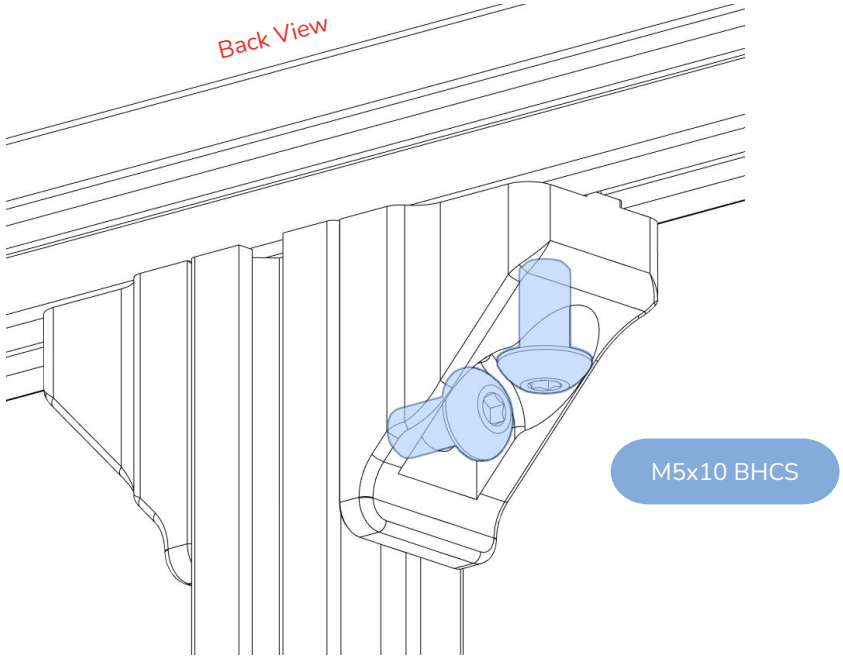
Adjust the position of the rear extrusion to match the dimensions shown on the right.

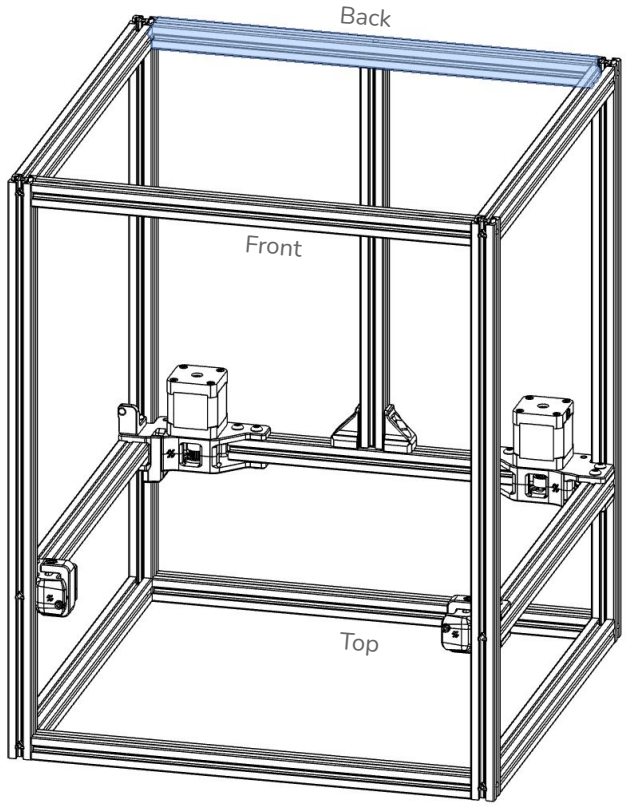
Ensure that the rear is parallel with the frame uprights.

OTHER FRAME SIZES

The distance is shown for a 250 spec frame. Add 25mm for a 300 or 50 for a 350 sized frame.


Back View


Back View
M5x10 BHCS



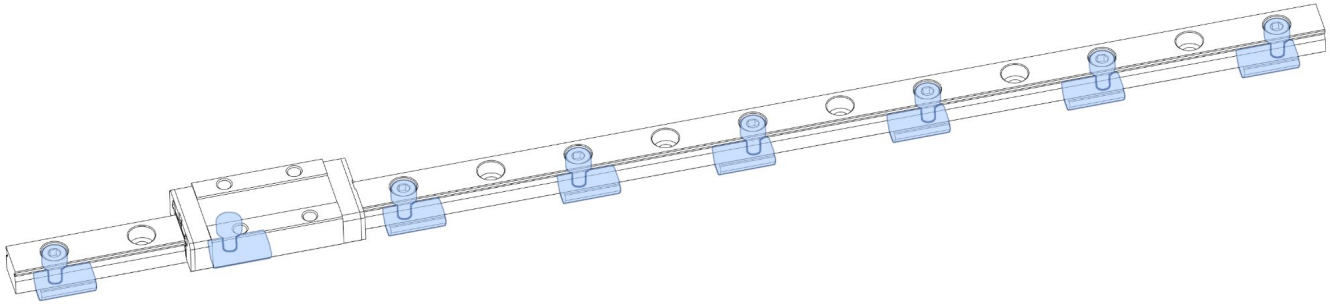
UPSIDE DOWN ASSEMBLY

For ease of assembly we recommend to flip the printer on its head for the next steps.

If you are building a 300mm machine you will need to start you screws from the second screw hole in the rails leaving the end holes open to allow space for T-nuts at later steps. Small reminder: The carriages are designed to slide along the rail easily. This unfortunately also includes sliding off the rails. Dropping the carriage likely irreparably damages it. ALWAYS put the little black rubber Carriage stoppers back on the rail (into the holes without screws) until your printer ist up and runnning!

HANDLE WITH CARE

The carriage can slide off the rail if not handled properly. Dropping the carriage will likely damage it.
Any marks, dents or nicks might cause the linear rail to misbehave in operation.



LINEAR RAILS - PREPARATION AND MOUNTING

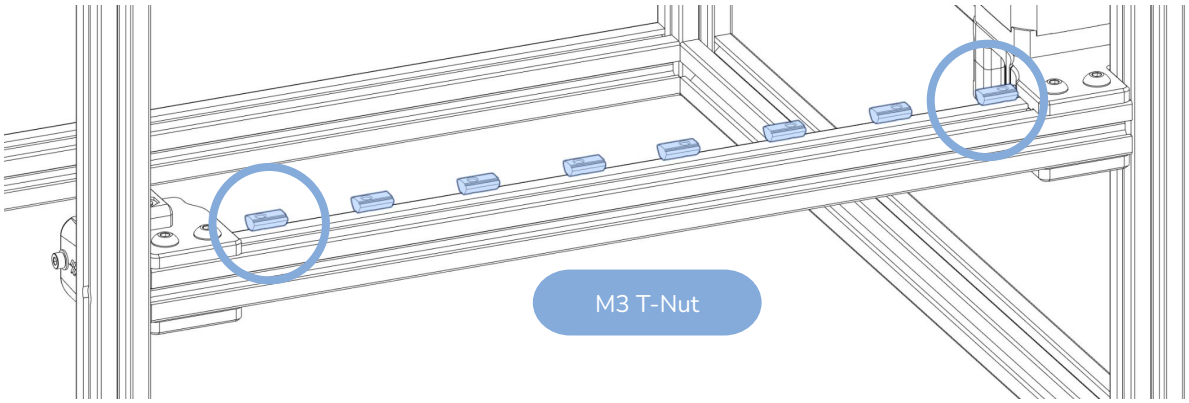
Most linear rails arrive with shipping oil. To ensure a smooth gliding motion and long service life, this oil needs to be removed and its rail carriage greased. See the VORON sourcing guide for a recommended list of lubricants. We attached
a link to a video guide to get you started.

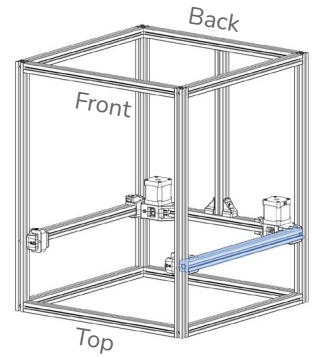
We opted to skip every other mounting hole in the linear rail when designing the mounting pattern for this printer. This cuts down on mounting hardware and still meets the requirements for our use case.

When tightening the screws tighten them from the center outward to ensure that the rail sits flush on the extrusion.

https://voron.link/agu0nes

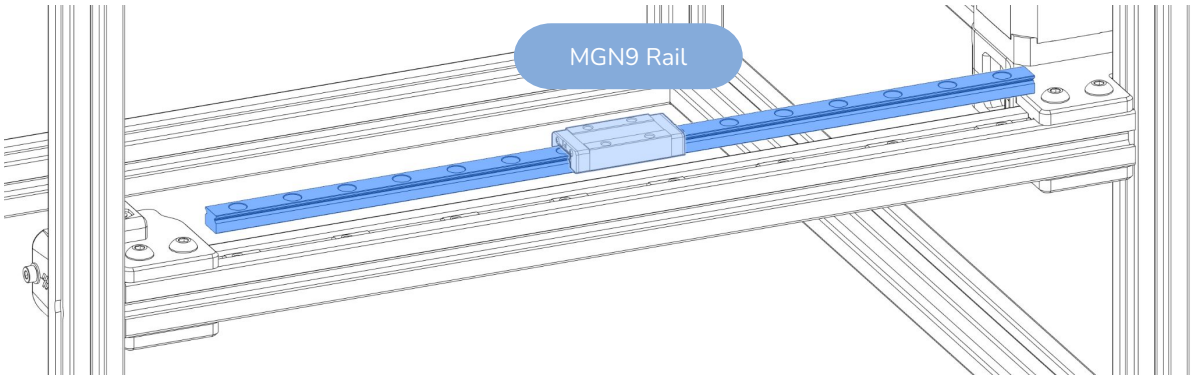
If you are building a 300mm machine you will need to start you screws from the second screw hole in the rails leaving the end holes open to allow space for T-nuts at later steps. Small reminder: The carriages are designed to slide along the rail easily. This unfortunately also includes sliding off the rails. Dropping the carriage likely irreparably damages it. ALWAYS put the little black rubber Carriage stoppers back on the rail (into the holes without screws) until your printer ist up and runnning!





T-NUT ORIENTATION

Insert the t-nuts as shown in highlight.
Every other hole in the rail will be left empty.

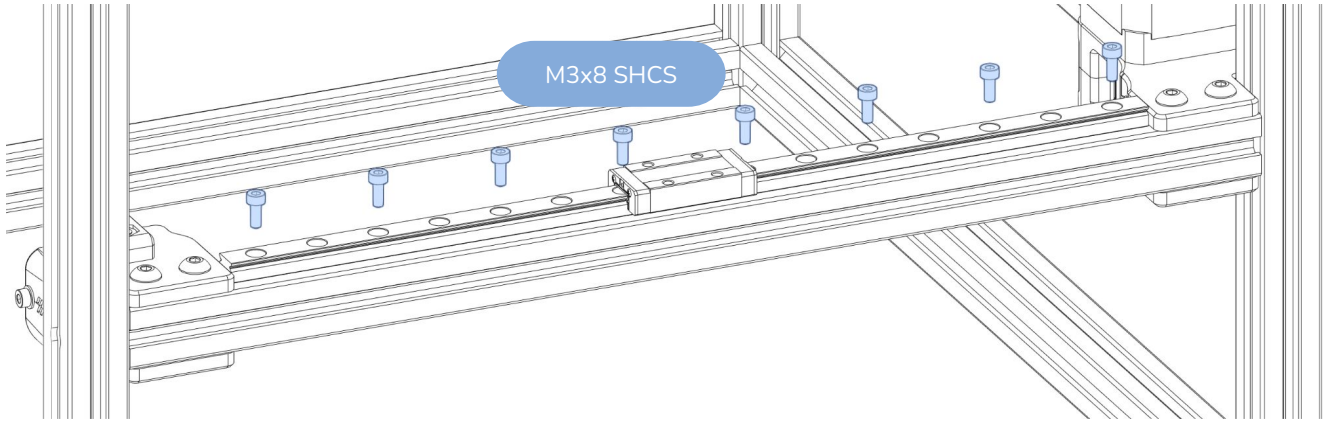


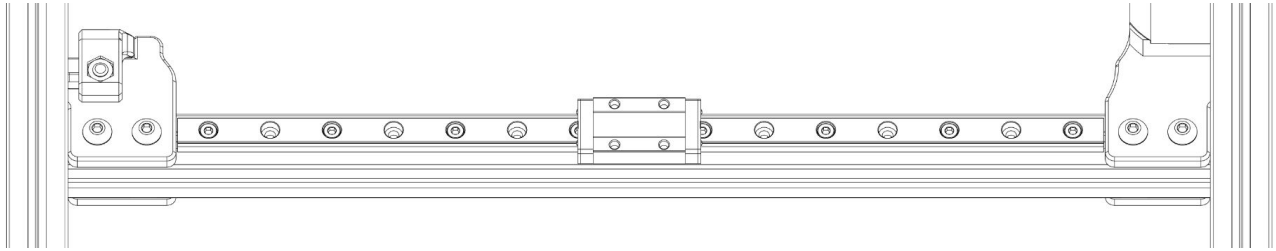
MIND THE CARRIAGE

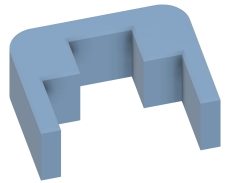
The carriages are designed to slide along the rail easily.
This unfortunately also includes sliding off the rails.

Dropping the carriage will likely damage it.

If you are building a 300mm machine you will need to start you screws from the second screw hole in the rails leaving the end holes open to allow space for T-nuts at later steps. Small reminder: The carriages are designed to slide along the rail easily. This unfortunately also includes sliding off the rails. Dropping the carriage likely irreparably damages it. ALWAYS put the little black rubber Carriage stoppers back on the rail (into the holes without screws) until your printer ist up and runnning!



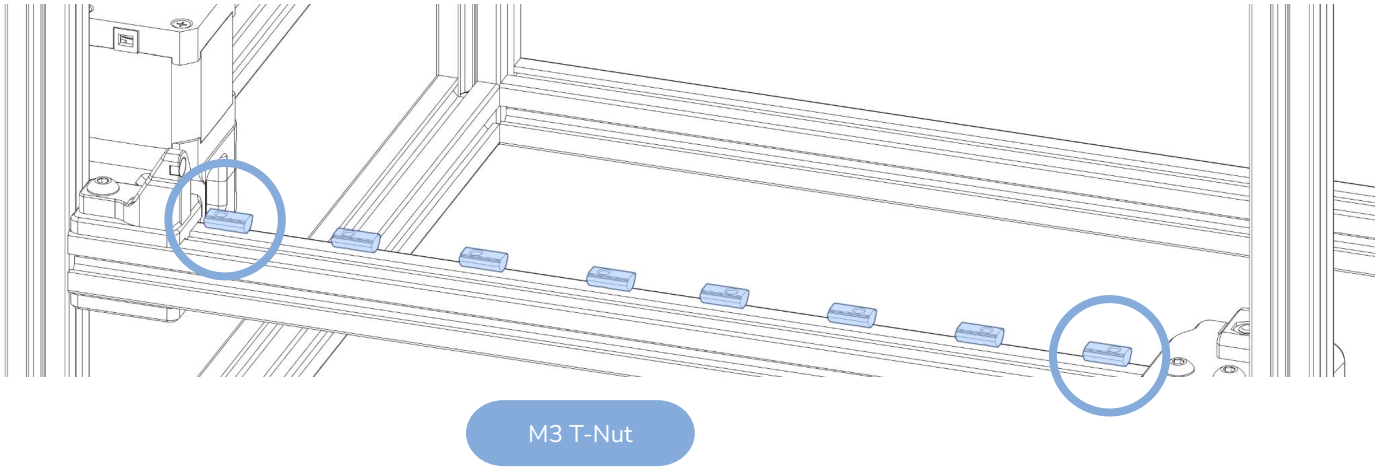


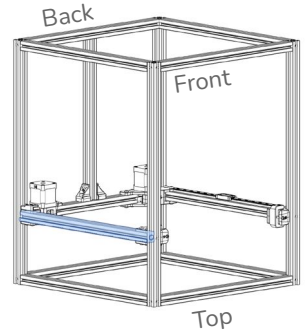


CENTERED RAIL INSTALLATION GUIDE

Use the MGN9 guides to position the rail in the center of the extrusion prior to tightening the screws.

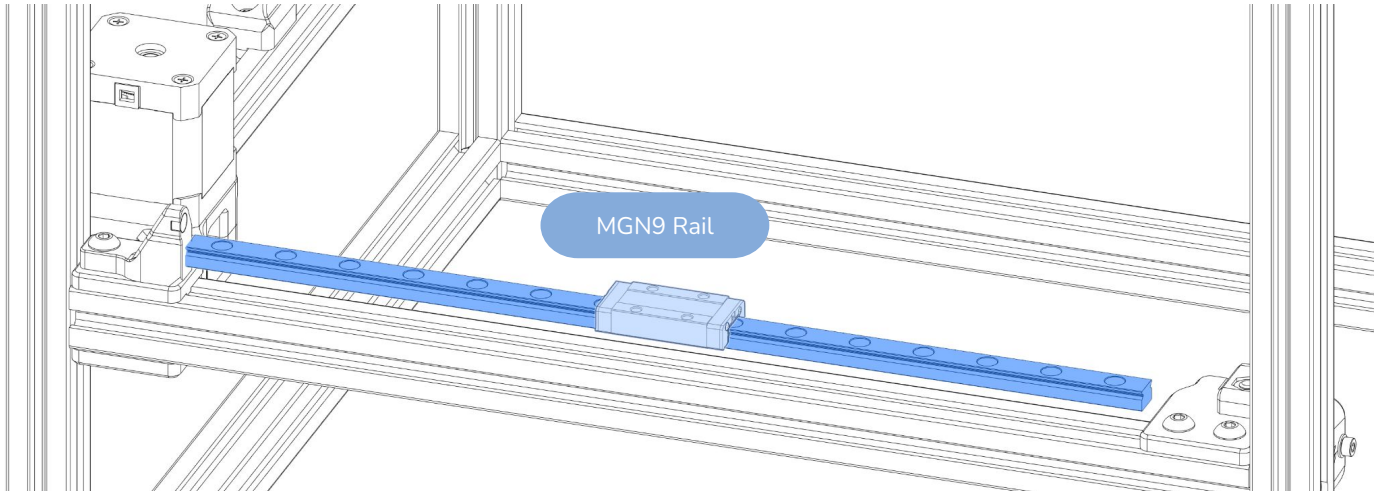
If you are building a 300mm machine you will need to start you screws from the second screw hole in the rails leaving the end holes open to allow space for T-nuts at later steps. Small reminder: The carriages are designed to slide along the rail easily. This unfortunately also includes sliding off the rails. Dropping the carriage likely irreparably damages it. ALWAYS put the little black rubber Carriage stoppers back on the rail (into the holes without screws) until your printer ist up and runnning!



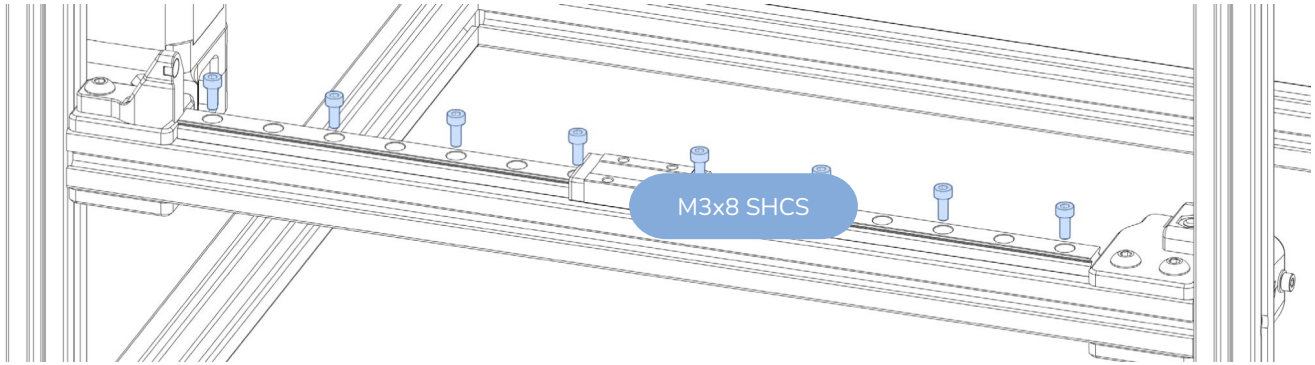


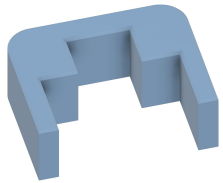
T-NUT ORIENTATION

Insert the t-nuts as shown in highlight.



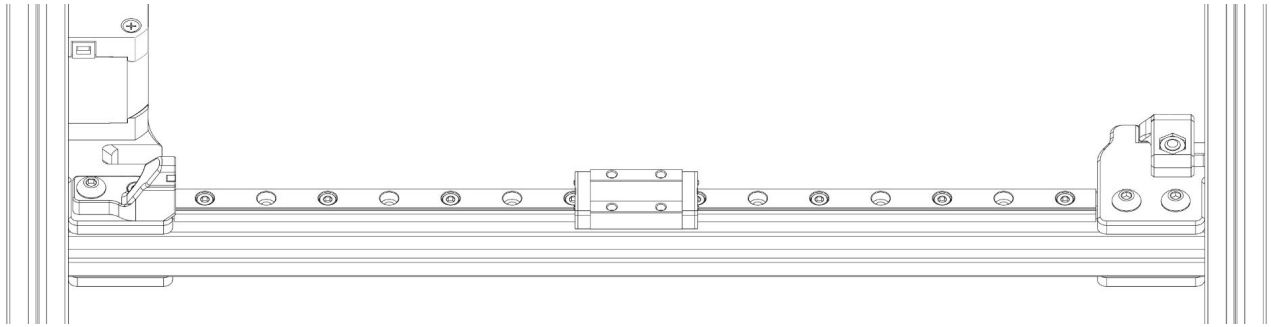
If you are building a 300mm machine you will need to start you screws from the second screw hole in the rails leaving the end holes open to allow space for T-nuts at later steps. Small reminder: The carriages are designed to slide along the rail easily. This unfortunately also includes sliding off the rails. Dropping the carriage likely irreparably damages it. ALWAYS put the little black rubber Carriage stoppers back on the rail (into the holes without screws) until your printer ist up and runnning!





CENTERED RAIL INSTALLATION GUIDE

Use the MGN9 guides to position the rail in the center of the extrusion prior to tightening the screws.



The first design released under the name VORON was the “VORON Geared Extruder”, on January 28, 2015.

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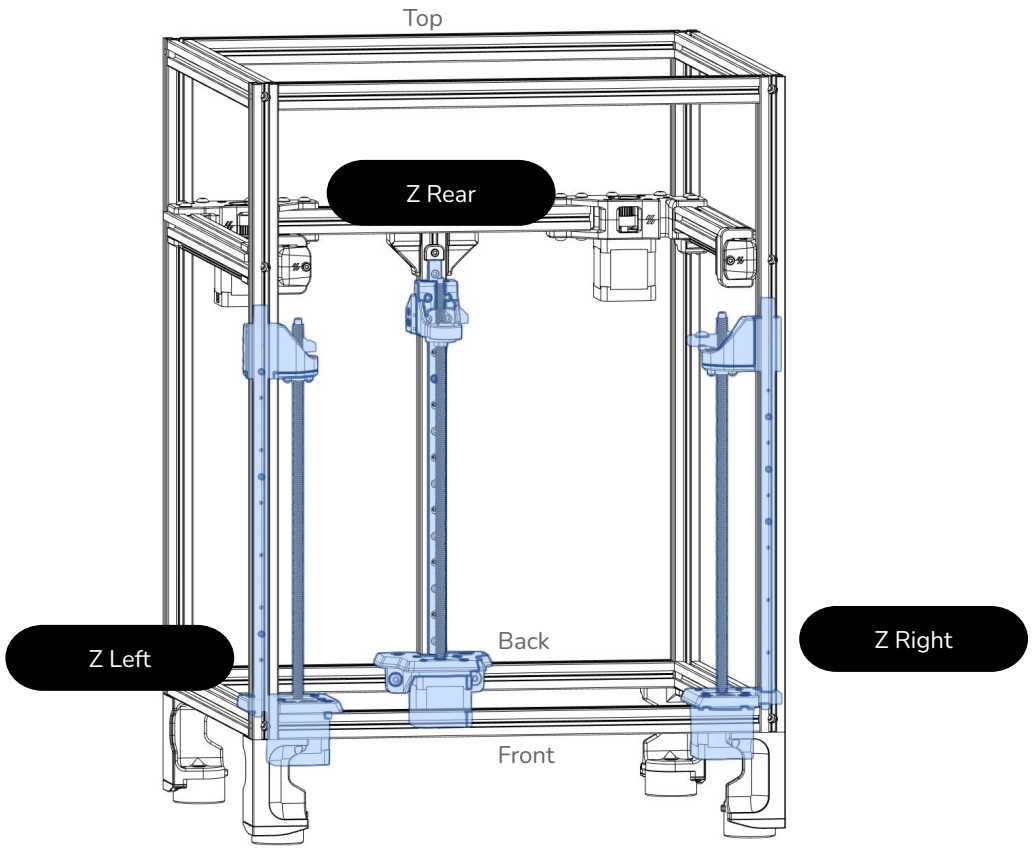
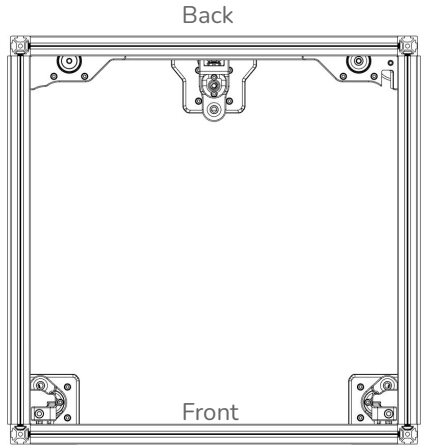
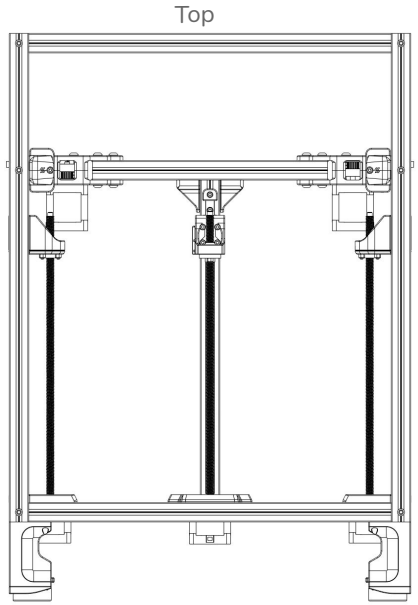
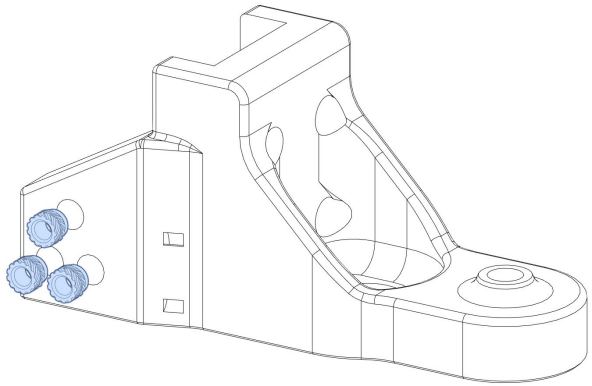

Top
Z Rear
Z Left
Back
Z Right
Front


Back
Front

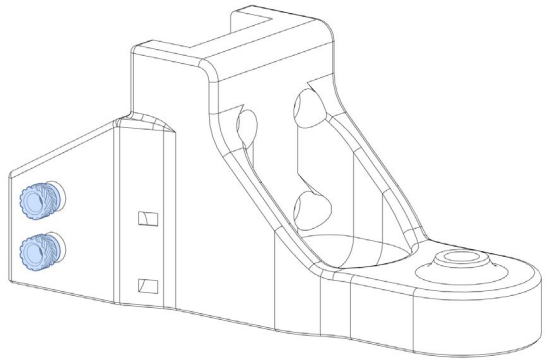

Top

Heat Set Insert



GENERIC CABLE CHAIN

The 3 hole pattern is usually found on generic cable chains.

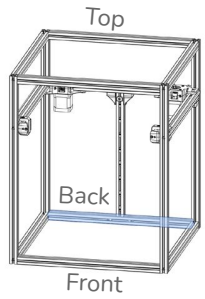


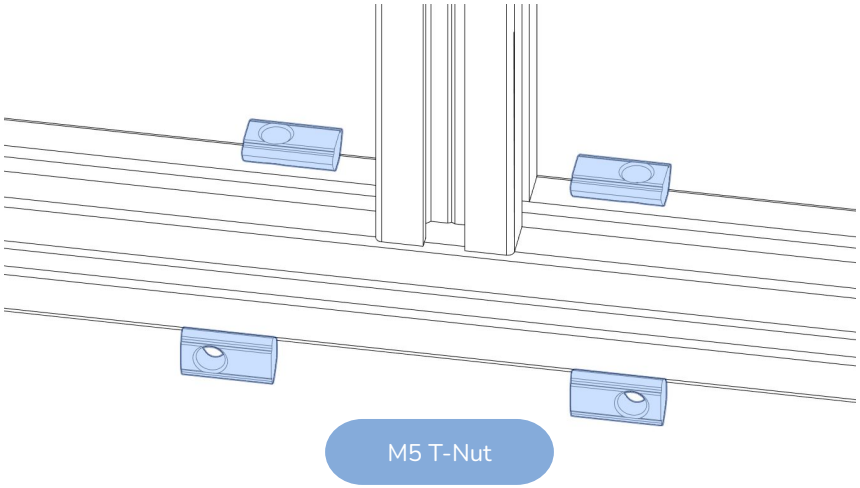
IGUS CABLE CHAIN

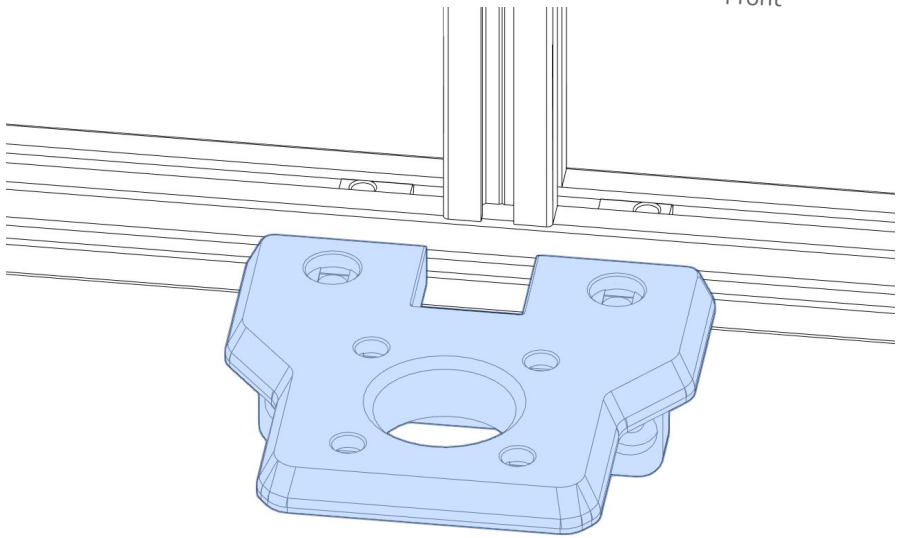
IGUS chain have 2 mounting holes.

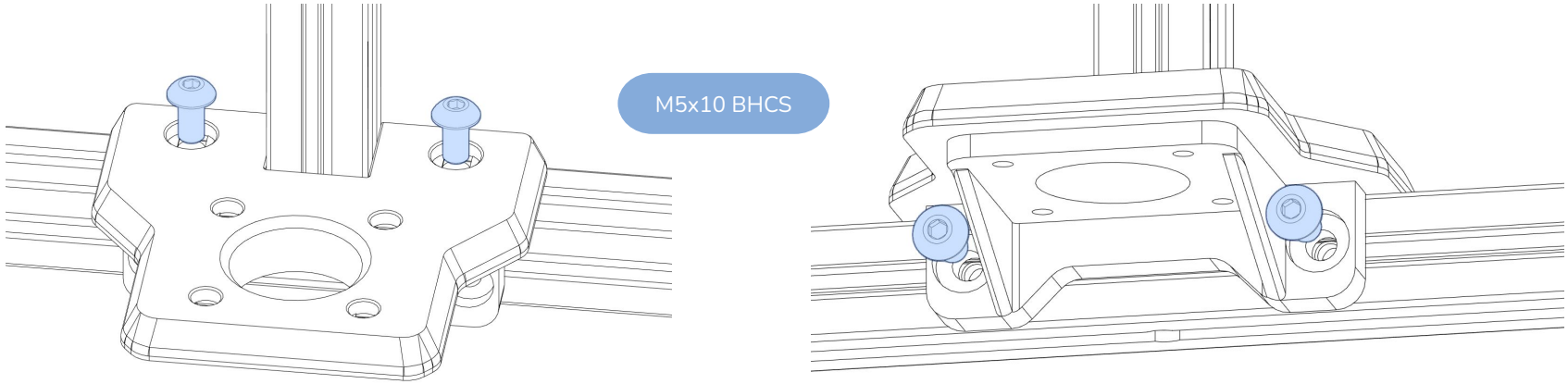
CABLE CHAIN MOUNTING PATTERN

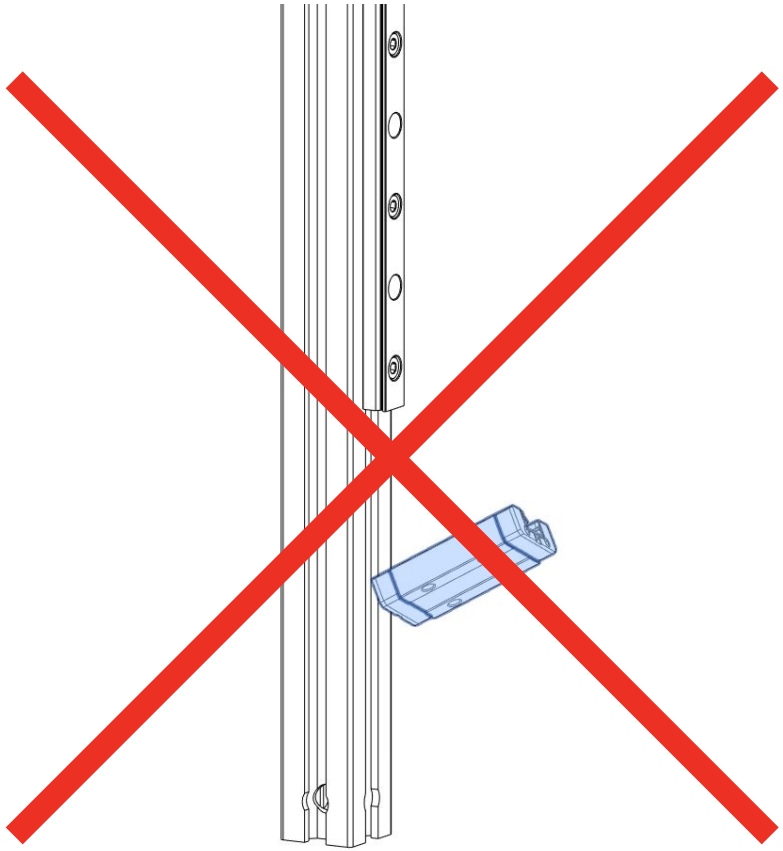
There are multiple mounting patterns for cable chains on the market. Pick the part that matches with the hole pattern on your mounting chain. The remainder of the manual will only show the “generic” pattern.


Top
Back
Front


M5 T-Nut




M5x10 BHCS



For illustration purposes only. Do not attempt to replicate.

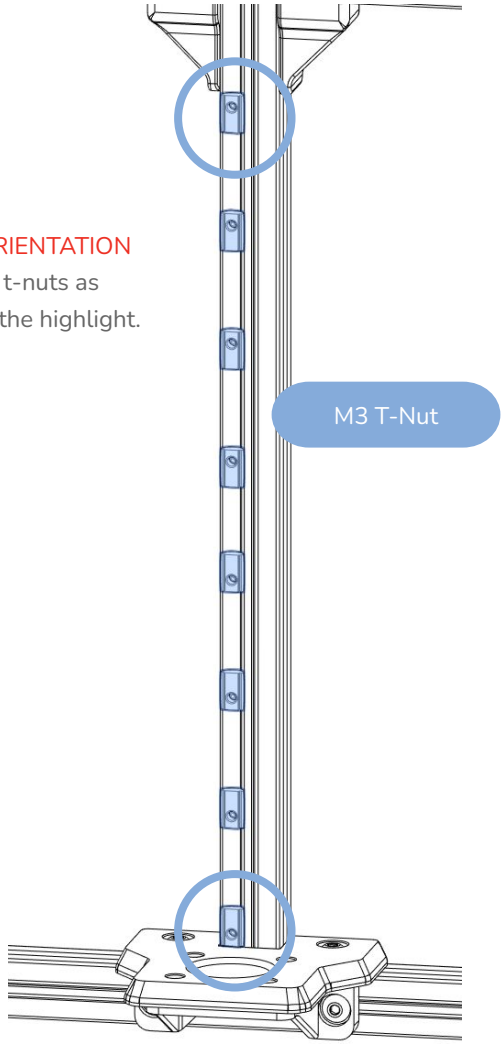
RAIL SAFETY

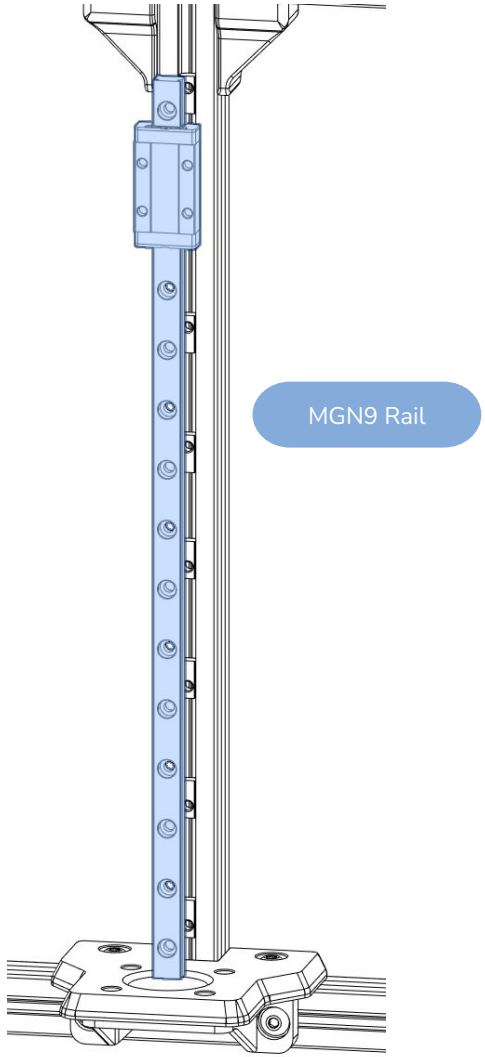
Mind the rail carriage during installation.

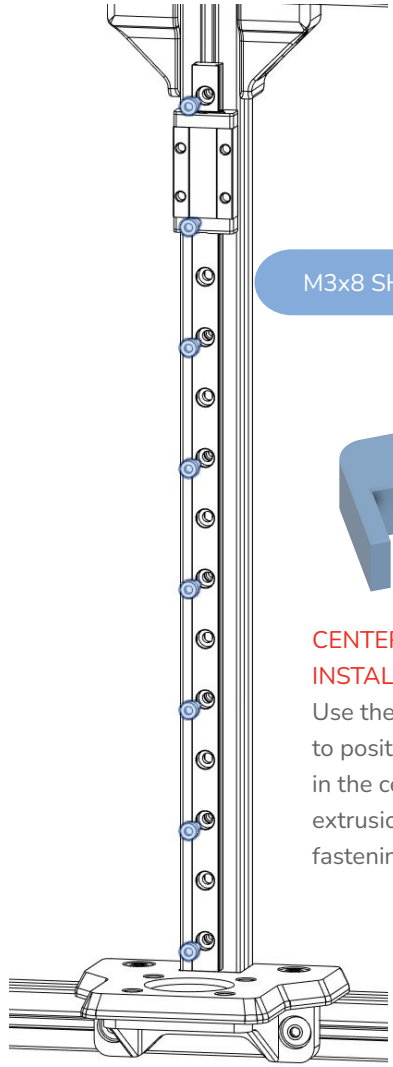
If your rails were delivered with plastic stoppers you can also temporarily reinstall them to prevent carriages from falling off their rails and spilling their bearing balls.

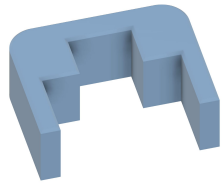
T-NUT ORIENTATION

Insert the t-nuts as shown in the highlight.



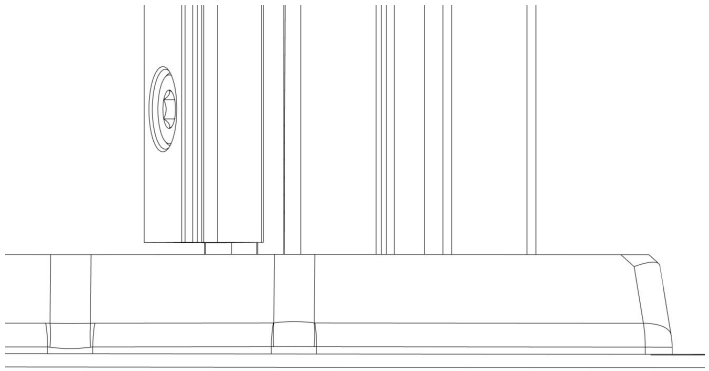






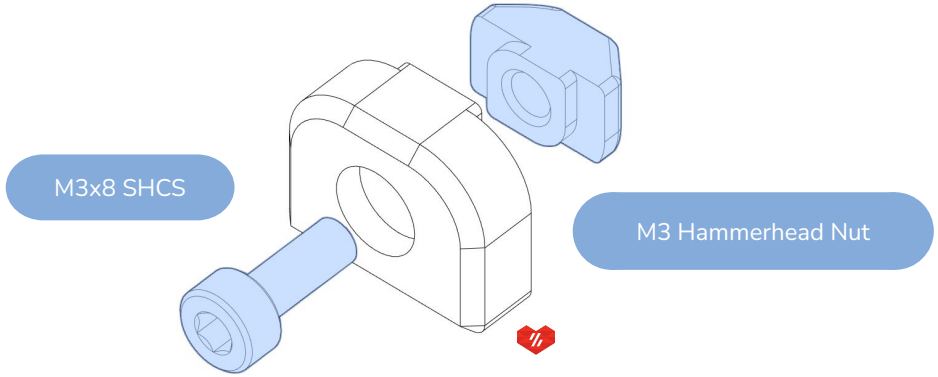
CENTERED RAIL INSTALLATION GUIDE

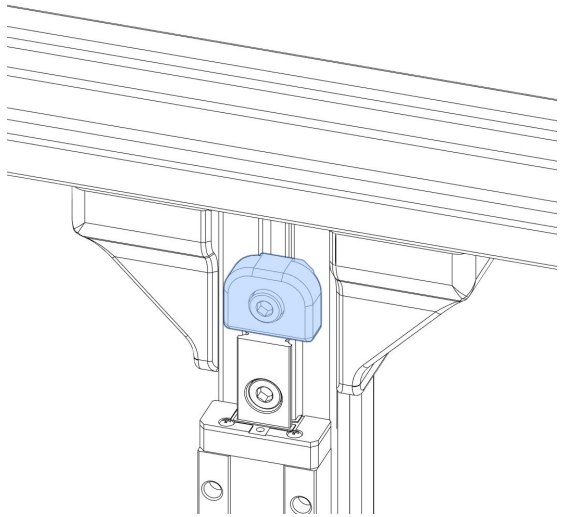
Use the MGN9 guides to position the rail in the center of the extrusion prior to fastening the screws.

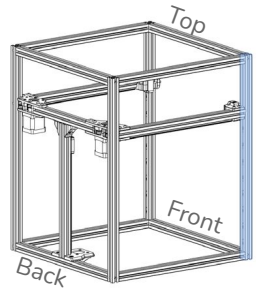


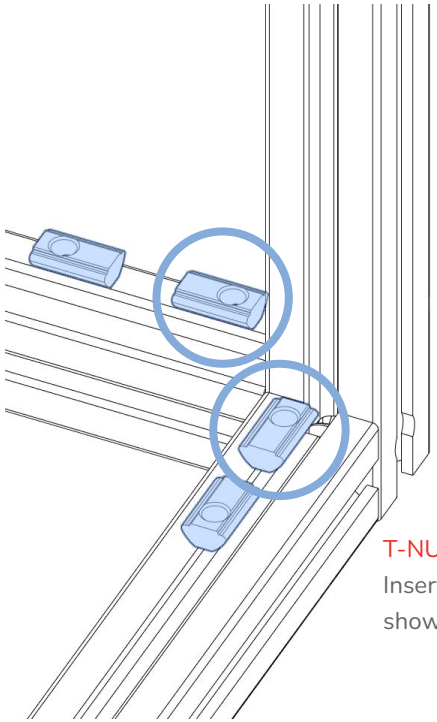
BOTTOM GAP

Leave a small gap between the printed part and the rail. 1-2 mm is fine.



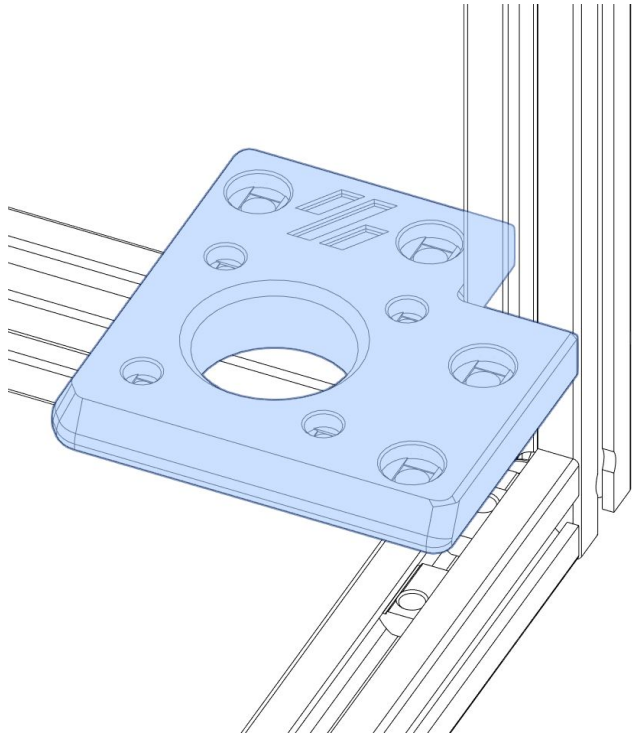


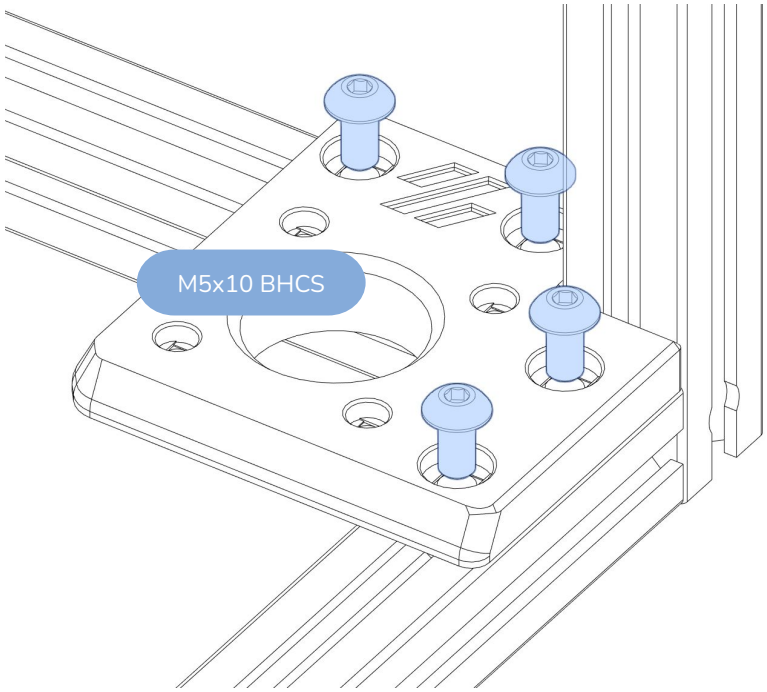


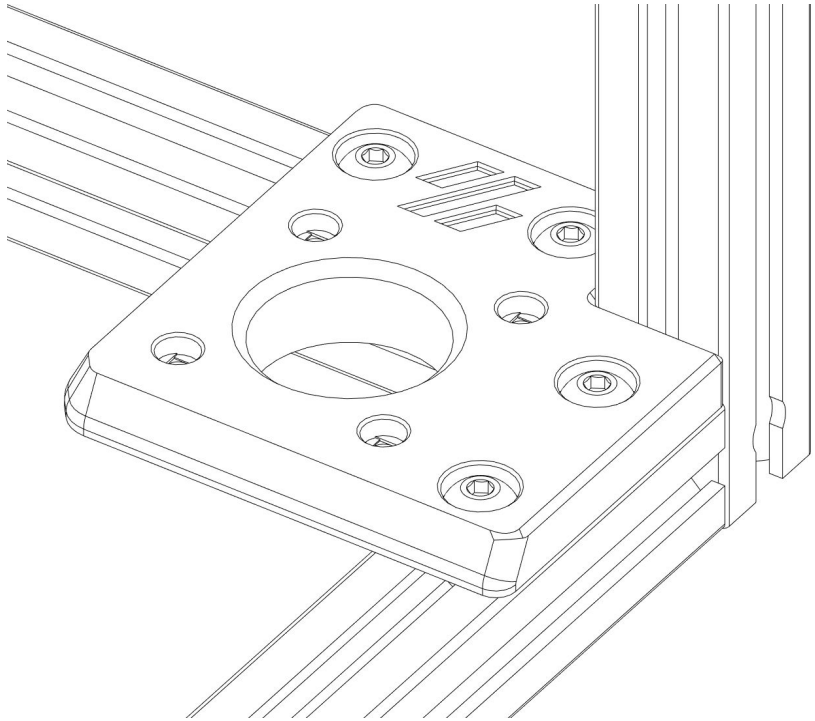


T-NUT ORIENTATION

Insert the t-nuts as shown in the highlight.

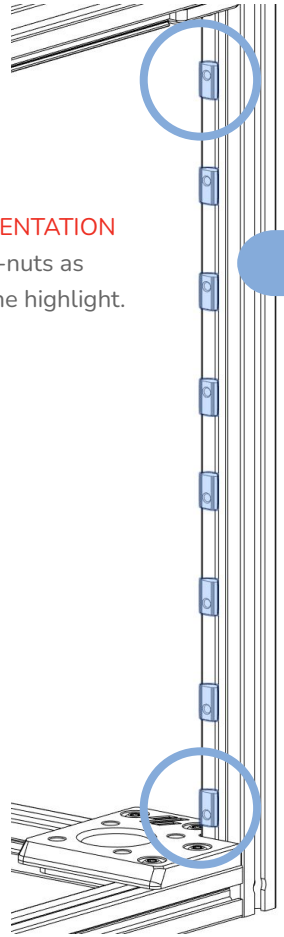



M5x10 BHCS

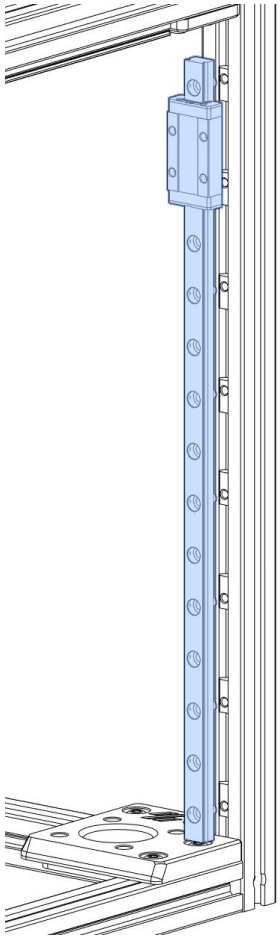


T-NUT ORIENTATION

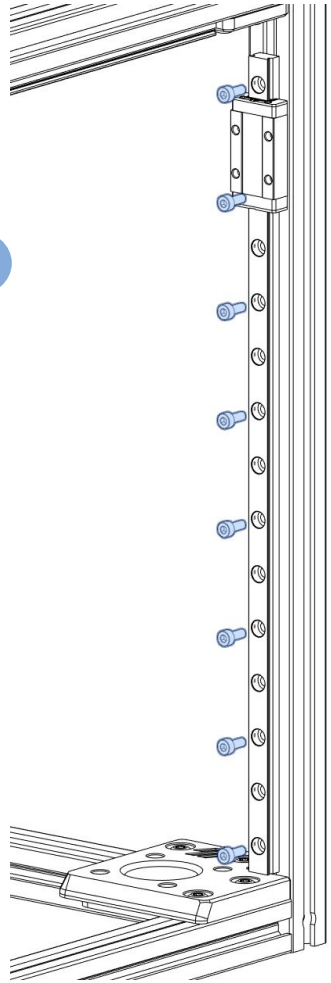
Insert the t-nuts as shown in the highlight.



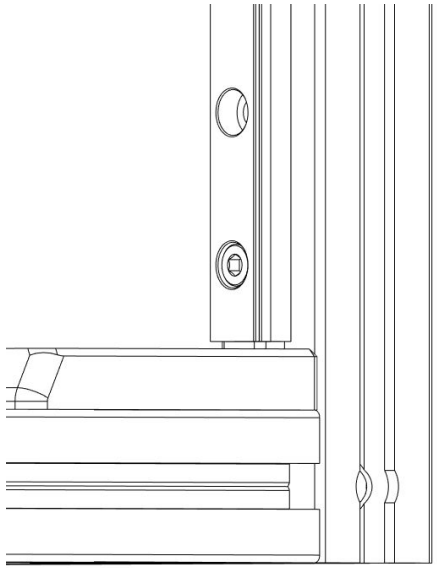
M3 T-Nut



MGN9 Rail

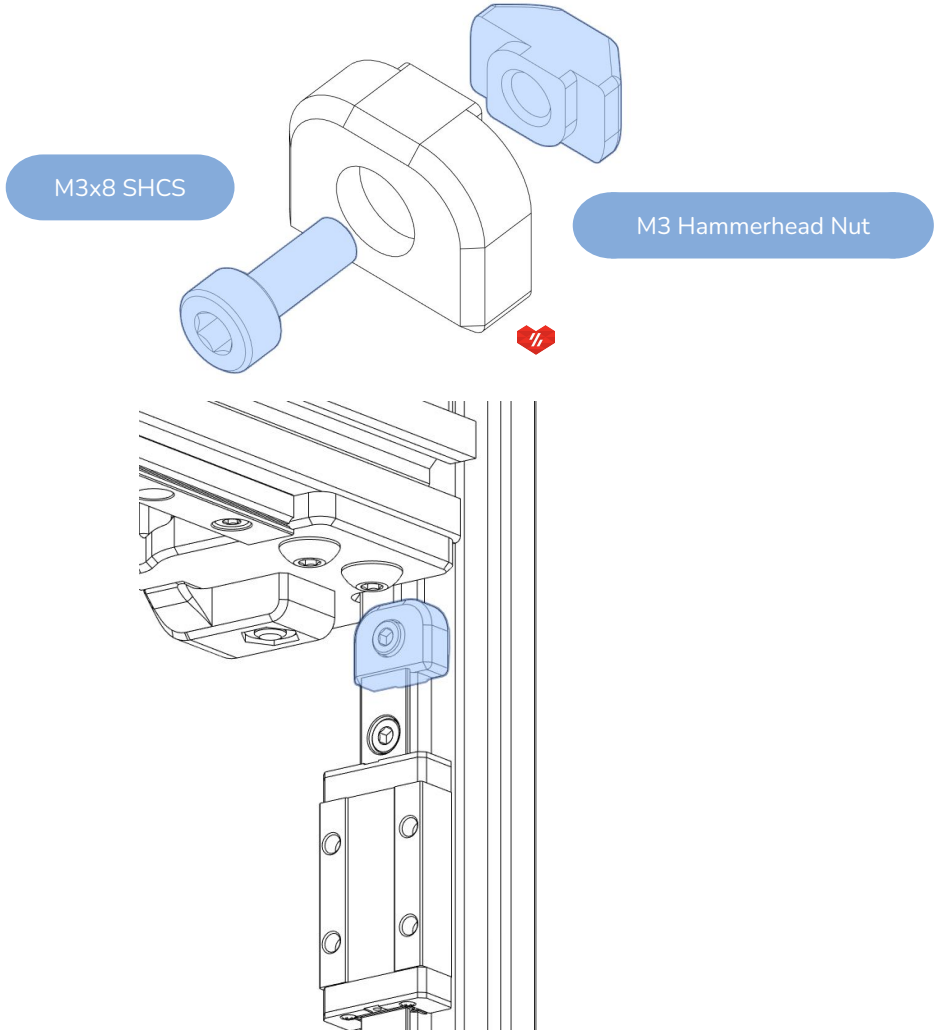


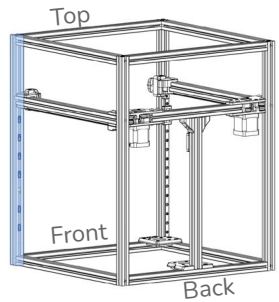
M3x8 SHCS

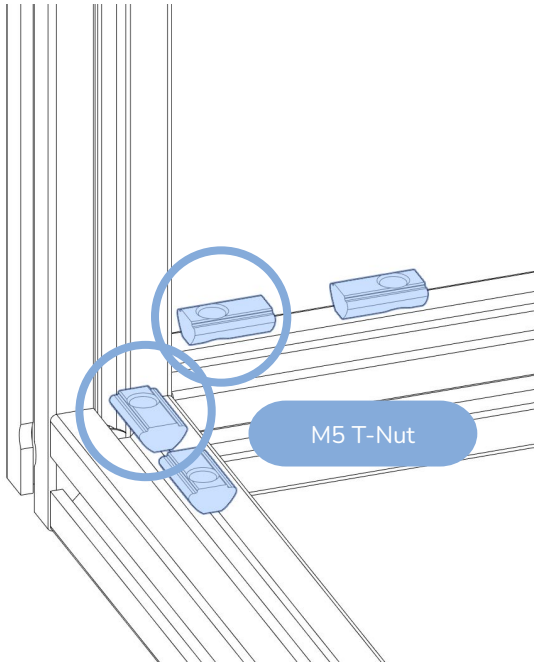


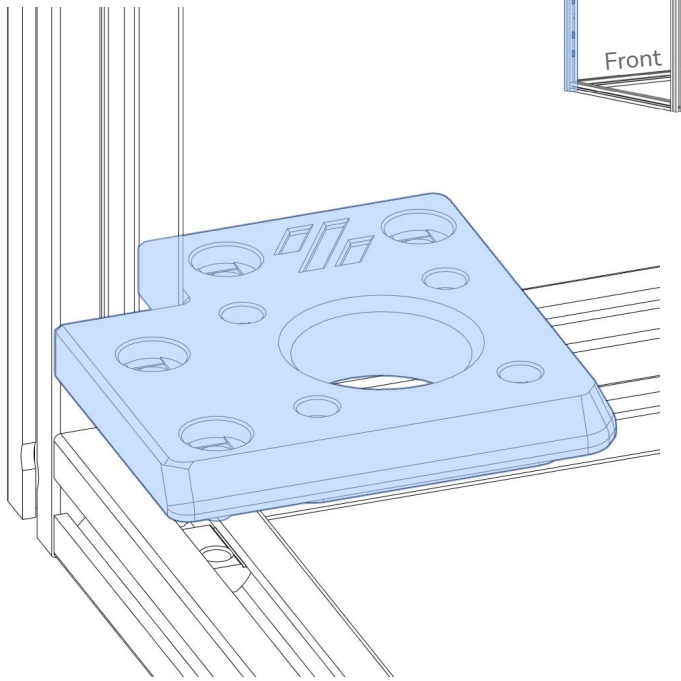
BOTTOM GAP

Leave a small gap between the printed part and the rail. 1-2 mm is fine.



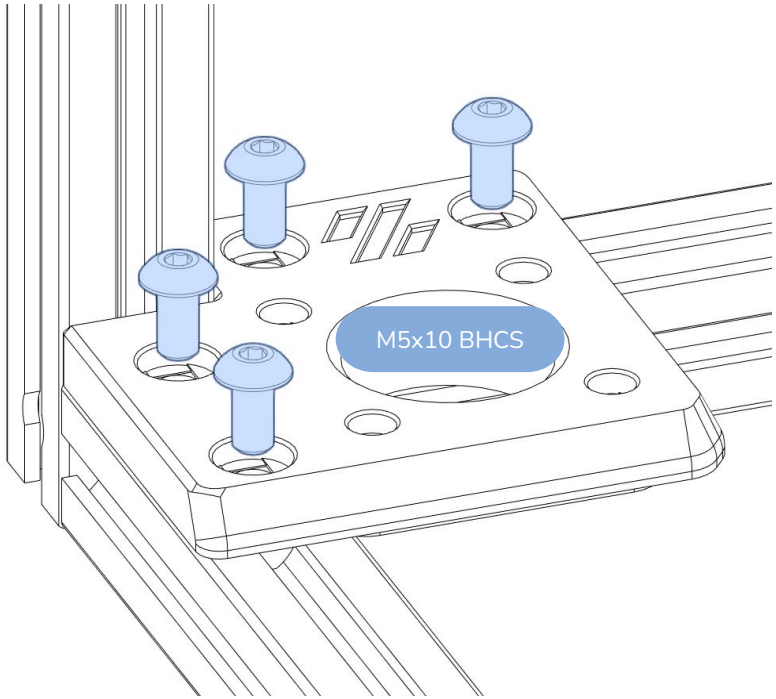


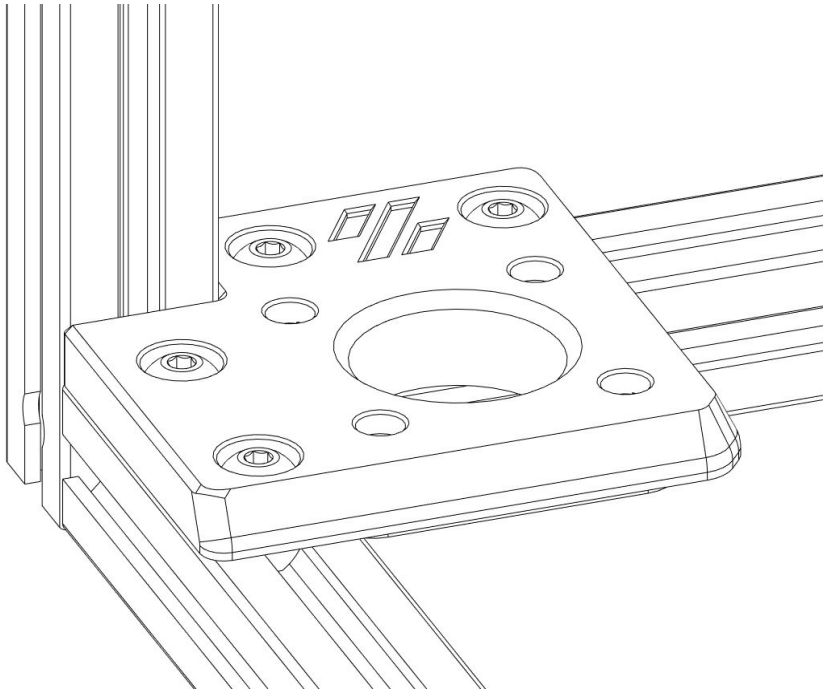


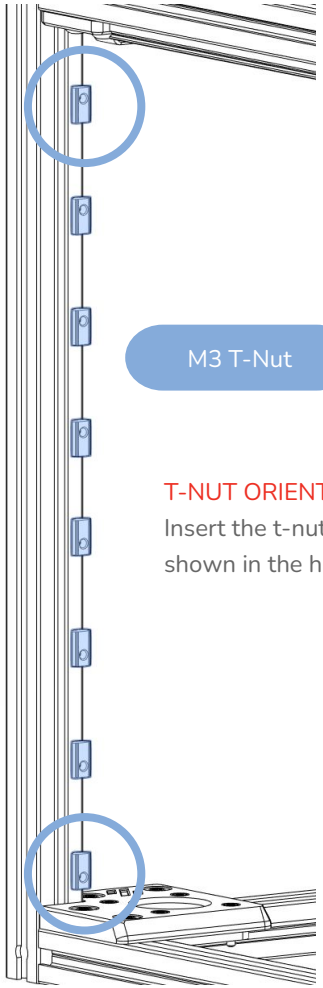


T-NUT ORIENTATION

Insert the t-nuts as shown in the highlight.

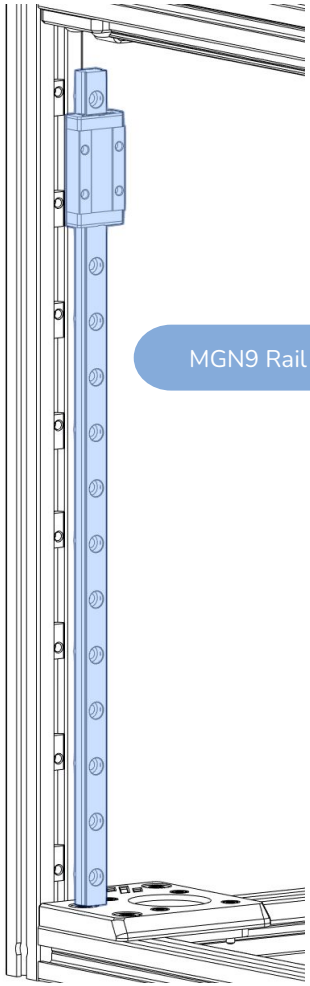

M5x10 BHCS

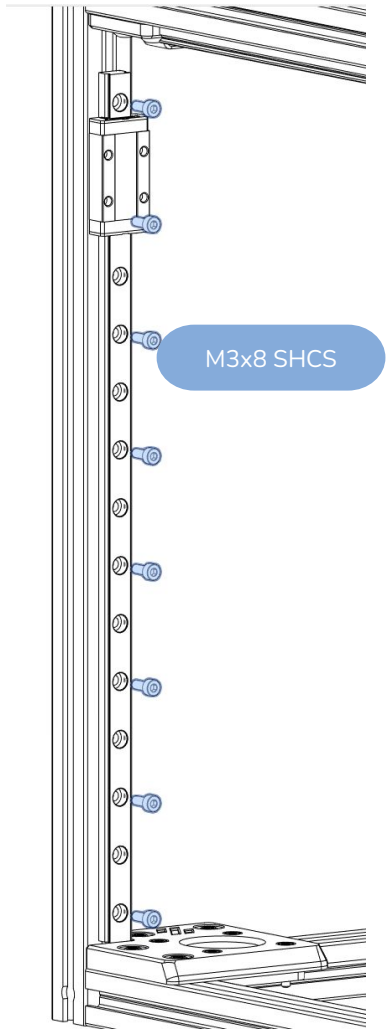


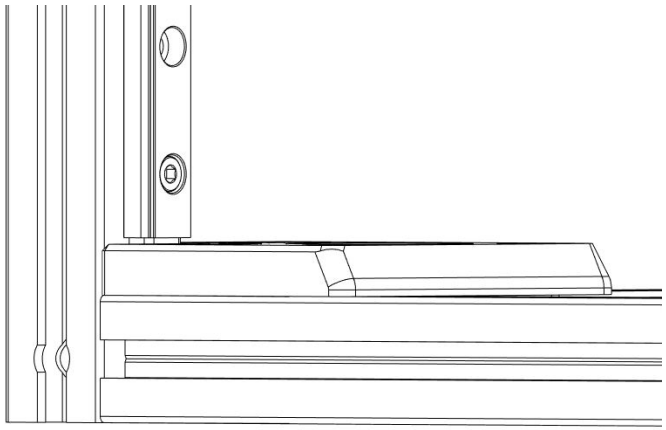


T-NUT ORIENTATION

Insert the t-nuts as shown in the highlight.

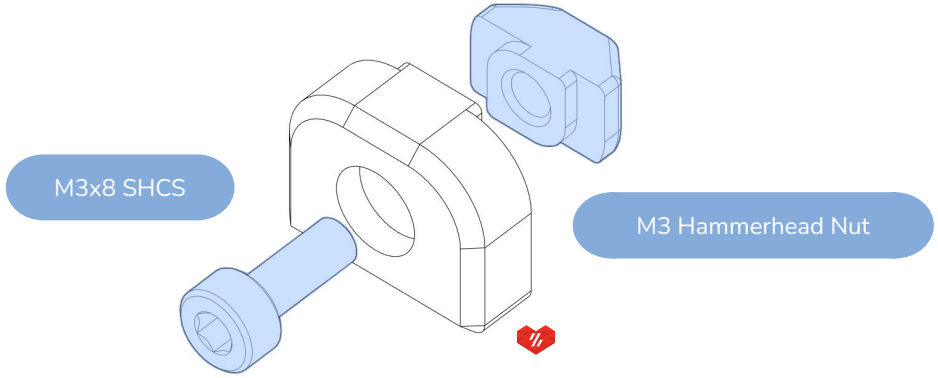


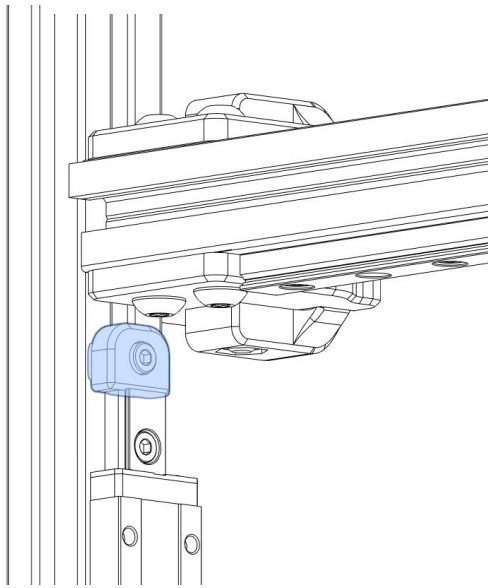


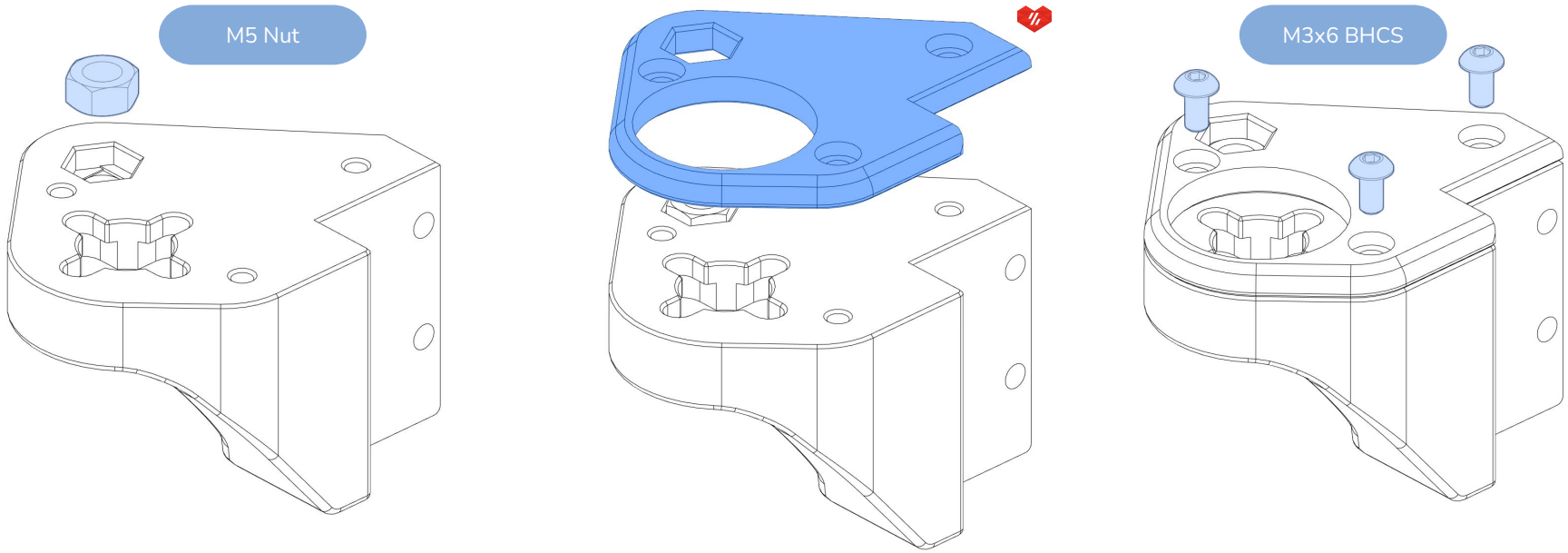


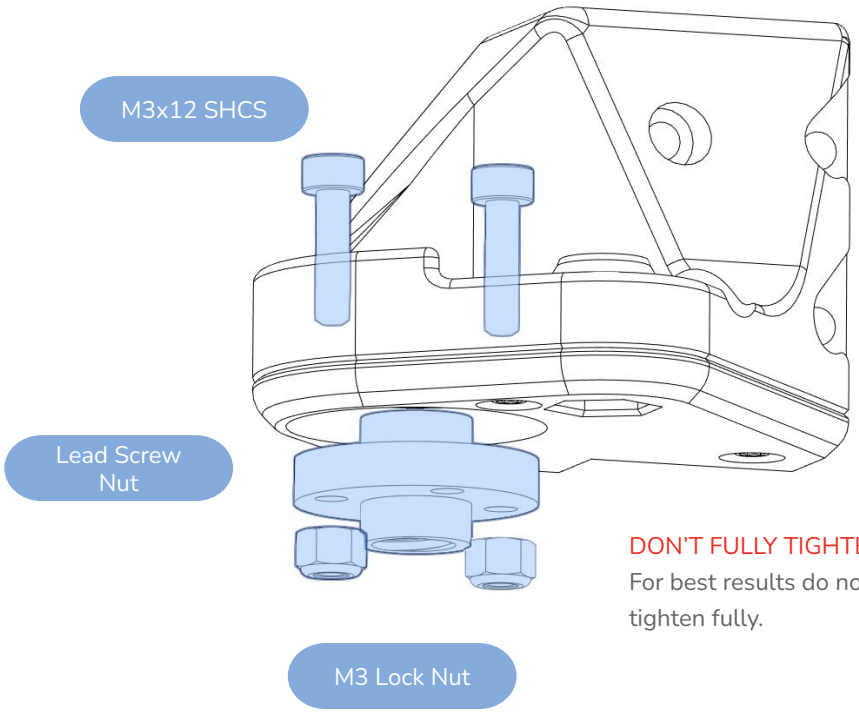
BOTTOM GAP

Leave a small gap between the printed part and the rail. 1-2 mm is fine.



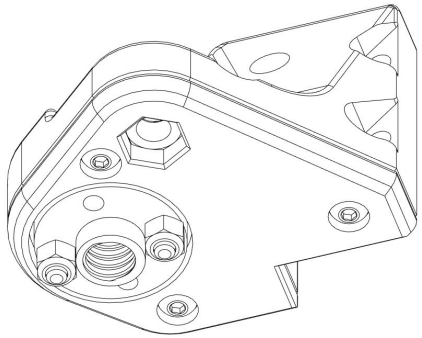


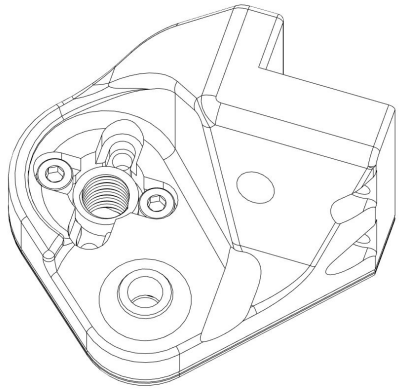

M5 Nut
M3x6 BHCS

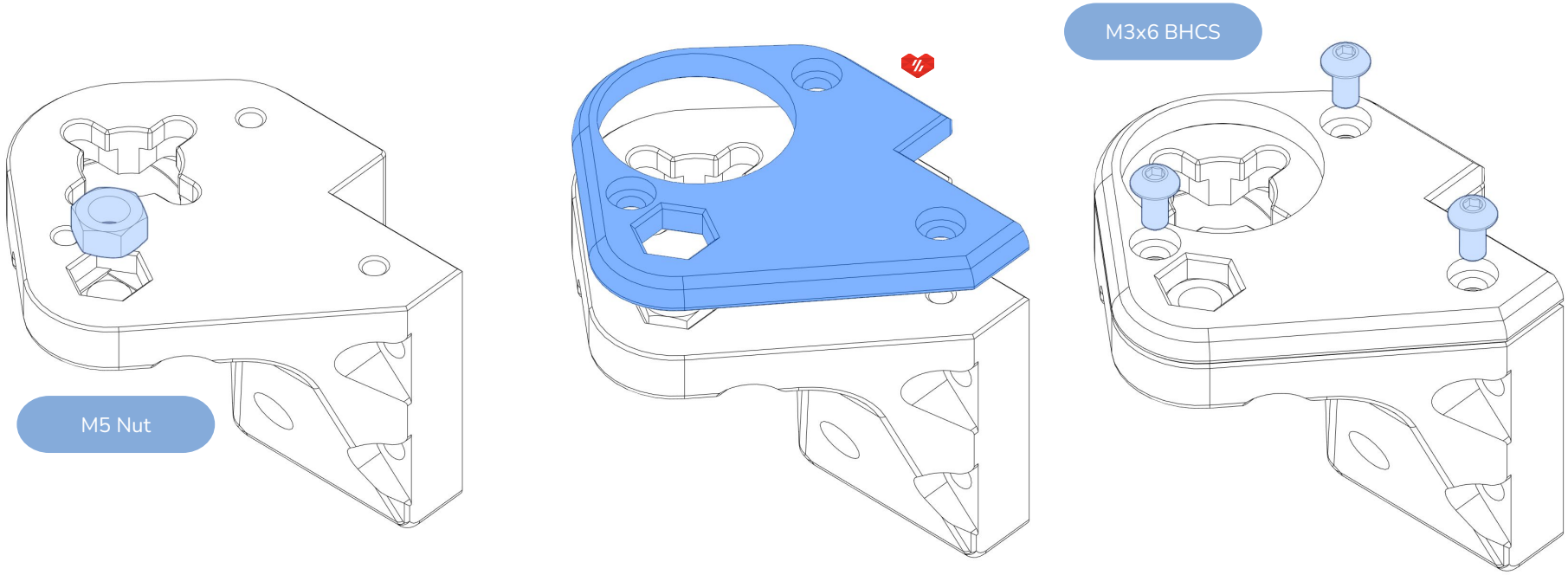


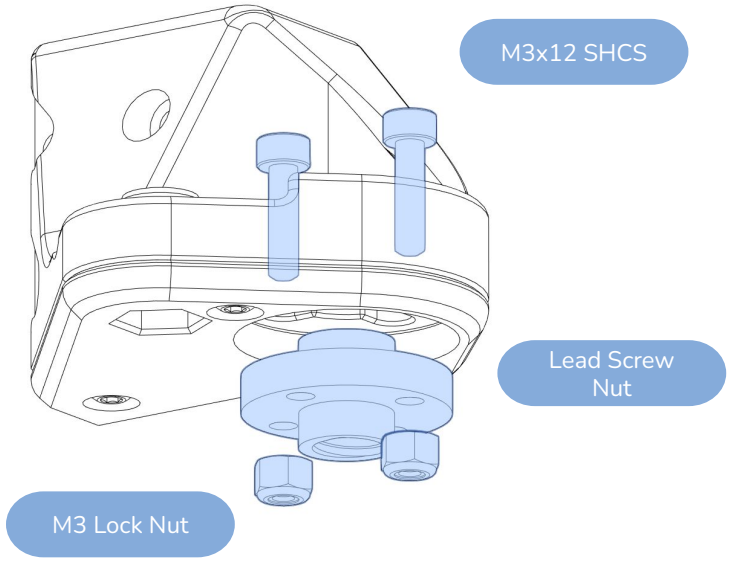
DON'T FULLY TIGHTEN

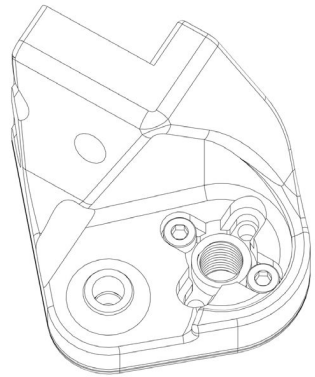
For best results do not tighten fully.

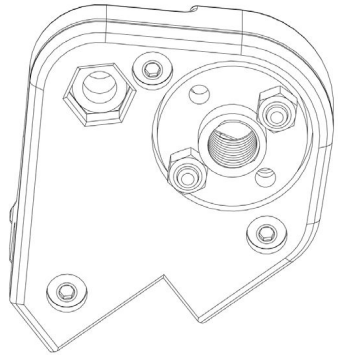





M5 Nut
M3x6 BHCS

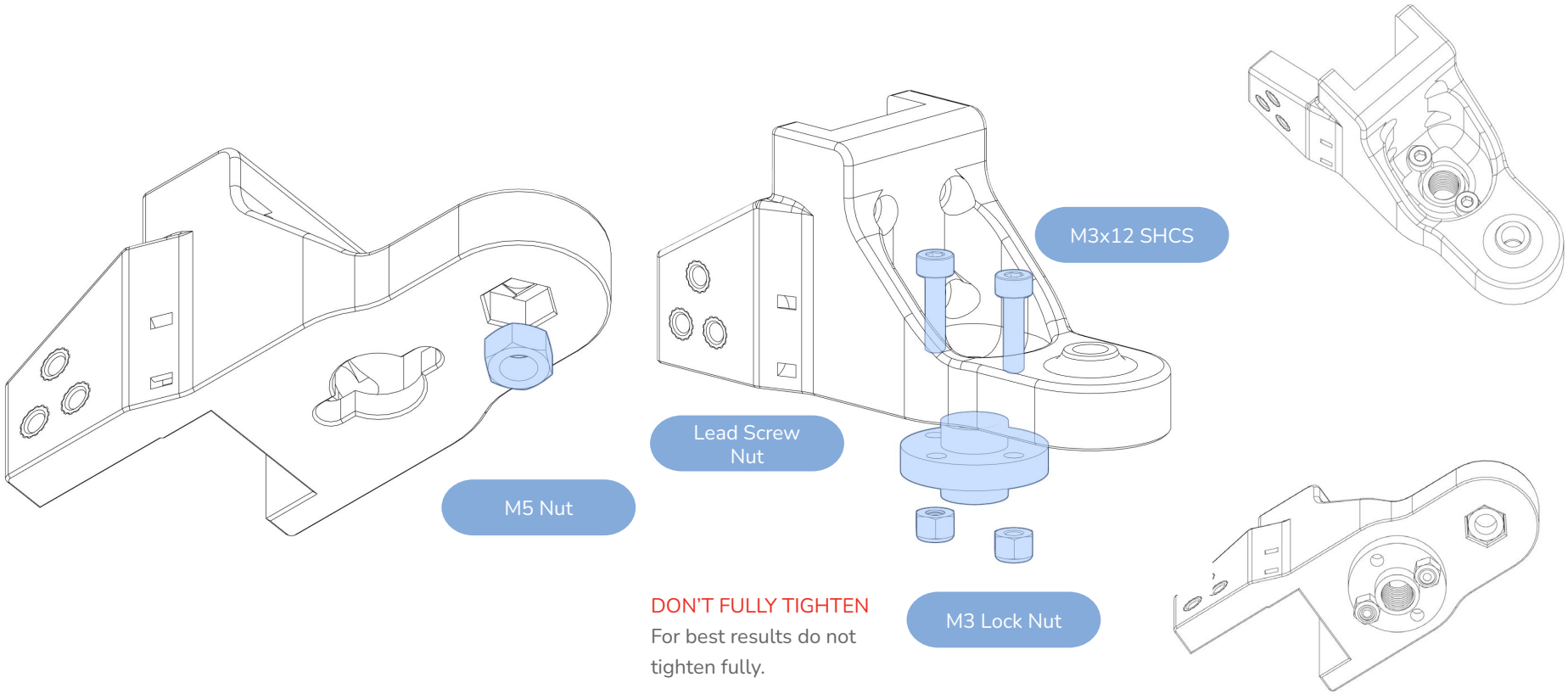






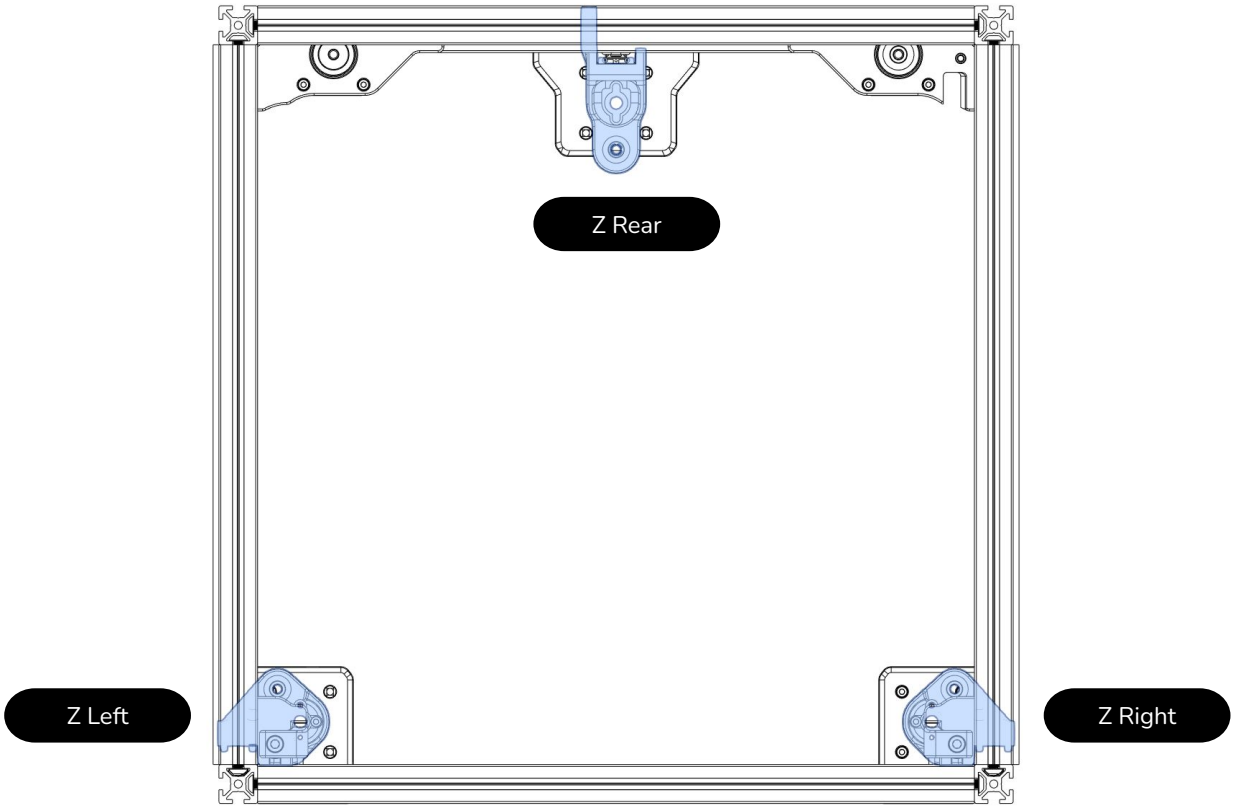
DON'T FULLY TIGHTEN

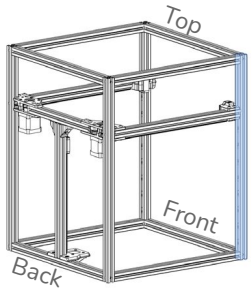
For best results do not tighten fully.

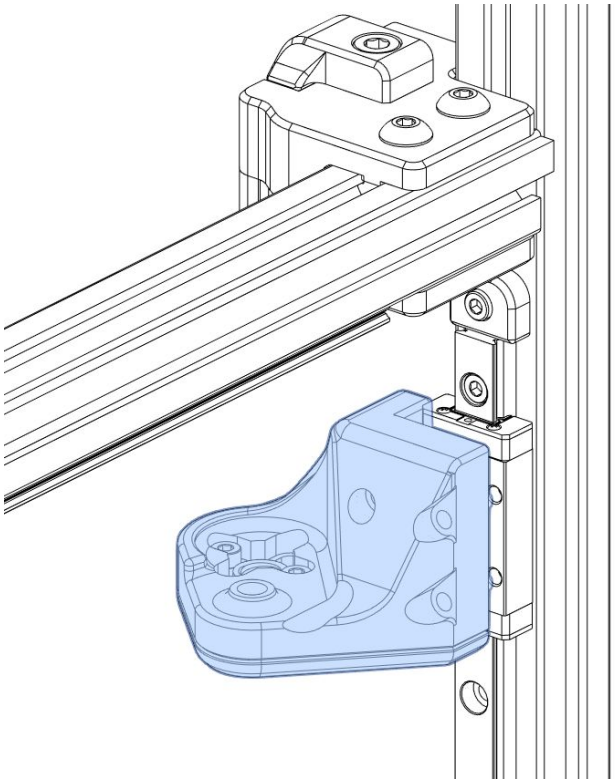


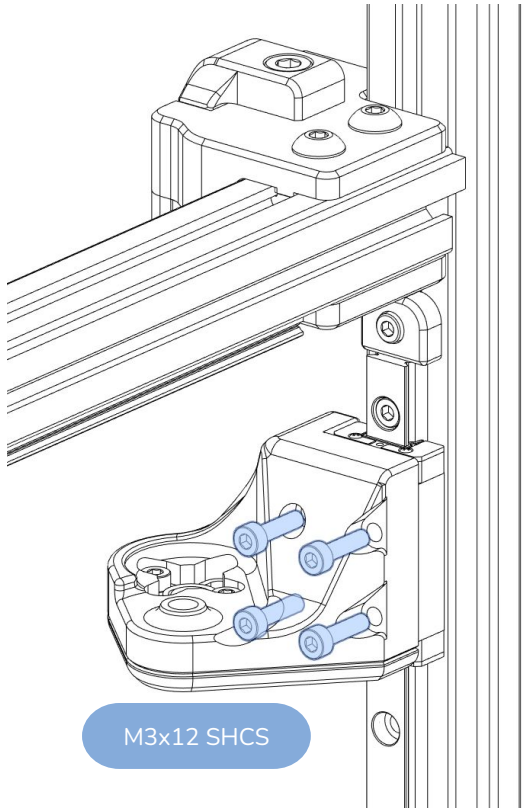
DON'T FULLY TIGHTEN

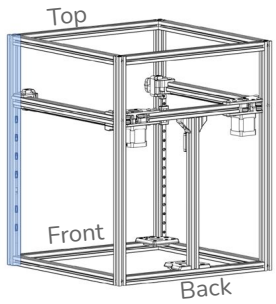
For best results do not tighten fully.

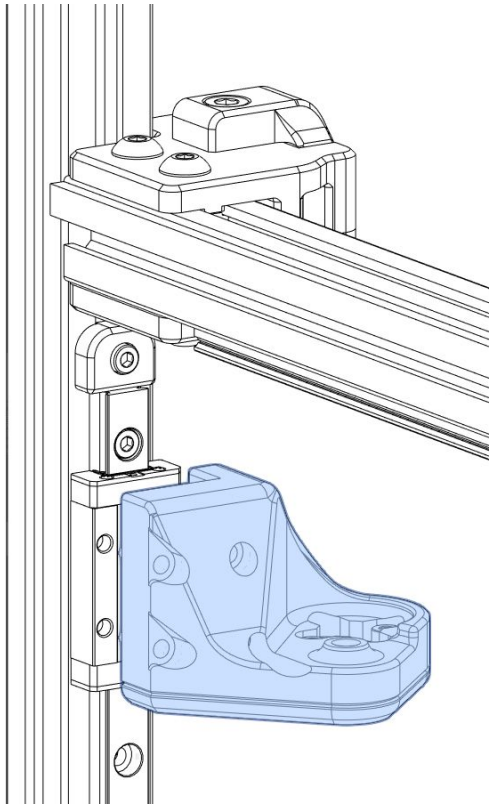

Z Rear
Z Left
Z Right

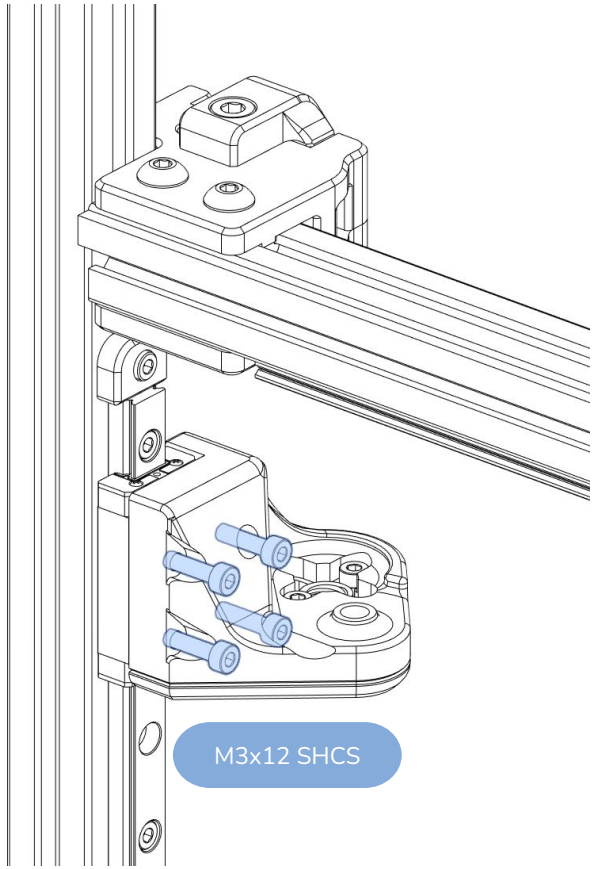

Top
Front
Back

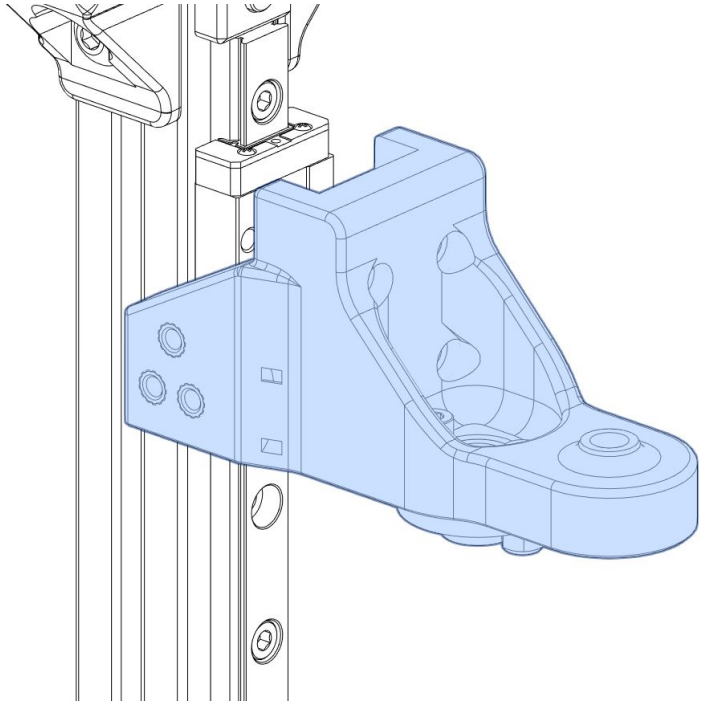


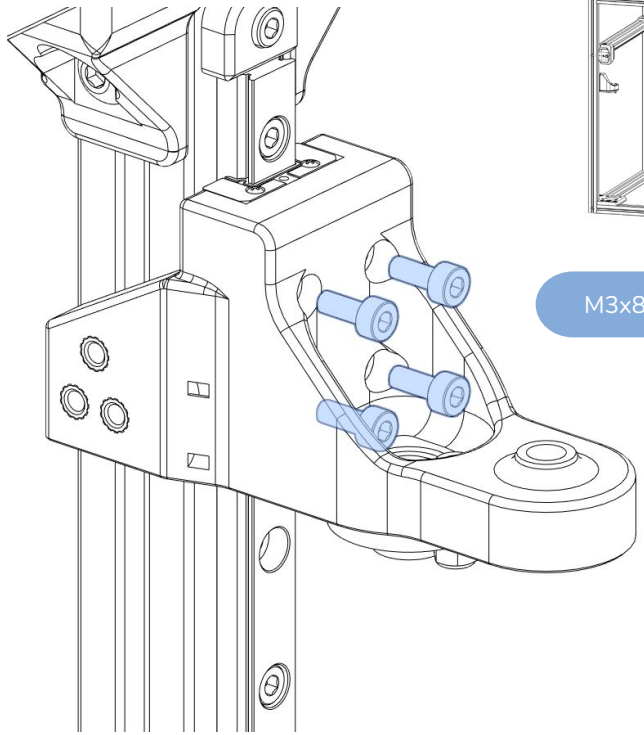

M3x12 SHCS

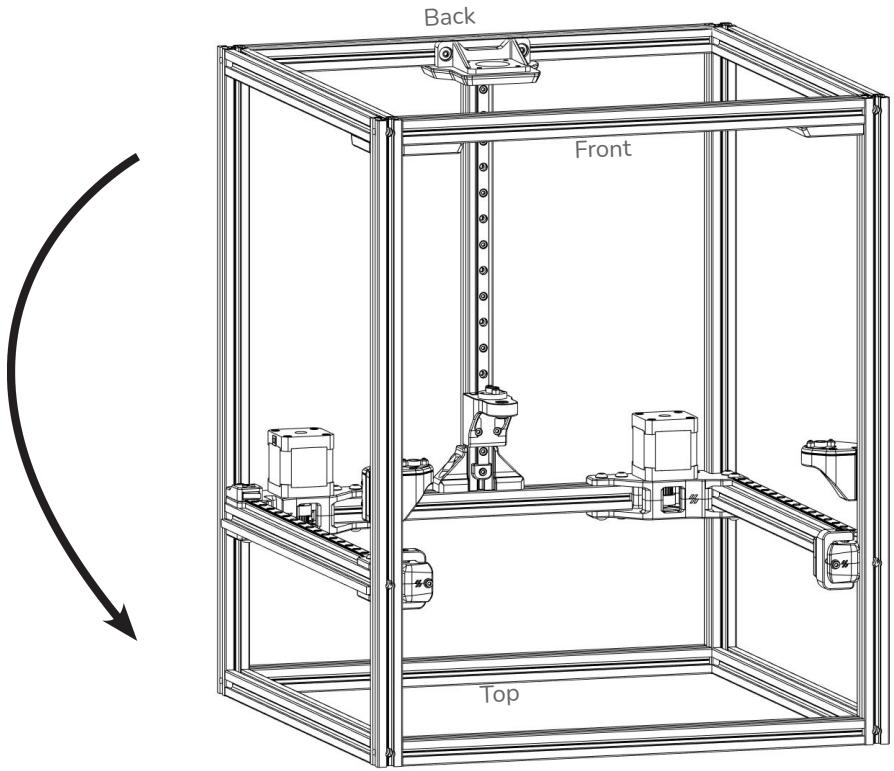

Top
Front
Back




M3x12 SHCS

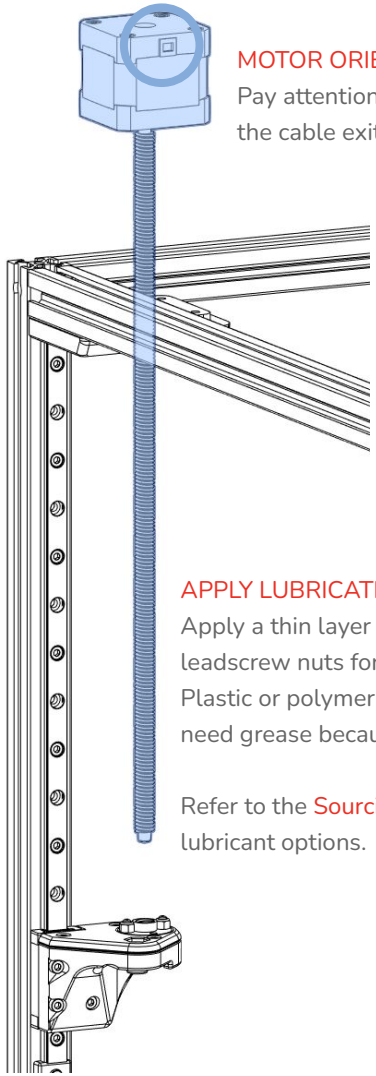



Top
Back
Front
M3x8 SHCS



FLIP UPSIDE DOWN

Turn the printer upside down for the next assembly steps.



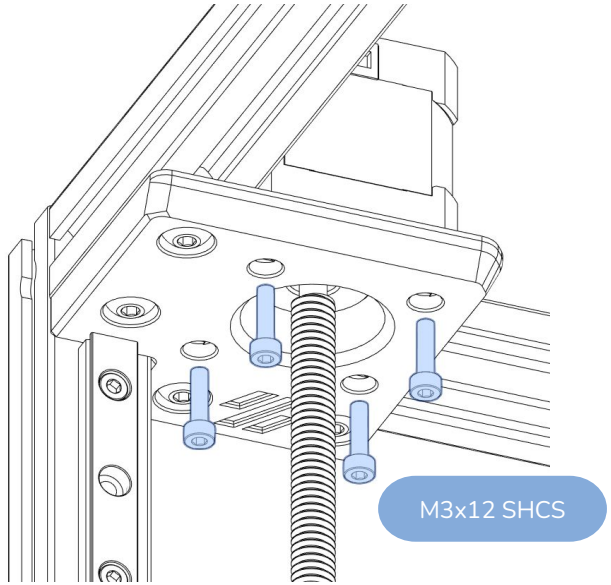
MOTOR ORIENTATION

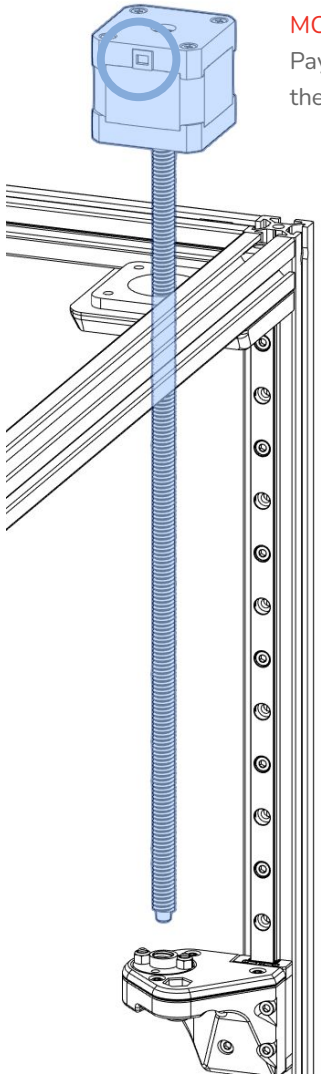
Pay attention to the orientation of the cable exit.

APPLY LUBRICATION

Apply a thin layer of grease to metal leadscrew nuts for smooth operation. Plastic or polymer leadscrew nuts don't need grease because they self-lubricate.

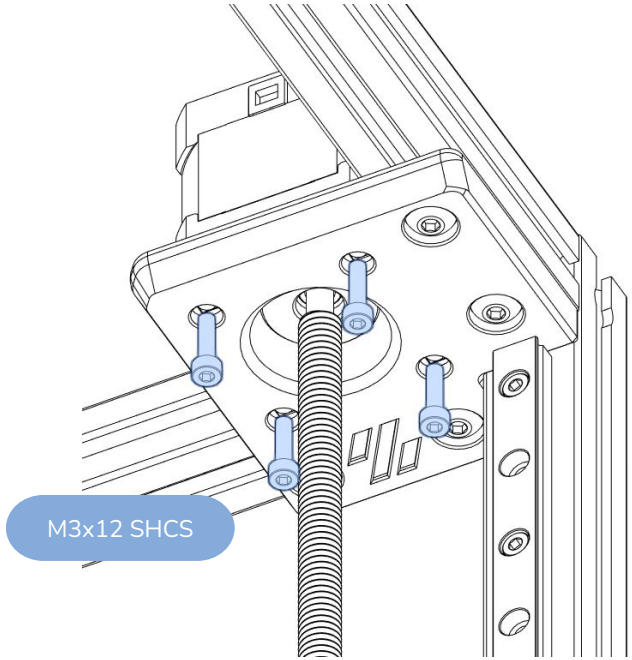
Refer to the Sourcing Guide for lubricant options.





MOTOR ORIENTATION

Pay attention to the orientation of the cable exit.



APPLY LUBRICATION

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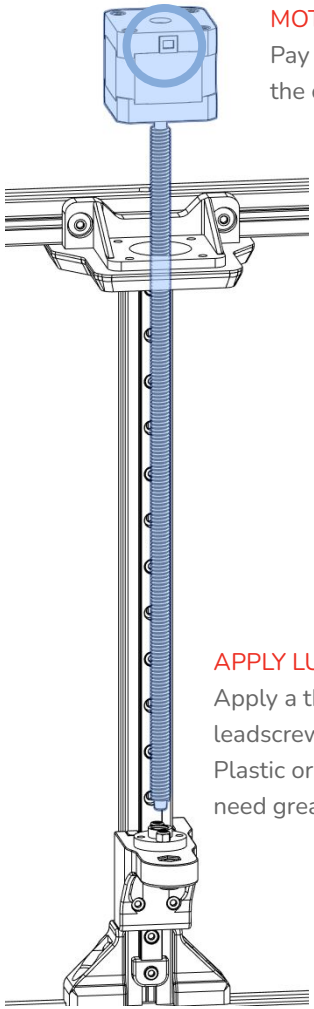
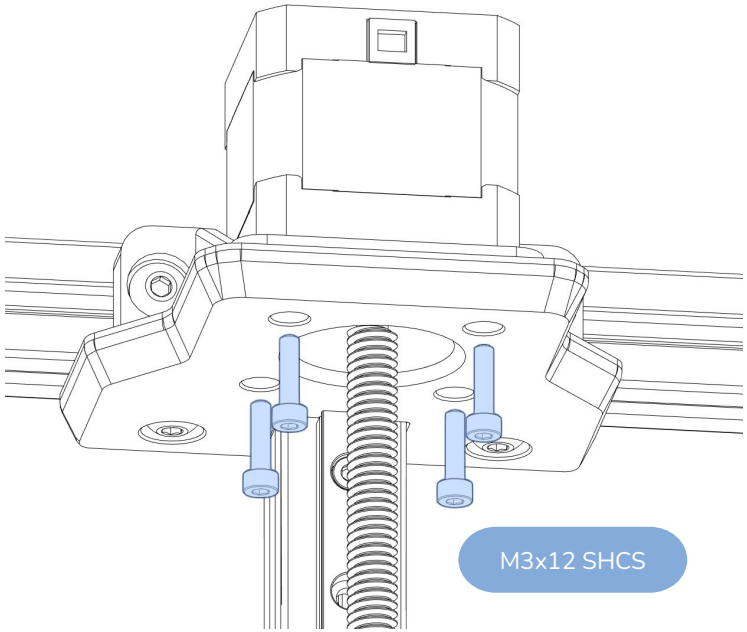


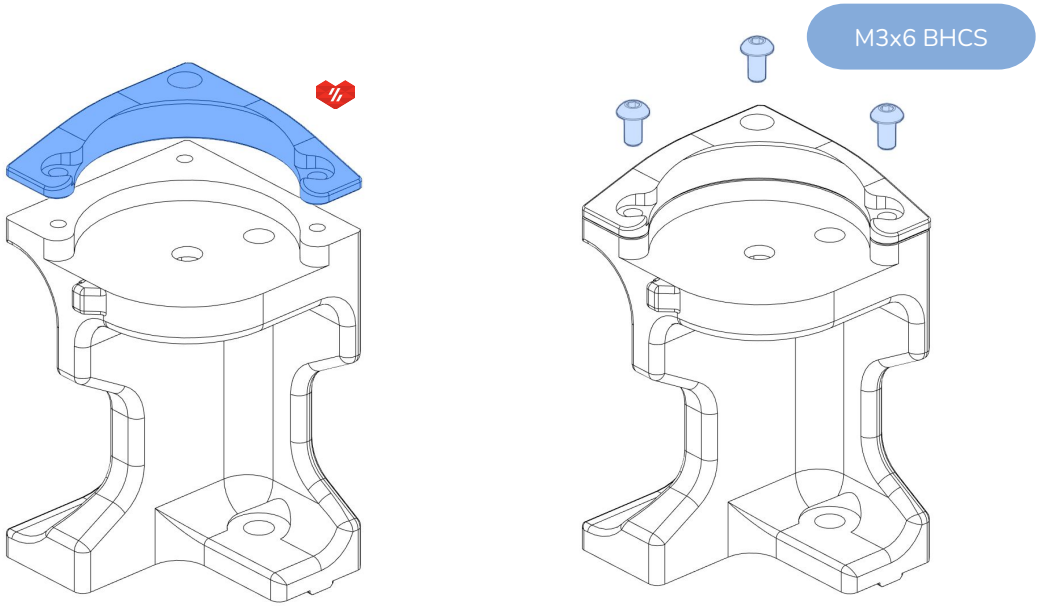
MOTOR ORIENTATION

Pay attention to the orientation of the cable exit.

APPLY LUBRICATION

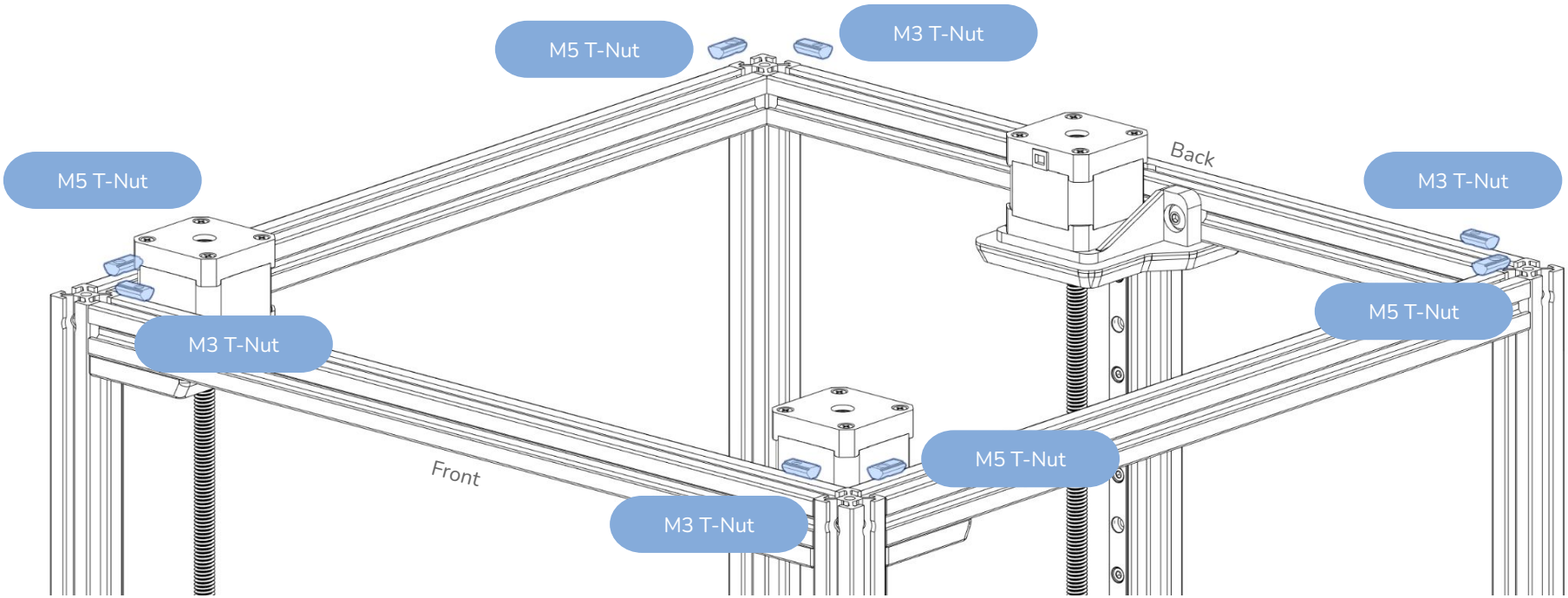
Apply a thin layer of grease to metal leadscrew nuts for smooth operation. Plastic or polymer leadscrew nuts don't need grease because they self-lubricate.

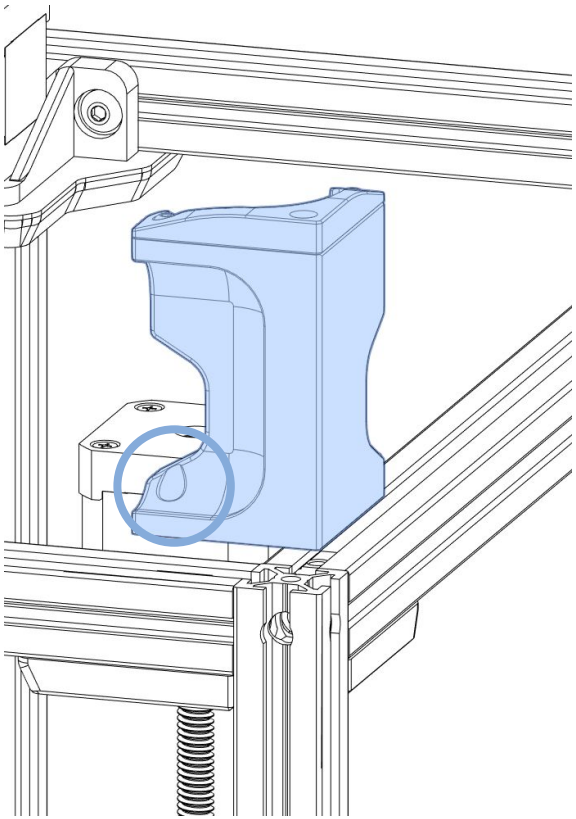


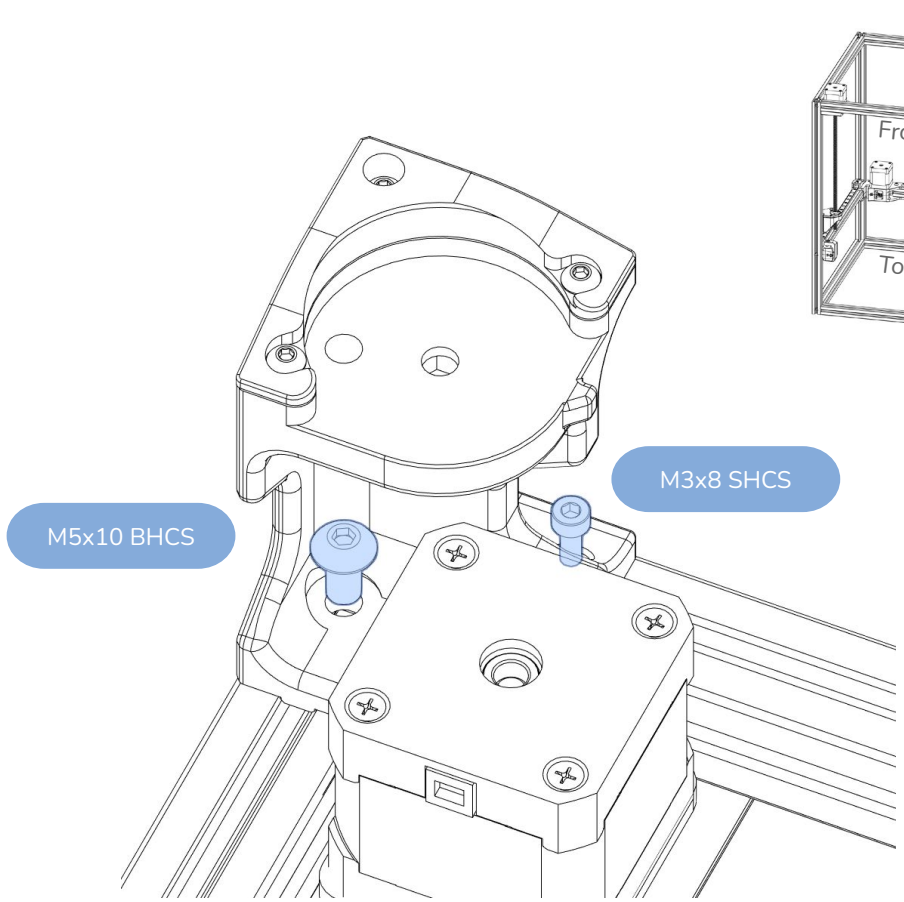


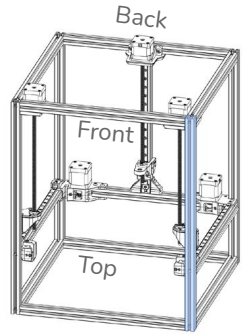
ASSEMBLE FOUR FEET

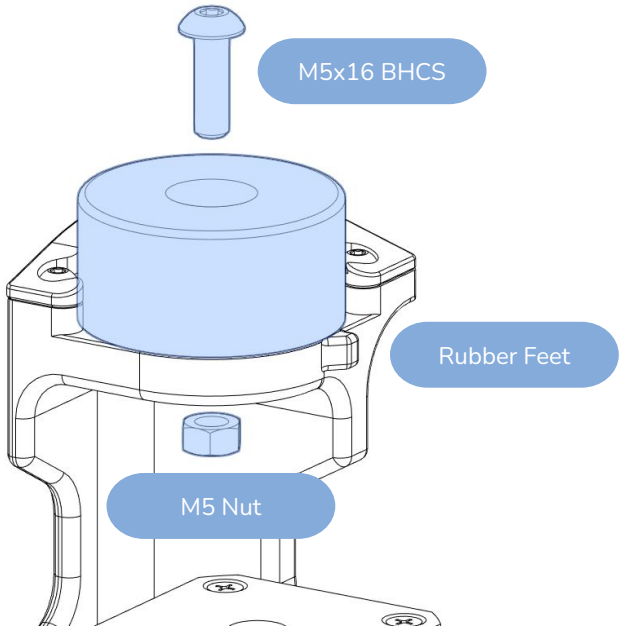
Repeat the instructions and assemble all four feet.

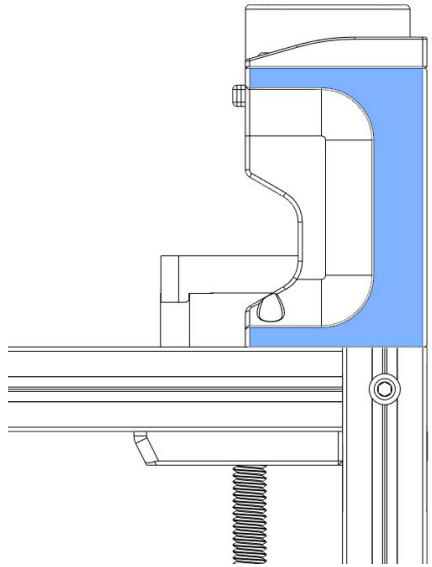

M5 T-Nut
M3 T-Nut
M5 T-Nut
Back
M3 T-Nut
M5 T-Nut
M3 T-Nut
Front
M5 T-Nut
M3 T-Nut




M5x10 BHCS
M3x8 SHCS

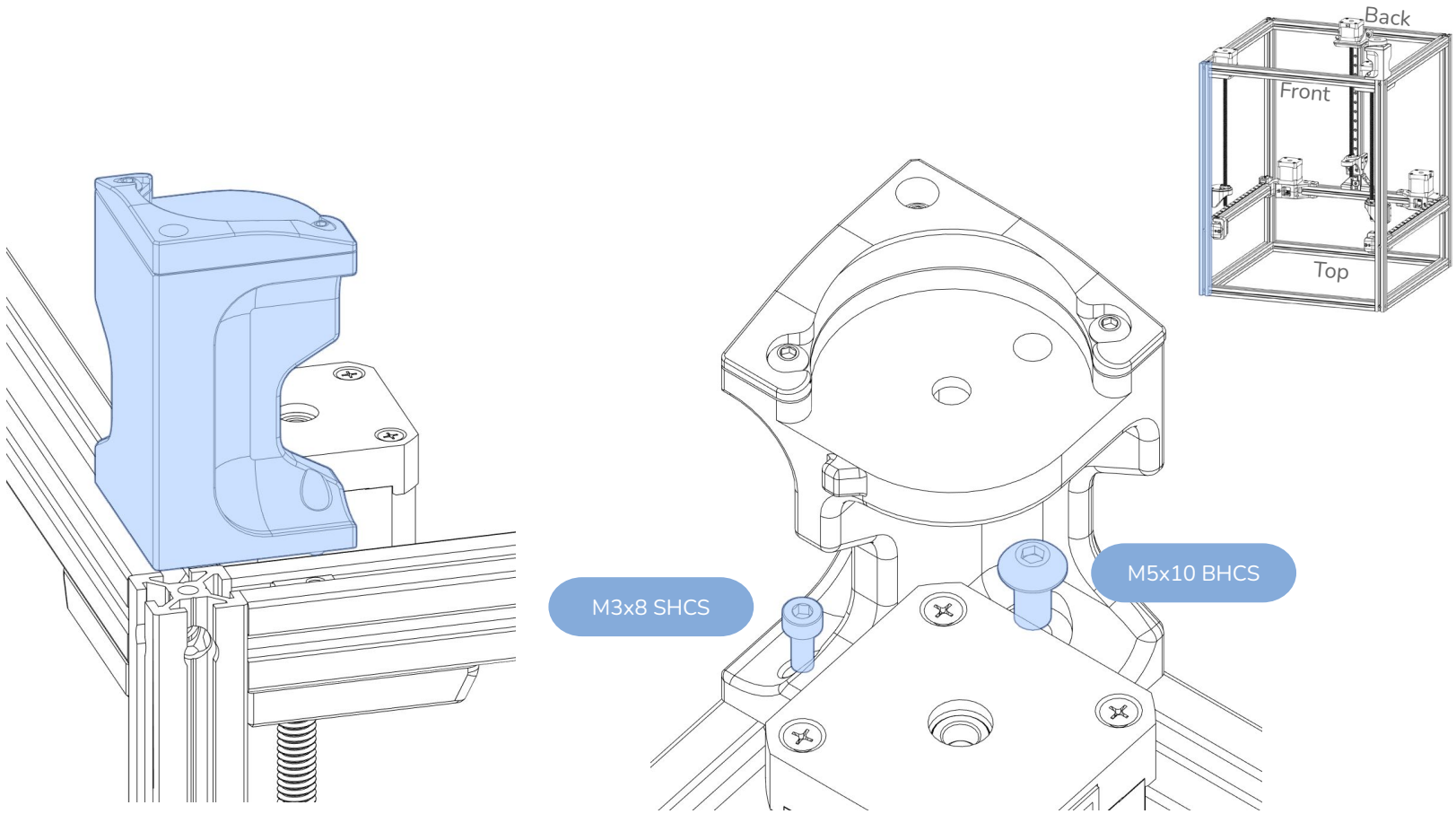

Back
Front
Top

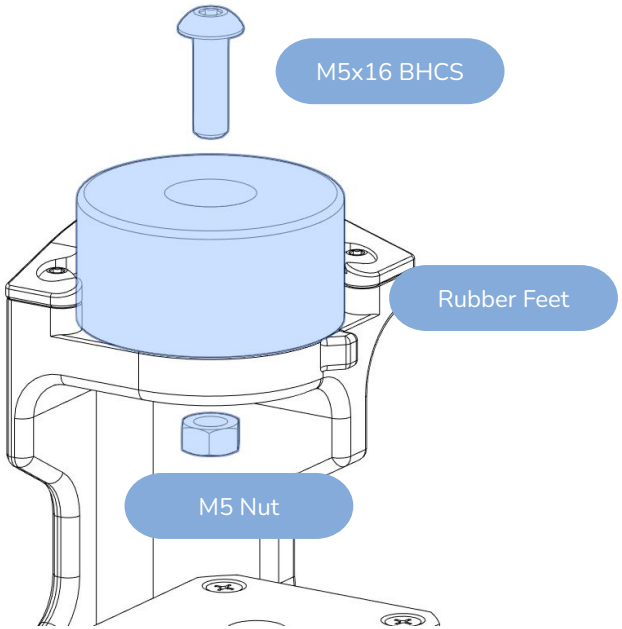


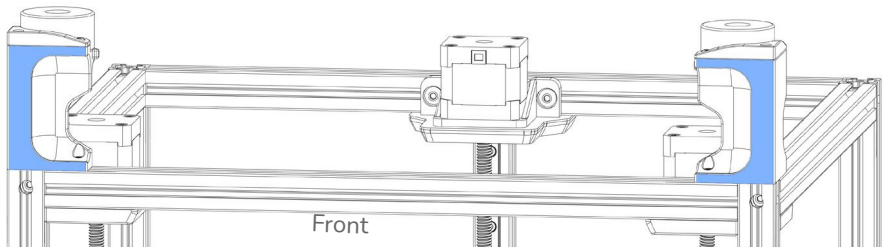


MIND THE PART ORIENTATION

The profile shown above are towards the front and rear of the printer.

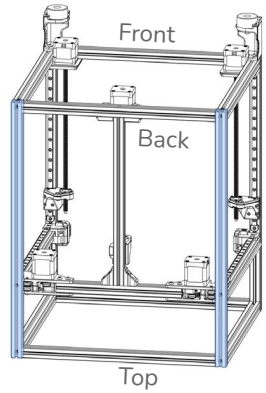

Back
Front
Top
M3x8 SHCS
M5x10 BHCS

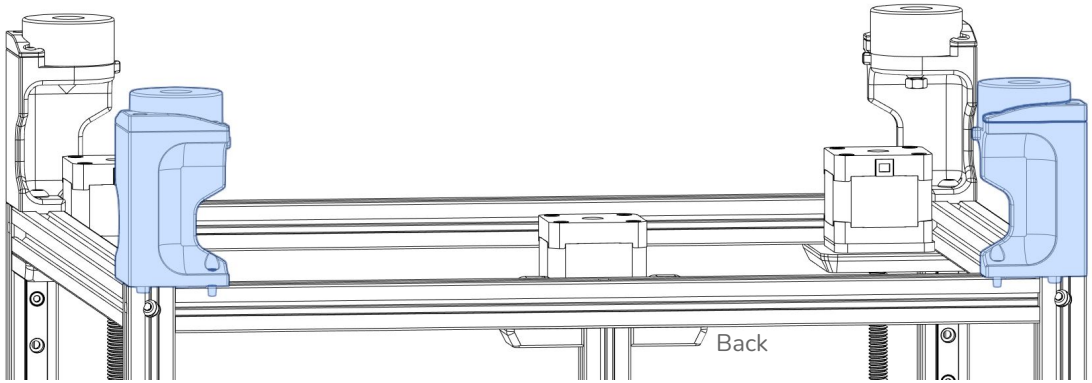




MIND THE PART ORIENTATION

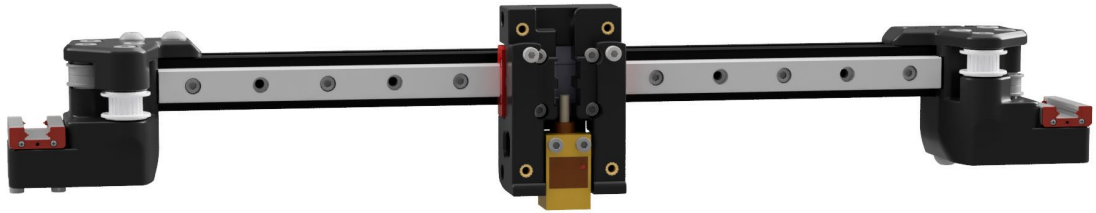
The profile shown above are towards the front and rear of the printer.

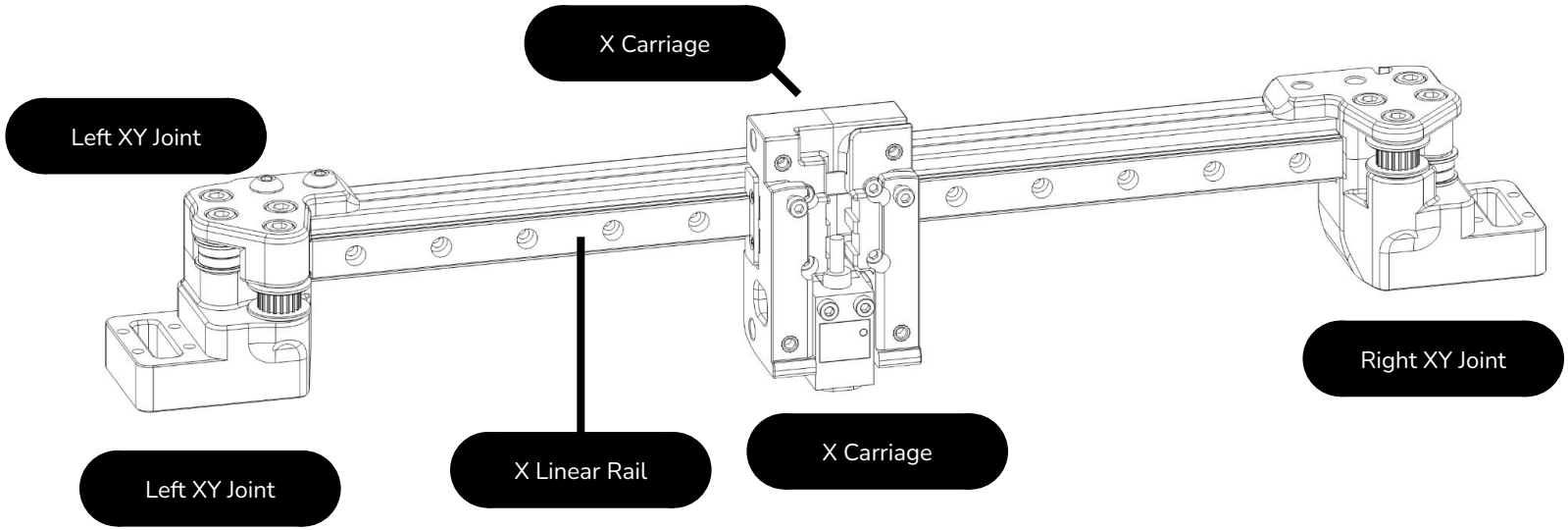




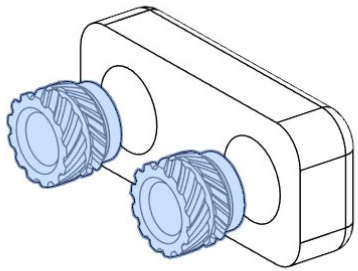
MIND THE PART ORIENTATION

The profile shown above are towards the front and rear of the printer.

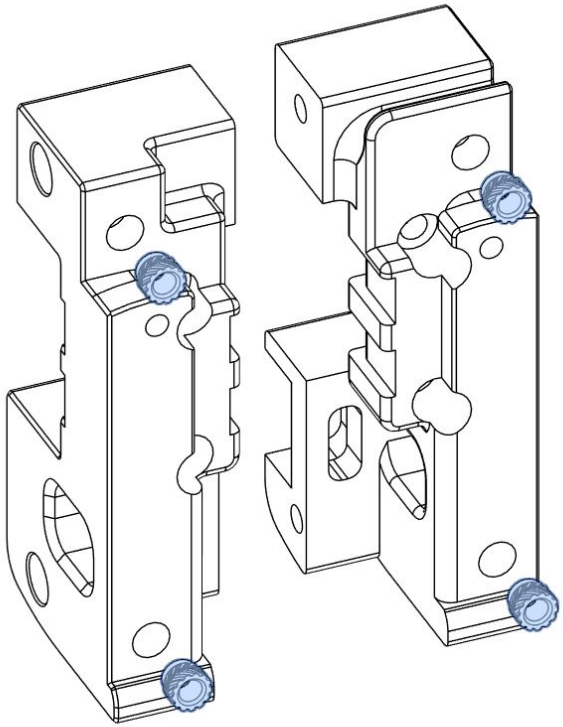


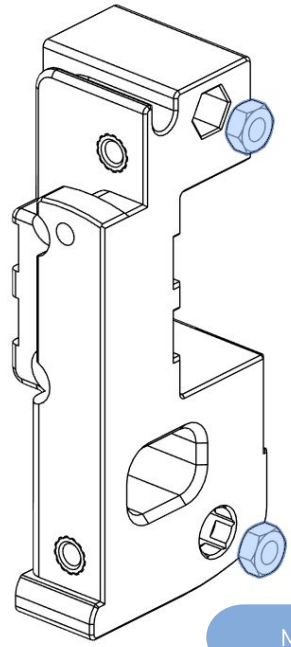

X Carriage
Left XY Joint
Right XY Joint
Left XY Joint
X Linear Rail
X Carriage

Skip this step, we use physical endstops and no hall effect ones.



Heat Set Insert



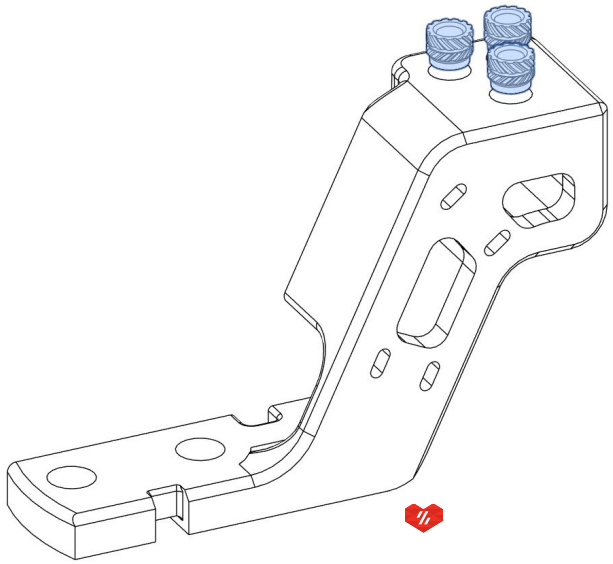


M3 Nut

Skip this step, we use physical endstops and no hall effect ones.

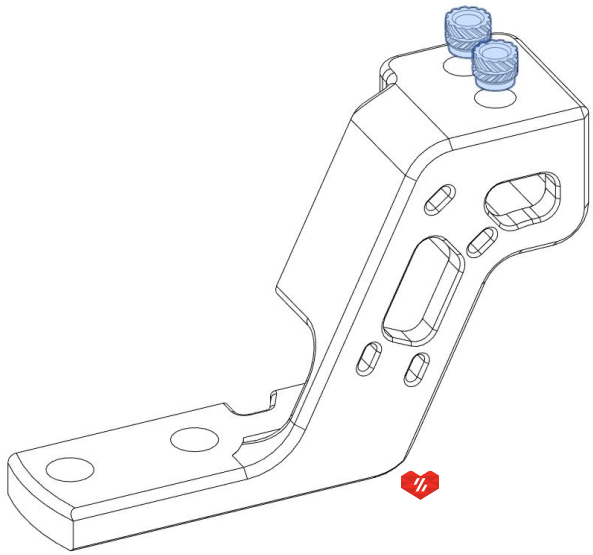
GENERIC CABLE CHAINS

The 3 hole pattern is usually found on generic cable chains.



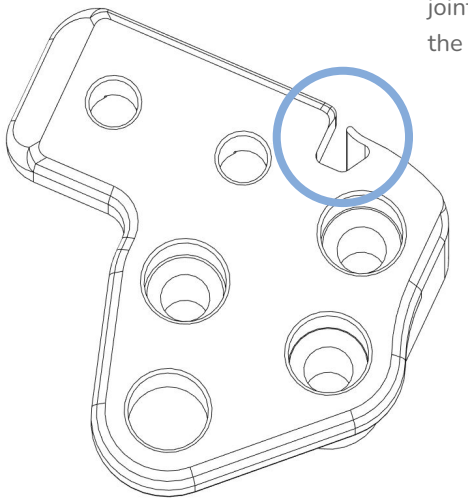
IGUS CABLE CHAINS

IGUS chains have 2 mounting holes.



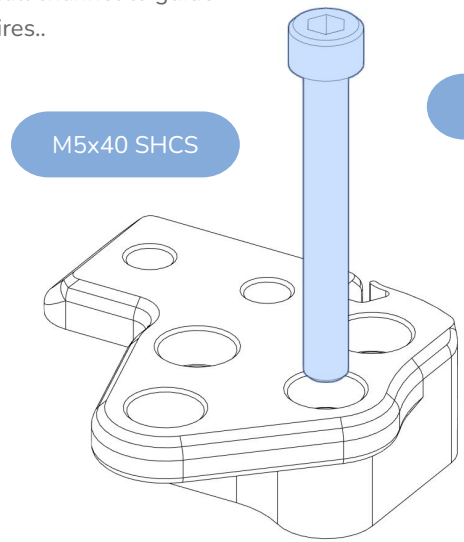
WHICH TO CHOOSE?

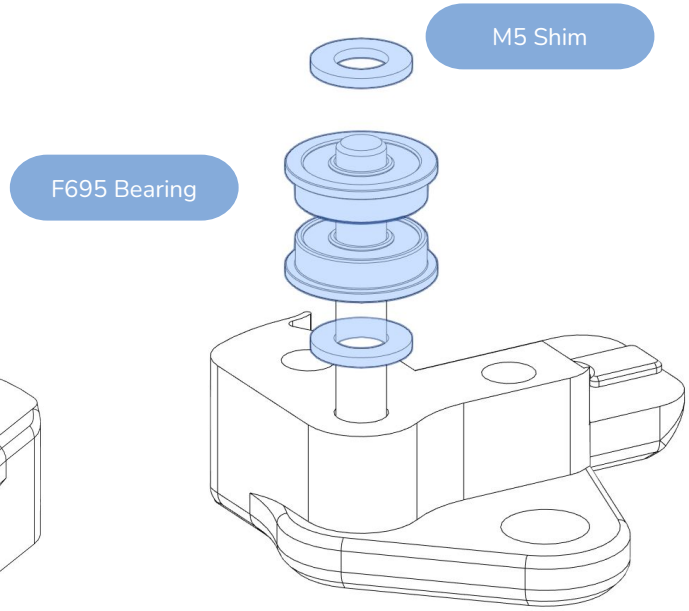
Pick the style that matches the mounting pattern of your cable chains.

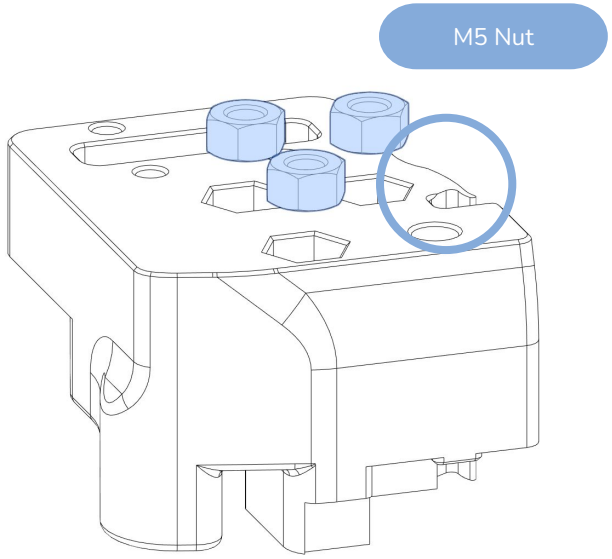


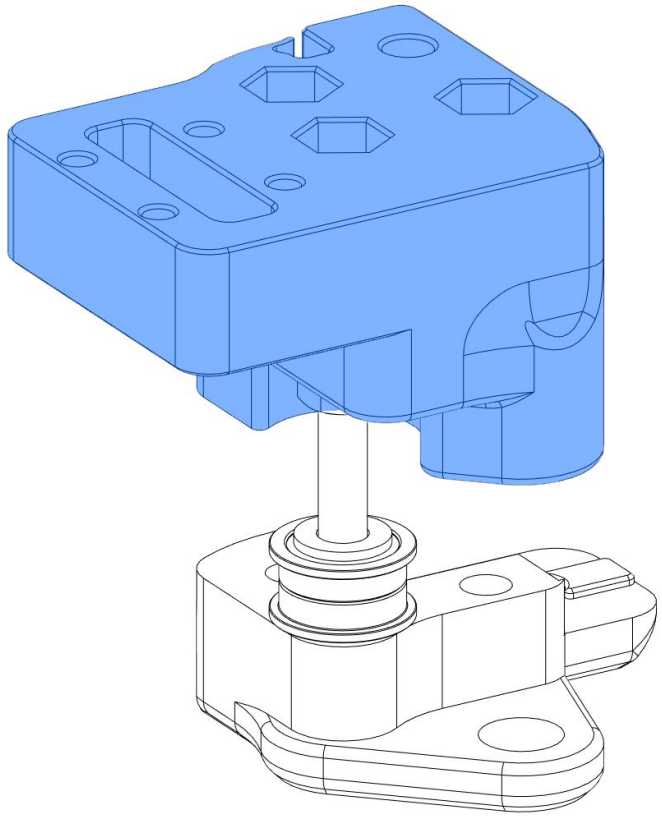
CABLE PATH

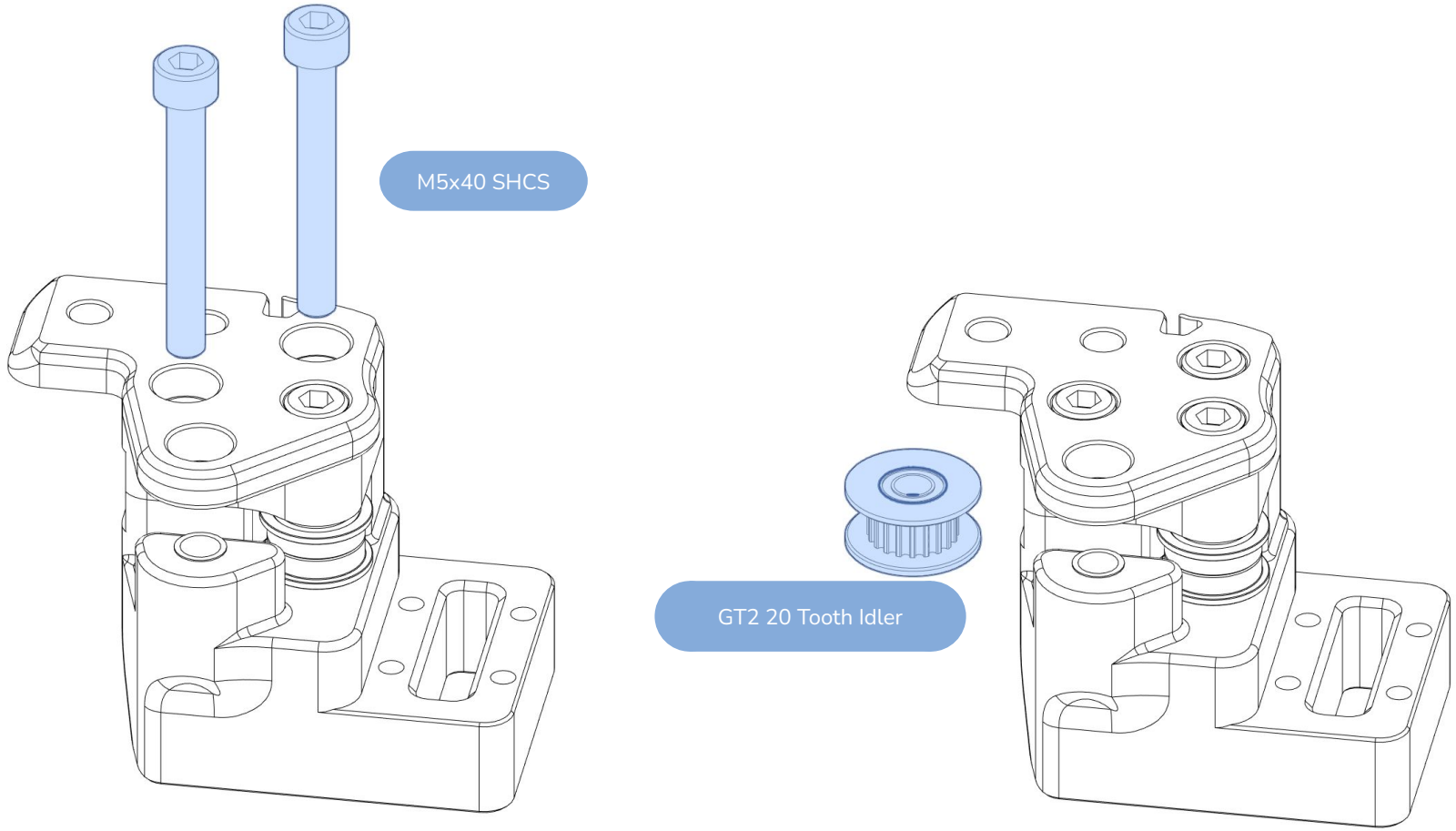
The printed parts for the right XY joint have a small channel to guide the endstop wires..

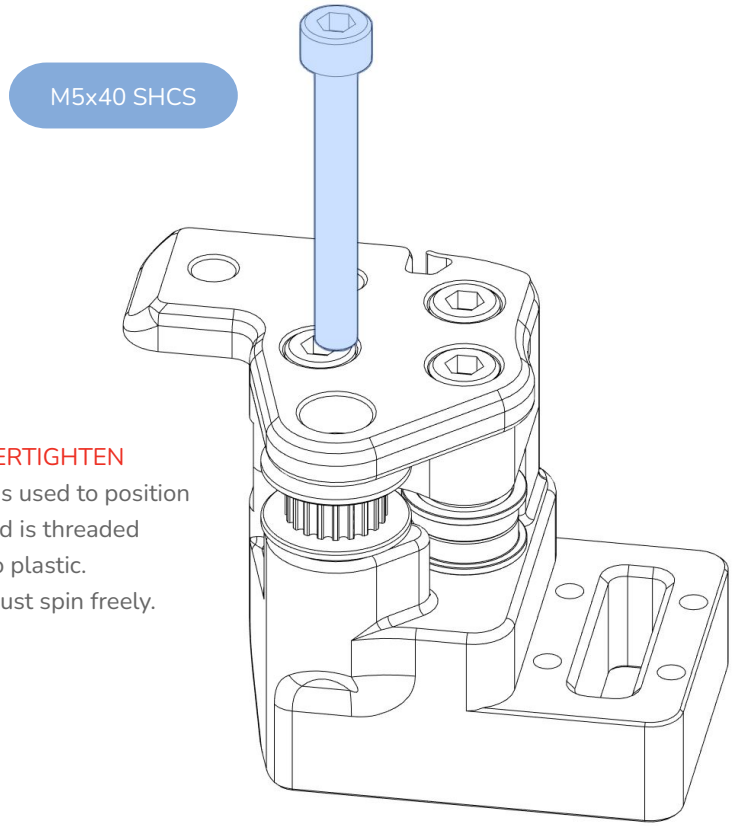





M5 Nut




M5x40 SHCS
GT2 20 Tooth Idler

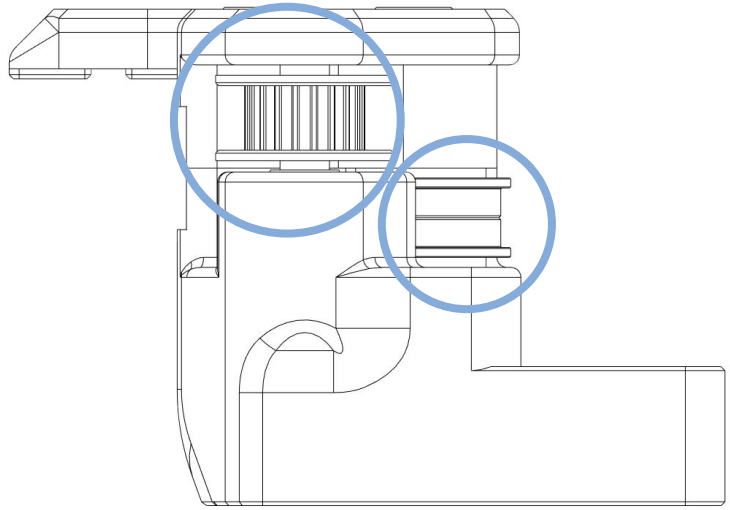


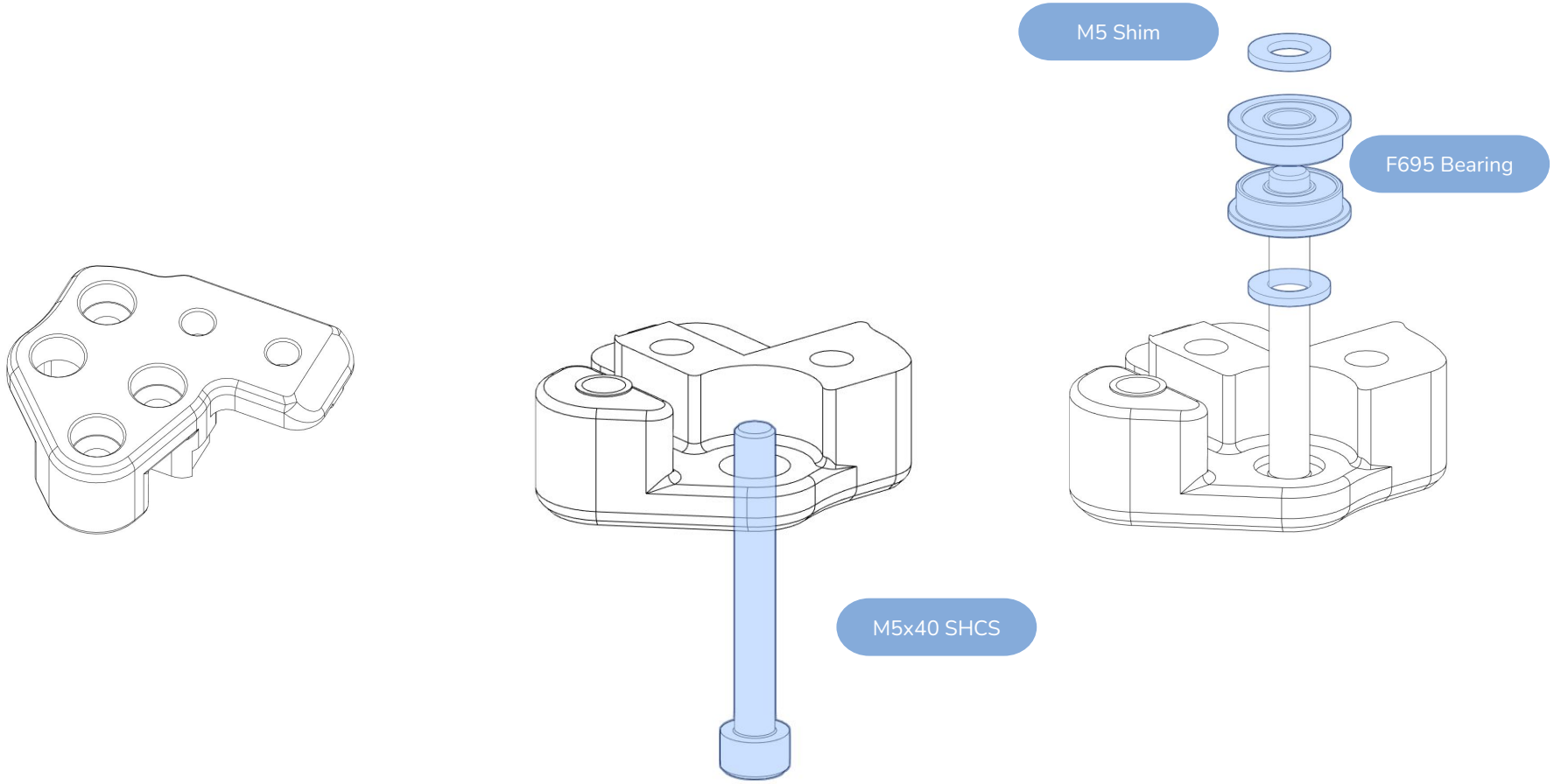
DON'T OVERTIGHTEN

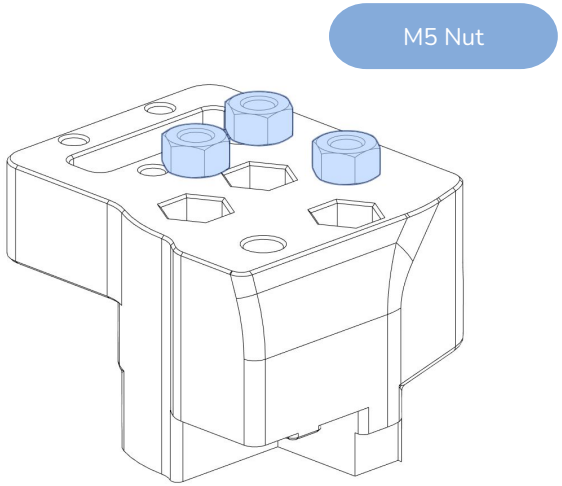
The screw is used to position the idler and is threaded directly into plastic.
The idler must spin freely.

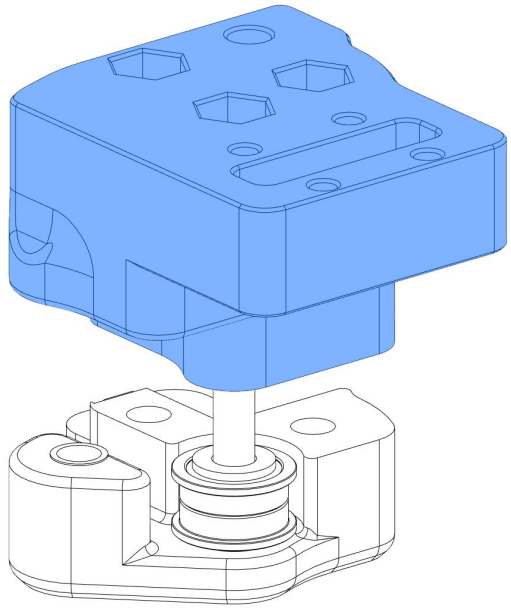
CHECK YOUR WORK

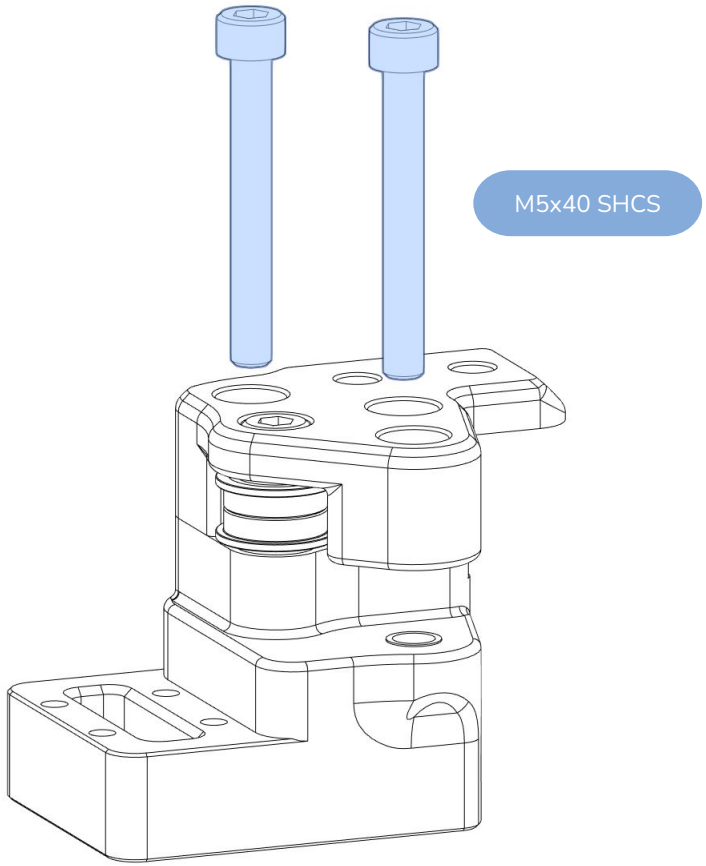
Compare your assembled part to the graphic shown here.

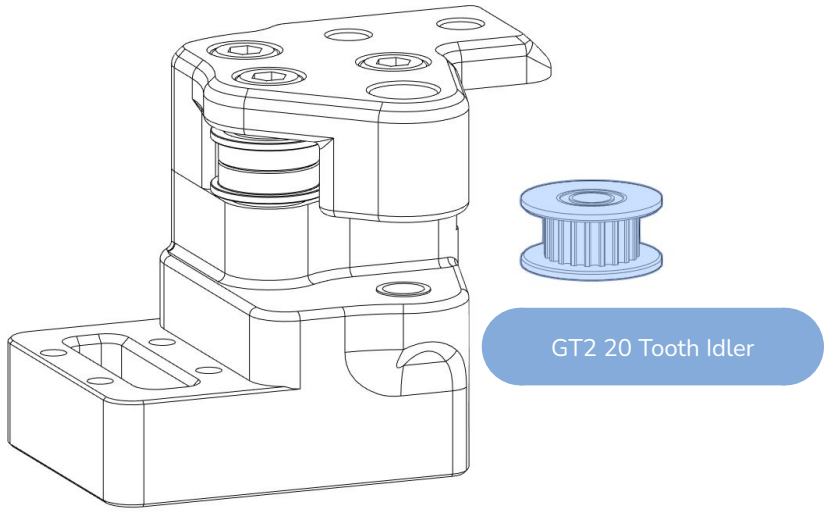


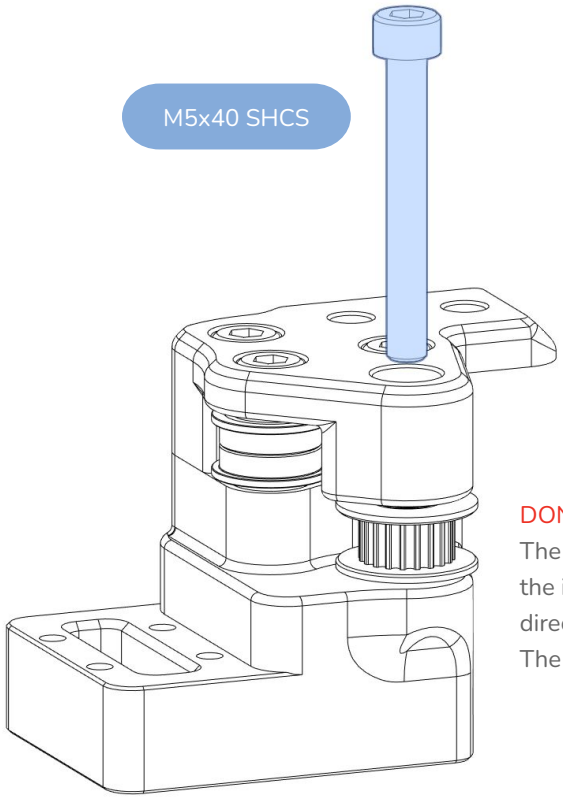

M5 Shim
F695 Bearing
M5x40 SHCS


M5 Nut



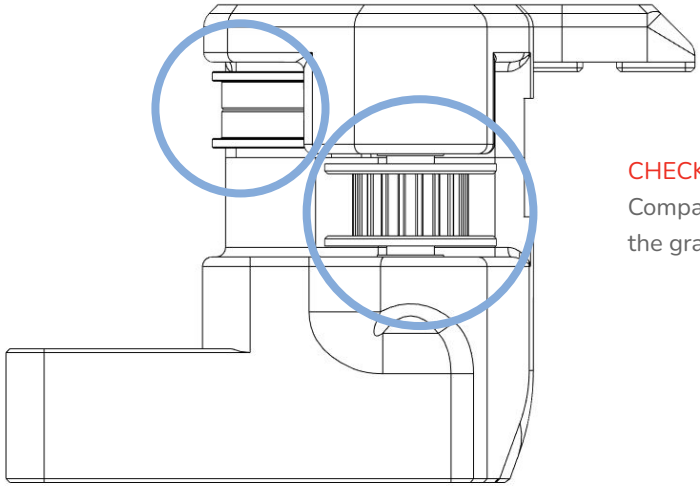

M5x40 SHCS


GT2 20 Tooth Idler



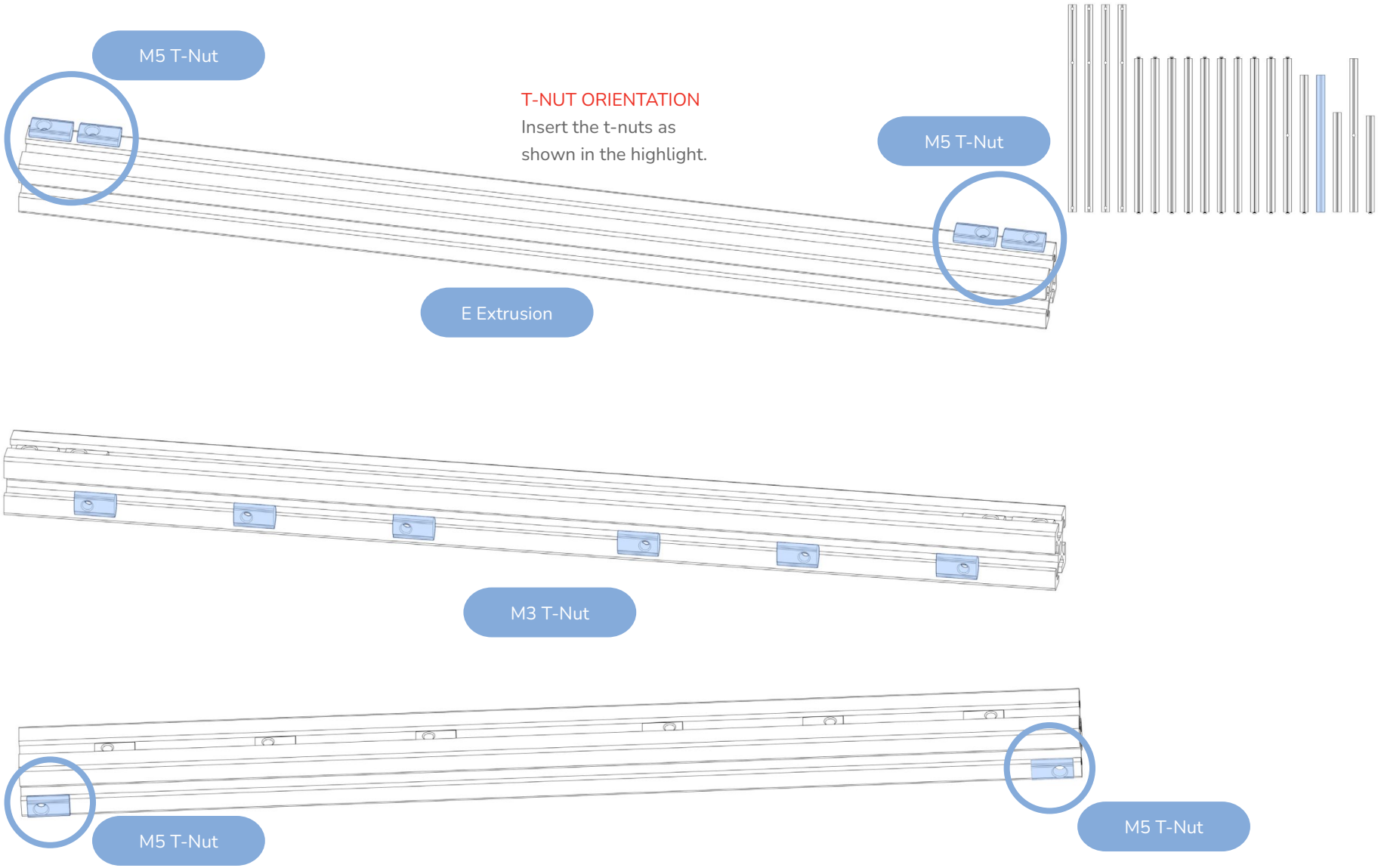
DON'T OVERTIGHTEN

The screw is used to position the idler and is threaded directly into plastic.
The idler must spin freely.

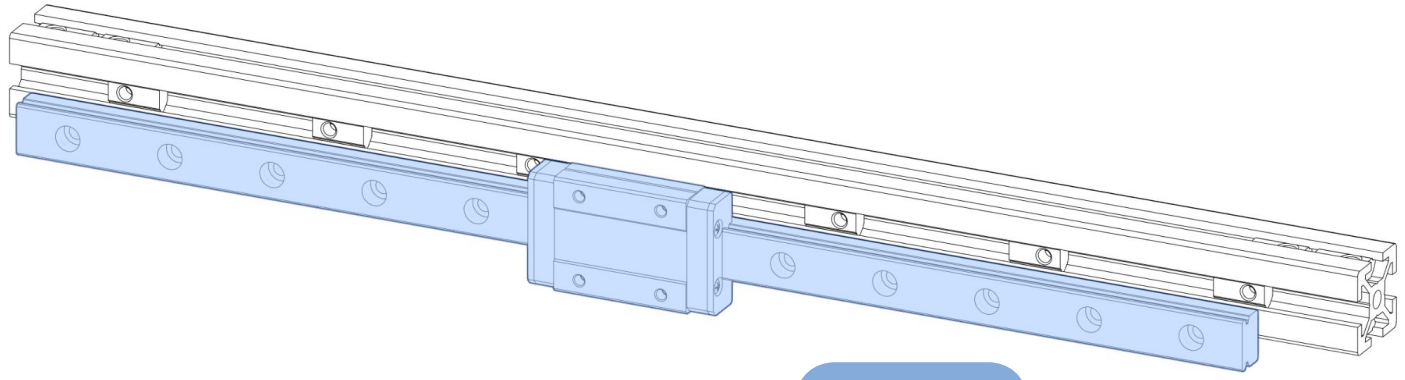


CHECK YOUR WORK

Compare your assembled part to the graphic shown here.


M5 T-Nut
T-NUT ORIENTATION
Insert the t-nuts as
shown in the highlight.
M5 T-Nut
E Extrusion
M3 T-Nut
M5 T-Nut
M5 T-Nut

Again, if you are building a bigger printer than the 250mm version, start at the outer holes and screw the rail down every second hole.
Small reminder (again): The carriages are designed to slide along the rail easily. This unfortunately also includes sliding off the rails. Dropping the carriage likely irreparably damages it. ALWAYS put the little black rubber Carriage stoppers back on the rail (into the holes without screws) until your printer ist up and runnning!

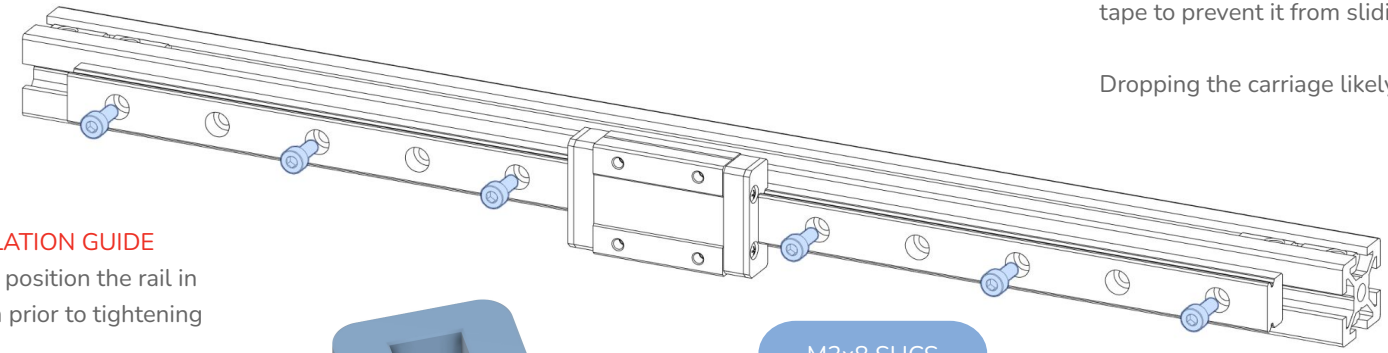


MGN12 Rail

MIND THE CARRIAGE

Temporarily secure the carriage with a piece of sticky tape to prevent it from sliding off the rail.

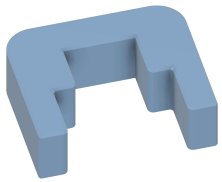
Dropping the carriage likely irreparably damages it.

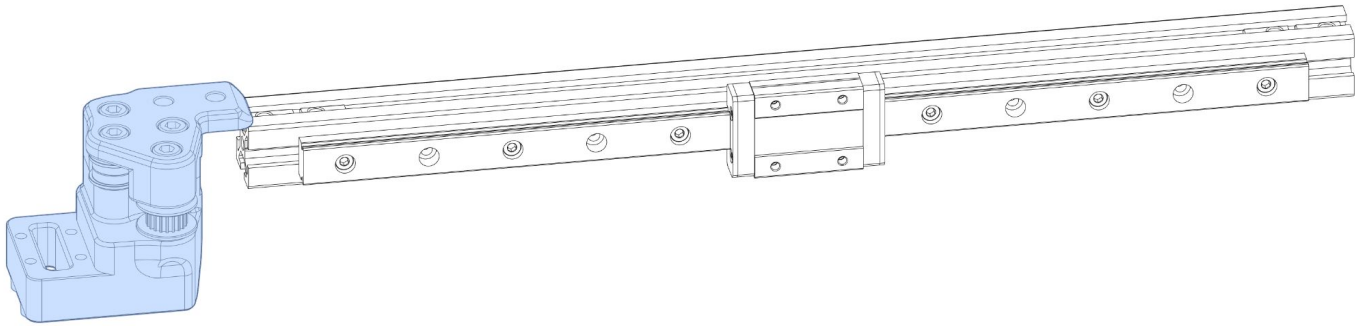


M3x8 SHCS

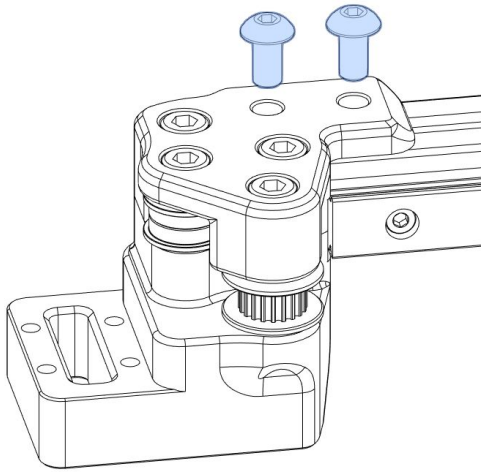
CENTERED RAIL INSTALLATION GUIDE

Use the MGN12 guides to position the rail in the center of the extrusion prior to tightening the screws.



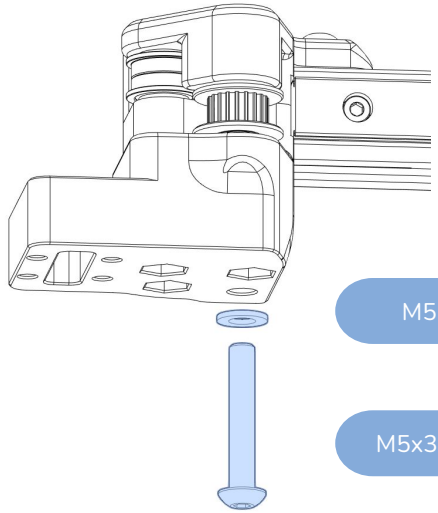


M5x10 BHCS



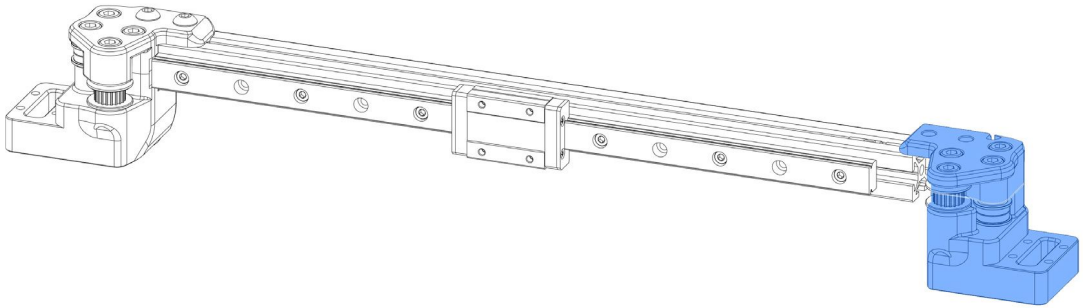
LEAVE LOOSE

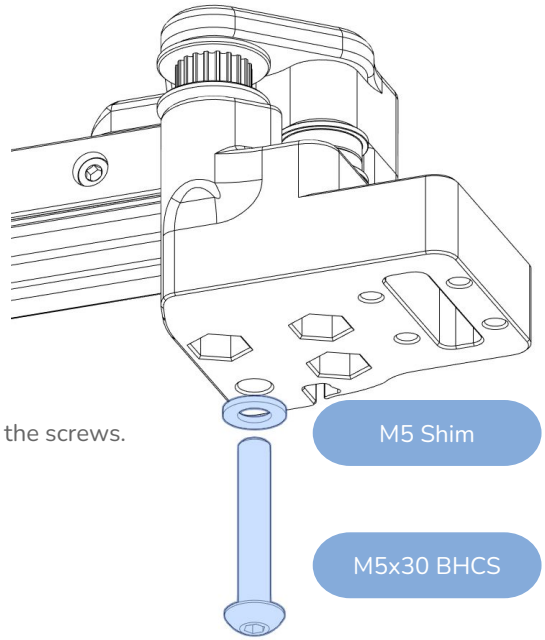
Lightly tighten the screws.



M5 Shim

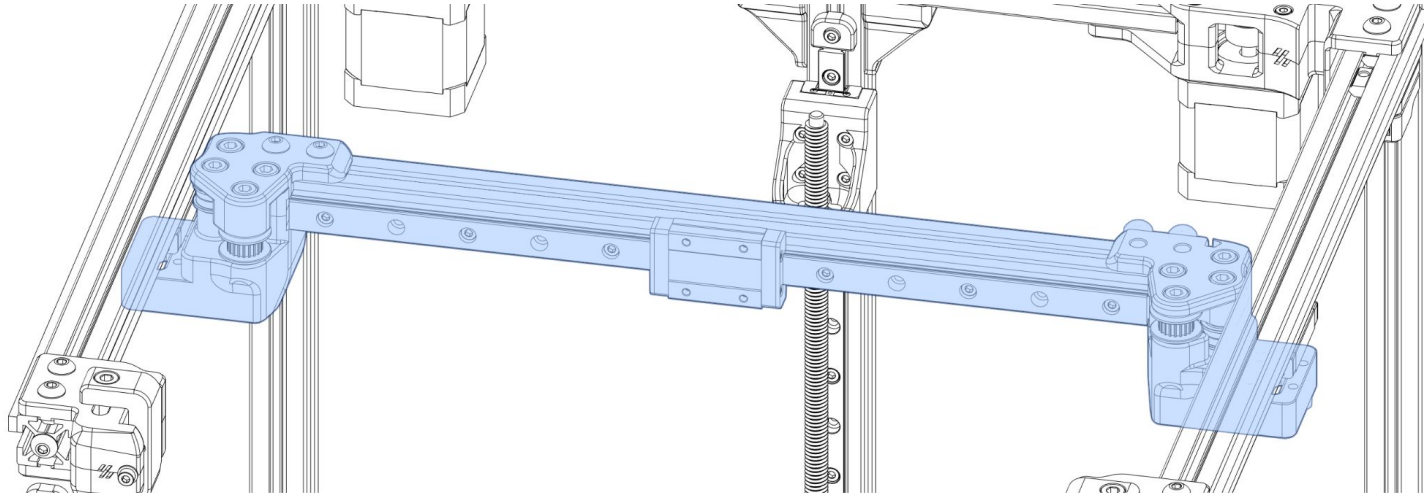
M5x30 BHCS

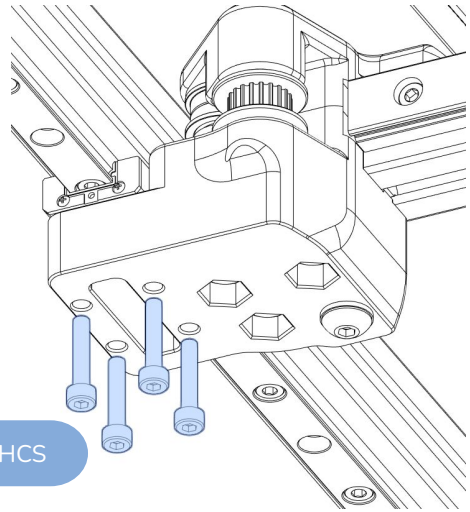




LEAVE LOOSE

Lightly tighten the screws.

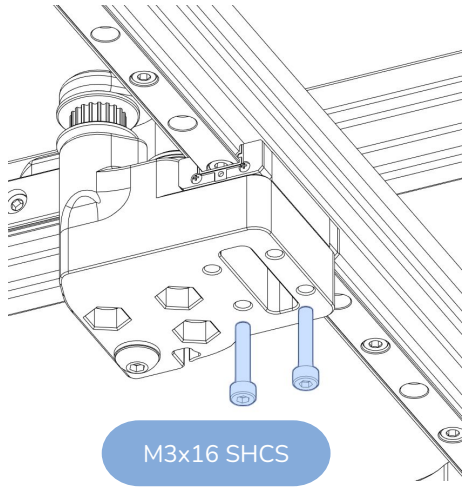




M3x16 SHCS

MISSING SOME SCREWS?

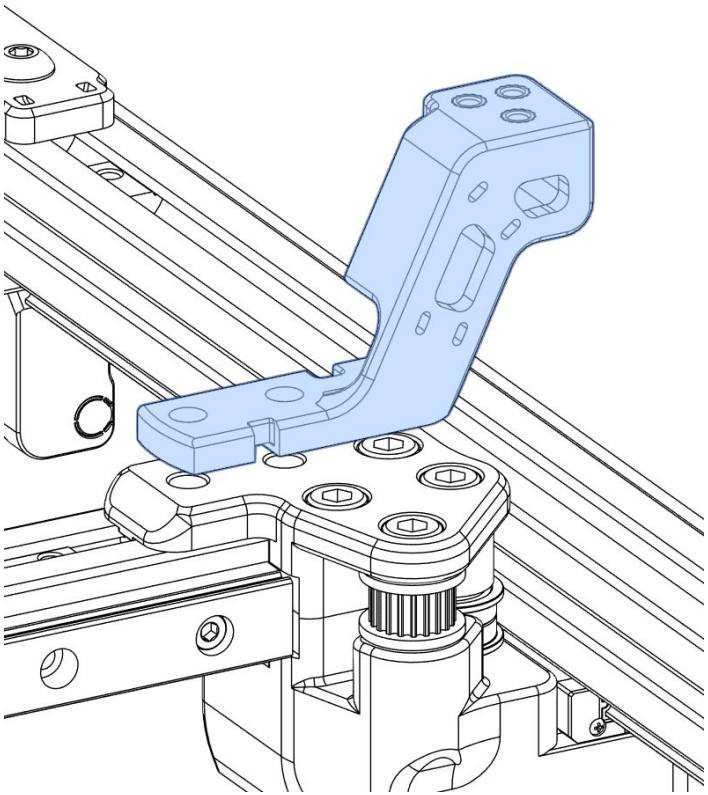
The other two screws will be installed while mounting the XY Endstop Pod.

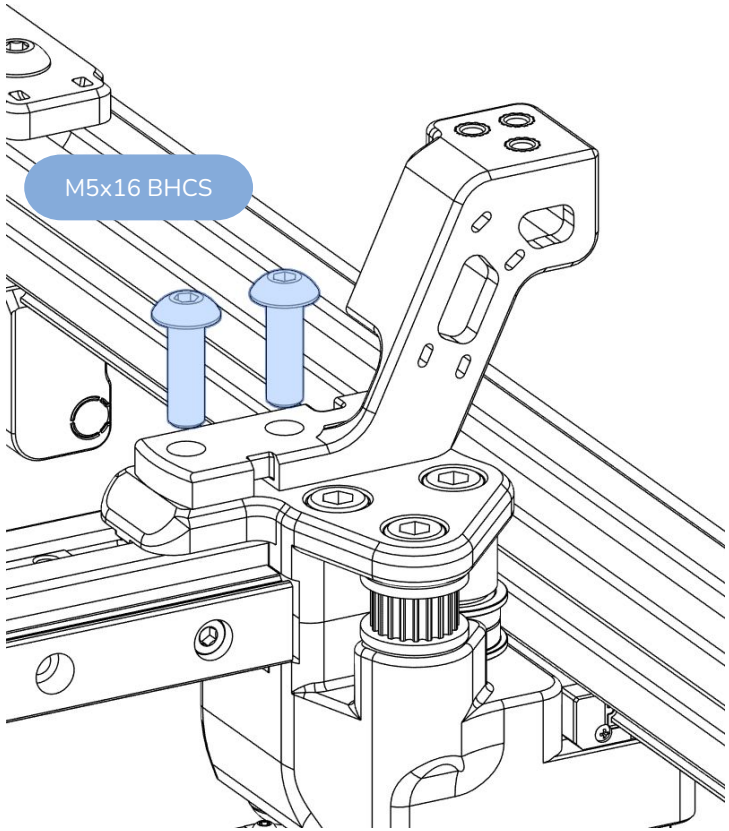


M3x16 SHCS

Install all 4 screws on the right XY Joint.

Leave this step out, you dont use a cable chain on X and Y.





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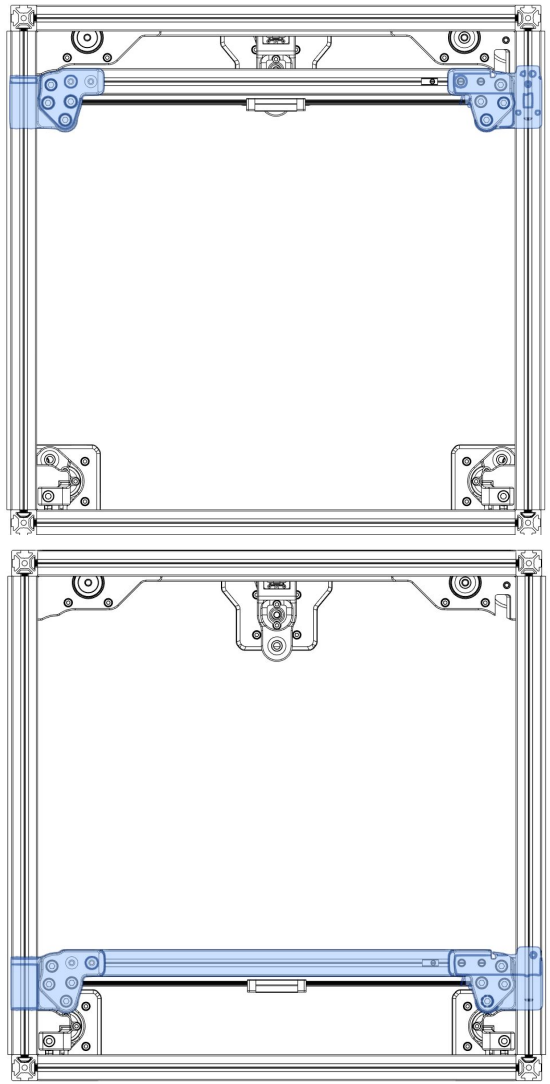
SQUARING THE GANTRY

Move the gantry all the way back until it hits the A and B drive on both sides.

Fully tighten all screws on the X axis.

https://voron.link/cekh81l



The VORON Legacy is a modernized design true to the spirit of the original VORON 1.0.

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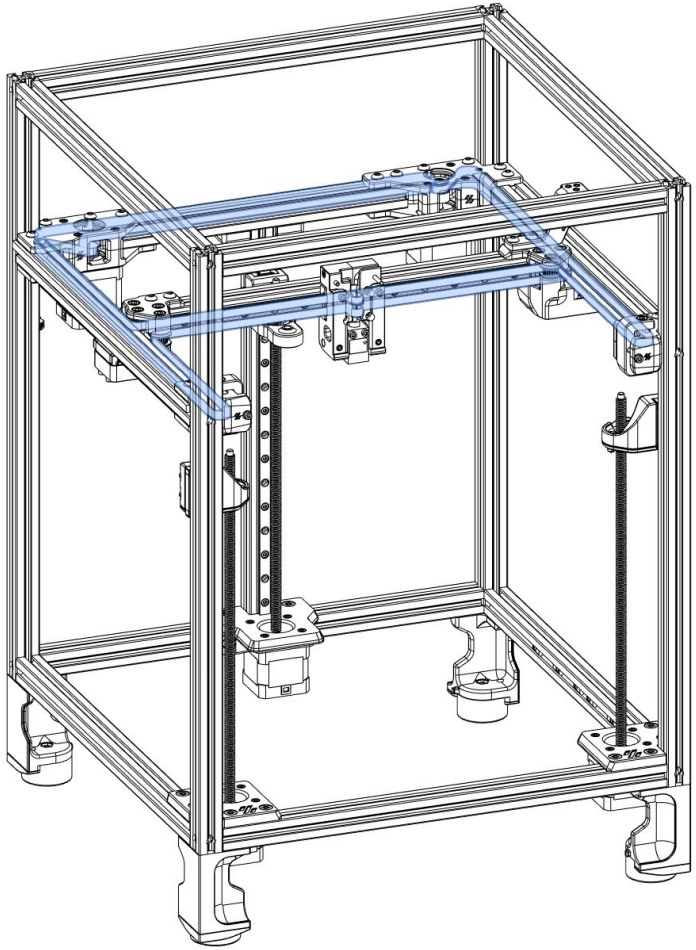
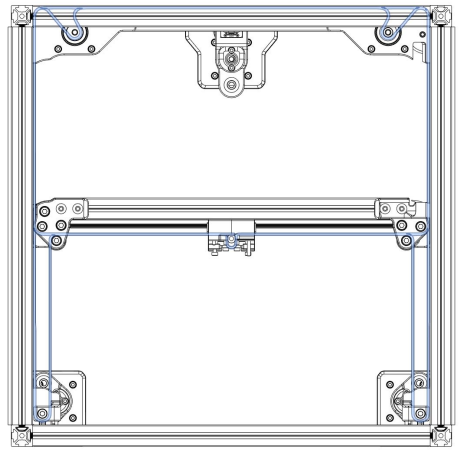


THE VORON BELT PATH

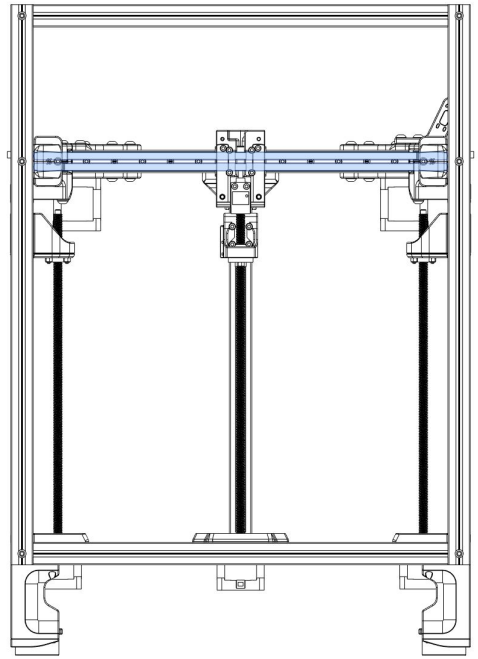
VORON printers use a belt path based on the popular CoreXY pattern.

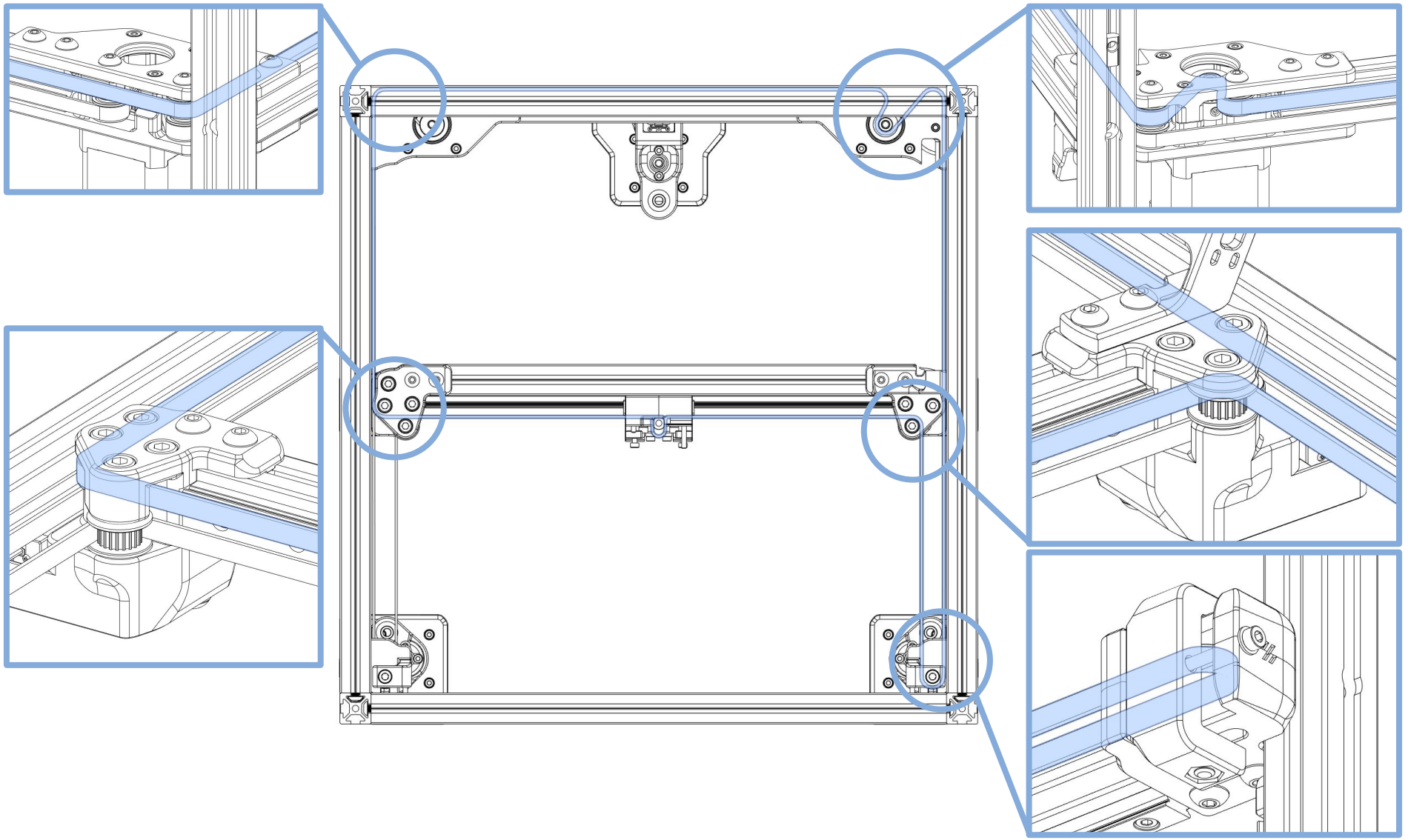
The individual belt paths are stacked on top of each other and the crossing often found in CoreXY designs is omitted. Compared to many other implementations, the motors are moved to a less intrusive position. To learn more about the principles behind CoreXY visit https://voron.link/ef72dd6.

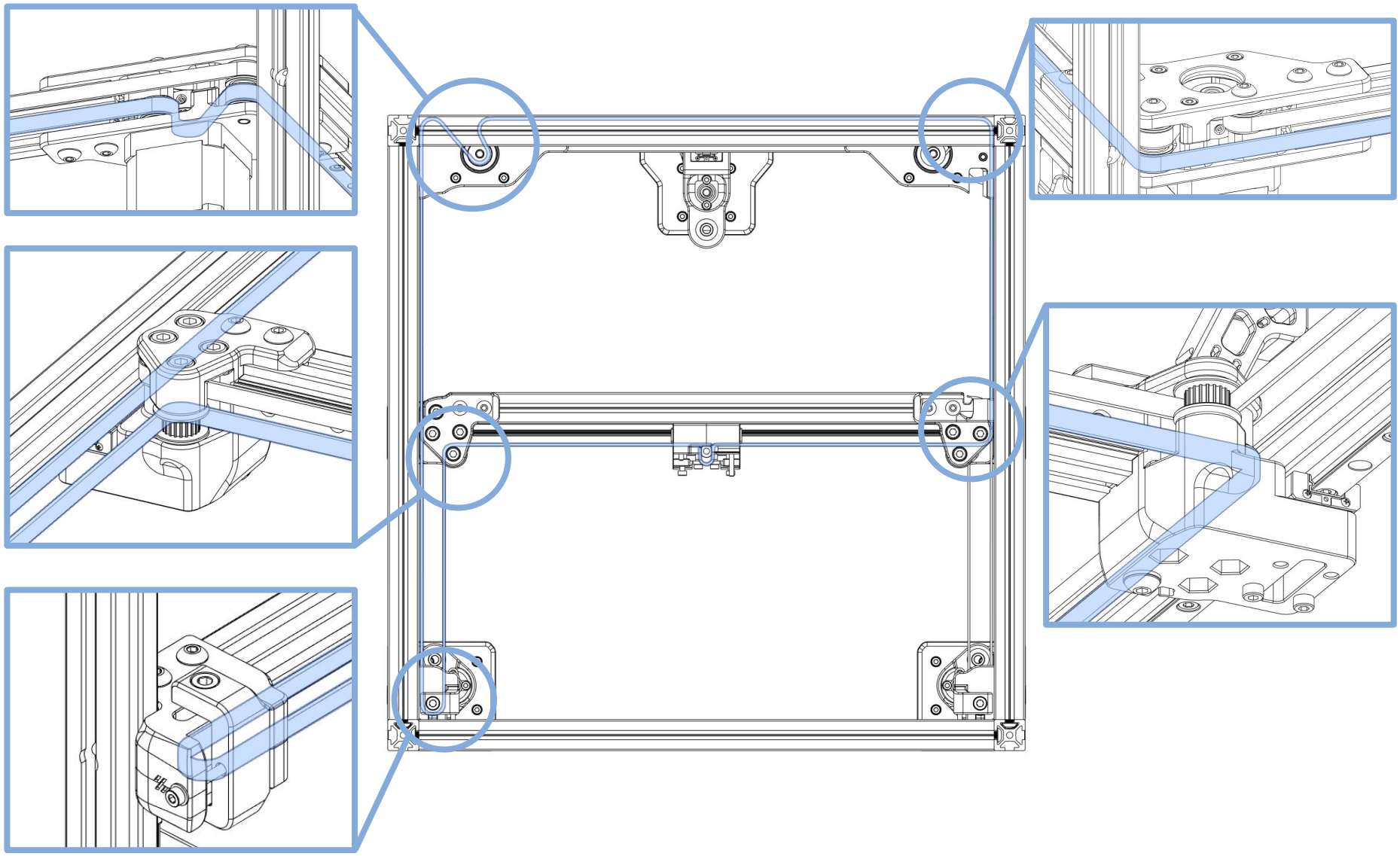


Equal belt tension is important to the proper function of a CoreXY motion system.

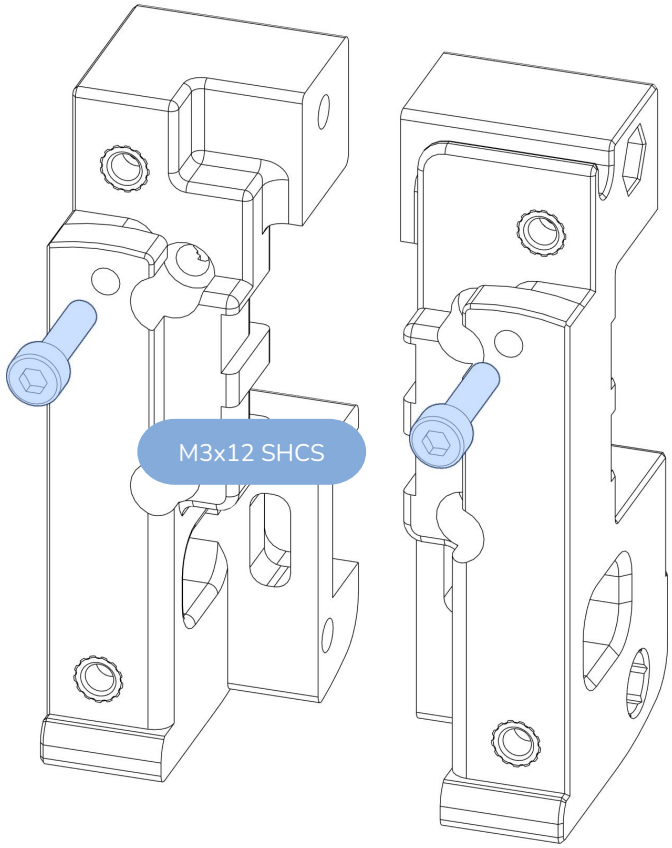
We recommend to run one belt to get the required length, remove the belt from the printer and cut the second belt to the exact same length.
As both belt paths have the same length this is an easy way of getting a consistent tension.

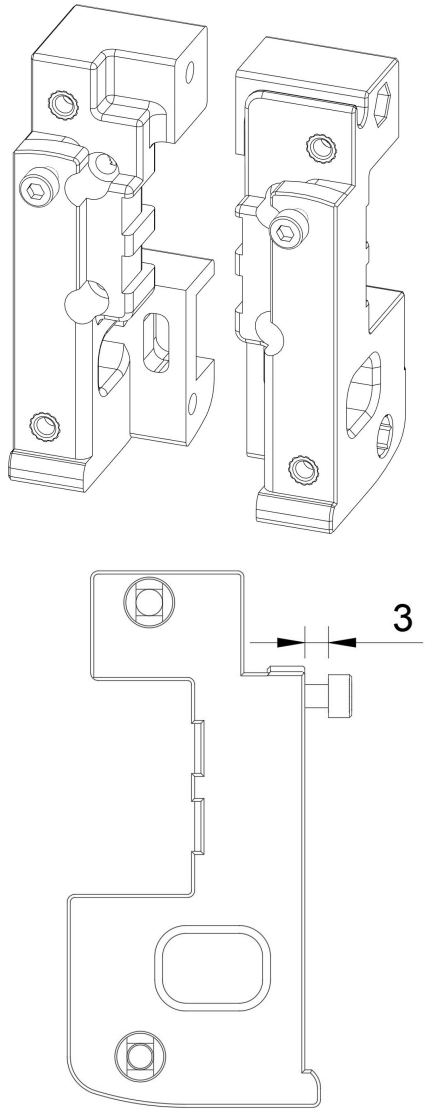




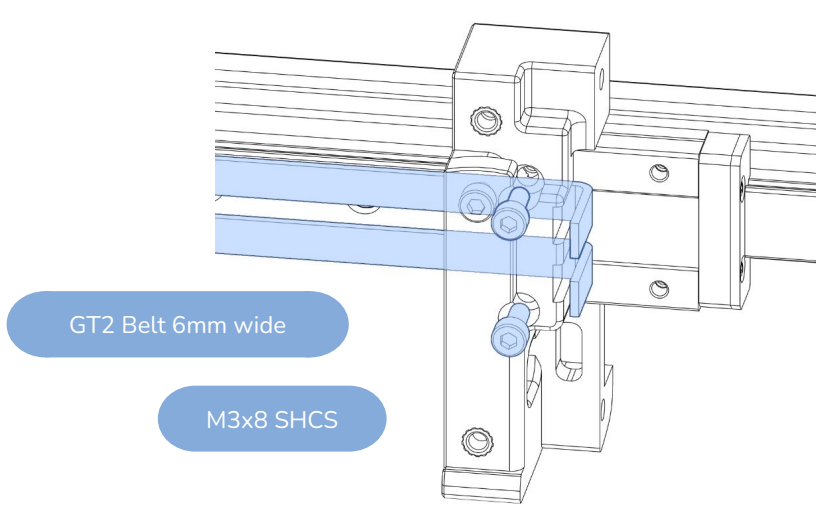


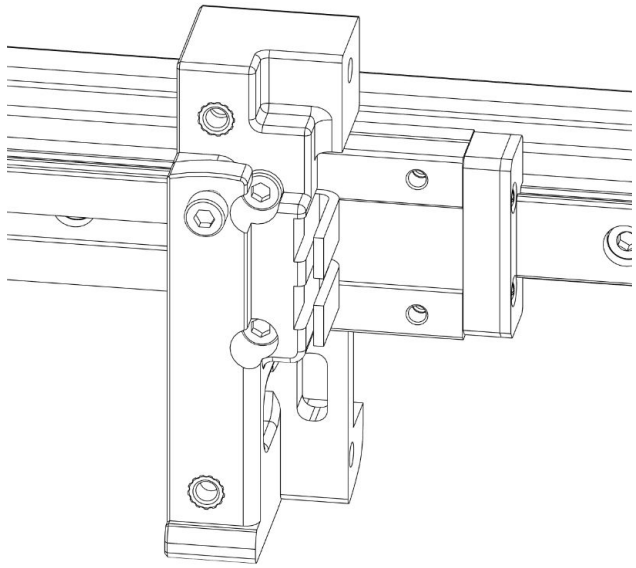
You use a different X carriage / Probe. Fix the belts with some tape and proceed with the belt Routing. After you routed the belts, fix them on the other side with some tape as well.





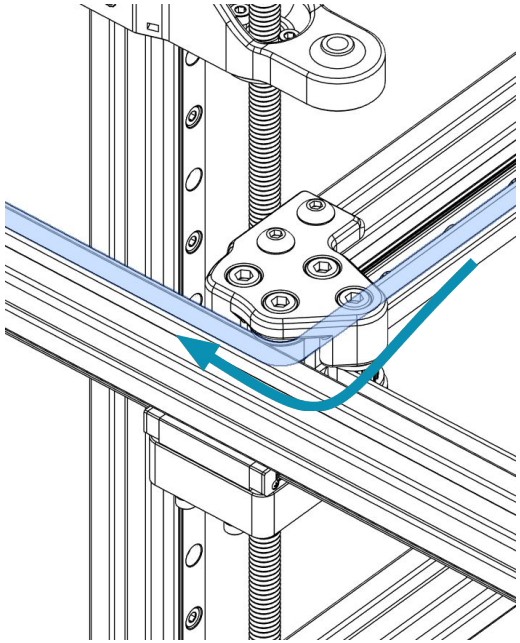
You use a different X carriage / Probe. Fix the belts with some tape and proceed with the belt Routing. After you routed the belts, fix them on the other side with some tape as well.

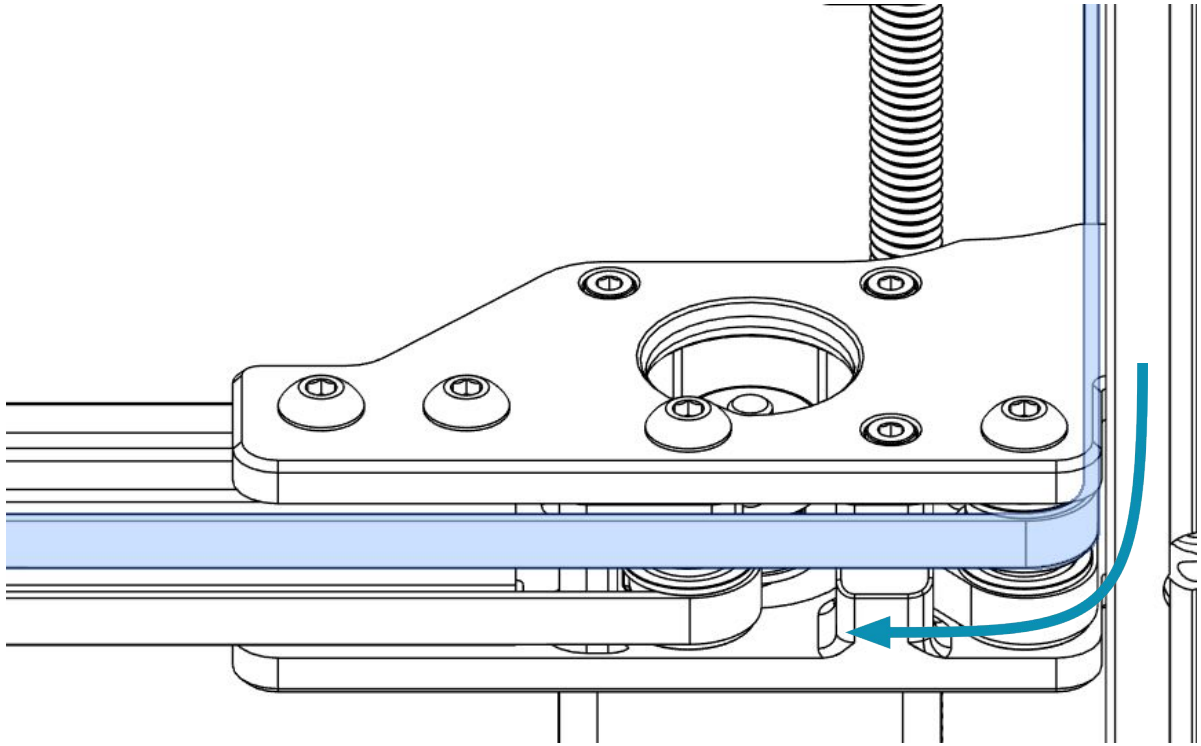




CLAMP BELTS

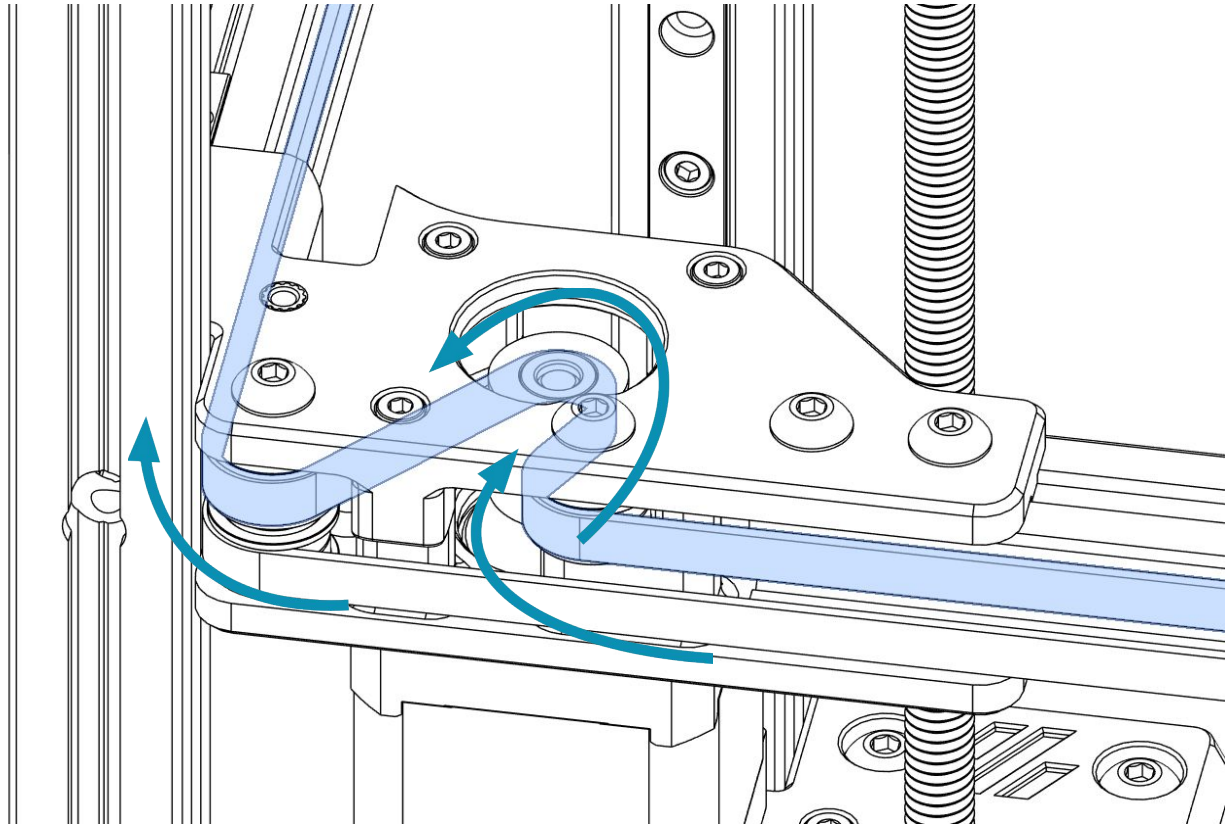
Clamp both A and B belts in place by installing the left X carriage part. The belt teeth face away from the extrusion.

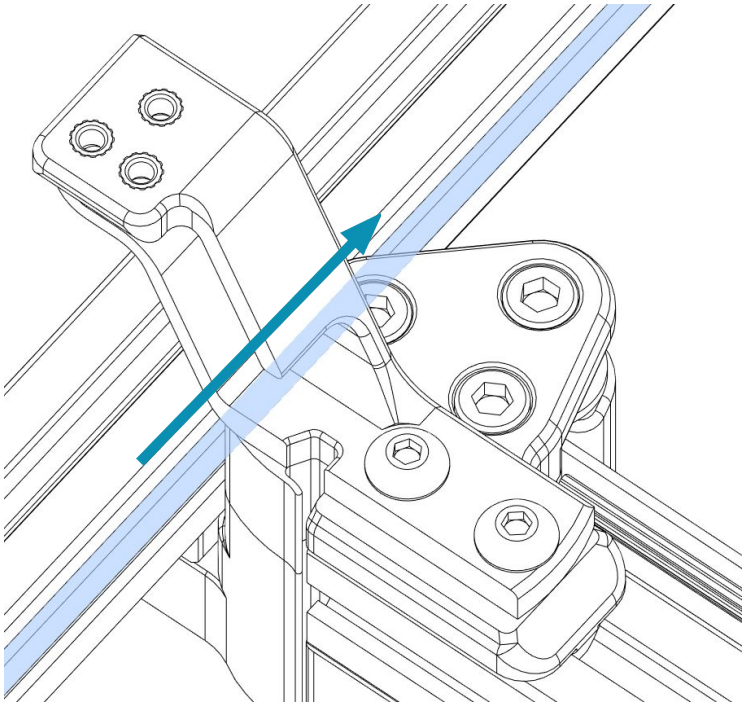


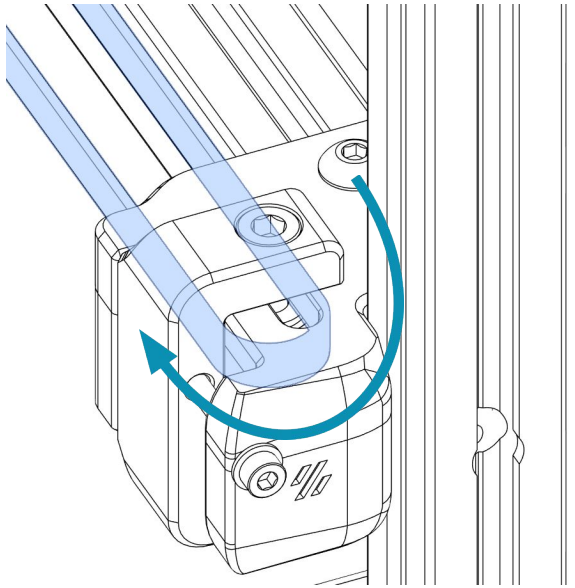


A BELT ROUTING

Follow the path pointed out by the arrows. Needle nose pliers, tweezers or similar tools can help in this step.

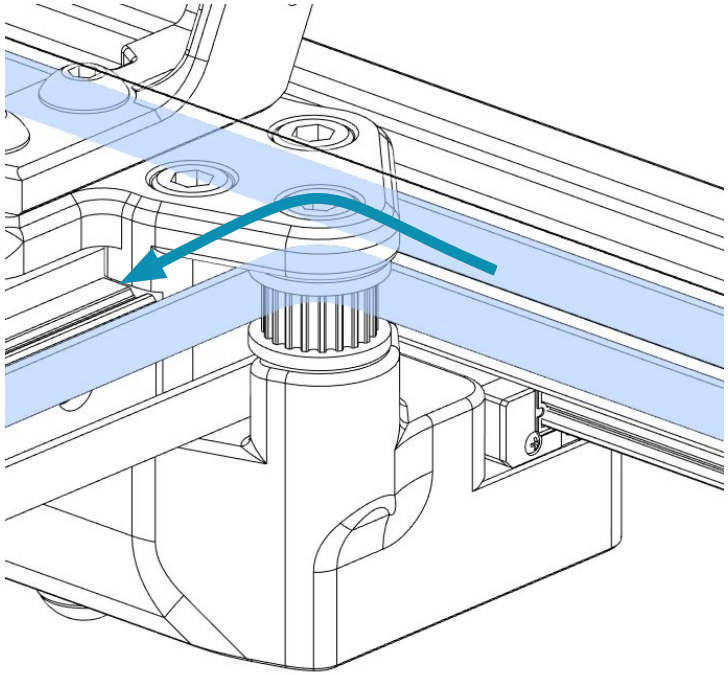


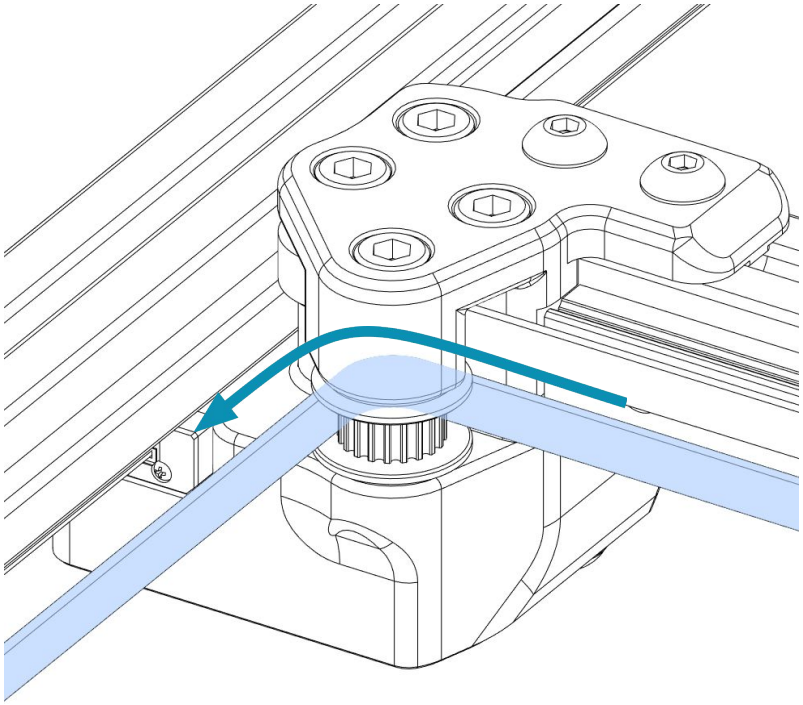




BELTING IDLERS

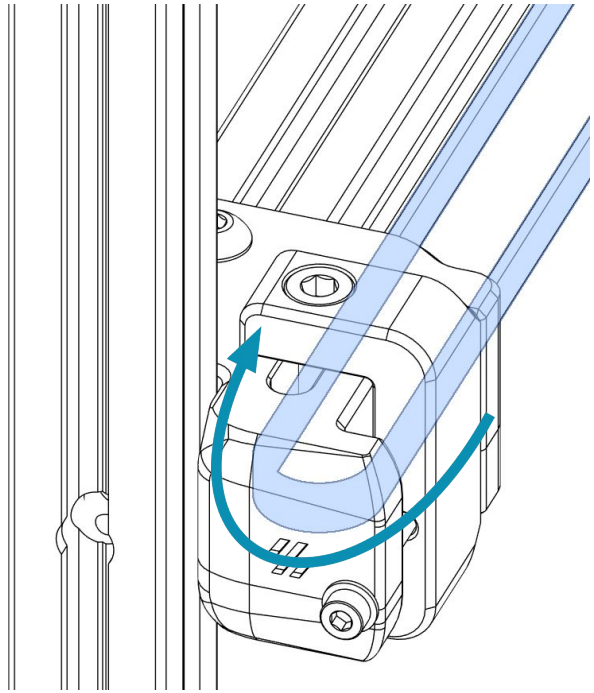
If you're having trouble guiding the belts around the bearing stack temporarily remove the M3x40 SHCS to get better access.





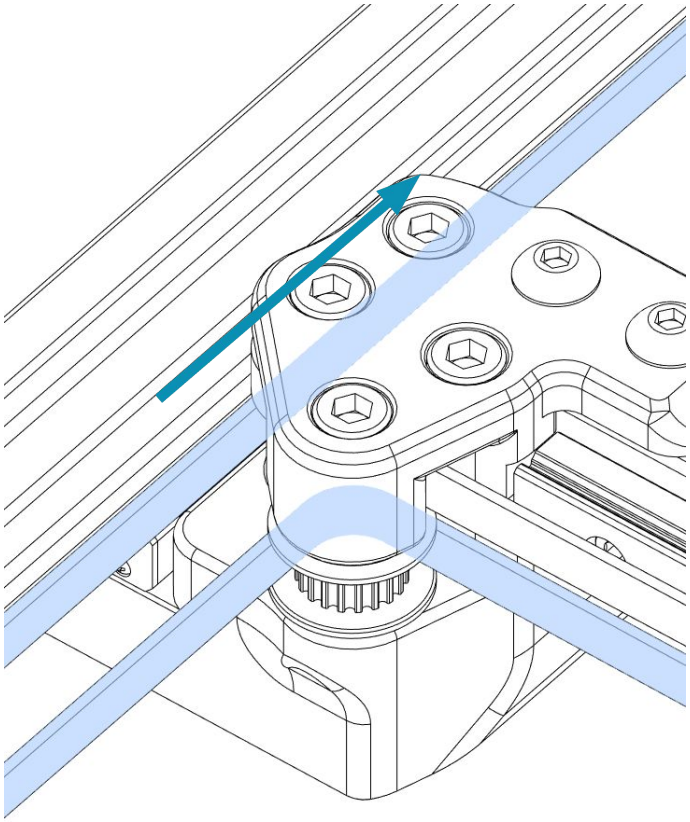
B BELT ROUTING

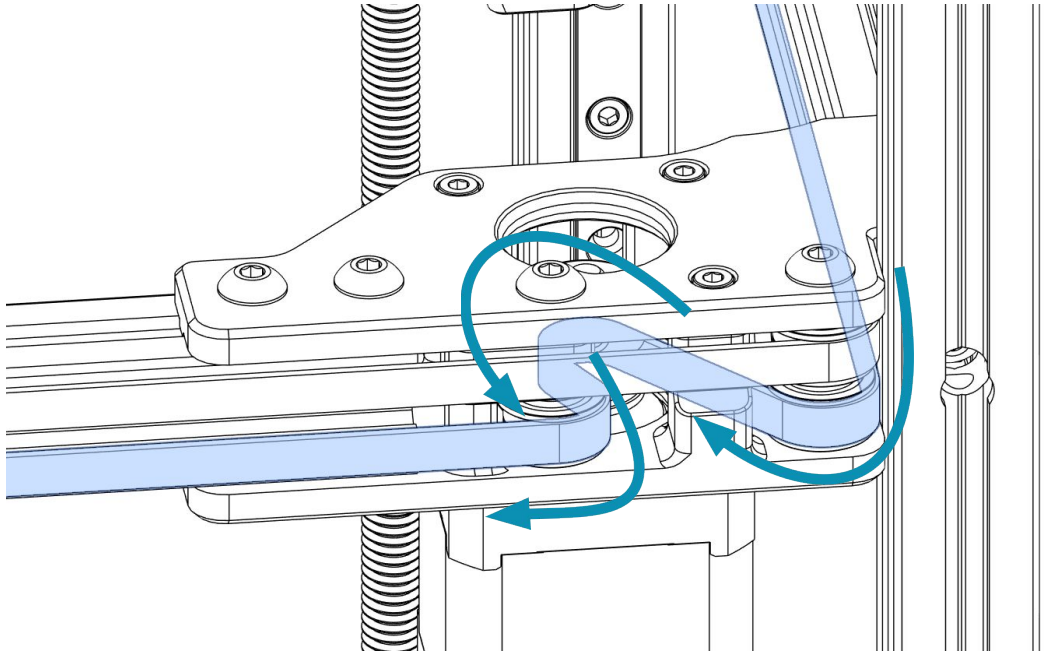
Follow the path pointed out by the arrows. Needle nose pliers, tweezers or similar tools can help in this step.

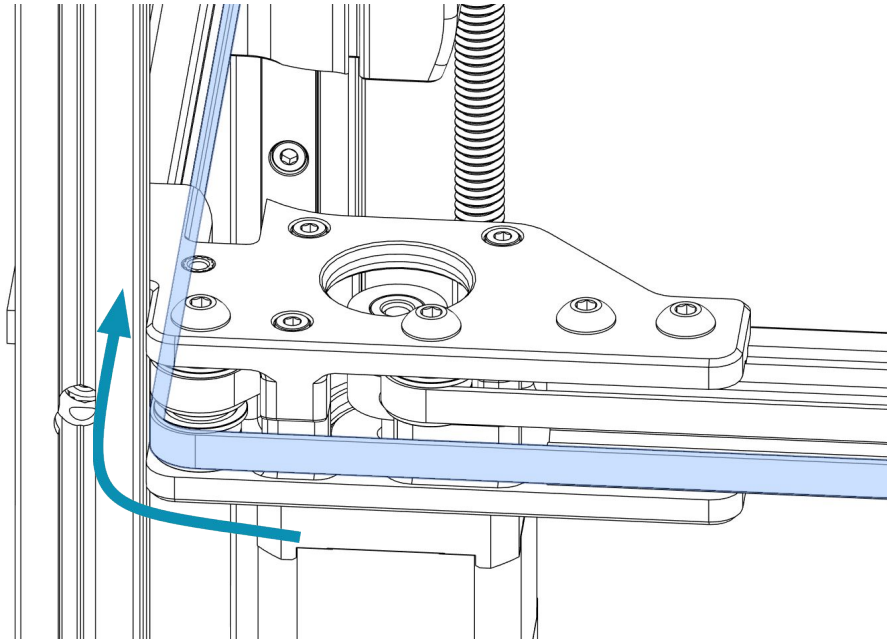


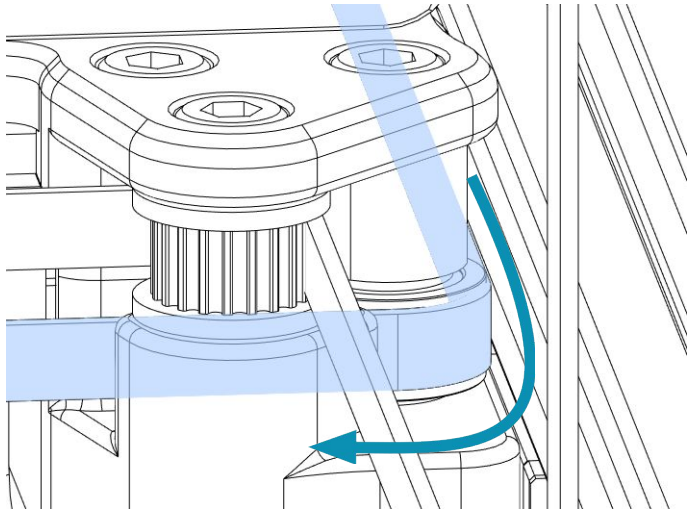
BELTING IDLERS

If you're having trouble guiding the belts around the bearing stack temporarily remove the M3x40 SHCS to get better access.

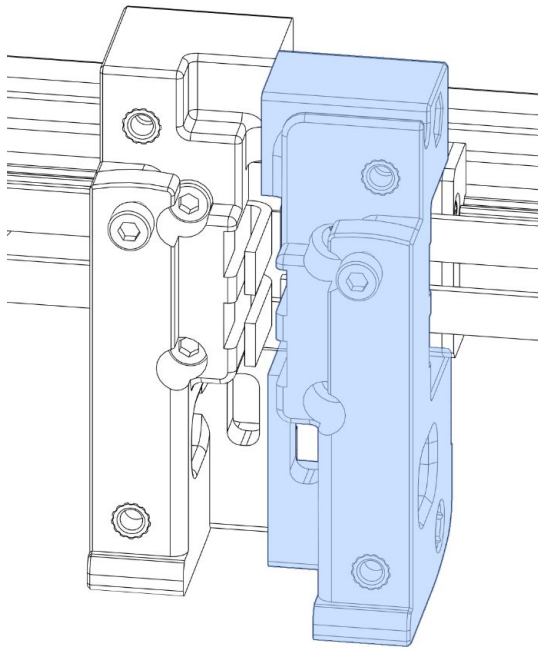








Skip these steps, you will do it later after tap is assembled.

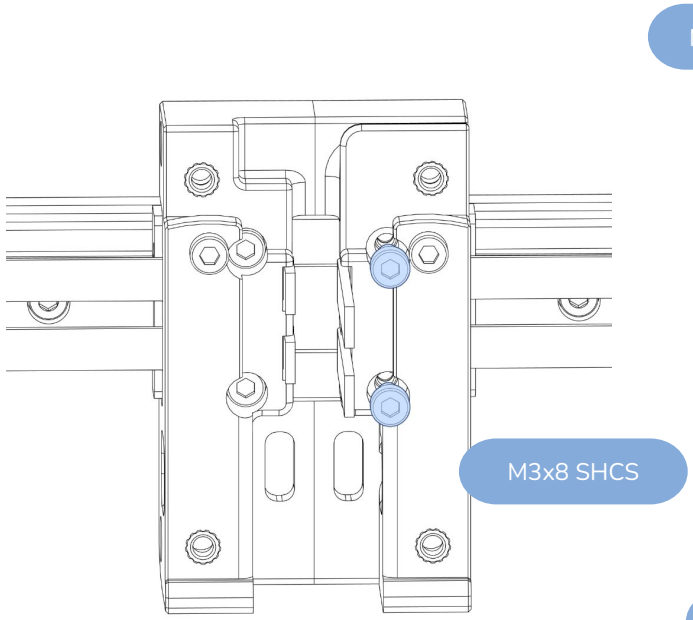


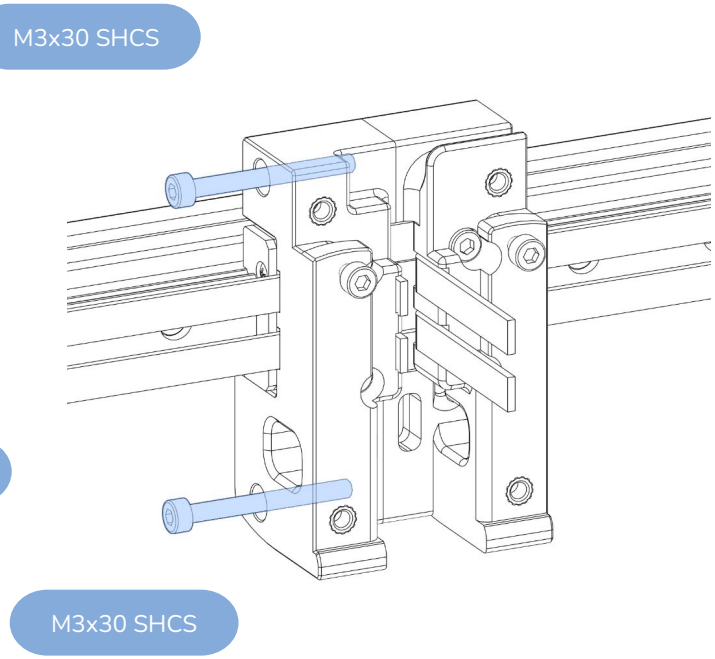
X CARRIAGE

Use the second part of the X carriage to capture the belt ends.

131

Skip these steps, you will do it later after tap is assembled.





FIX BELTS

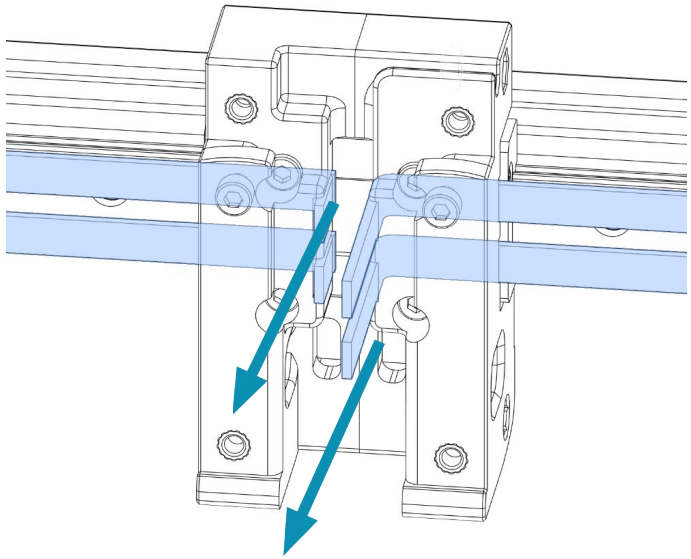
Lightly tighten the screws.
The belt must still be able to move.

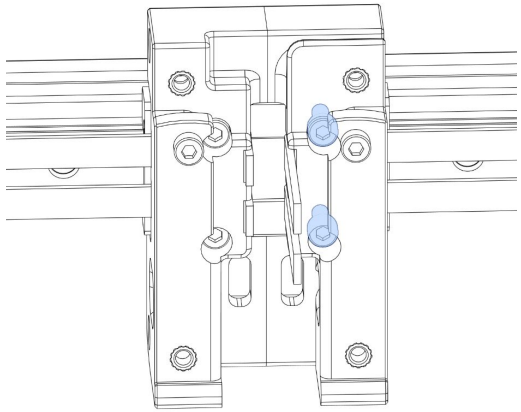
LEAVE LOOSE

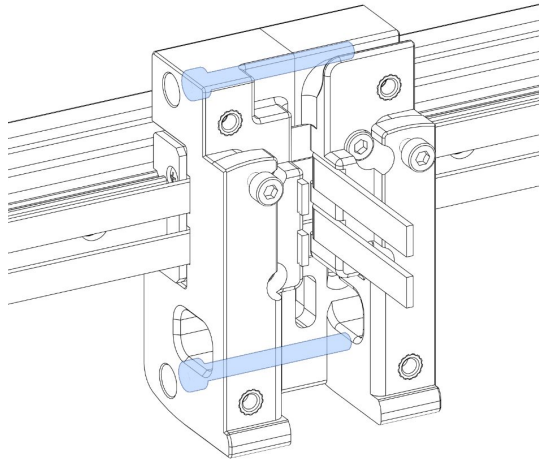
Lightly tighten the screws.

132

Skip these steps, you will do it later after tap is assembled.







PULL TIGHT

Grab both belt ends with a pair of pliers and pull the belt tight.

As both belts are cut to the exact same total length and the belt paths are equal in length in this design, make sure the same length of belt protrudes from the carriage.

TIGHTEN SCREWS

Fully tighten the carriage screws.

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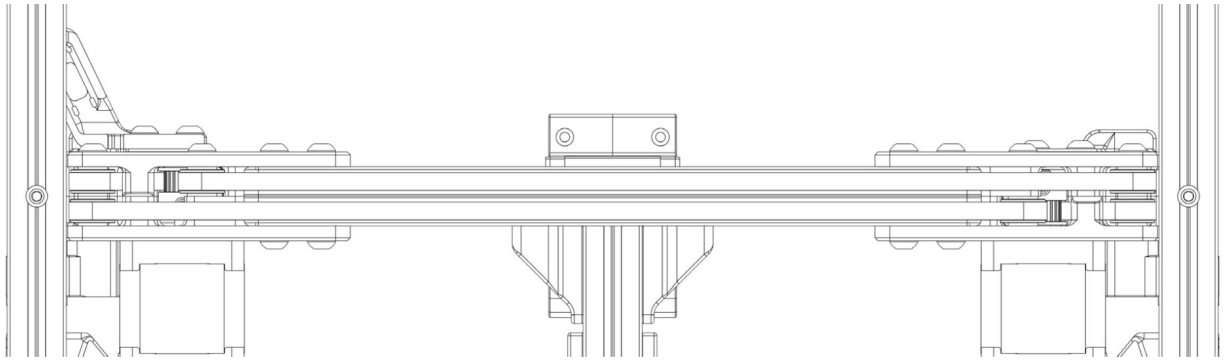
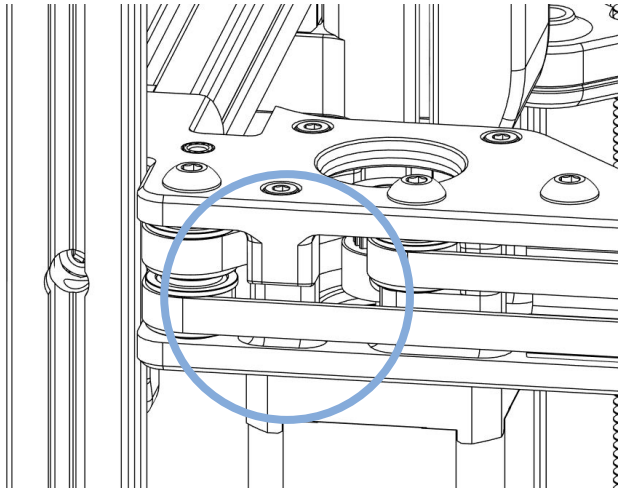
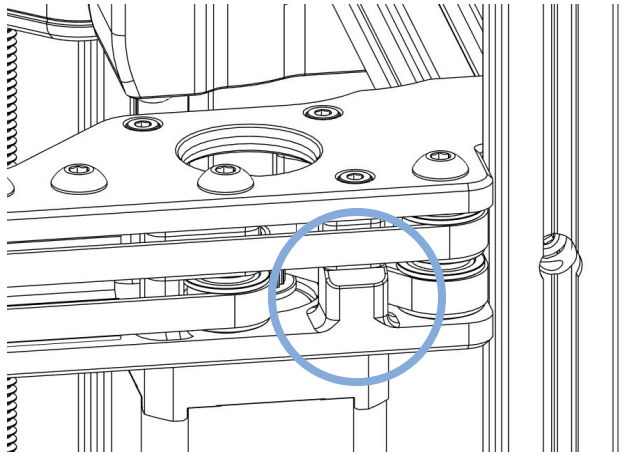


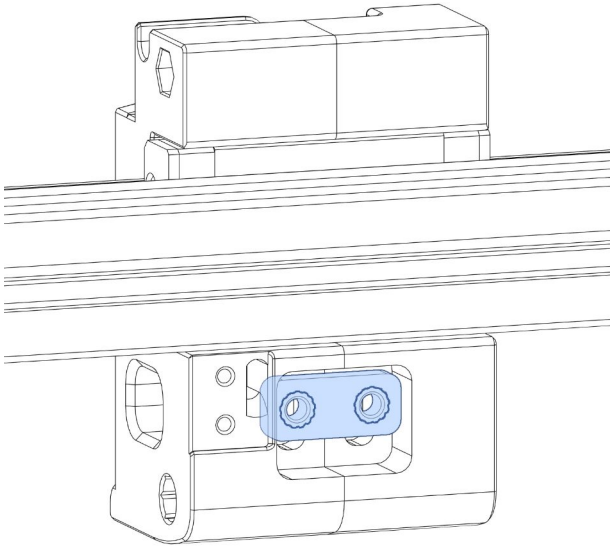


CHECK YOUR WORK

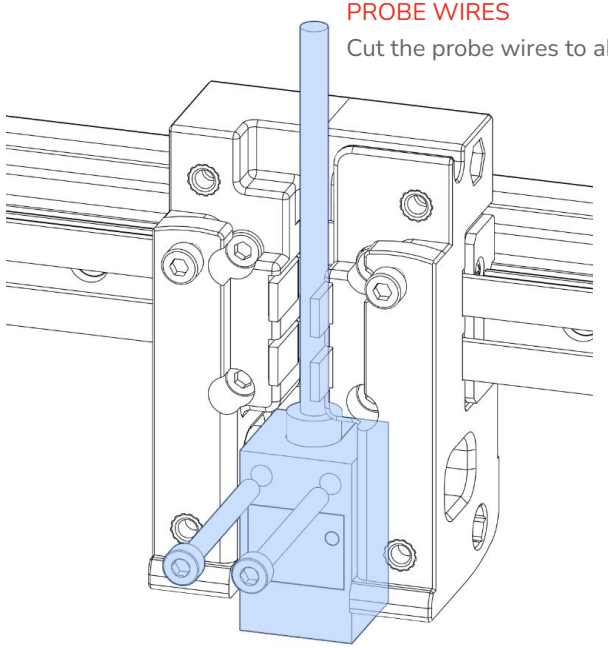
Make sure that the belt is not riding on the plastic parts.



Skip these steps since you don't use the stock X carriage but instead you use tap.



M3x30 SHCS



PROBE WIRES

Cut the probe wires to about 150mm.

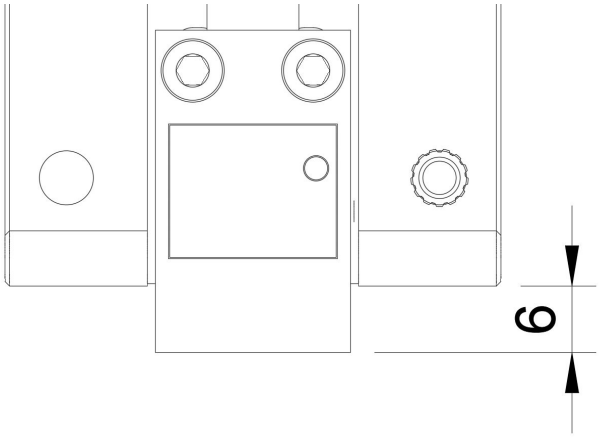
OTHER PROBE TYPES

The picture shows the recommended Omron TL-Q5MC probe.

Other probes with a similar form factor and characteristics might work as well. A design for a PINDA probe adapter is included in the released files.

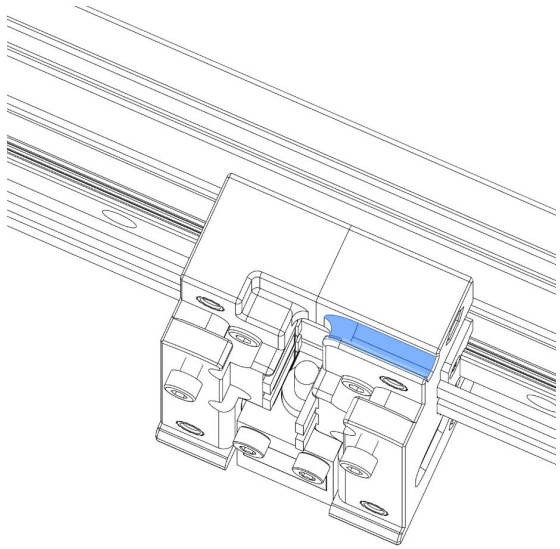
135

Skip these steps since you don't use the stock X carriage but instead you use tap.



ADJUST PROBE POSITION

The position can be fine-tuned later. Set an initial position of about 6mm below the plastic part.

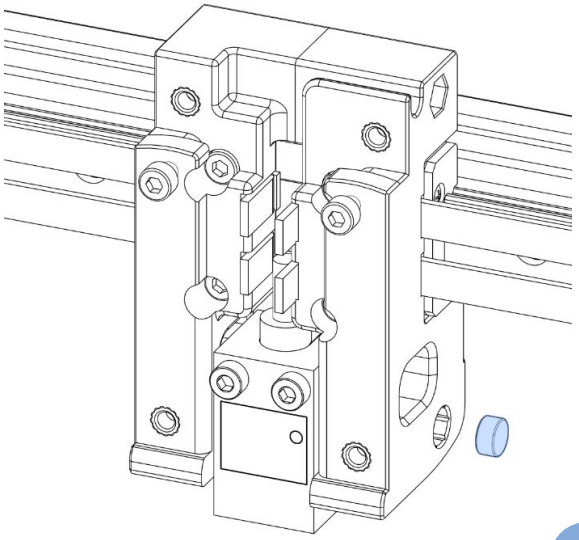


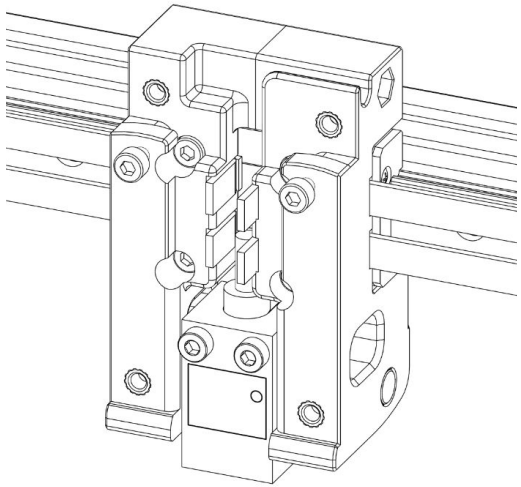
CHANNEL FOR PROBE CABLE

Guide the probe cable into the highlighted slot.

136

Skip these steps since you don't use the stock X carriage but instead you use tap.





Magnet

OPTION: HALL EFFECT ENDSTOP

If you are using a Hall Effect Endstop insert a 3x6 magnet into the highlighted position during calibration. See: https://voron.link/hxd3cv0.

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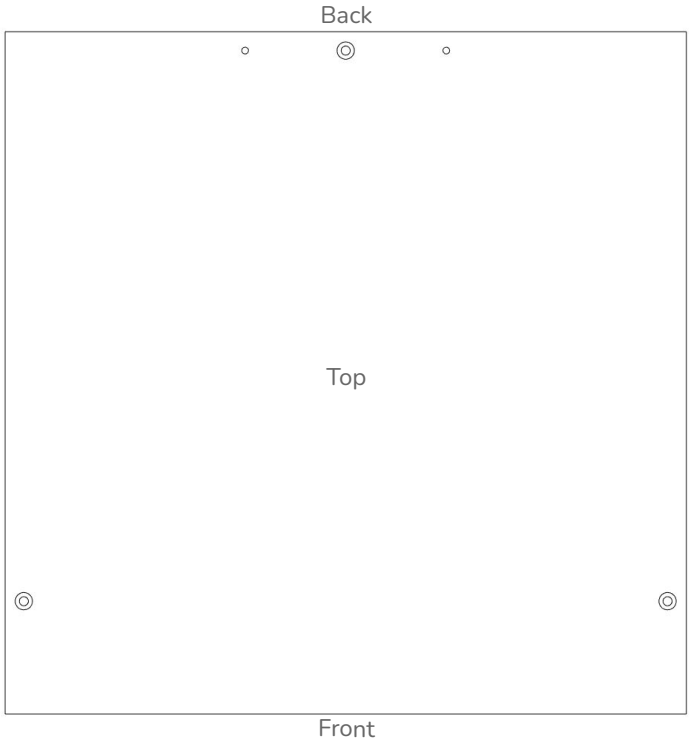
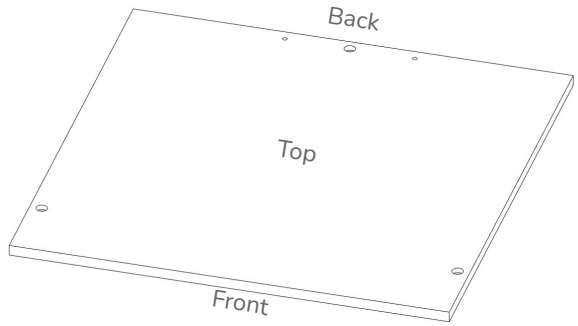
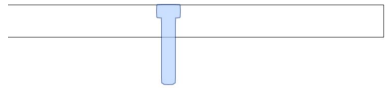


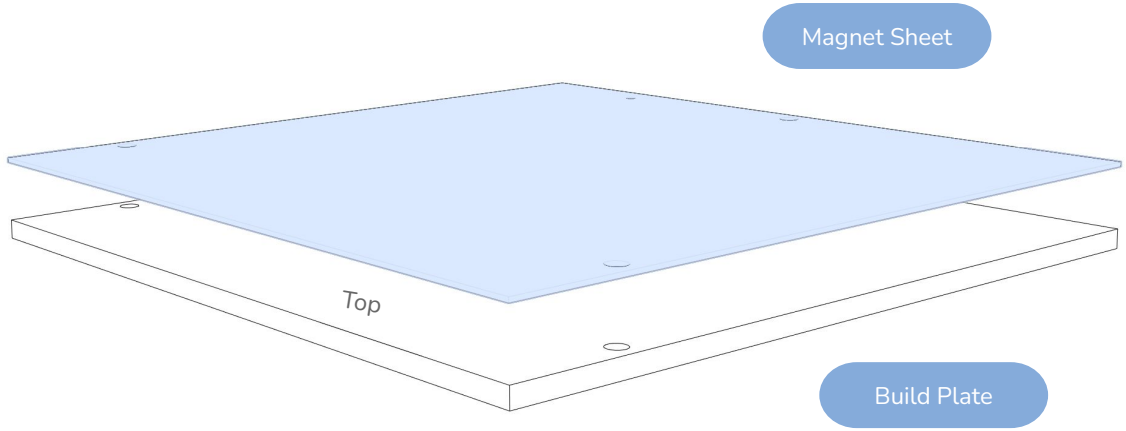


WHICH SIDE IS WHICH?

The top of the plate has mounting holes with bores that allow screw heads to sit flush/below the surface.

The plate has additional tapped holes to secure the PE connection and a thermal fuse, those are on the back side of the plate.





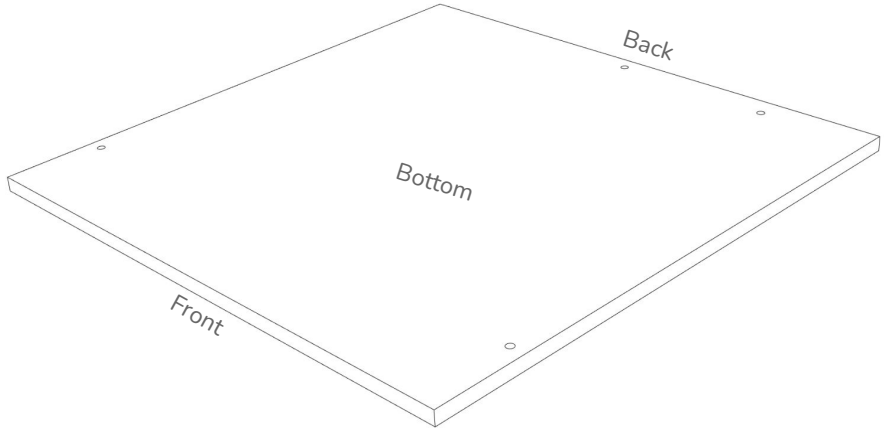
MAGNET APPLICATION

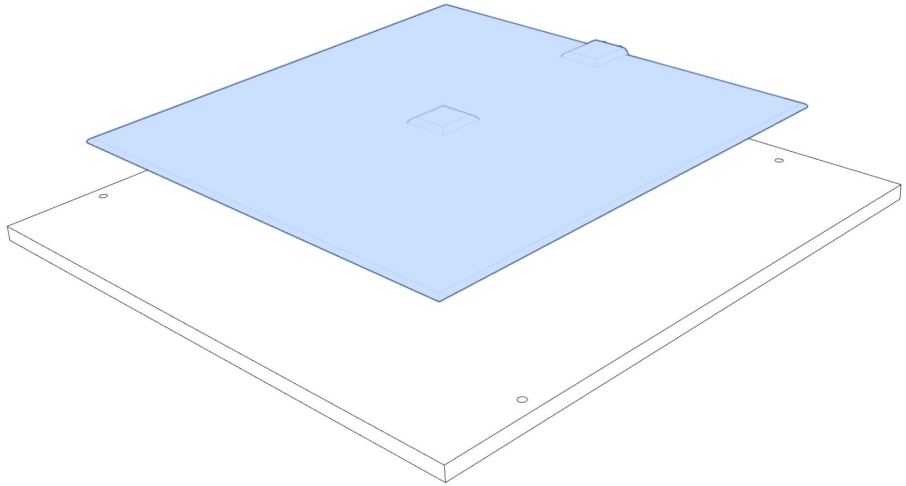
Clean the plate with isopropyl alcohol or similar cleaner prior to applying the magnet.

Use the edge of a plastic object or a small roller to firmly press the magnet on the plate to get a good bond.

If you have never done this before we recommend you watch the linked guide.

https://voron.link/rm6tpld

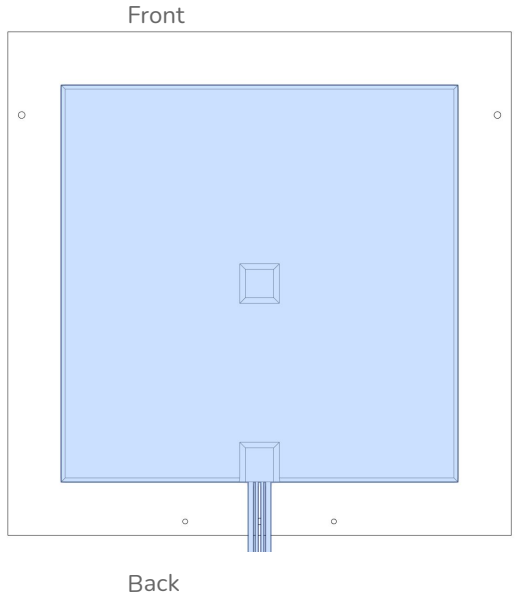


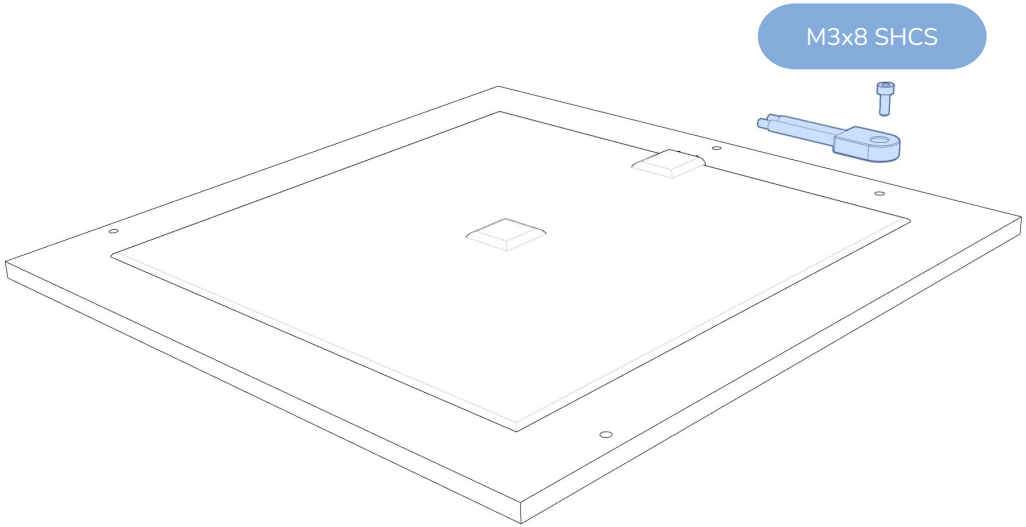


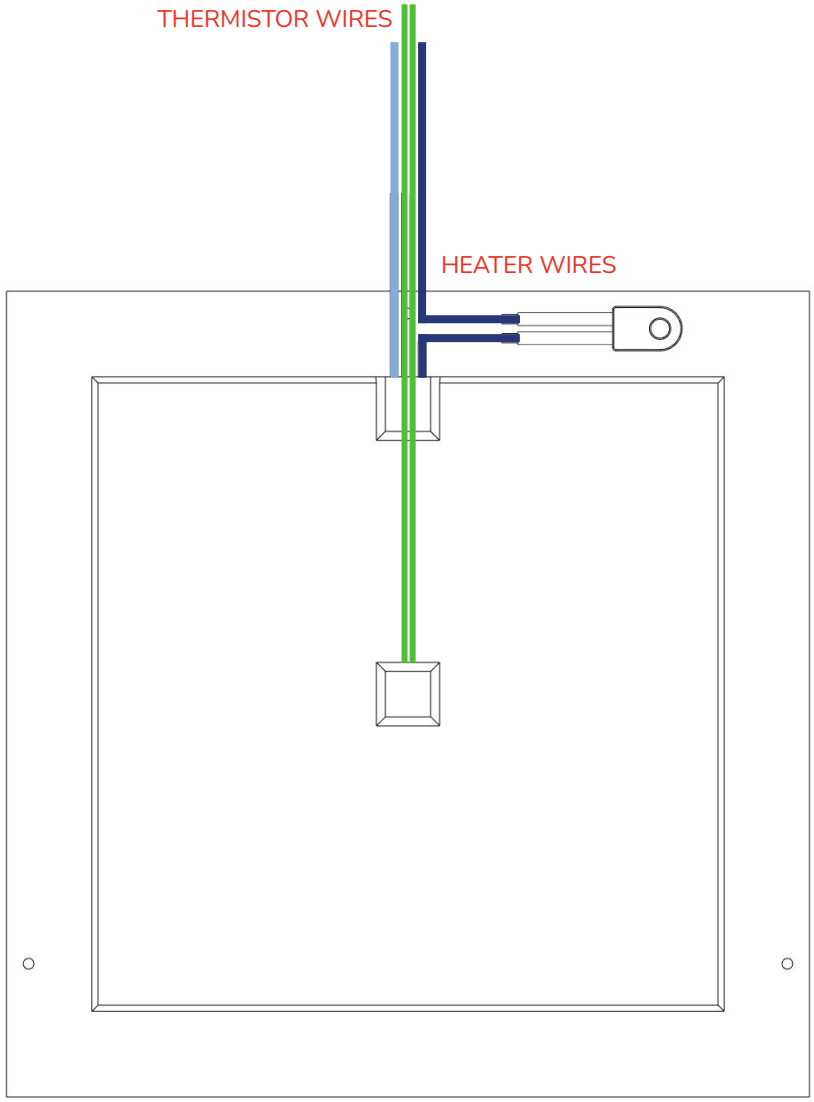
HEATER APPLICATION

The heater is installed in the same fashion as the magnet.

Center it on the underside of the build plate and make sure to firmly press it on the build plate.





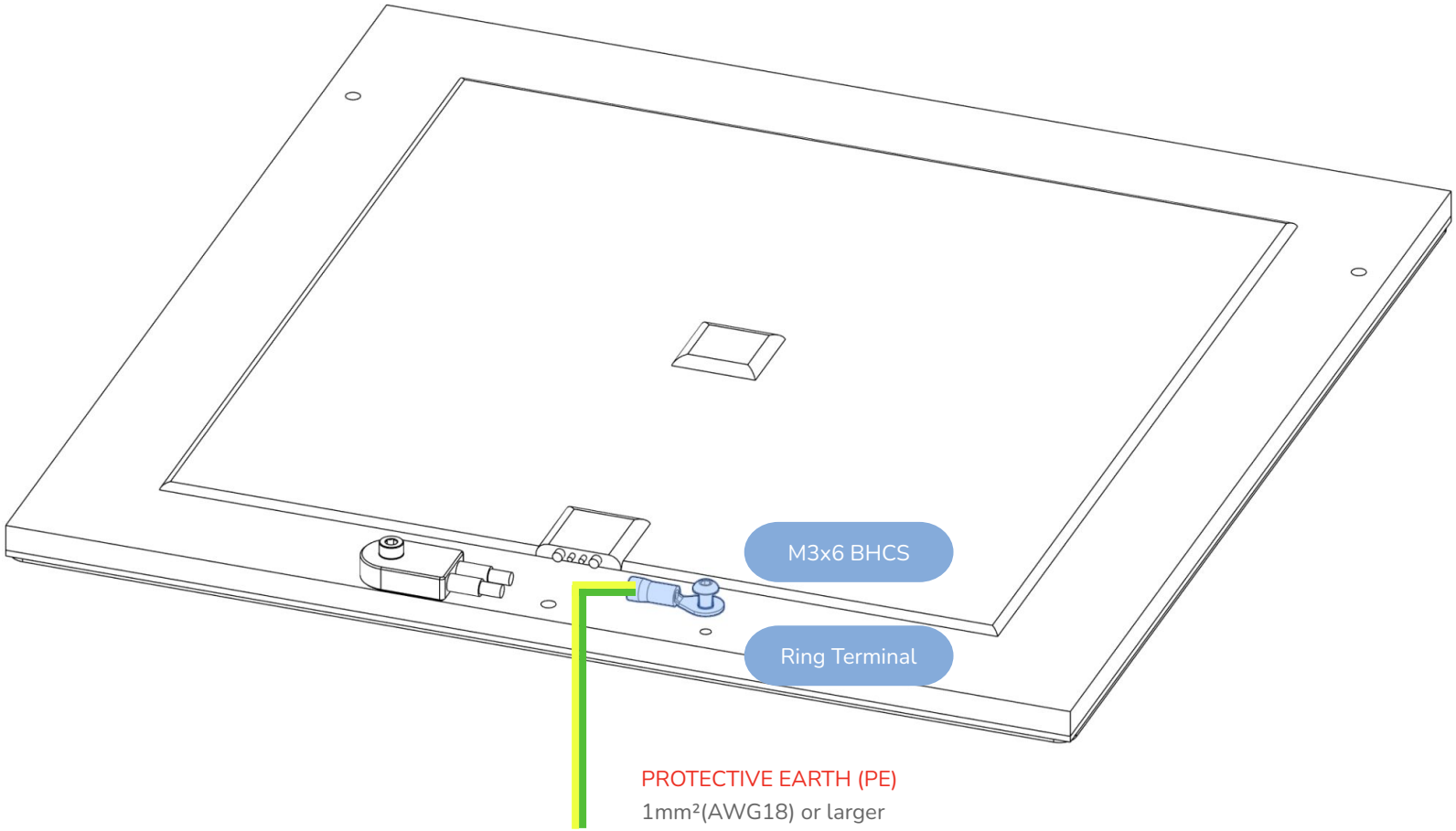


THERMAL FUSE

While not required to operate the printer, a thermal fuse attached to the build plate adds an additional layer of protection against potentially dangerous malfunctions.

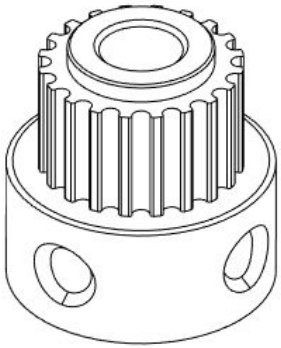
The thermal fuse is wired in-line with the heater wires.

Depending on the tapped holes in the plate you may need to use a shorter screw.


M3x6 BHCS
Ring Terminal
PROTECTIVE EARTH (PE)
1mm²(AWG18) or larger

You don't use this endstop. Your endstop is Tap.

GT2 20 Tooth Pulley



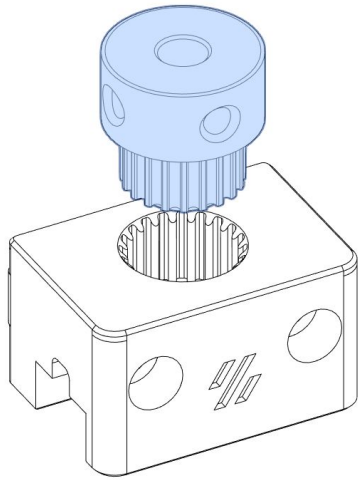
REMOVE FLANGE & SET SCREWS

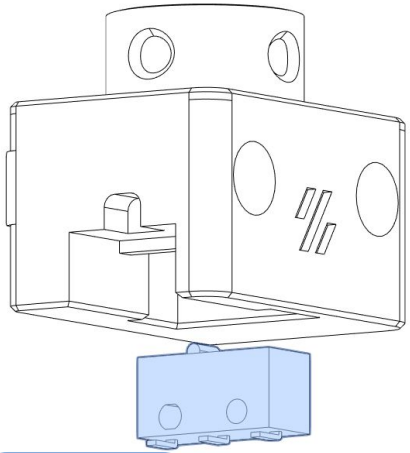
Use a bottle opener or some pliers to remove the top flange.

https://voron.link/ict0j6x

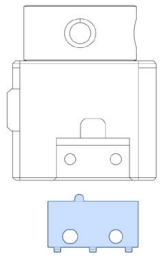
PRESS FIT

Apply the required force to fully seat the pulley in the printed part.

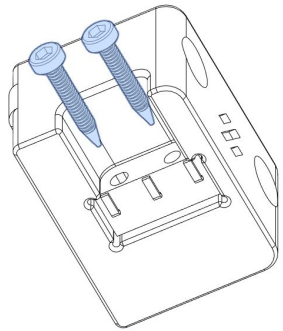




Microswitch



M2x10 Self Tapping



SWITCH WITHOUT LEVER

This part requires a switch without lever to be installed in the shown orientation.

You can remove the lever from microswitches by gently pressing on the lever's hinge point.

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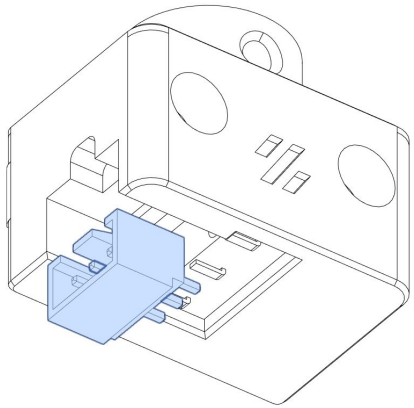
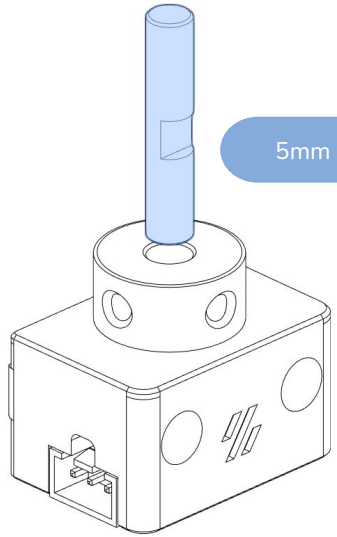
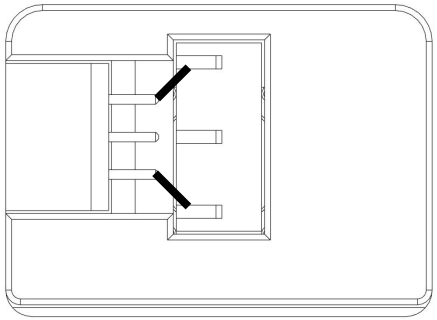
You don't use this endstop. Your endstop is Tap.





PREVENTING MISHAPS

You can add a notch to the Z endstop point and capture it with a set screw to prevent it from falling out.



SOLDER CONNECTOR

Solder a connection from the outer two terminals of the microswitch to the connector.

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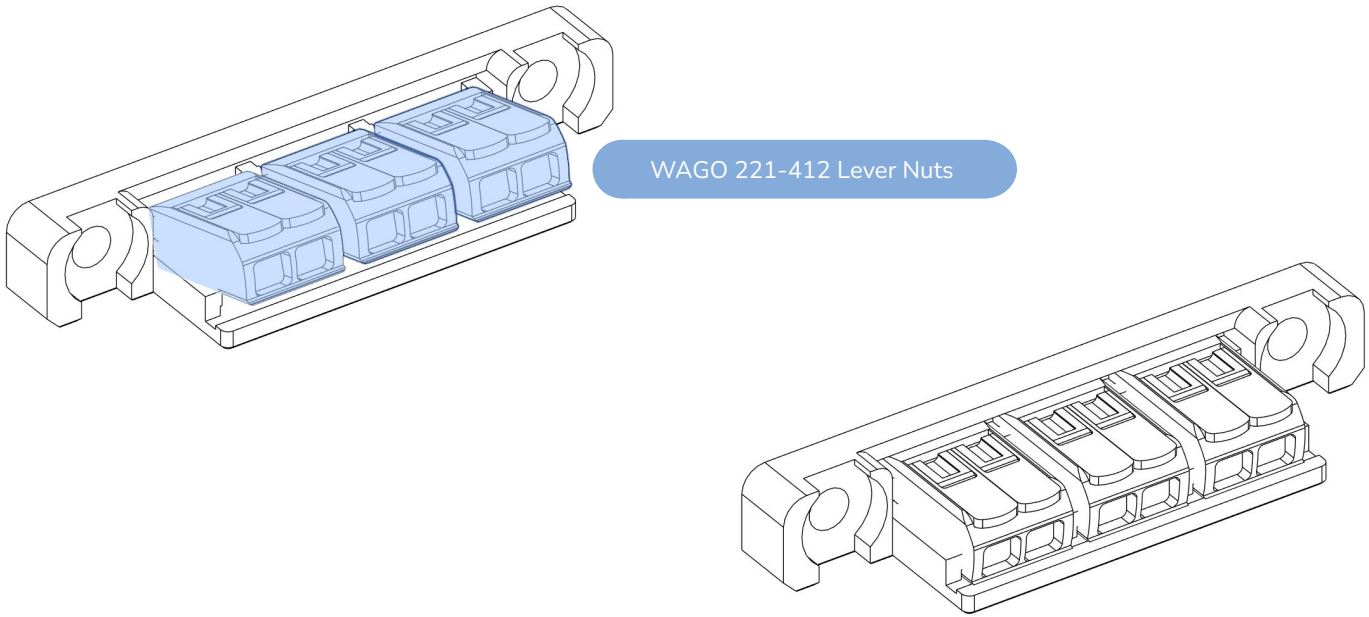
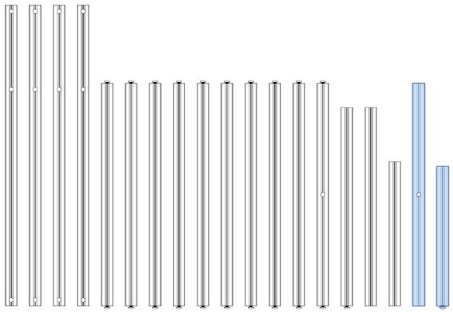
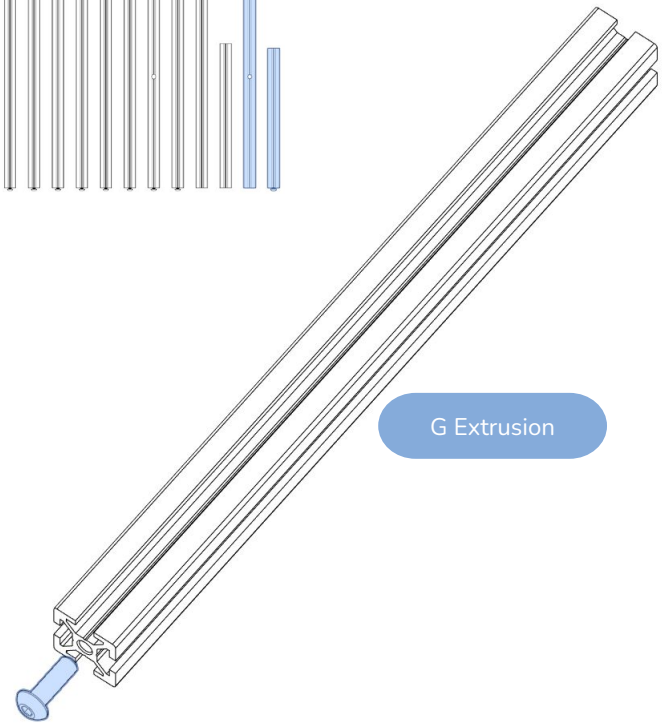

WAGO 221-412 Lever Nuts

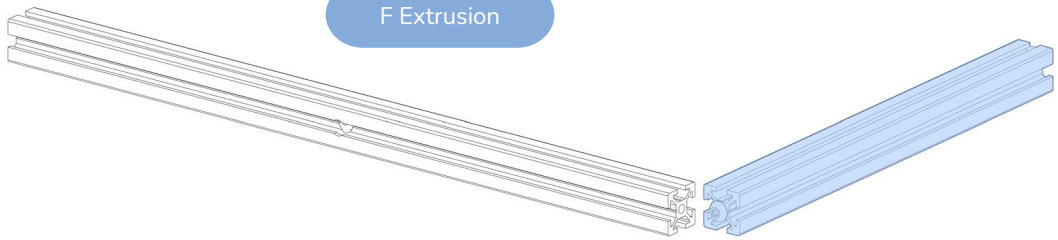


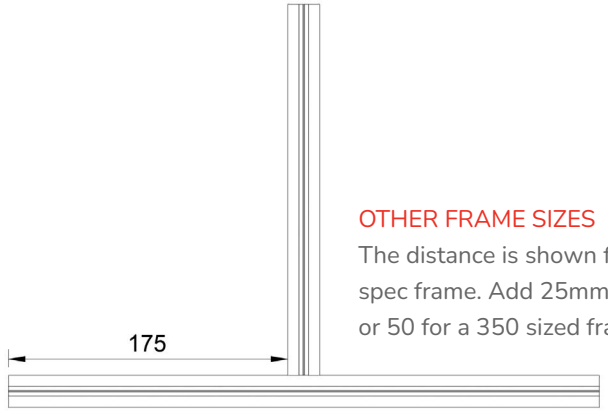


G Extrusion

M5x16 BHCS

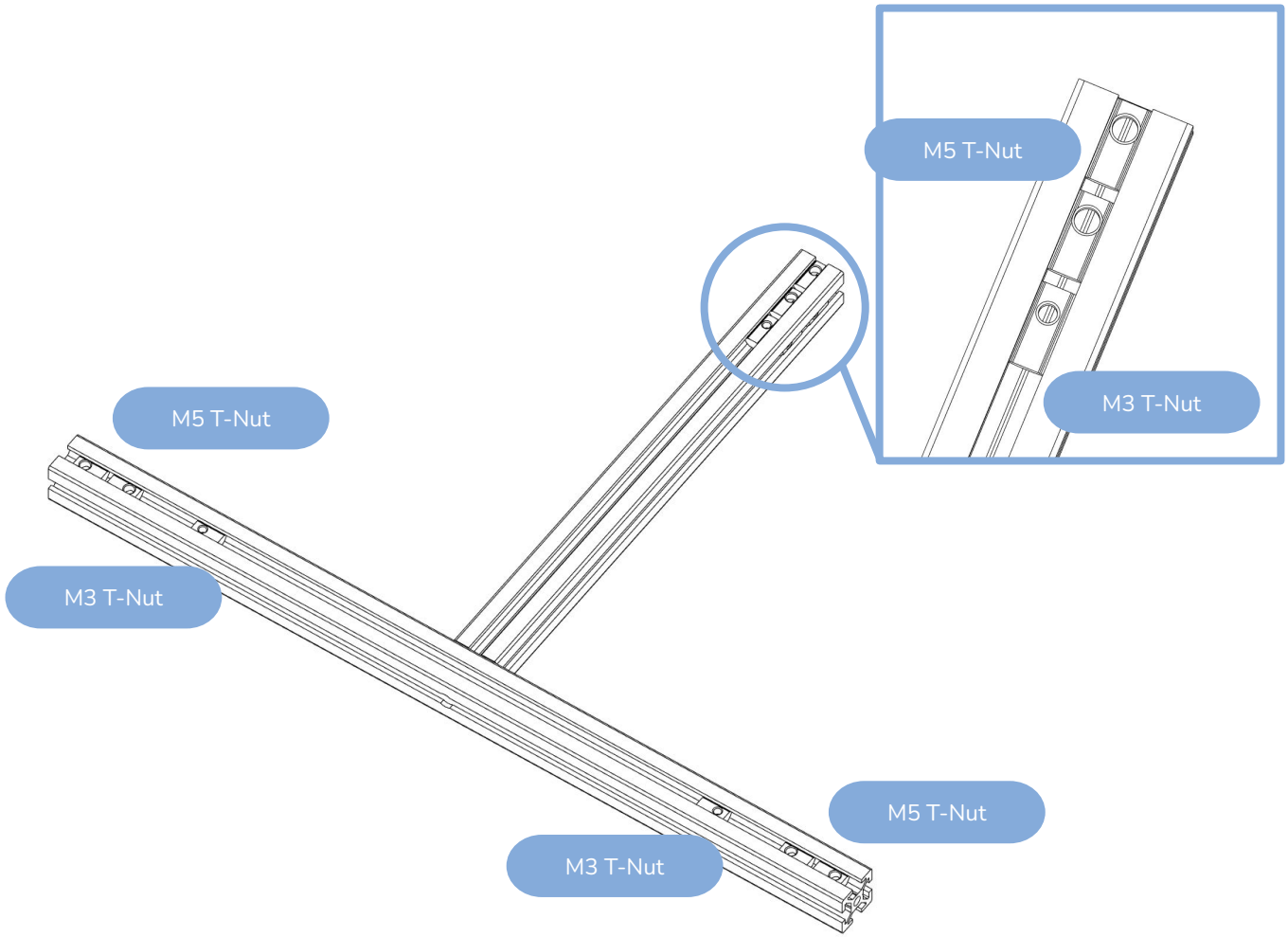
F Extrusion

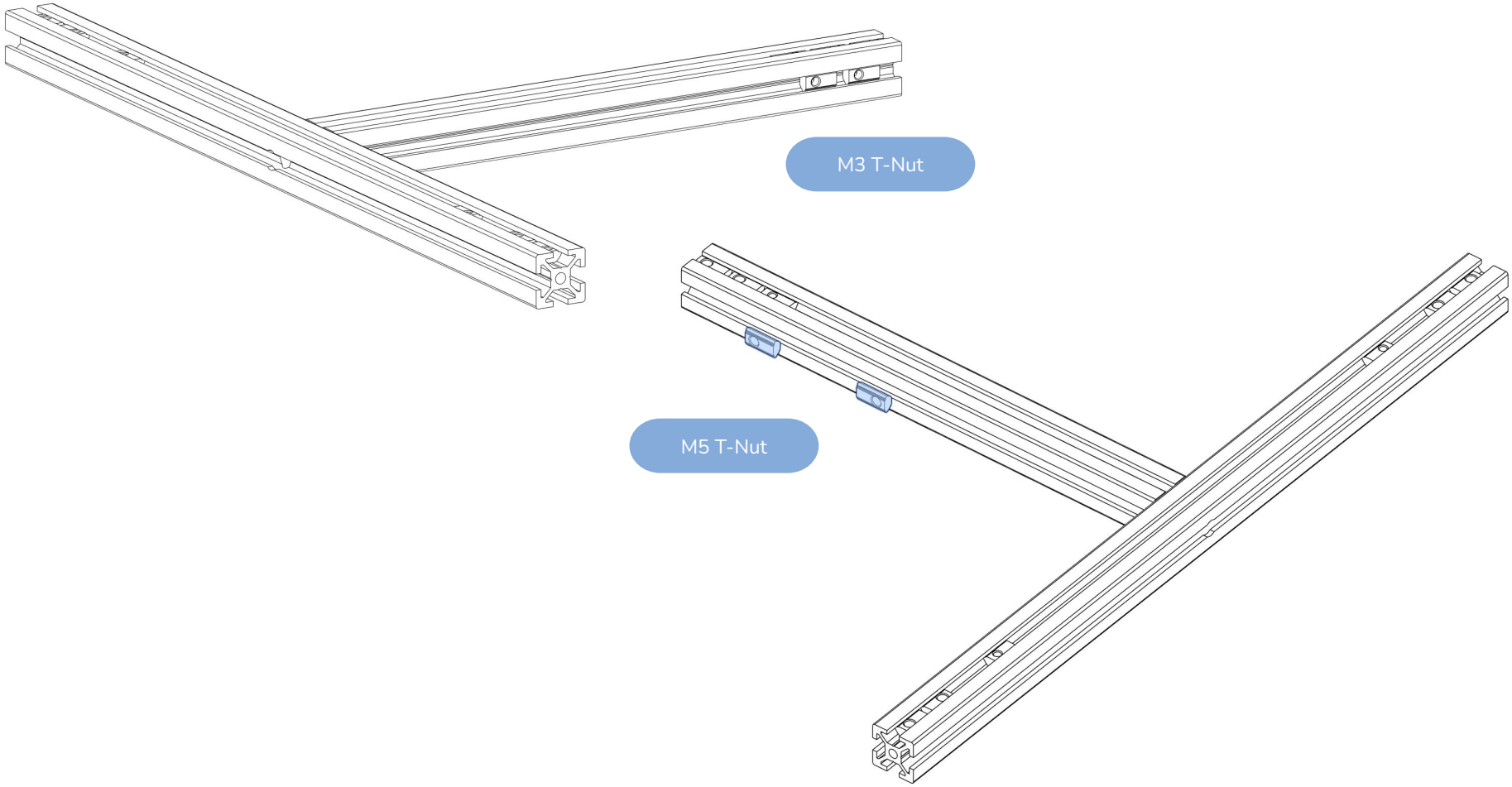


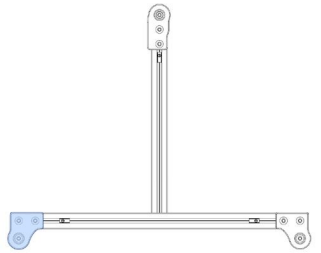


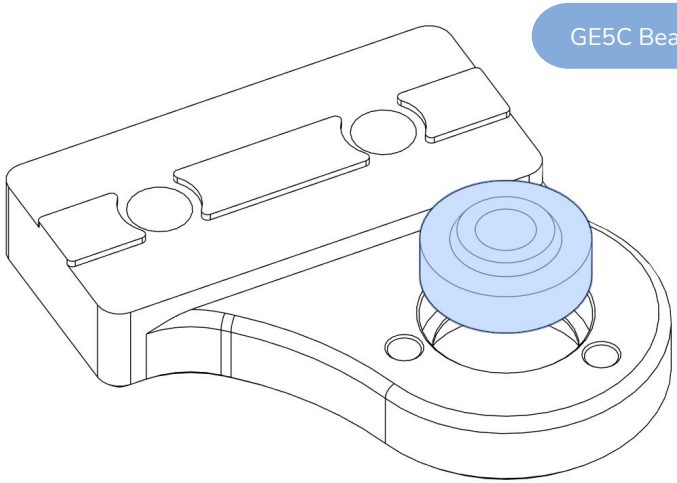
OTHER FRAME SIZES

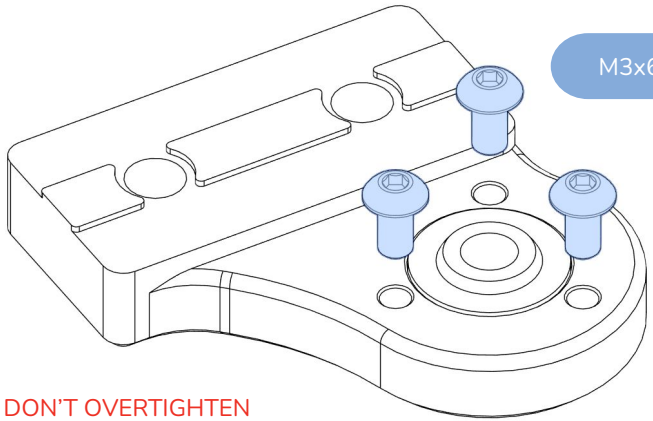
The distance is shown for a 250 spec frame. Add 25mm for a 300 or 50 for a 350 sized frame.


M5 T-Nut
M3 T-Nut
M5 T-Nut
M3 T-Nut
M5 T-Nut
M3 T-Nut


M3 T-Nut
M5 T-Nut

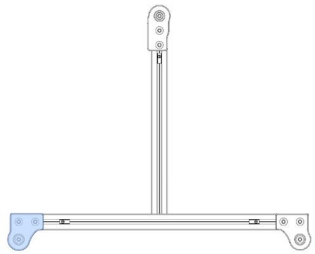


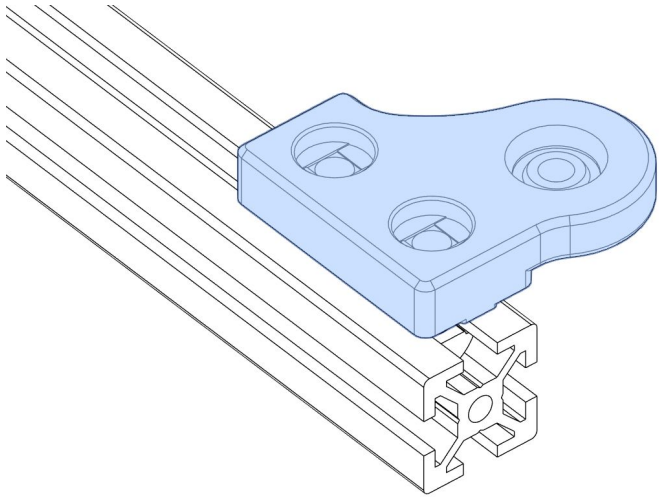


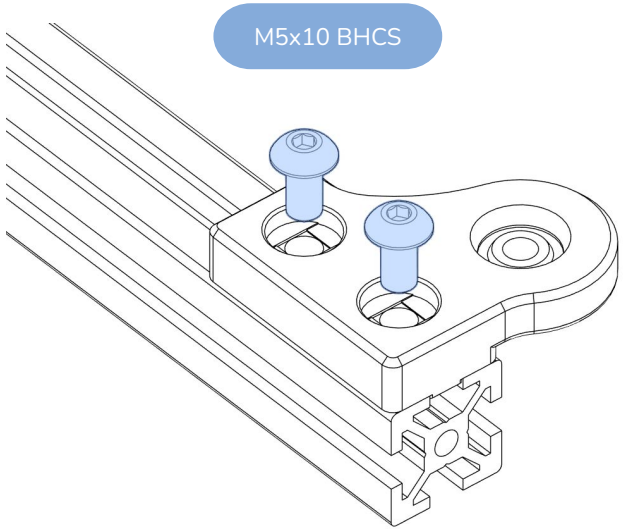


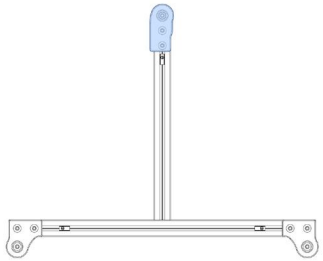
DON'T OVERTIGHTEN

The screws are used to position the bearing and are threaded directly into plastic.

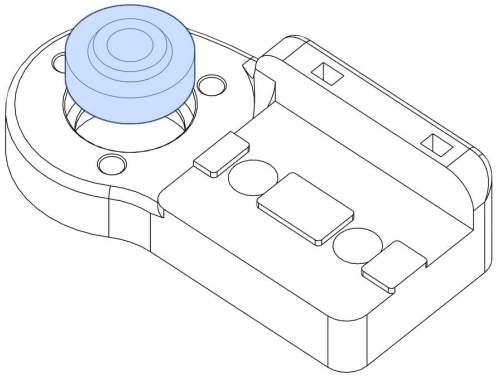




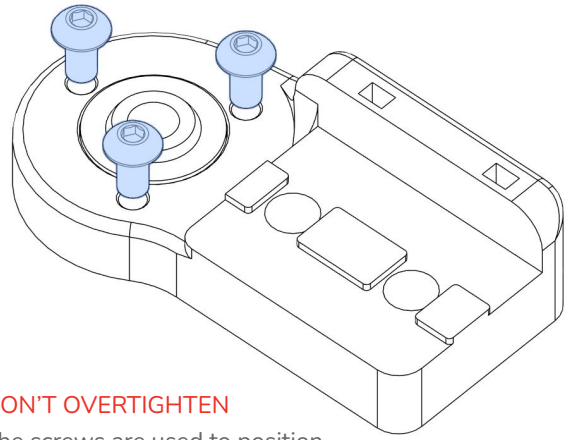

M5x10 BHCS



GE5C Bearing

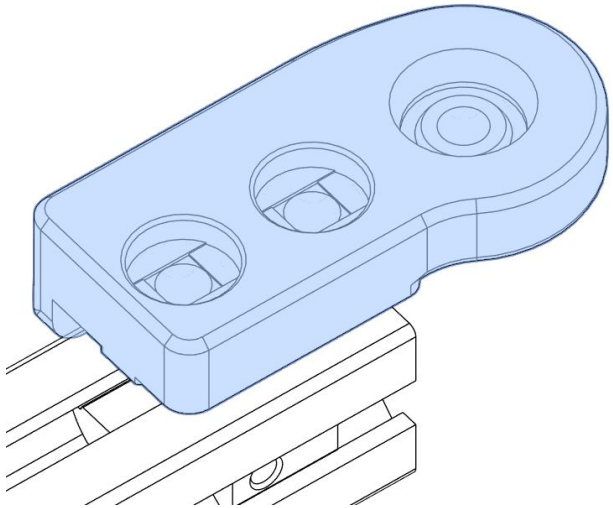


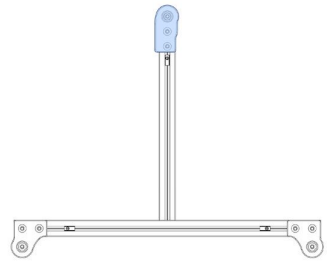
M3x6 BHCS



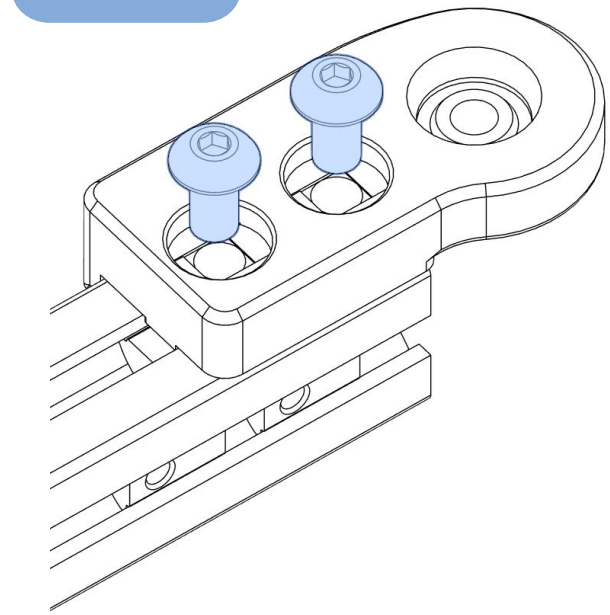
DON’T OVERTIGHTEN

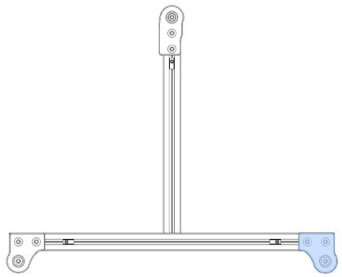
The screws are used to position the bearing and are threaded directly into plastic.

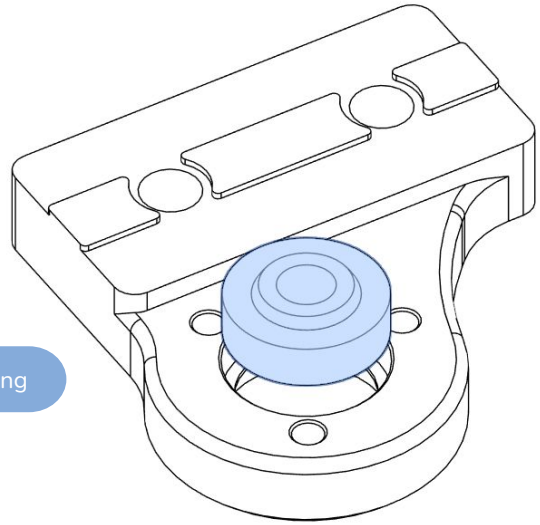


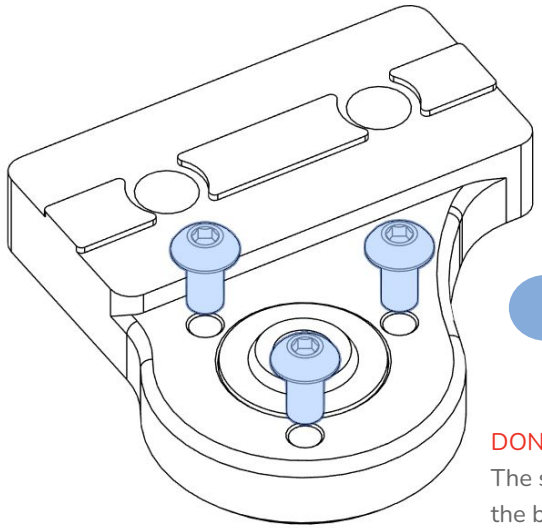


M5x10 BHCS



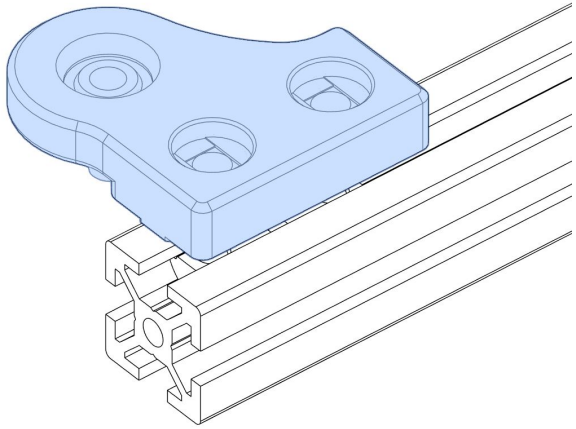


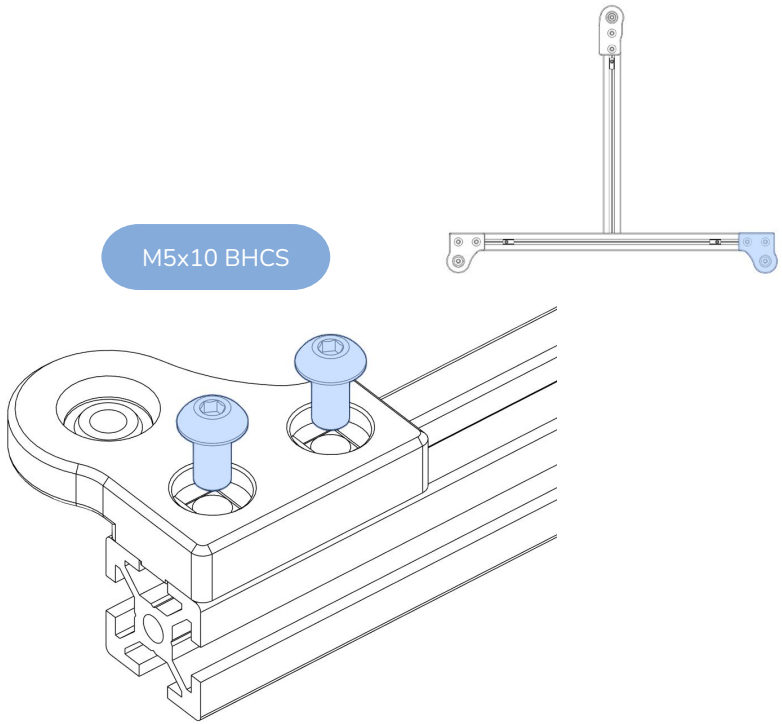


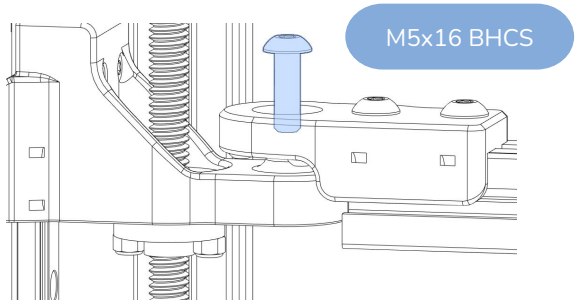


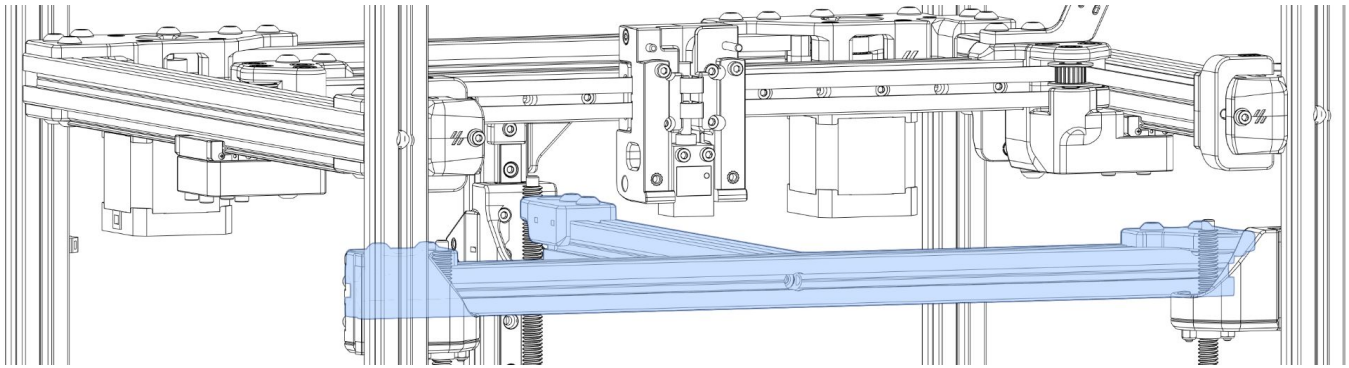
DON'T OVERTIGHTEN

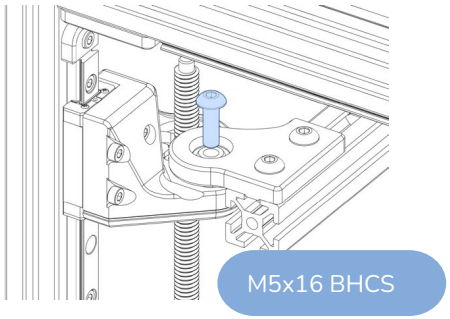
The screws are used to position the bearing and are threaded directly into plastic.

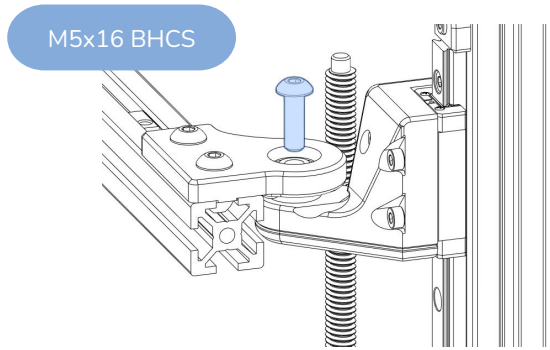


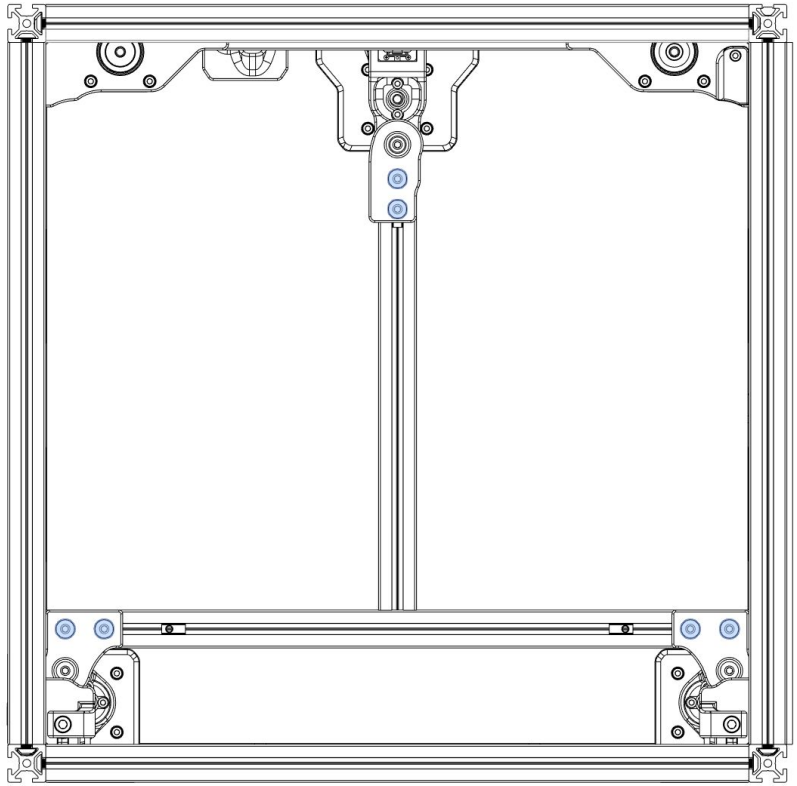

M5x10 BHCS


M5x16 BHCS



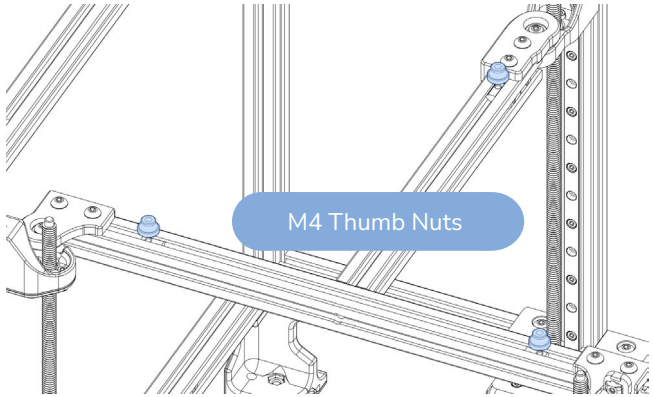

M5x16 BHCS

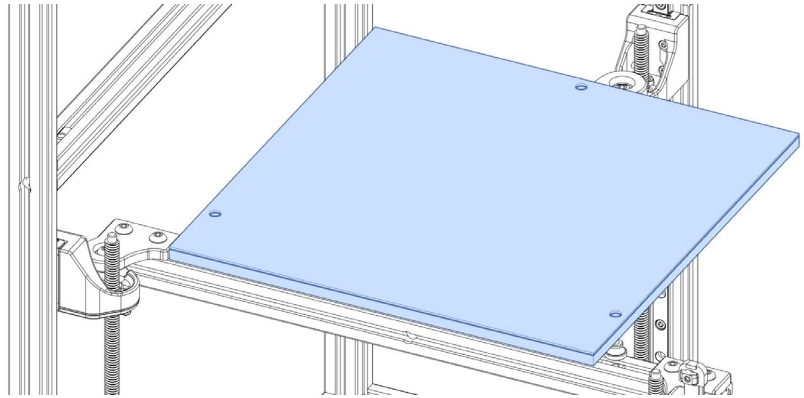

M5x16 BHCS

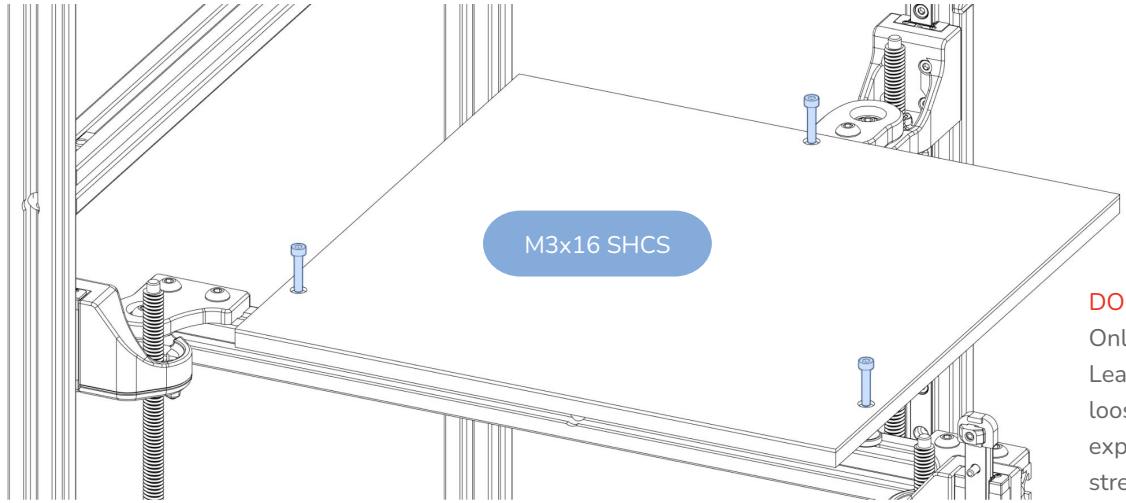


LOOSEN AND RETIGHTEN

Slightly loosen the screws that hold the bed frame to the printed parts and gently shake the bed frame before retightening them.







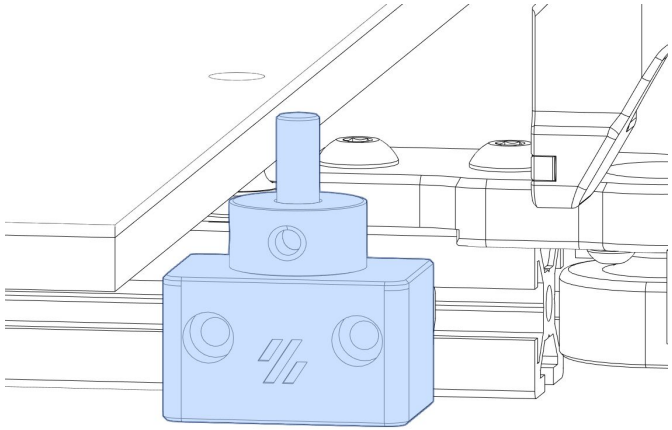
VARIABLE SCREW SIZE

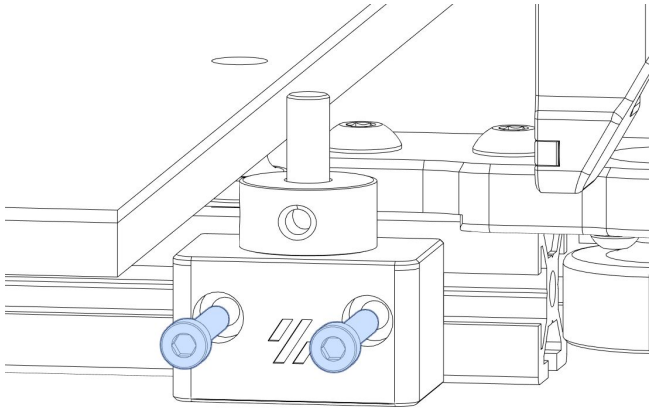
If you are using a thicker bed (9.5mm) you may need to change the M3x16 screws out for M3x20

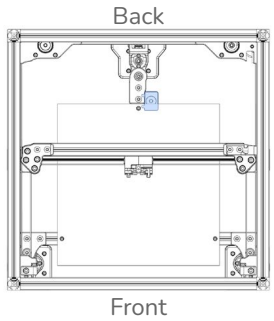
DON'T TIGHTEN FULLY

Only tighten one screw fully. Leave the remaining screws slightly loose. This will allow for thermal expansion without putting additional stress on the plate.

You don't use this endstop. Your endstop is Tap.



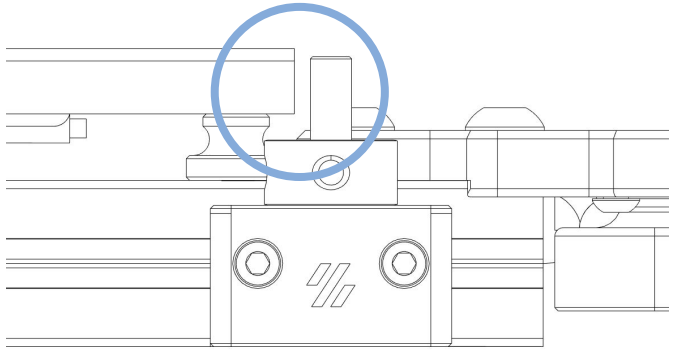




M3x20 SHCS

ADJUST Z ENDSTOP POSITION

The shaft of the Z Endstop must not touch the print bed. Adjust the position if required.



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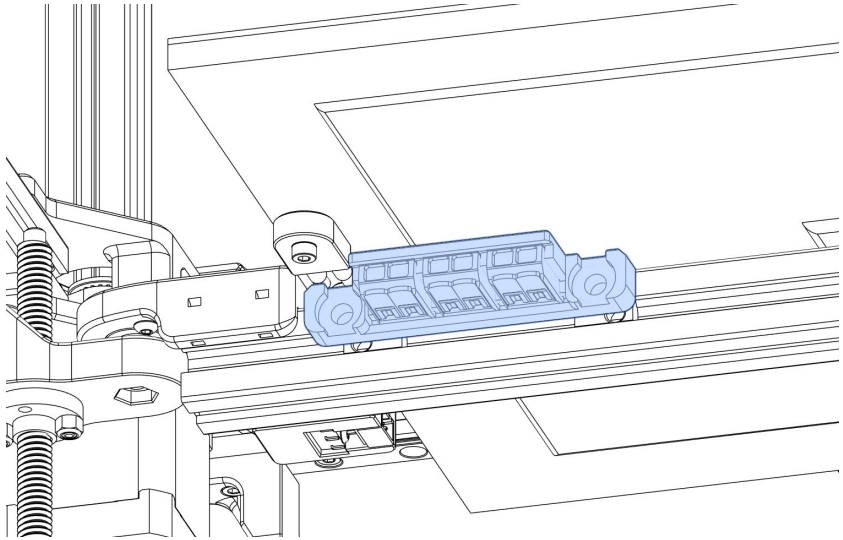
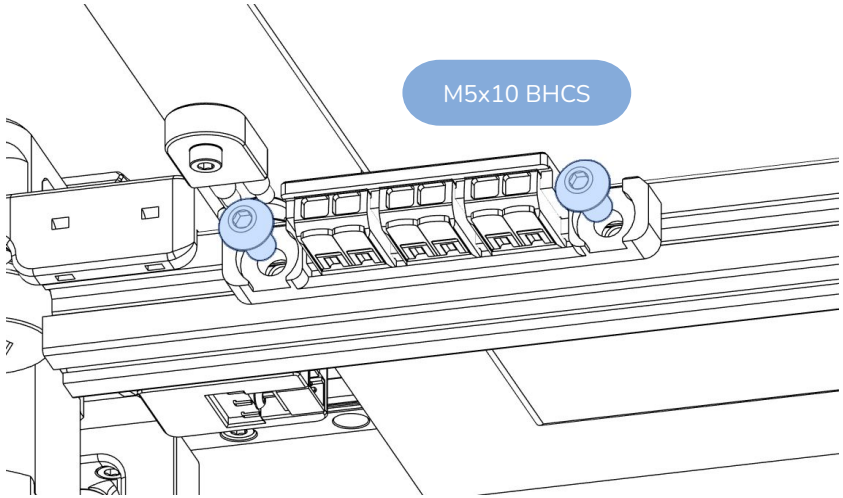
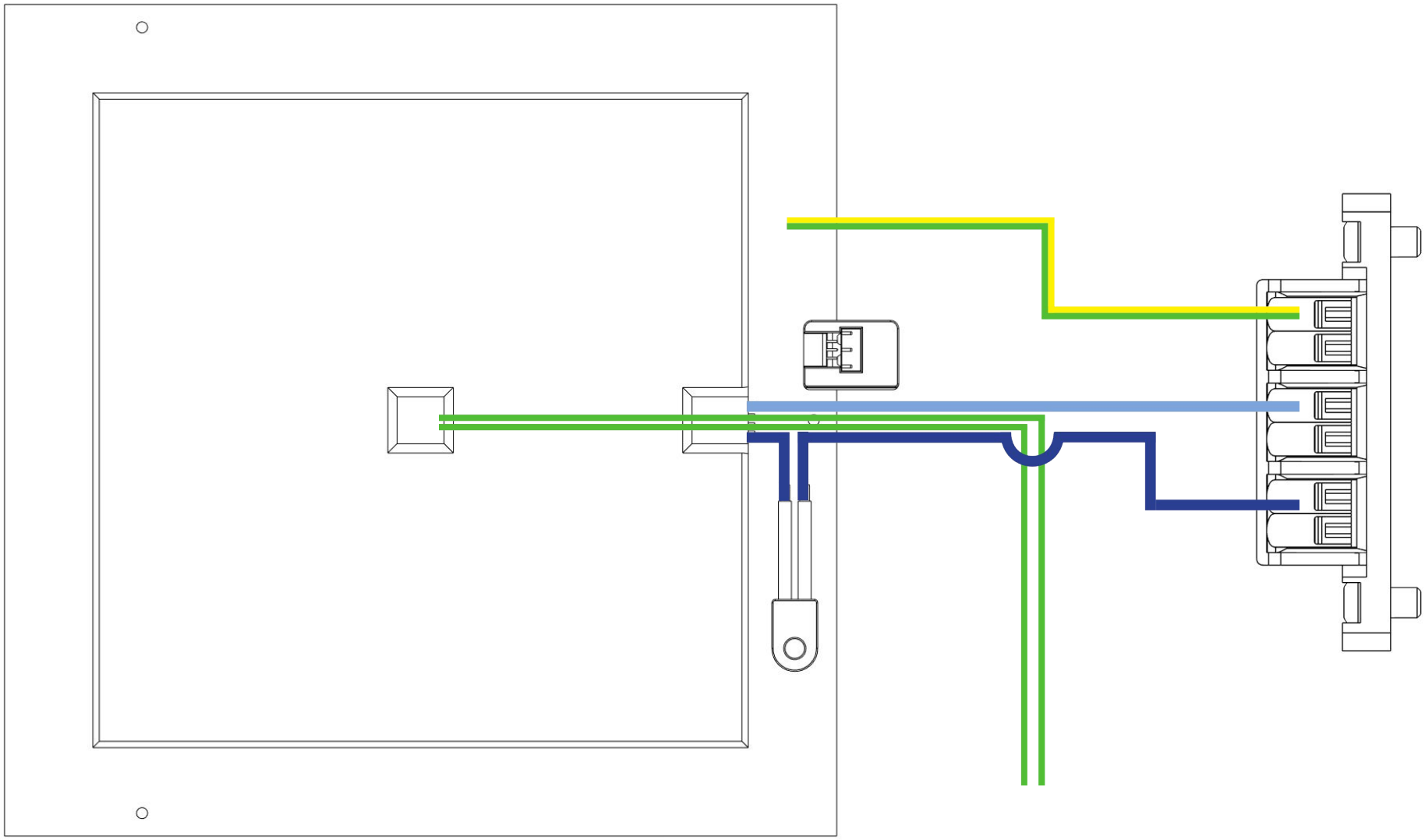



M5x10 BHCS



Switch to the [Stealthburner manual](https://github.com/VoronDesign/Voron-Stealthburner/tree/main/Manual).

Switch to the [Stealthburner manual](https://github.com/VoronDesign/Voron-Stealthburner/tree/main/Manual).

UNIVERSAL TOOLHEAD

This printer uses the Stealthburner toolhead, which is compatible with several of the printers in the VORON lineup. To keep things organized, Stealthburner's files are maintained separately. Follow the Stealthburner assembly manual to build your toolhead, and return here to proceed.

https://voron.link/6hbi9n3

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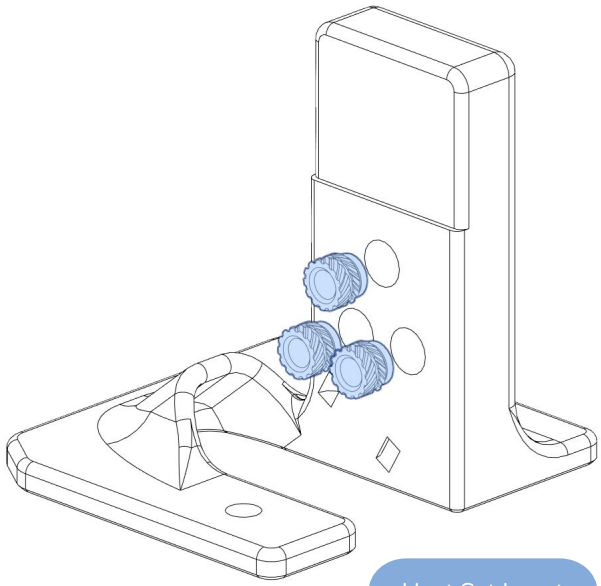
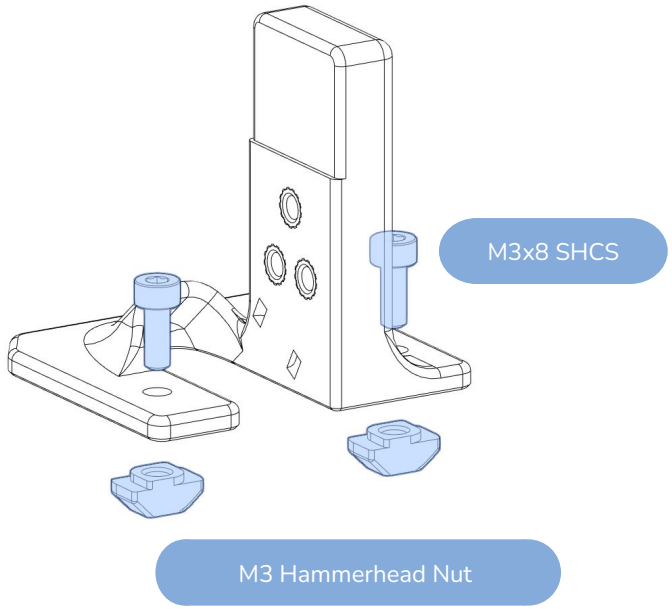
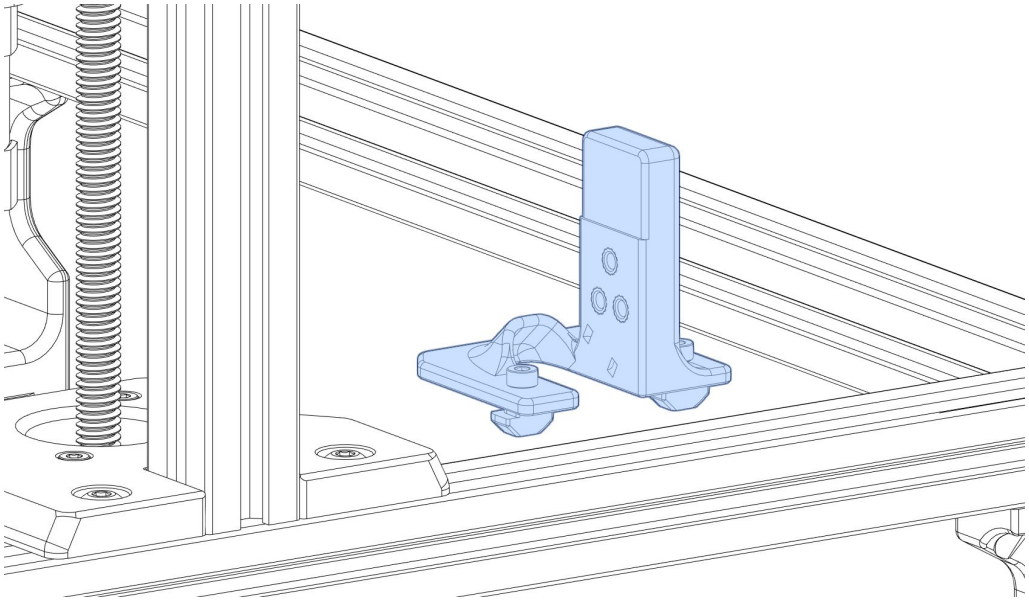
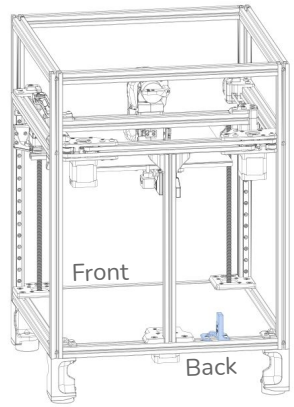
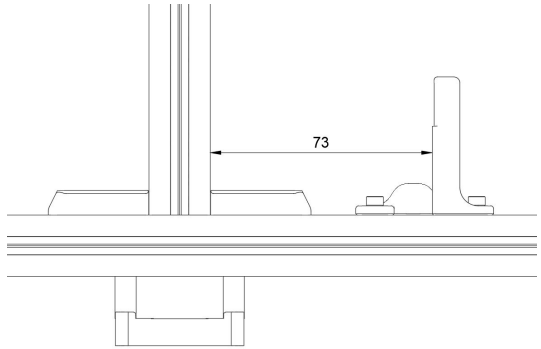
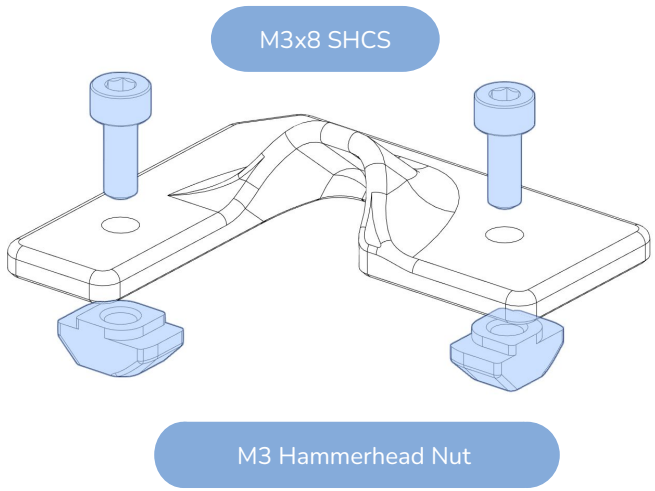

Heat Set Insert

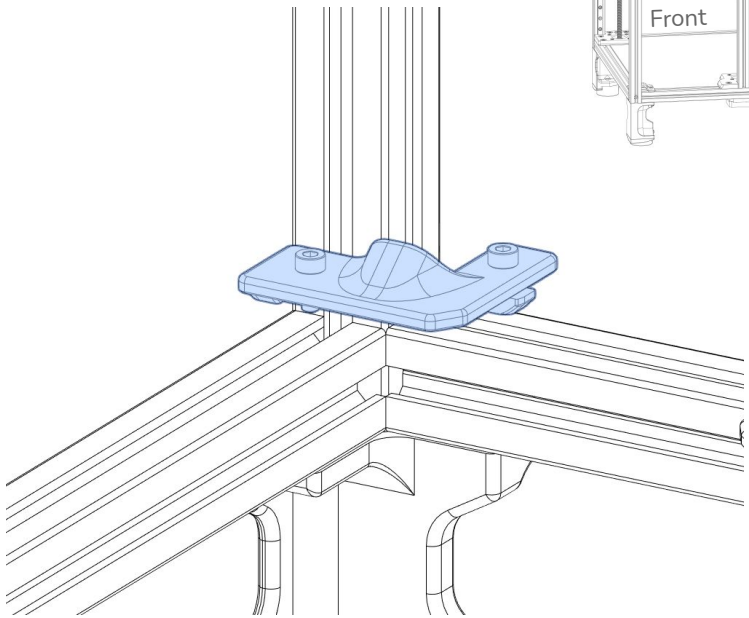

M3x8 SHCS
M3 Hammerhead Nut

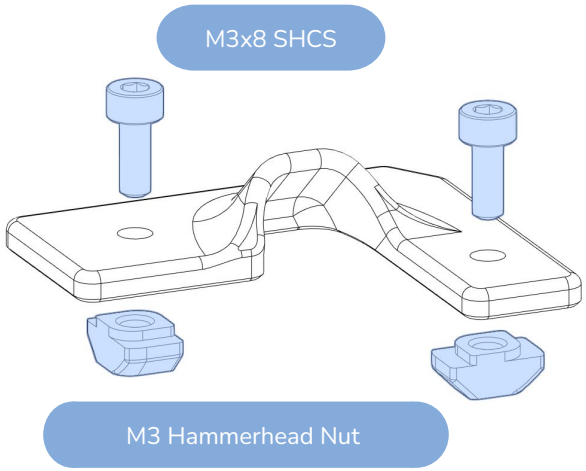


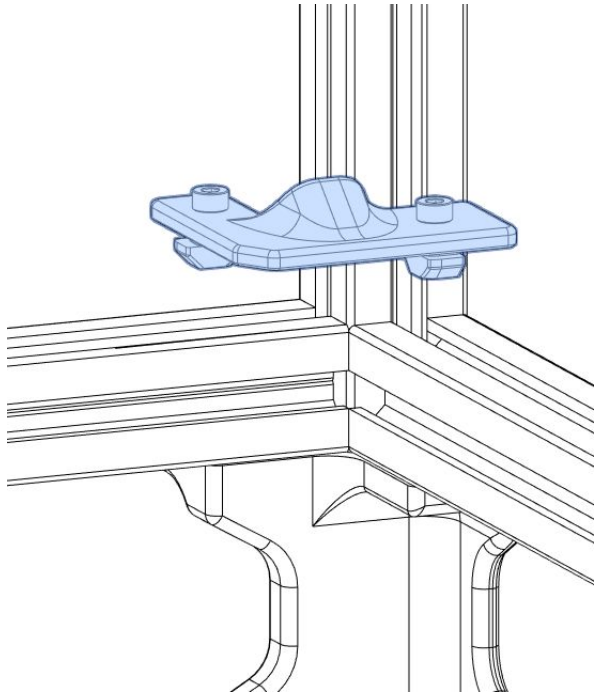

Front
Back

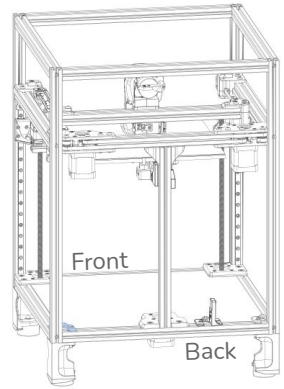

73

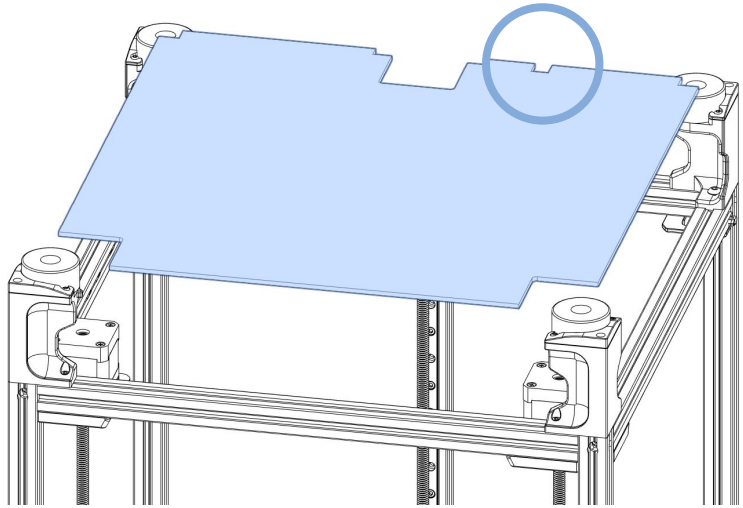

M3x8 SHCS
M3 Hammerhead Nut


Front
Back


M3x8 SHCS
M3 Hammerhead Nut

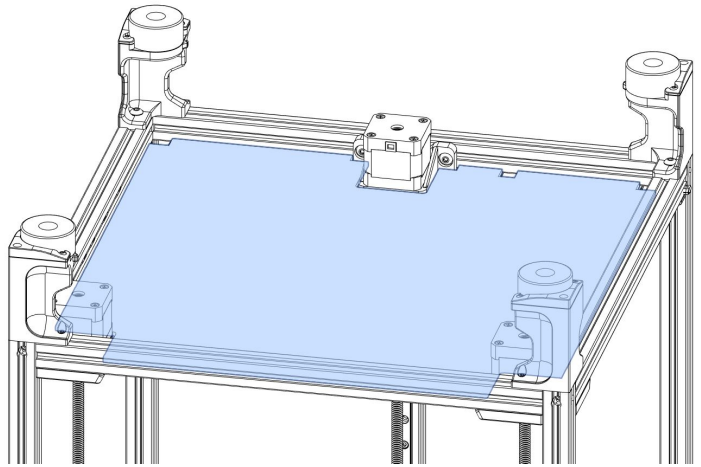


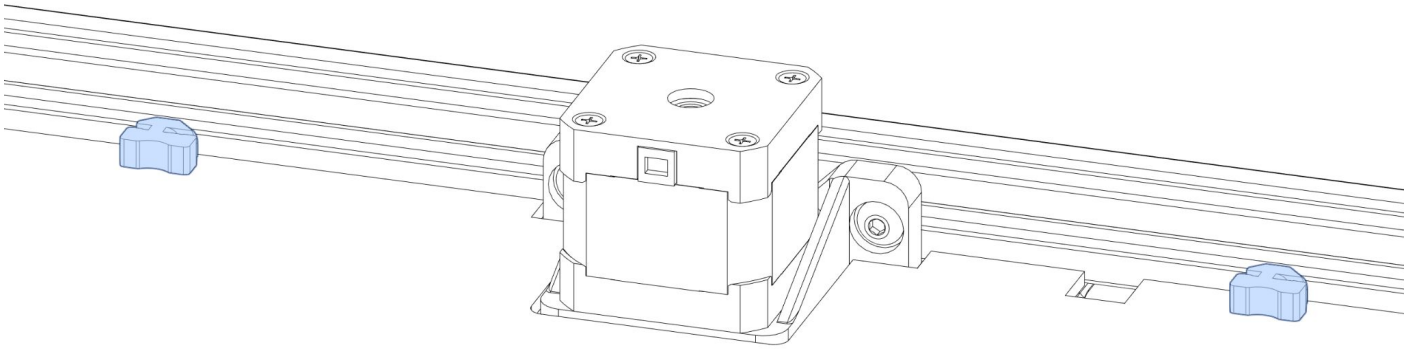

Front
Back

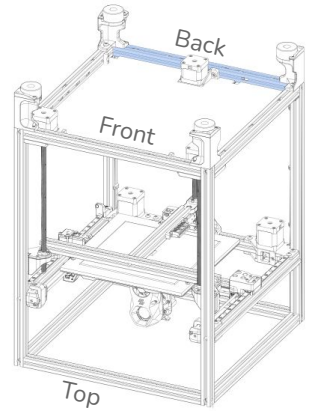


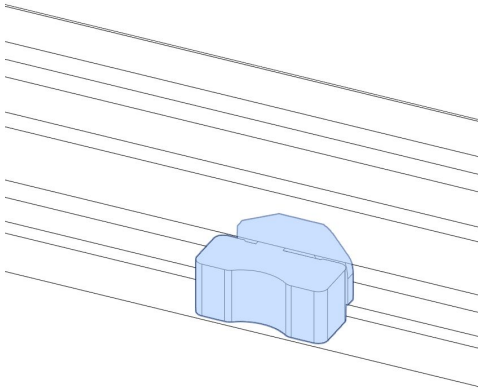
PANEL ORIENTATION

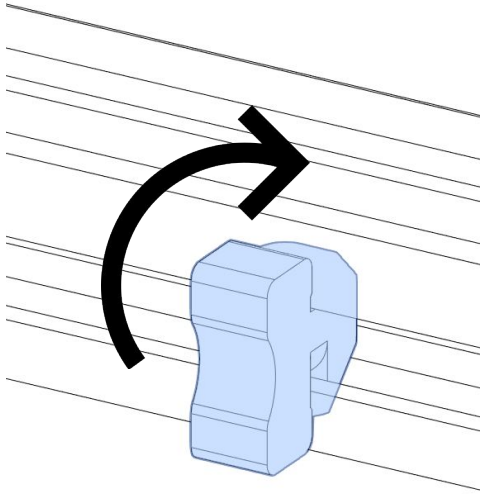
Align the notch to the back and in line with the Z chain anchor.





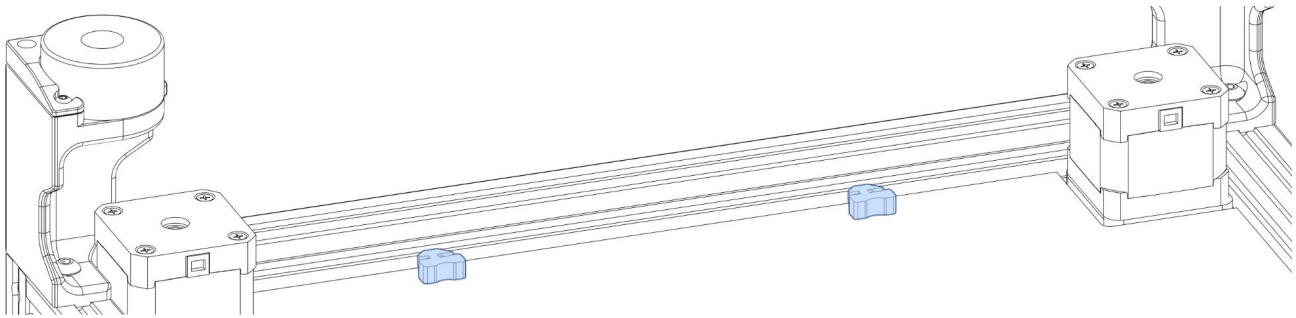


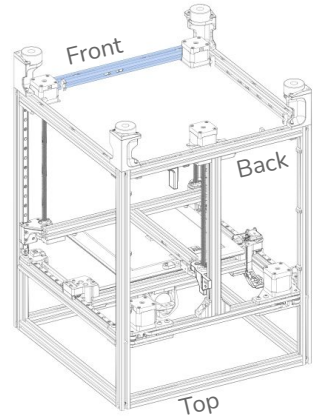


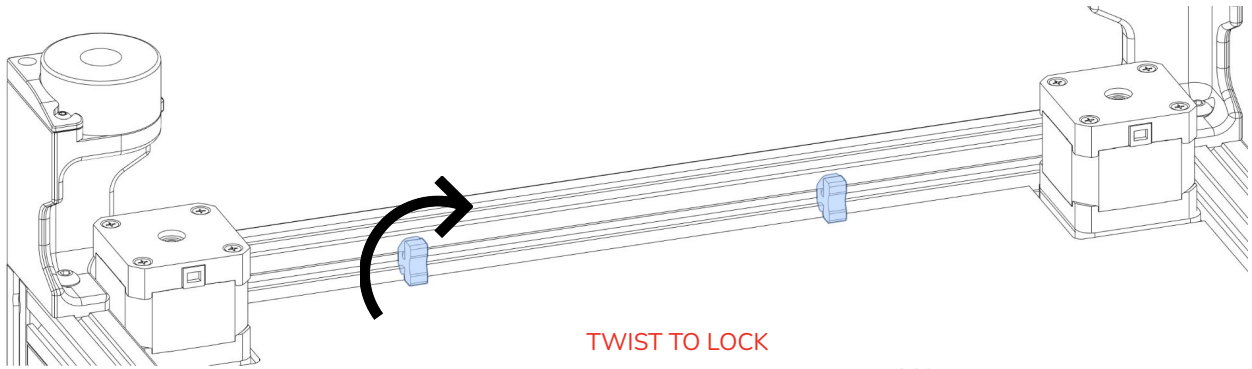


TWIST TO LOCK

Rotate the deck supports clockwise by 90° to lock them in place.

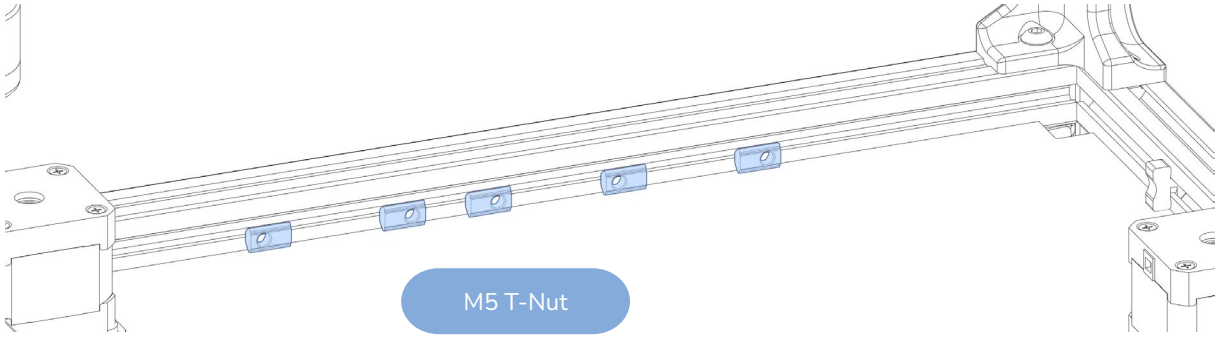


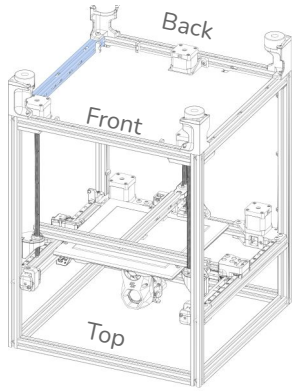


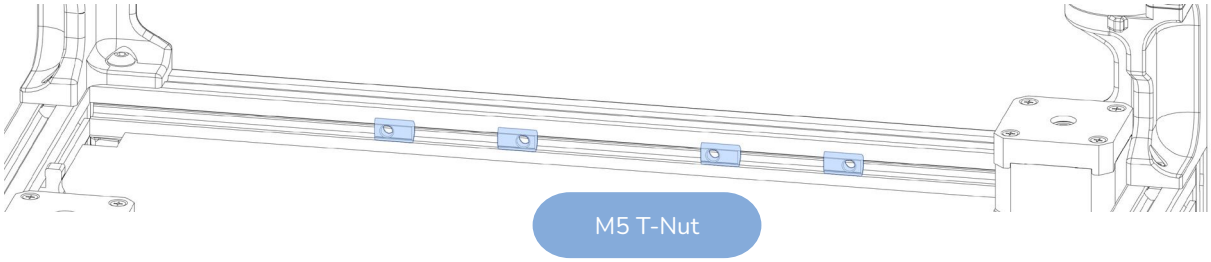


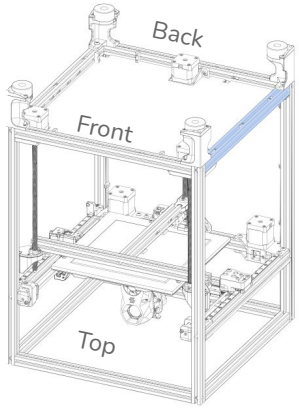
TWIST TO LOCK

Rotate the deck supports by 90° to lock them in place.

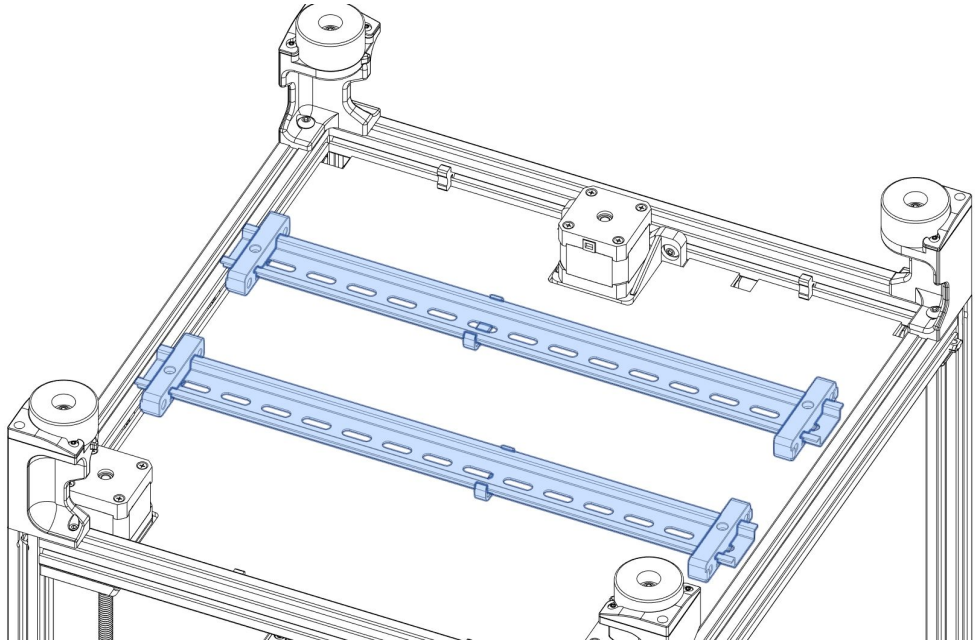

M5 T-Nut


Back
Front
Top


M5 T-Nut


Back
Front
Top

Rotate the din rails 90° compared to the original layout.



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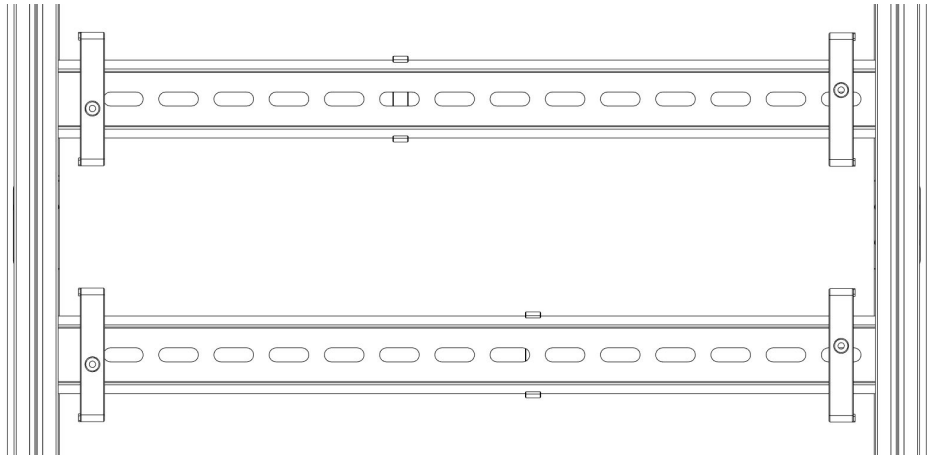
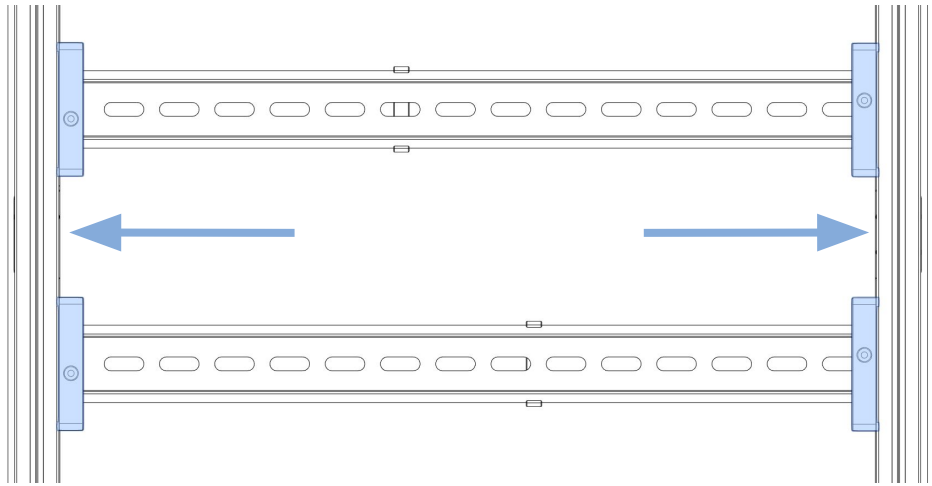
SPACING

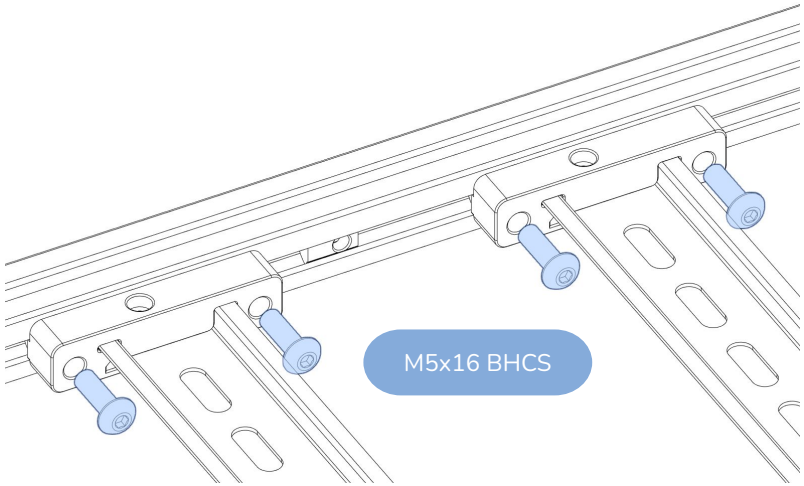
Roughly center the DIN rails and space them about 80mm apart.

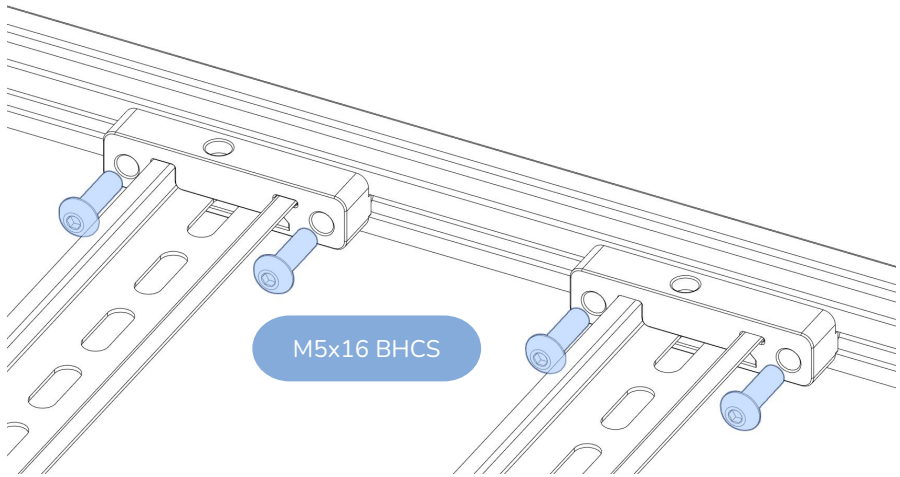


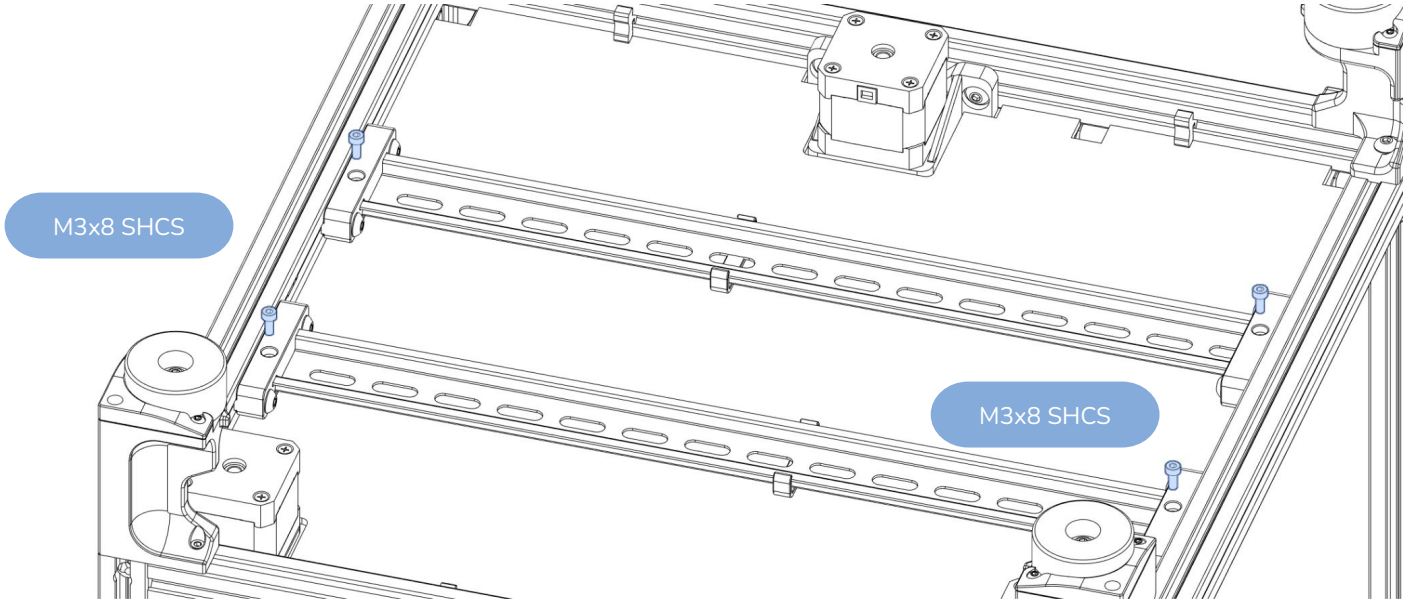
SLIDE INTO POSITION

Slide the mounts all the way to the extrusions.



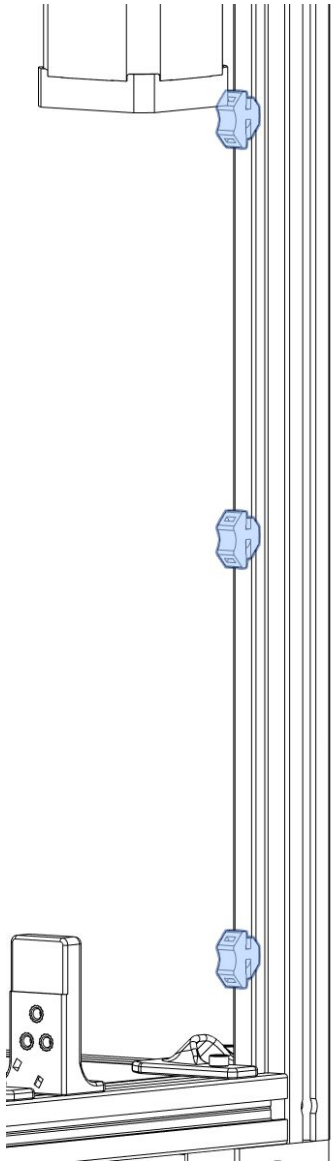

M5x16 BHCS

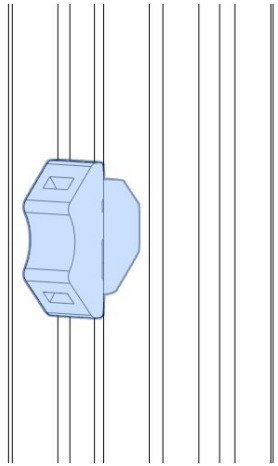

M5x16 BHCS

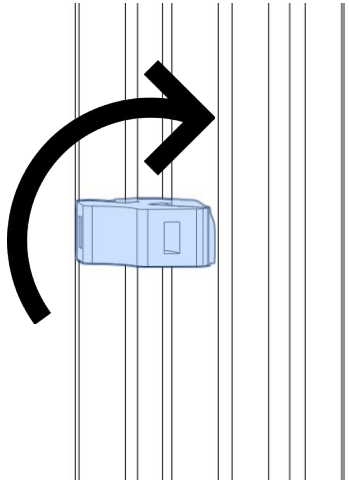


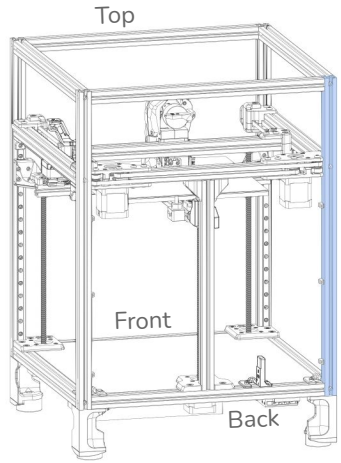
DON'T OVERTIGHTEN

The screws are used to keep the DIN rails from sliding and are threaded directly into plastic.



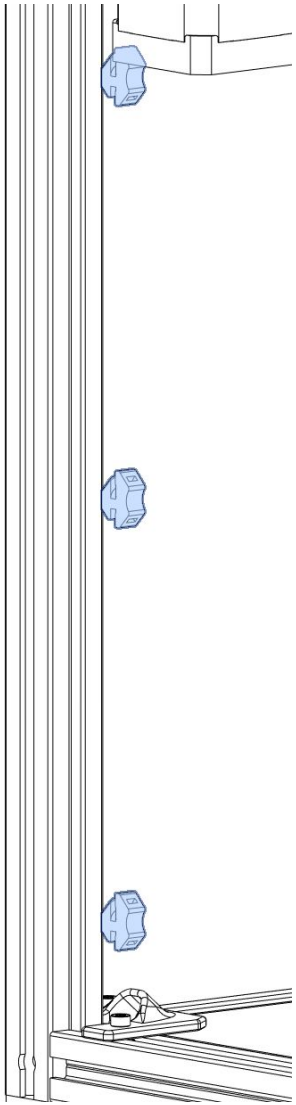


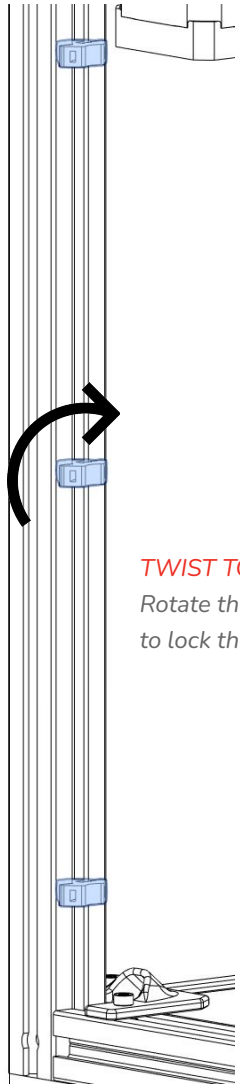


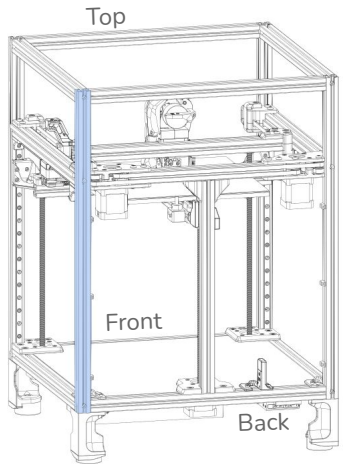


TWIST TO LOCK

Rotate the zip tie loops clockwise by 90° to lock them in place.







TWIST TO LOCK

Rotate the zip tie loops clockwise by 90° to lock them in place.

VORON Trident was released on August 16, 2021.

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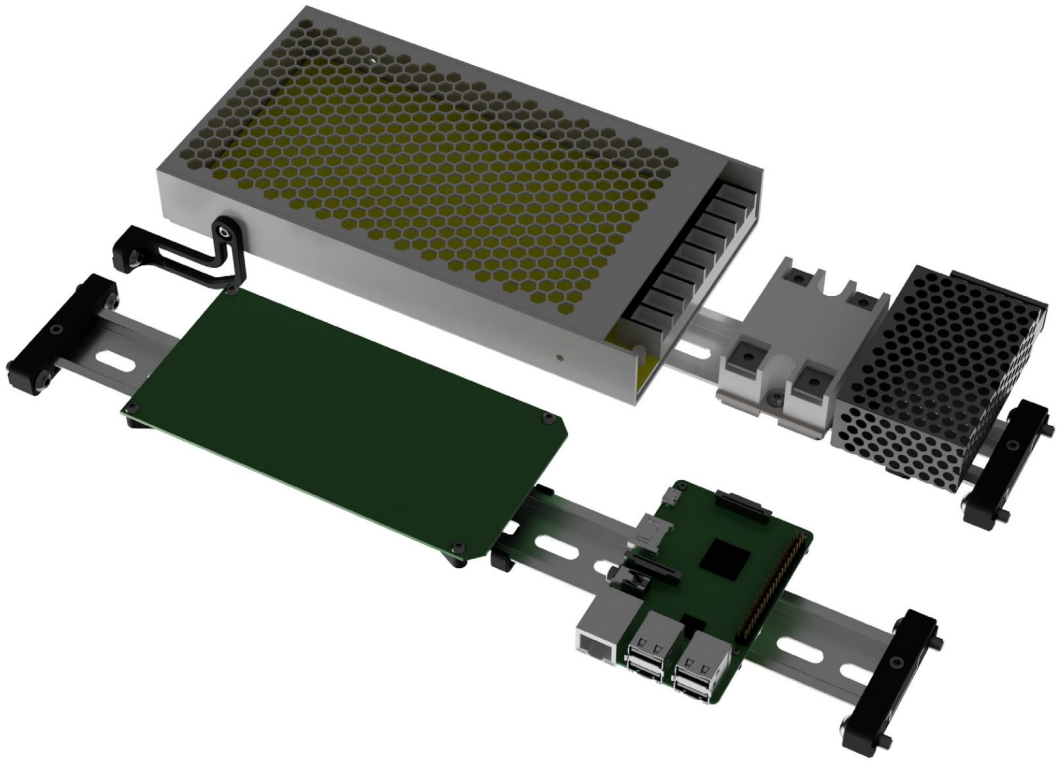
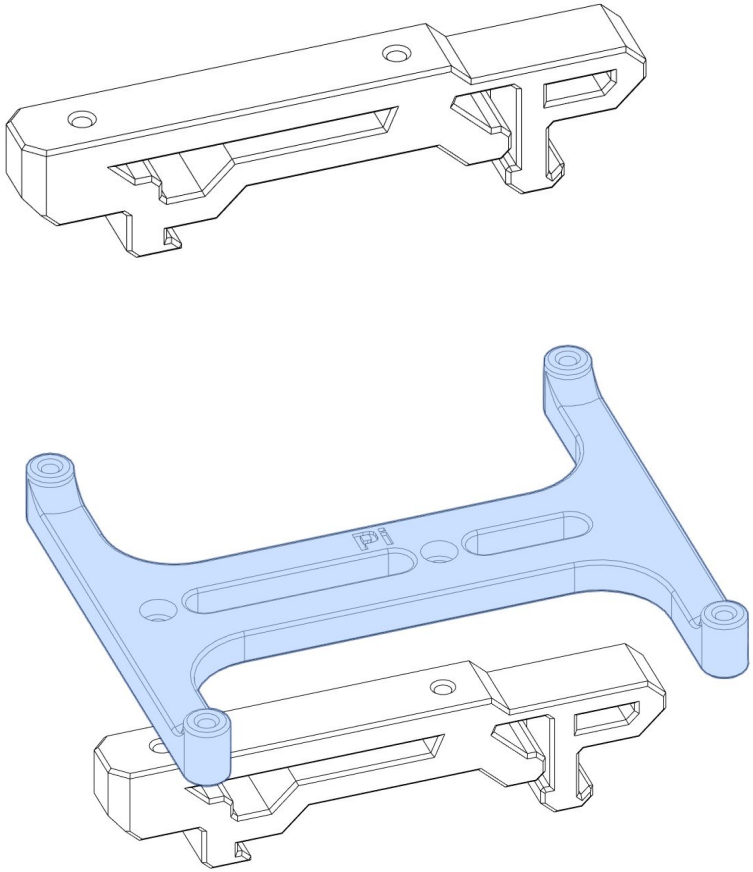
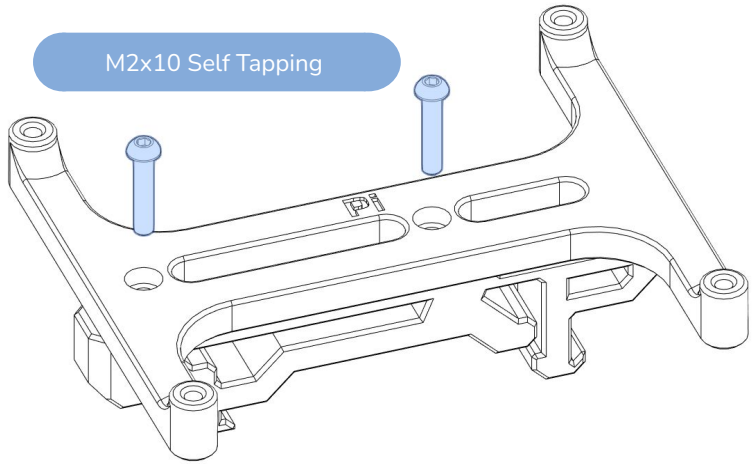
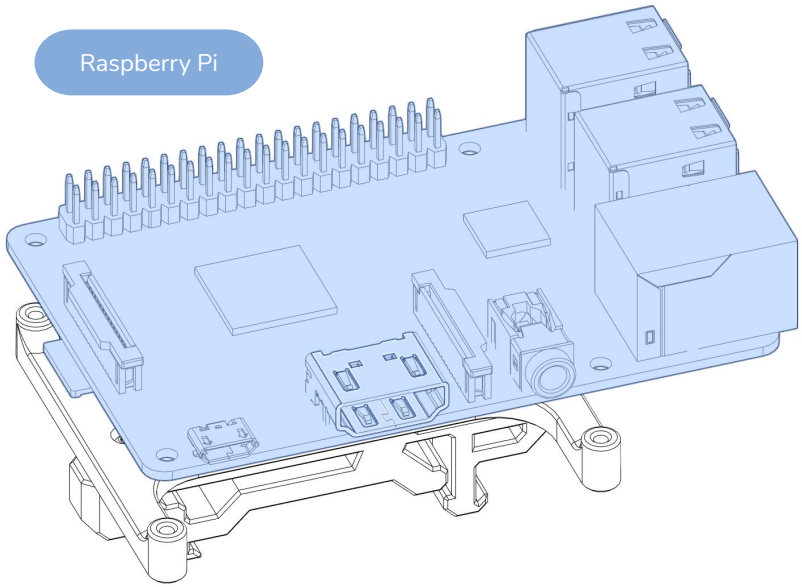


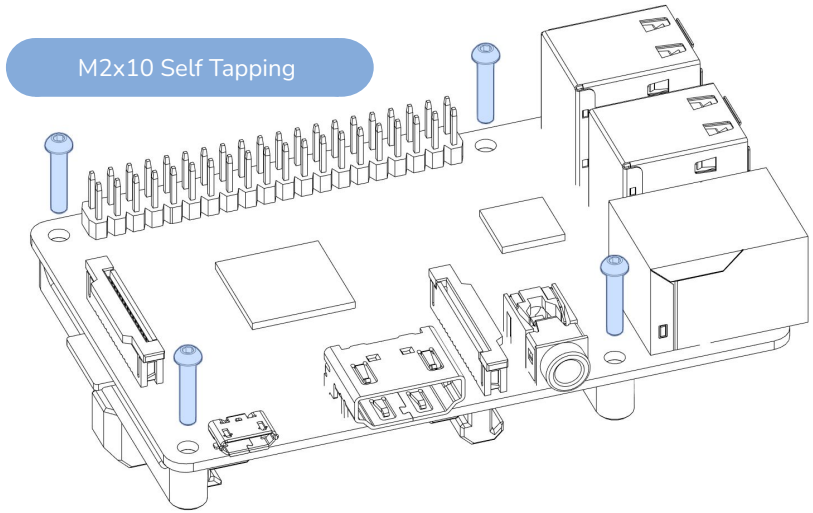
Skip; Raspberry Pi mounting and 5v PSU mounting is not needed. Instead, mount the CB1 / CM4 on the Manta.



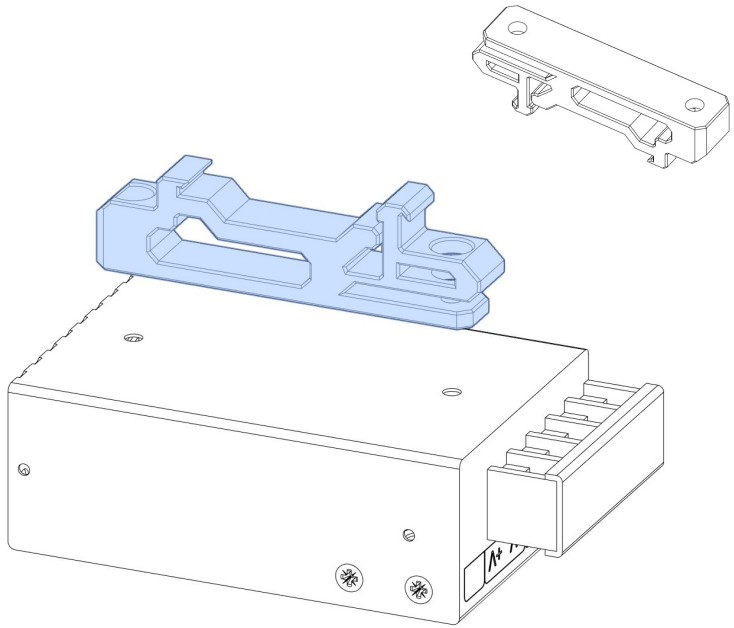


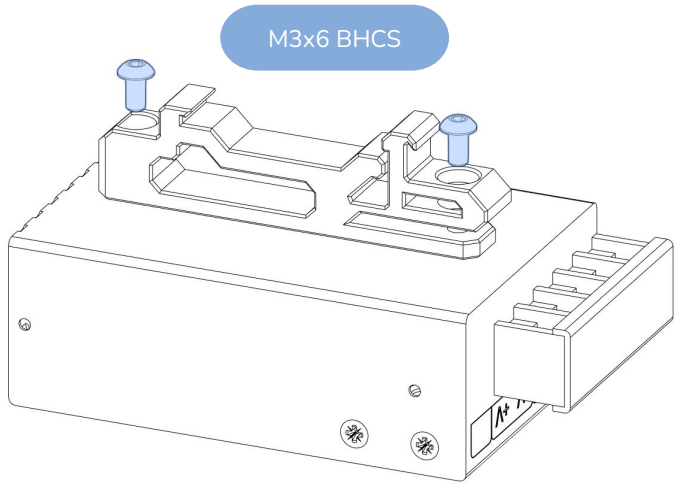
Skip; Raspberry Pi mounting and 5v PSU mounting is not needed. Instead, mount the CB1 / CM4 on the Manta.

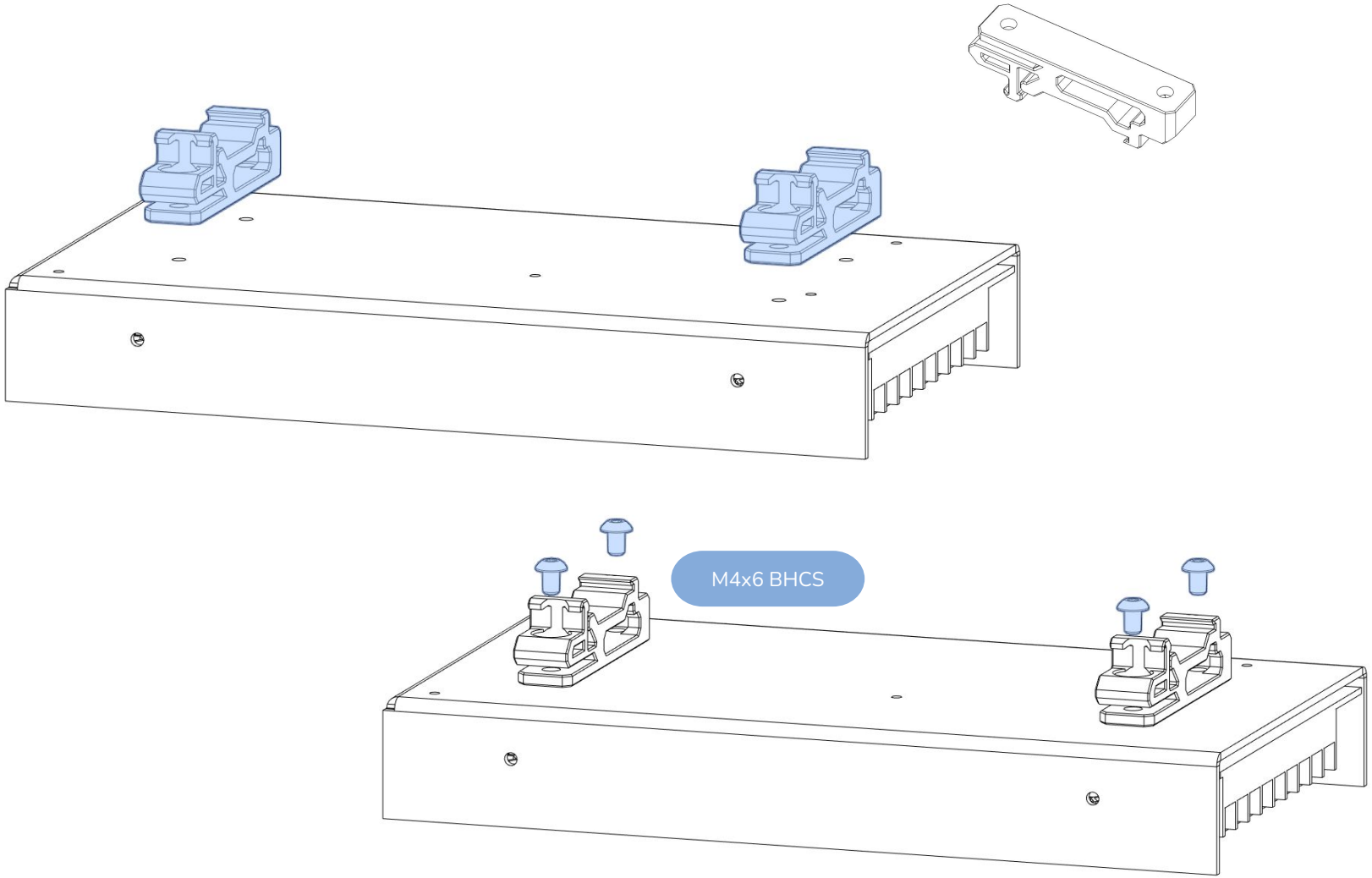


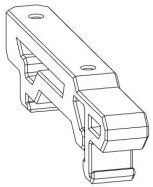


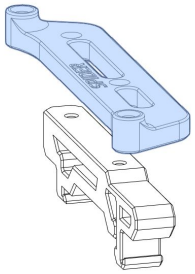
Skip; Raspberry Pi mounting and 5v PSU mounting is not needed. Instead, mount the CB1 / CM4 on the Manta.

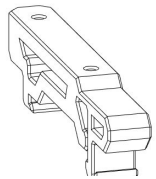


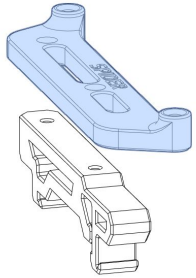


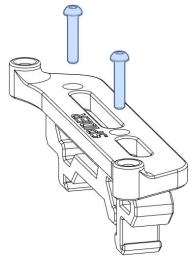

M4x6 BHCS



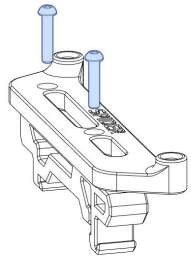






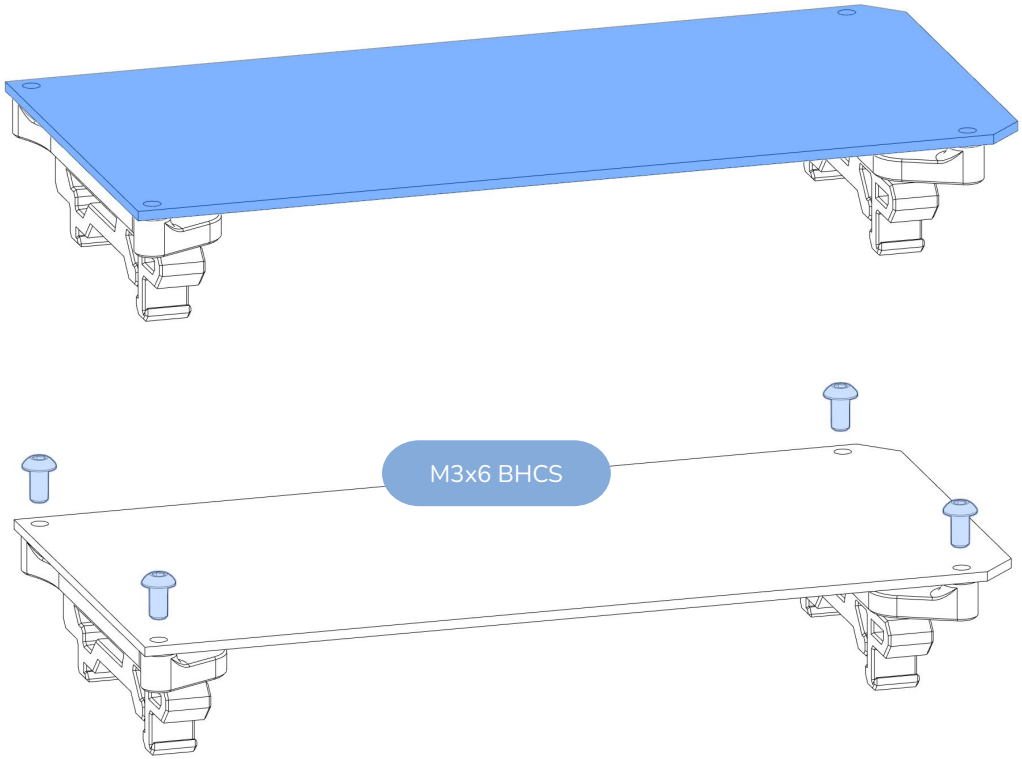


M2x10 Self Tapping



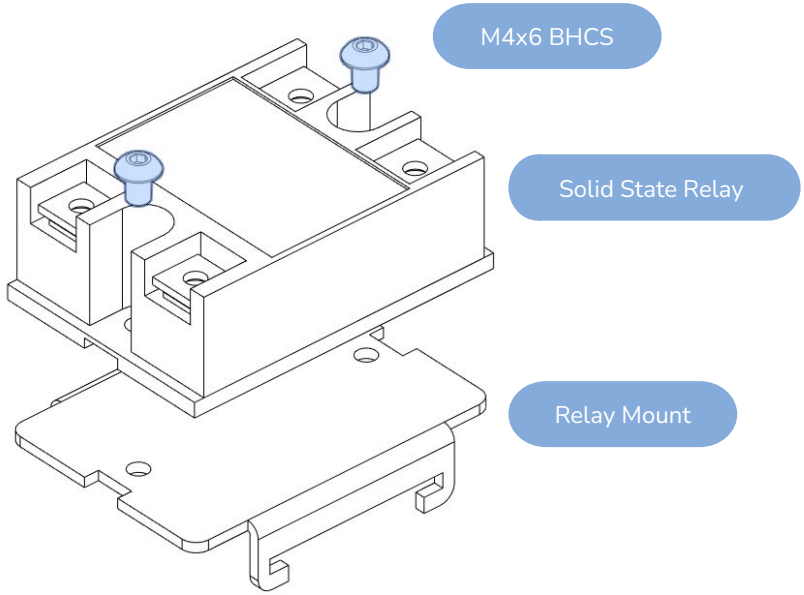
AVAILABLE MOUNTS

We also provide mounts for other controller boards. They are assembled in a similar manner.



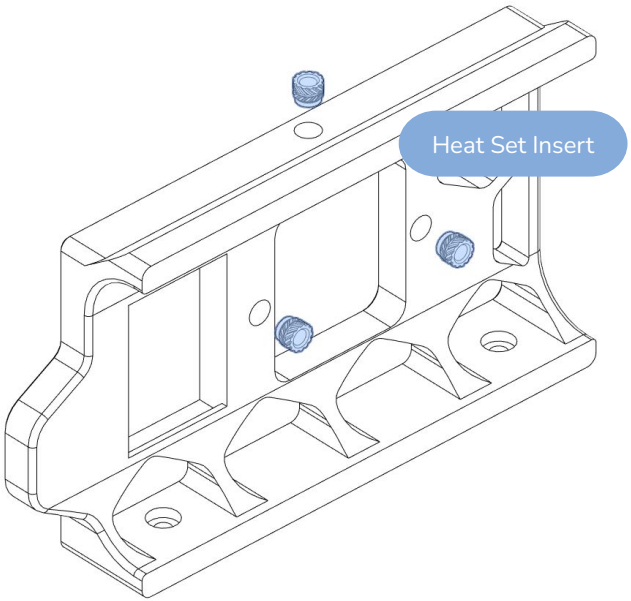
WHY DOES IT LOOK THAT WAY?

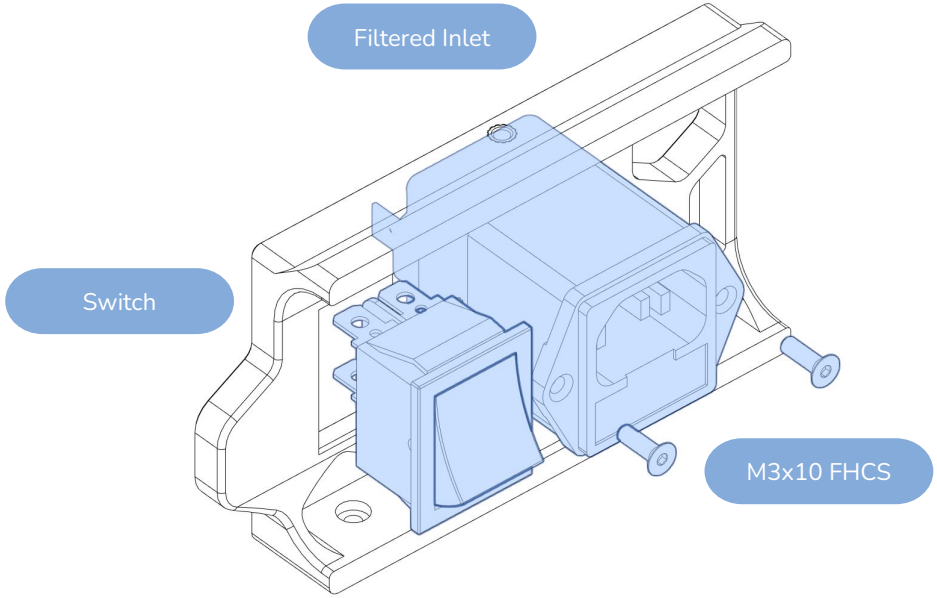
We used a dummy to keep the file size
CAD manageable. The wiring section v
featured image.



WHERE CAN I FIND THE RELAY MOUNT?

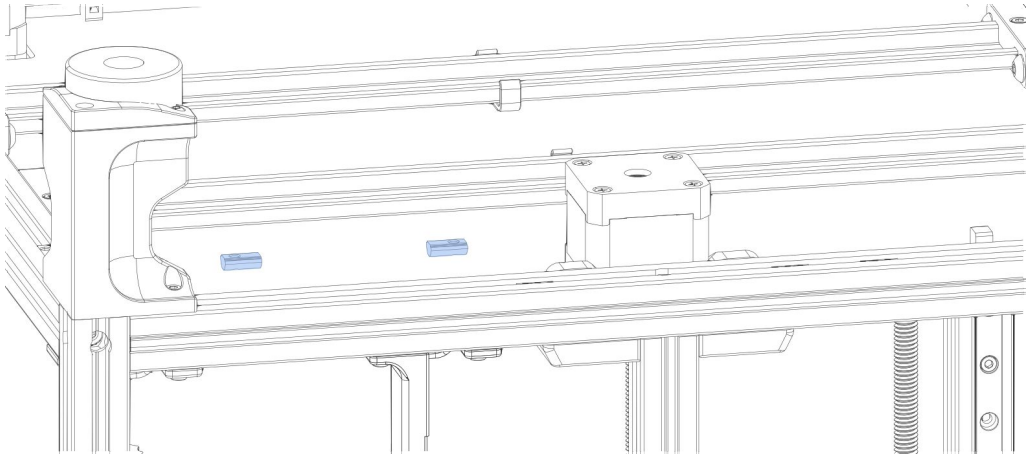
The SSR mount is an off the shelf part. Look for a metal bracket in your pile of parts. There is no printed mount.

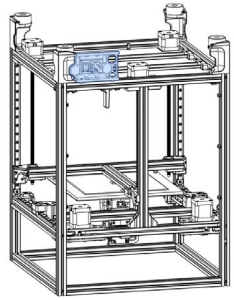


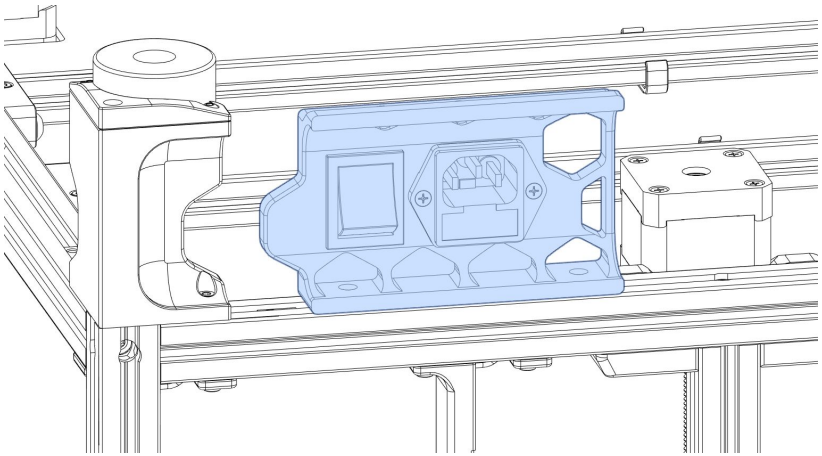


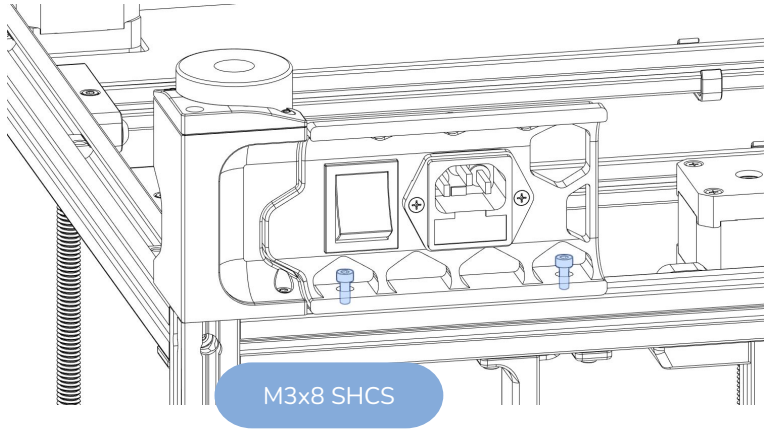
ALTERNATE INLETS

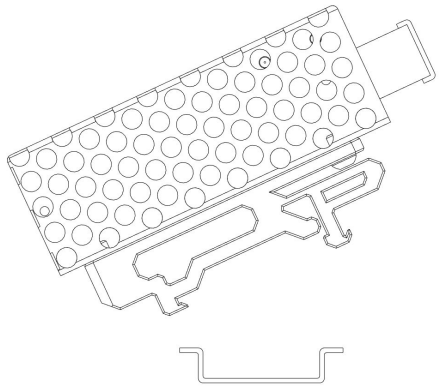
This page displays the filtered inlet. For combo inlets, the printed parts and inlet will differ.

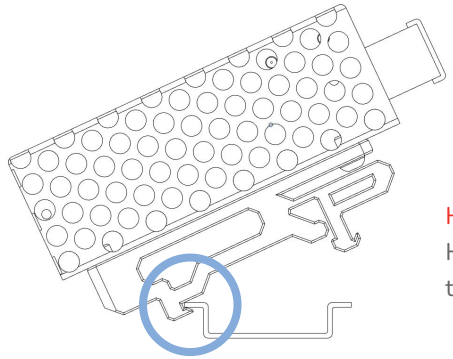






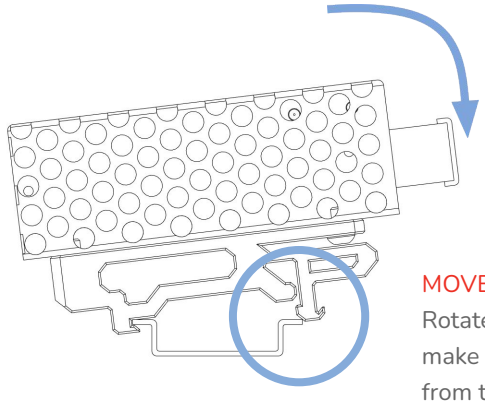

M3x8 SHCS





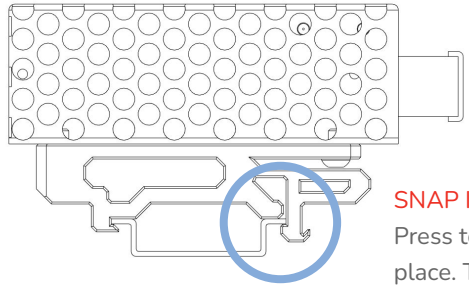
HOOK FIXED SIDE

Hook the fixed side of the printed mount to the side of DIN rail.



MOVE INTO POSITION

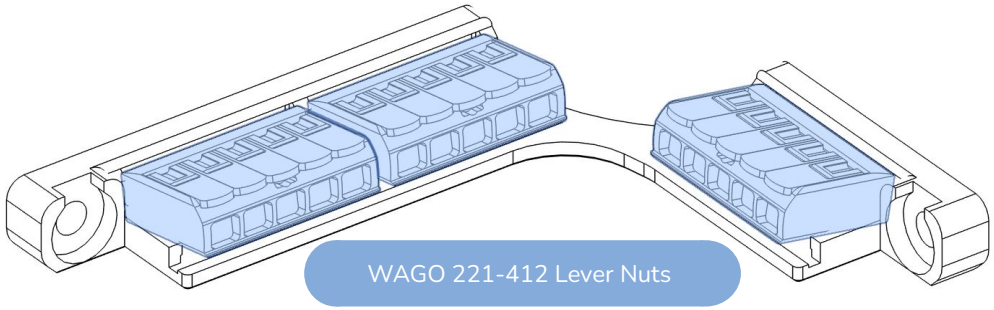
Rotate the part into place, make sure it does not unhook from the fixed side.

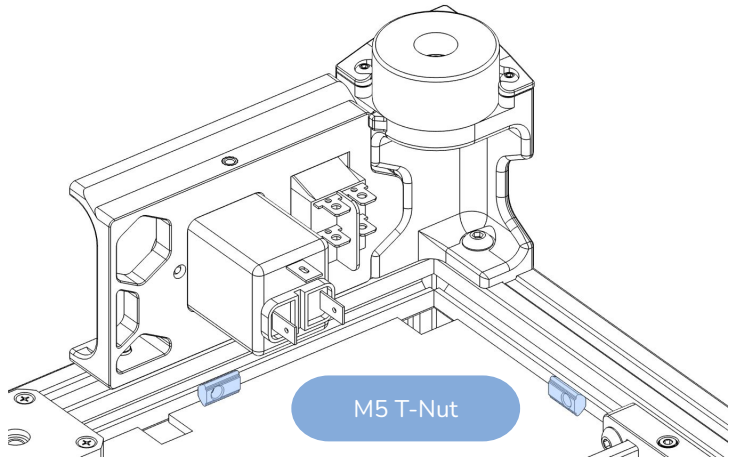


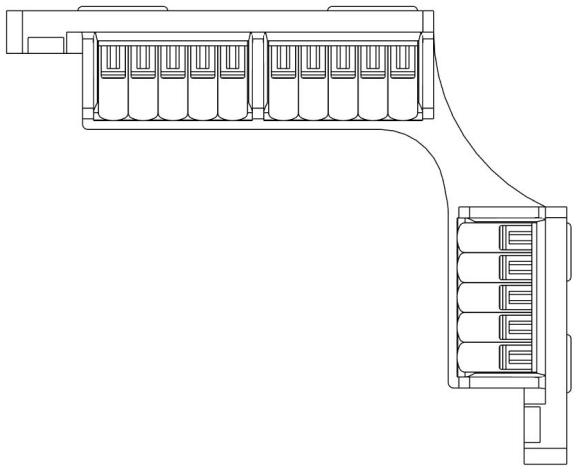
SNAP INTO PLACE

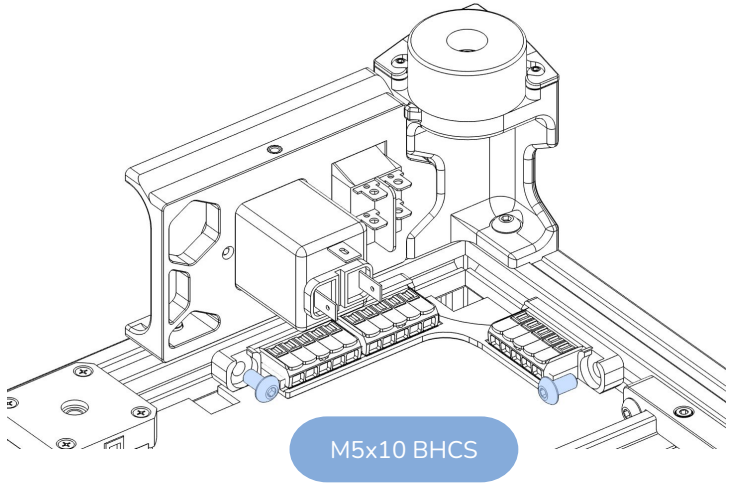
Press to snap the free side into place. The part should now sit securely on the DIN rail.

You use the 3in6 Power Distributor instead.









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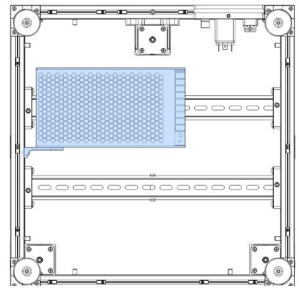
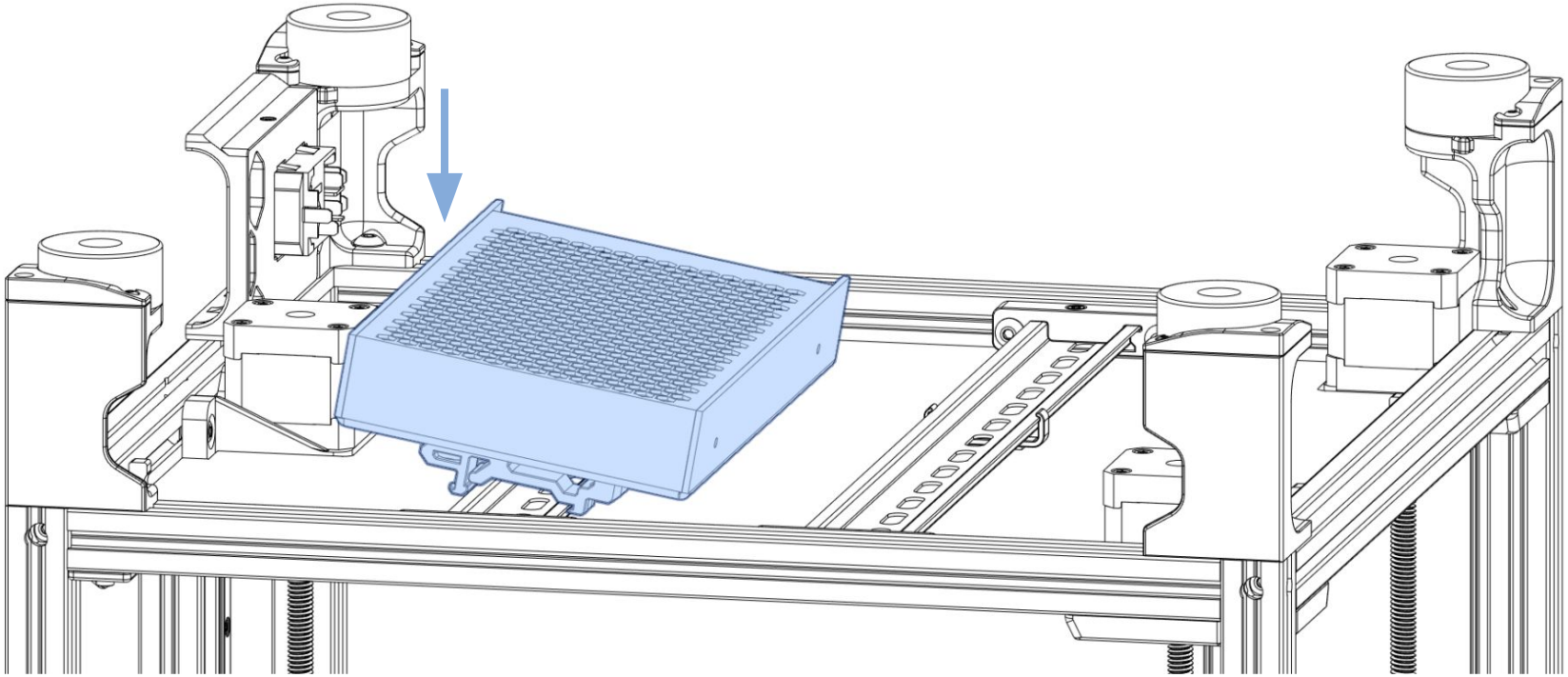
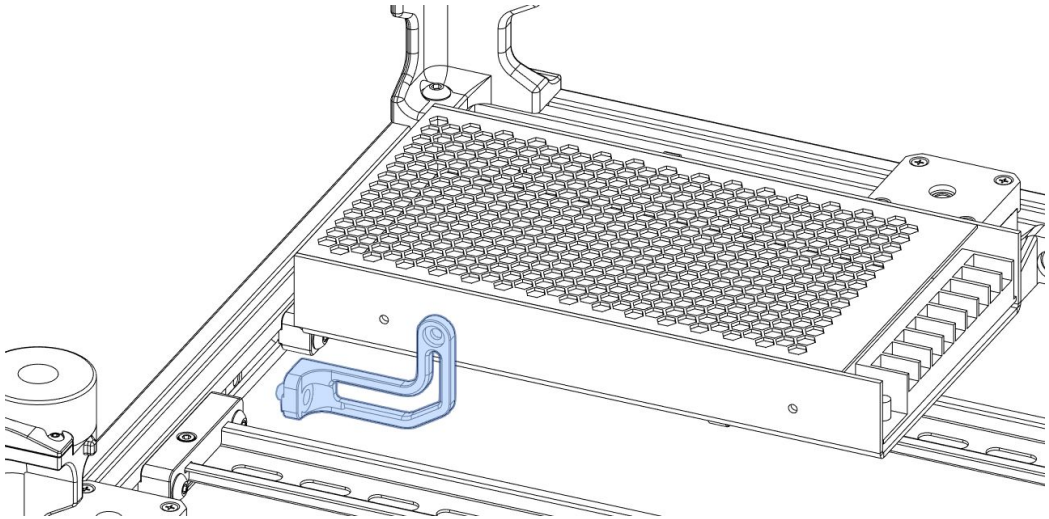
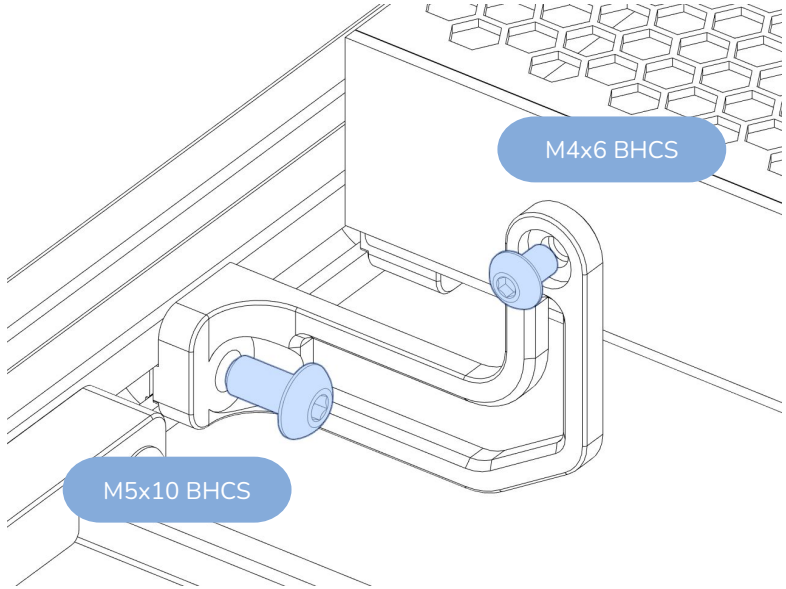
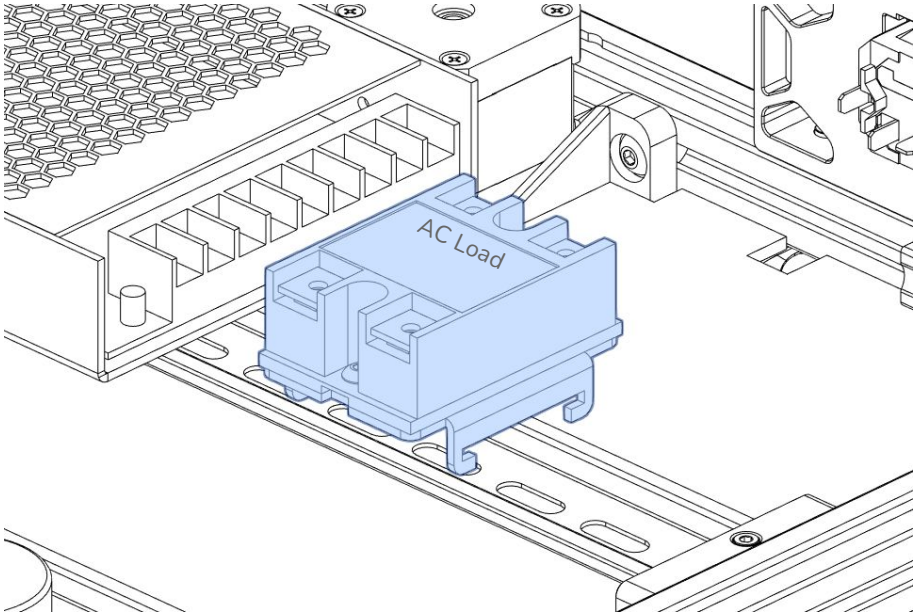
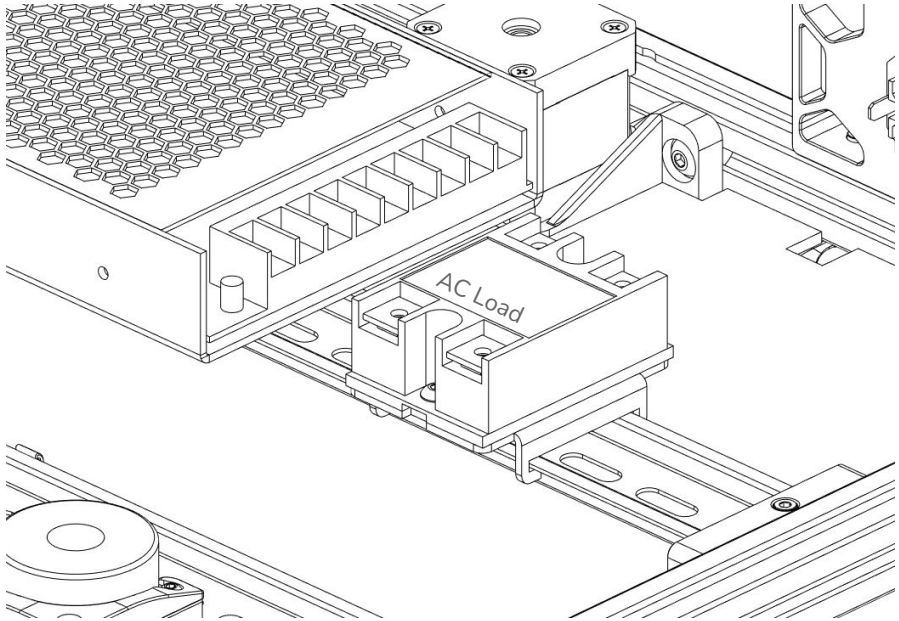


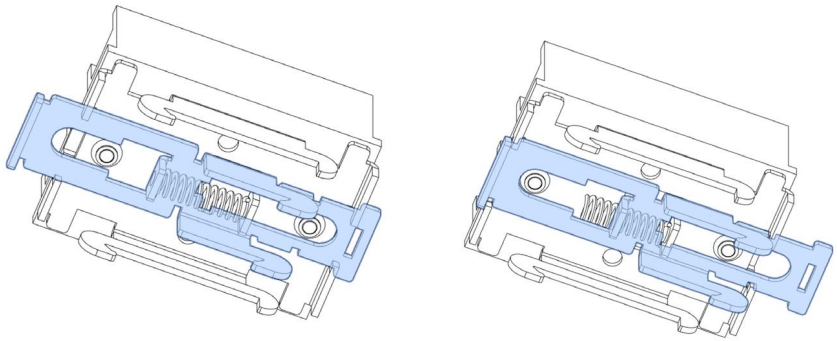





M4x6 BHCS
M5x10 BHCS



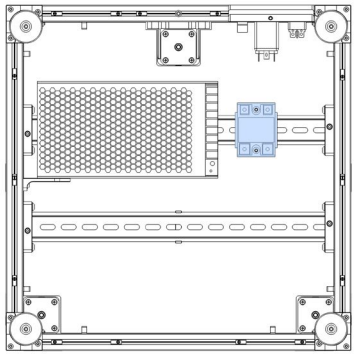




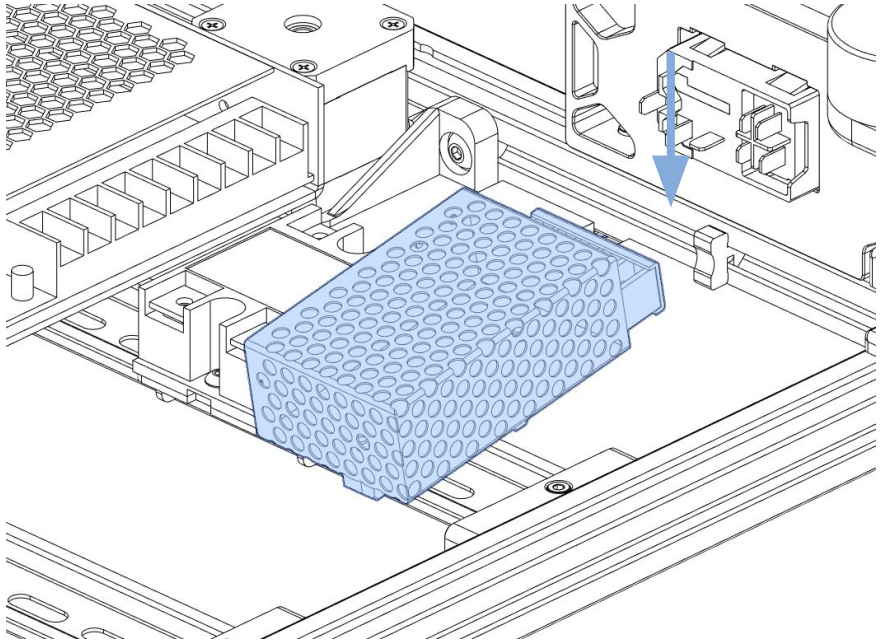
SPRING LOADED

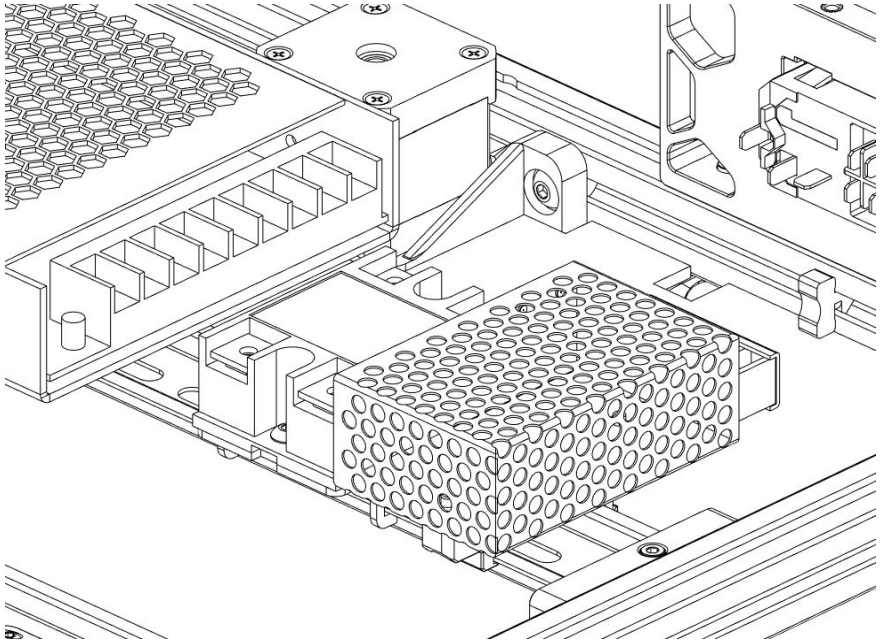
Use a flat head screwdriver to pull the latch open. It will lock open.

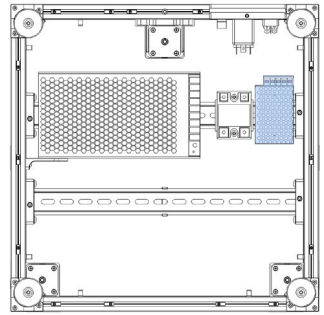
Be careful when releasing the latch, it will snap back into place. Mind your fingers.



another small reminder, you don't use a 5v PSU.







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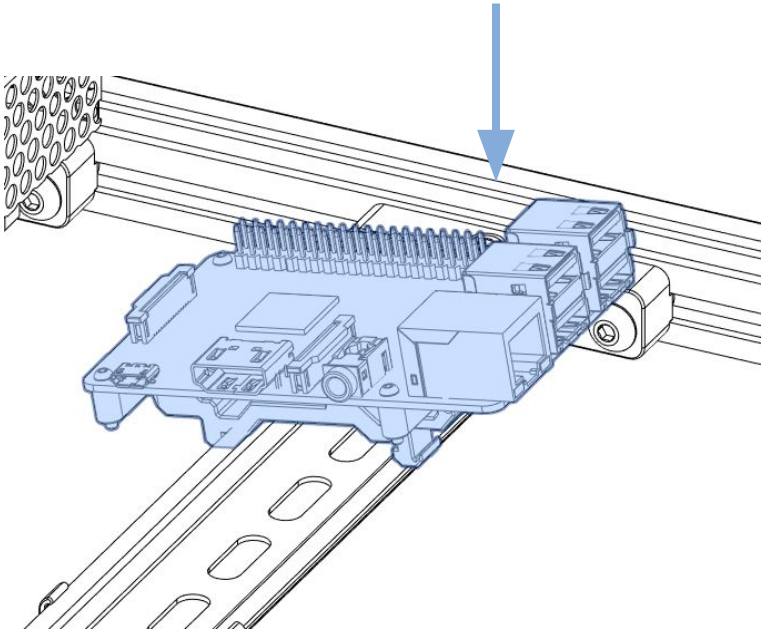
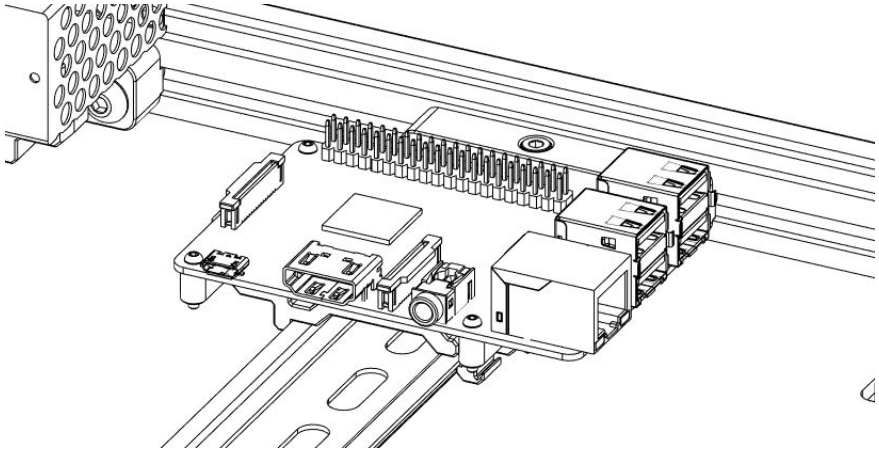
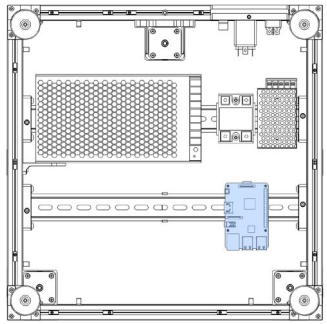
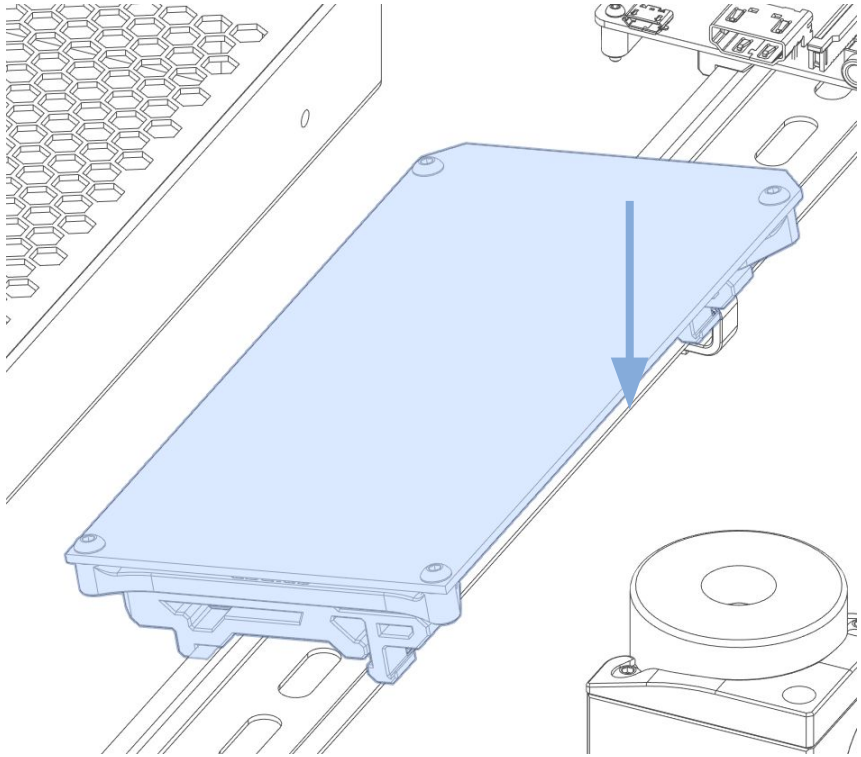
small reminder, you dont use a Raspberry Pi.

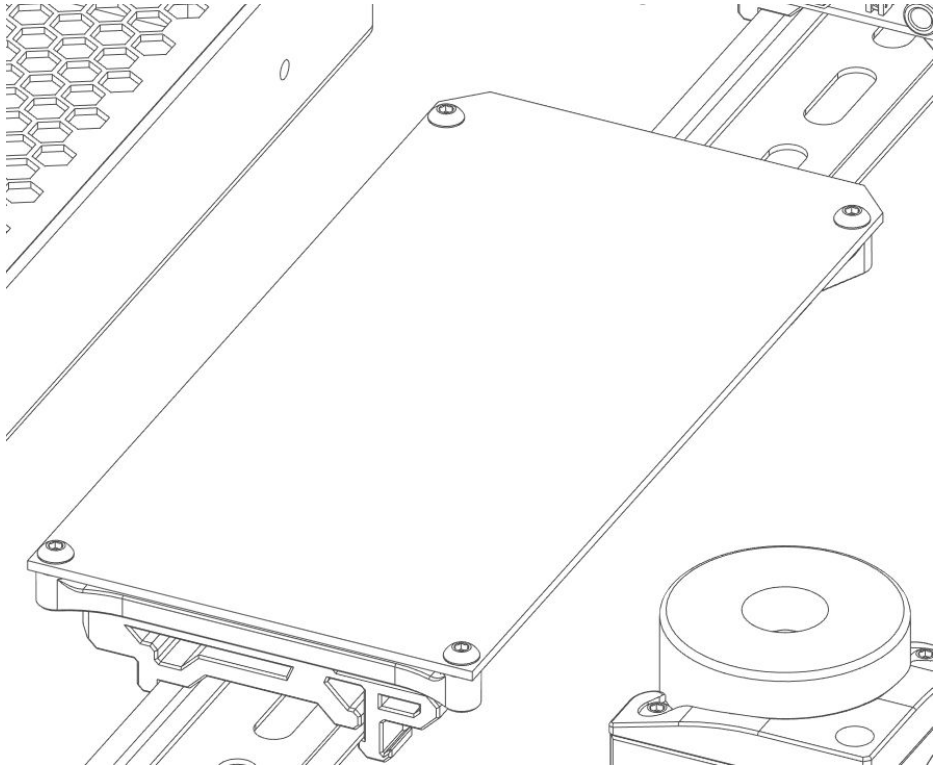


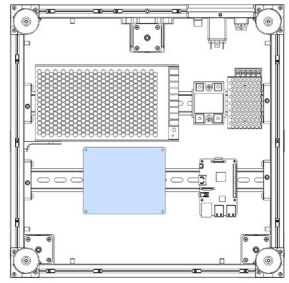




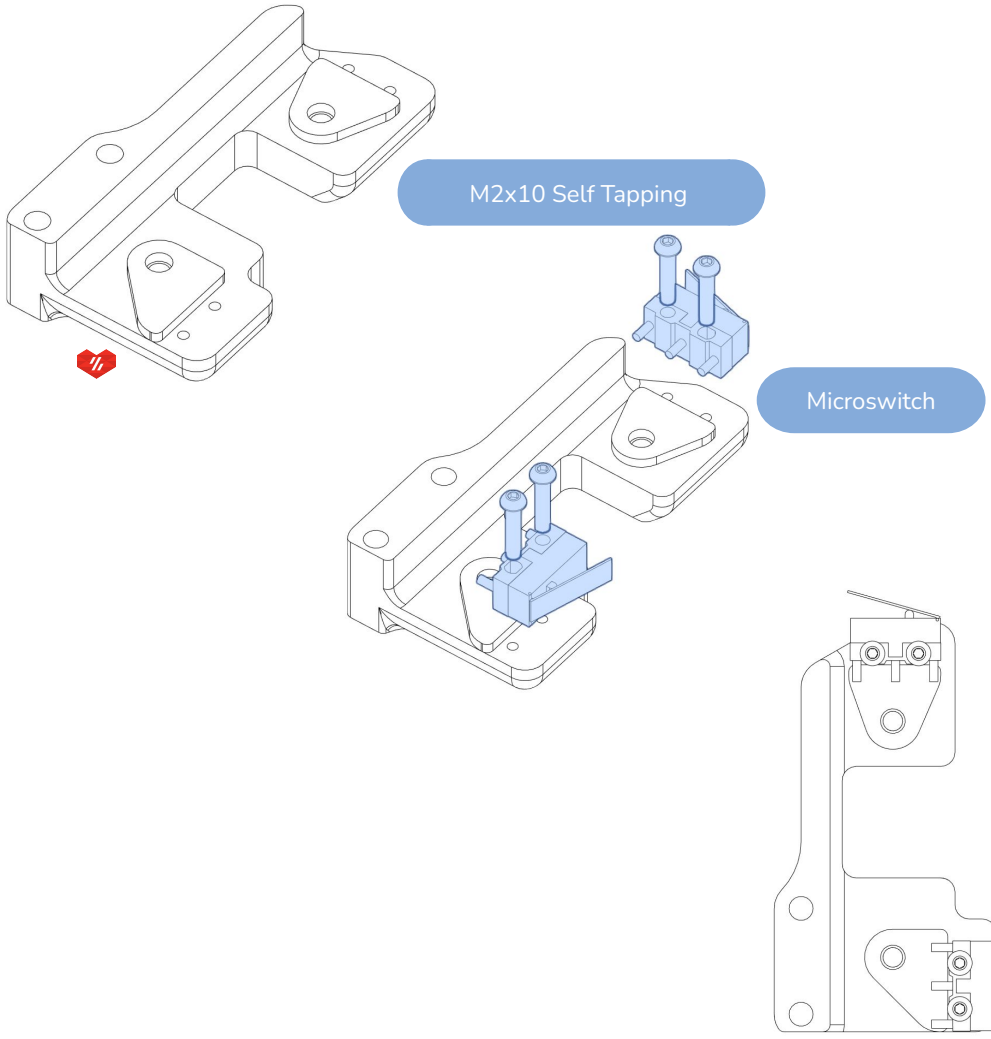
196

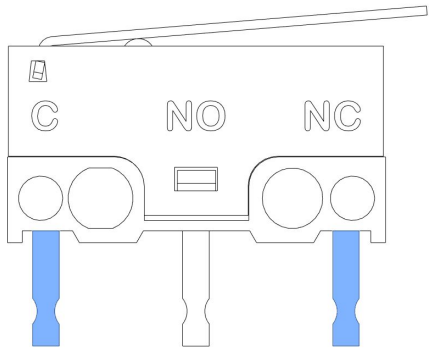






Skip; TAP is your Z-Endstop, your X Endstop sits on Tap and your Y endstop will be mounted on the A Motor mount later.





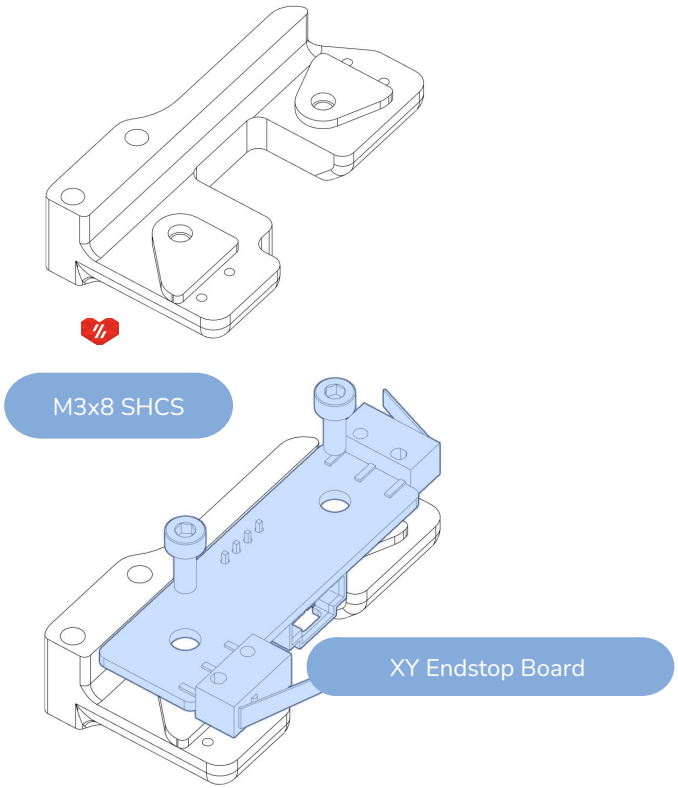
ENDSTOP SWITCHES FOR X AND Y

Endstops are wired in a “Normally Closed” configuration. On microswitches those are the 2 outer terminals indicated by C and NC.

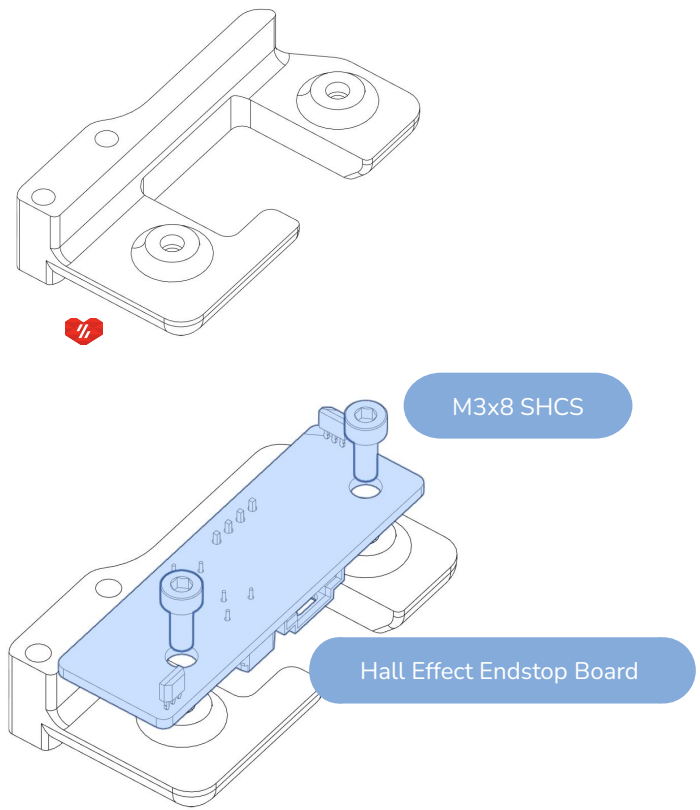
Prepare the switches for X and Y by soldering 150mm of wire to each of the outer terminals.

Skip; TAP is your Z-Endstop, your X Endstop sits on Tap and your Y endstop will be mounted on the A Motor mount later.

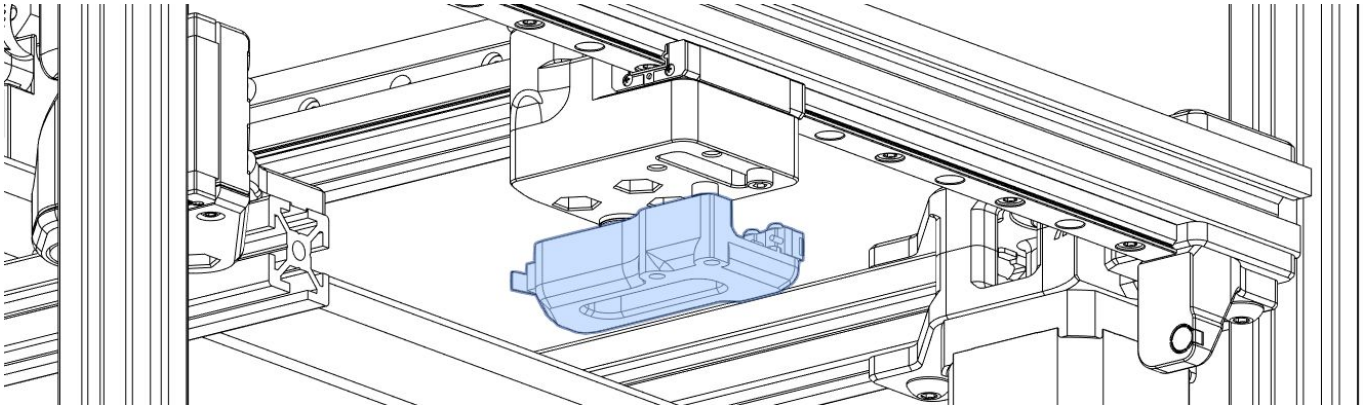
OPTION: XY ENDSTOP BOARD

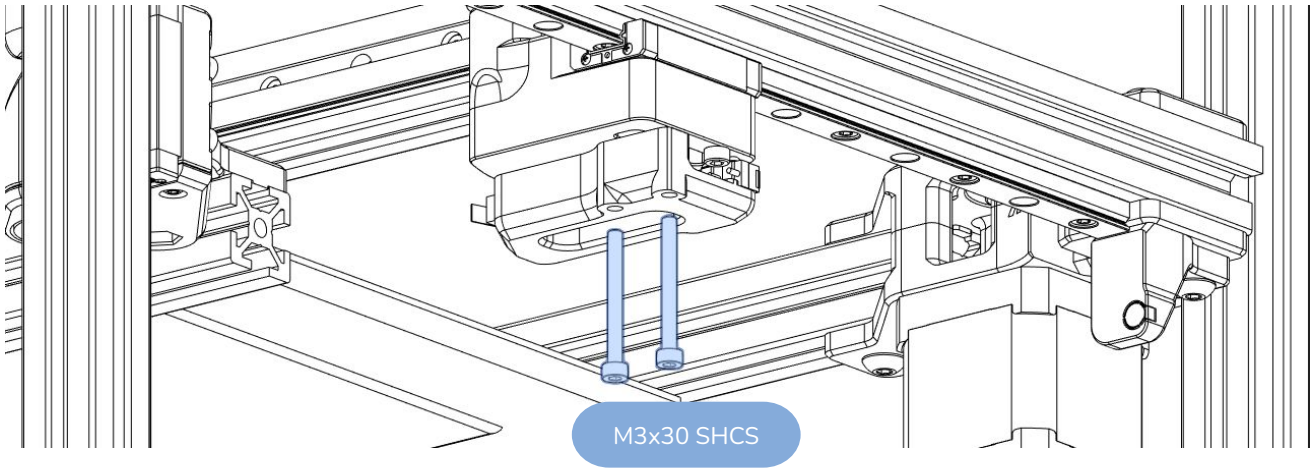


OPTION: HALL EFFECT ENDSTOP BOARD



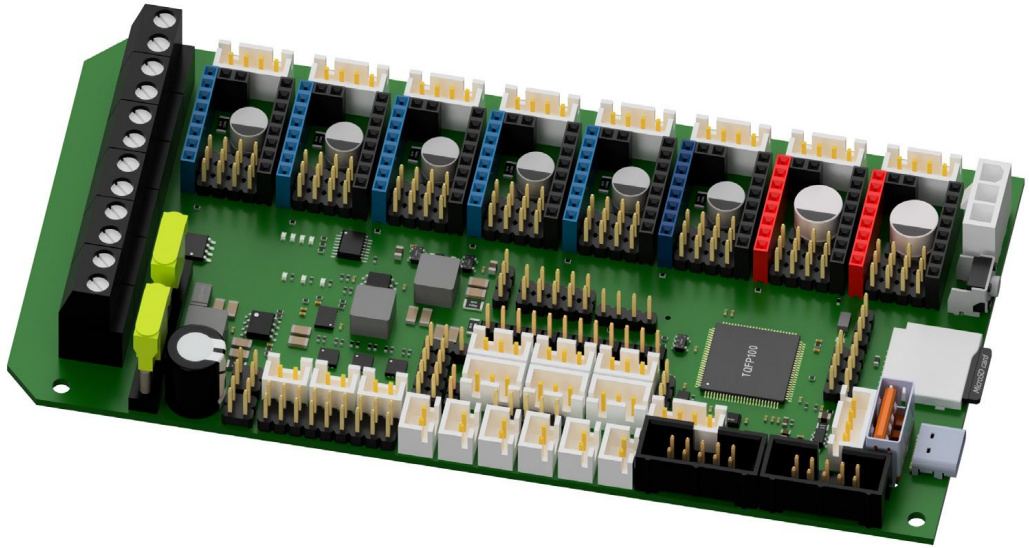
Skip; TAP is your Z-Endstop, your X Endstop sits on Tap and your Y endstop will be mounted on the A Motor mount later.





Around 1100 images have been created for use in this manual. Slightly below 700 made it into the final document.

Skip these steps, instead configure your Manta like this:
![M8P_v2 Jumper / Driver marking](https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/M8P_config.pdf)
Place a Jumper on all yellow marked pins and place the TMC2209s with attached heatsink on all driver ports with a blue circle after installing the jumpers.



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Place a Jumper on all yellow marked pins and place the TMC2209s with attached heatsink on all driver ports with a blue circle after installing the jumpers.

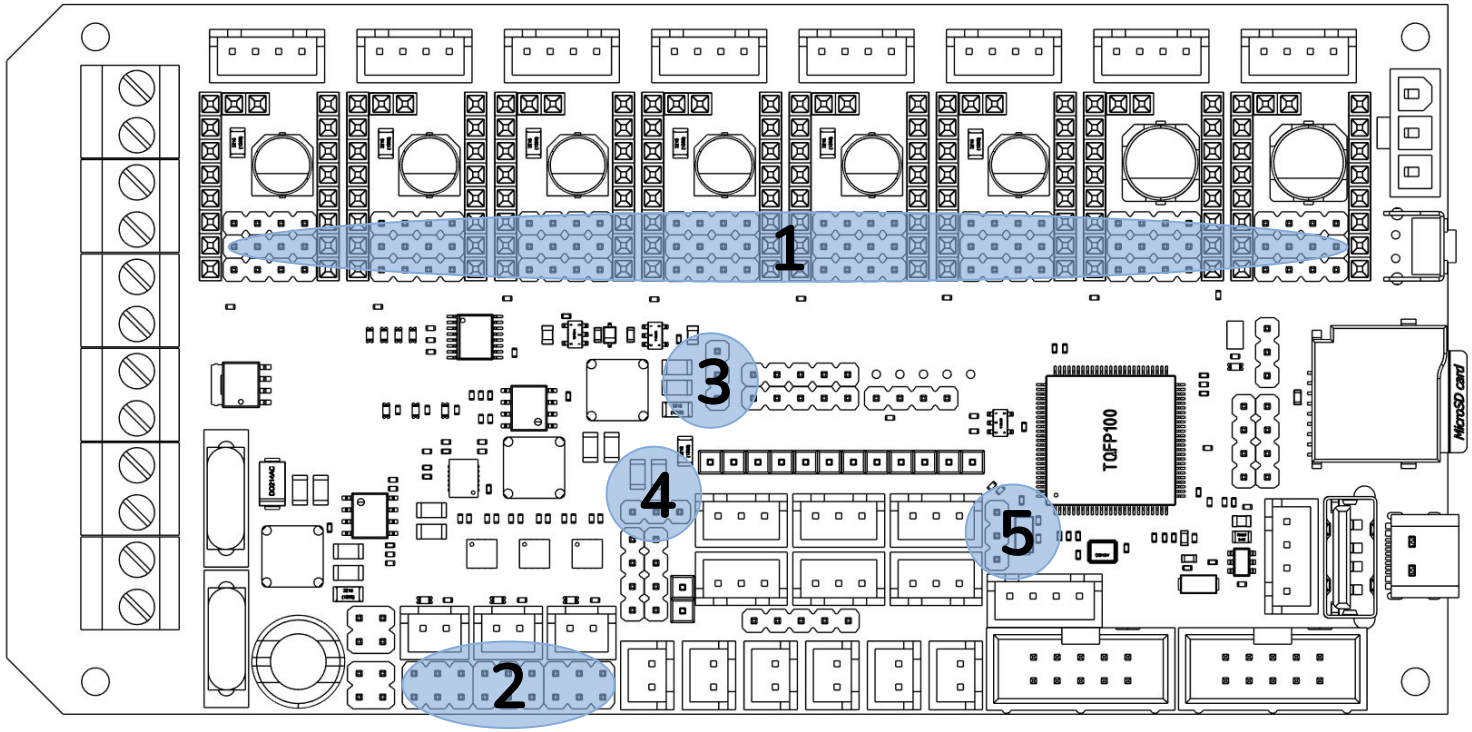
CONTROLLER BOARD

The assembly manual will outline the wiring for a Fysetc Spider V2.2 board. You can find additional documentation and alternative configurations on docs.vorondesign.com.

JUMPERS

Several jumpers need to be configured on the controller board. Begin by removing all the JUMPERS from the controller board (MCU).

1) Remove the jumpers in the “driver sockets”.
2) Remove all the jumpers on the “Fan Voltage Selection”
3) Remove the “USB 5V power supply” jumper
4) Remove the “LED Voltage Selection” jumper
5) Remove the “Probe Voltage Selection” jumper

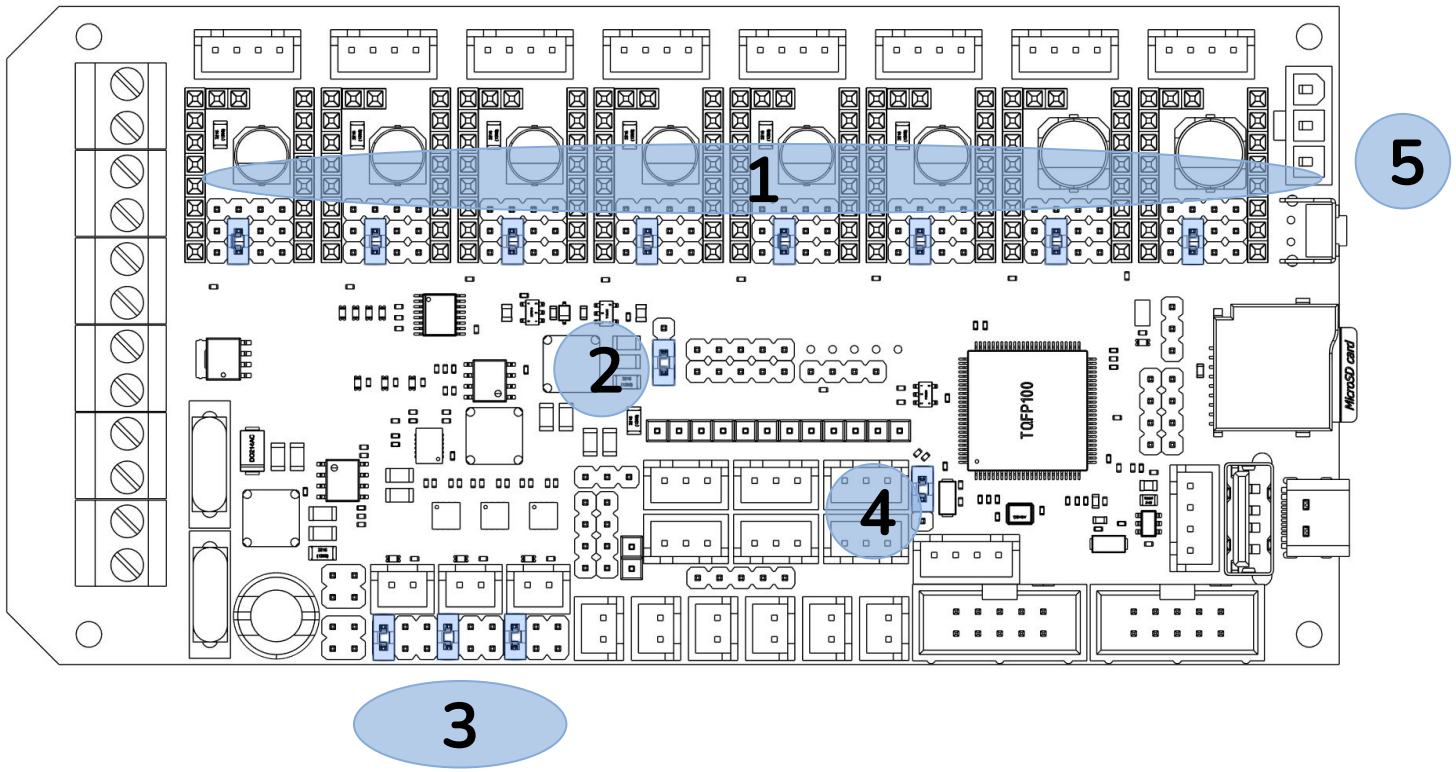


Skip these steps, instead configure your Manta like this:
![M8P_v2 Jumper / Driver marking](https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/M8P_config.pdf)
Place a Jumper on all yellow marked pins and place the TMC2209s with attached heatsink on all driver ports with a blue circle after installing the jumpers.

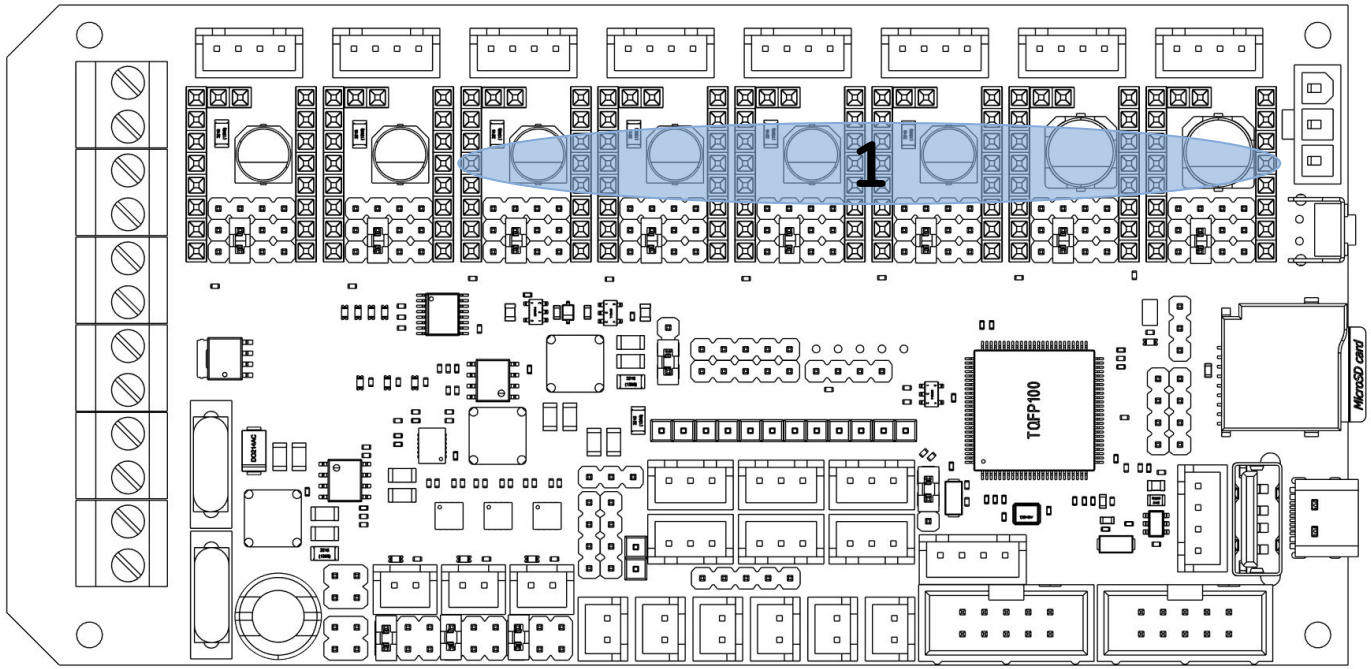
JUMPERS

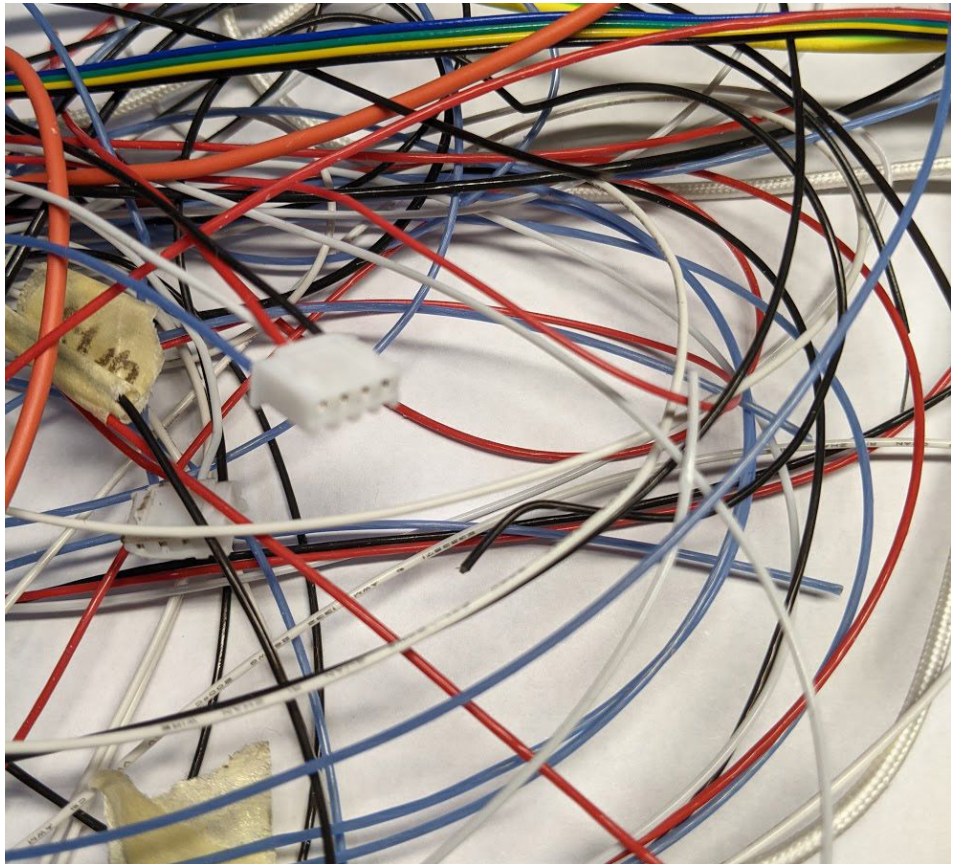
Several jumpers need to be set on the MCU.
Add the following JUMPERS to the controller board (MCU).

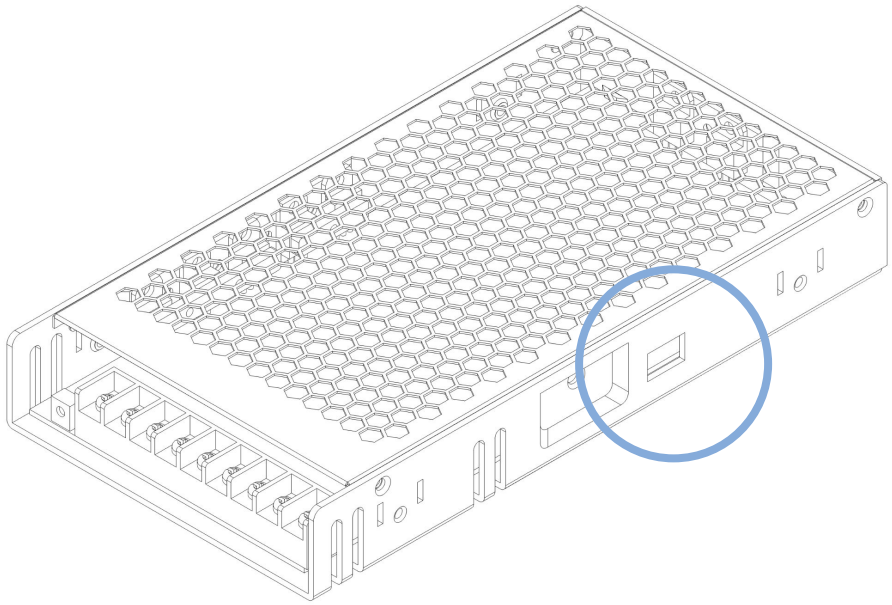
1) Set the jumpers in the “driver sockets” as shown to set TMC2209 UART mode.
2) Ensure the Power Selection header is set to the lower position (DC5V).
3) Set the Jumpers for the “Fan Voltage Selection” header so they match your fan's voltage. Shown here are the settings for 24VDC.
4) Set the jumper in “Probe Voltage Selection” header to 24VDC.
5) Set the included wire jumper to the 24VDC position.



Skip these steps, instead configure your Manta like this:
![M8P_v2 Jumper / Driver marking](https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/M8P_config.pdf)
Place a Jumper on all yellow marked pins and place the TMC2209s with attached heatsink on all driver ports with a blue circle after installing the jumpers.







INPUT VOLTAGE SWITCH

Check the input voltage switch of the power supply. It is located in the highlighted area behind the metal mesh.

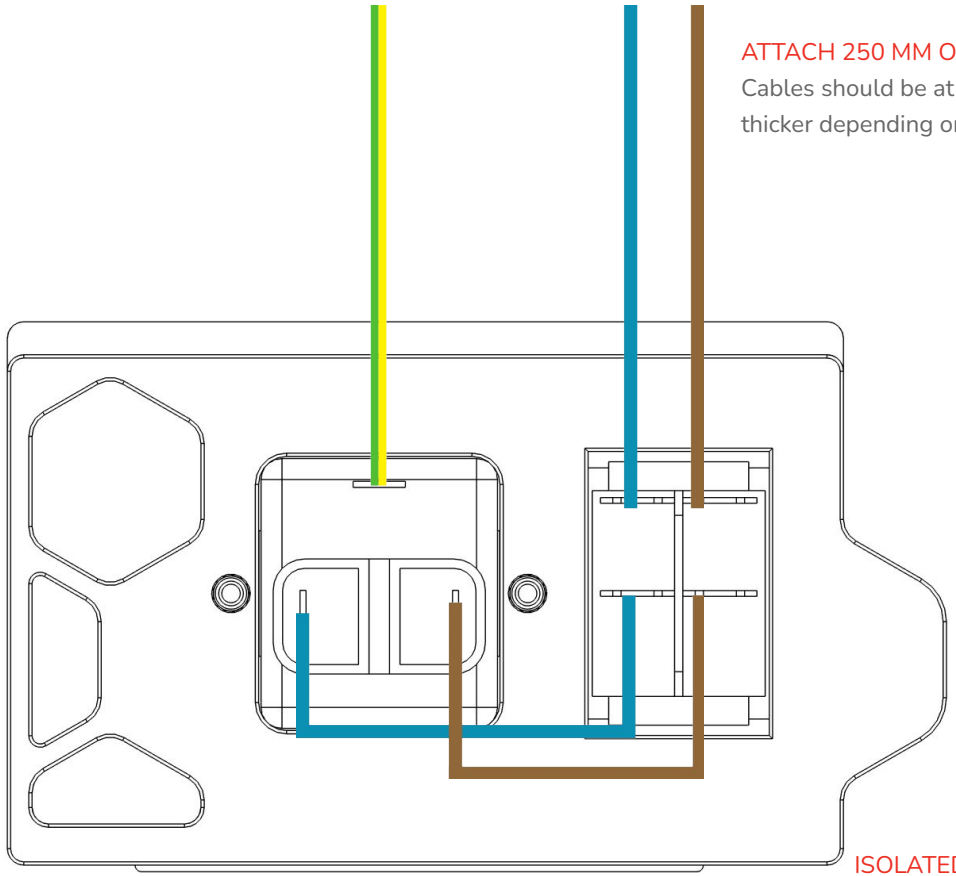
Make sure the selection matches your local mains voltage. Refer to the Mean Well LRS-200 datasheet for possible settings (https://voron.link/e0szdyh).

MAINS INLET WIRING

We show the wiring in the IEC colour scheme. Depending on your region the colour scheme and wiring standards will differ.

Mains wiring should only be done by qualified personnel trained in local regulations and safety standards. Depending on your local regulations you may be forbidden from wiring the mains side and/or putting the printer into operation; seek professional assistance.

Failure to observe those could result in bodily harm.

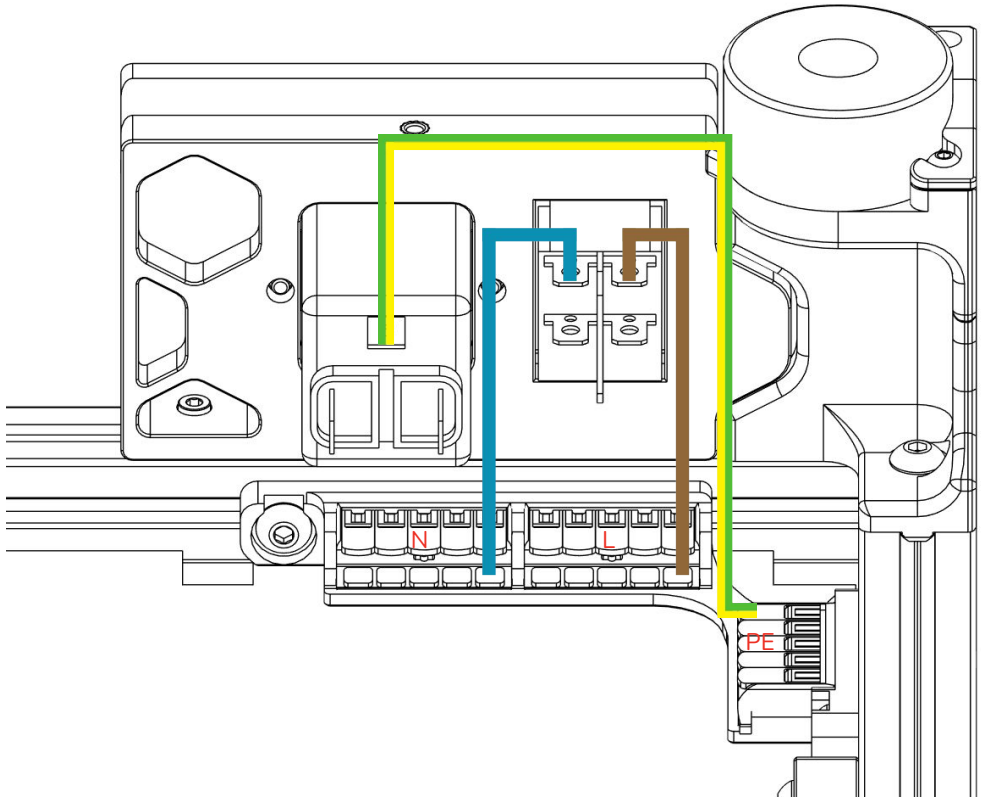


ATTACH 250 MM OF WIRE

Cables should be at least 0.75mm² (AWG18) or thicker depending on local regulations.

ISOLATED CONNECTORS ONLY

Make sure that all mains connectors are properly isolated and meet the applicable safety standards



ISOLATED CONNECTORS ONLY

Make sure that all mains connectors are properly isolated and meet the applicable safety standards

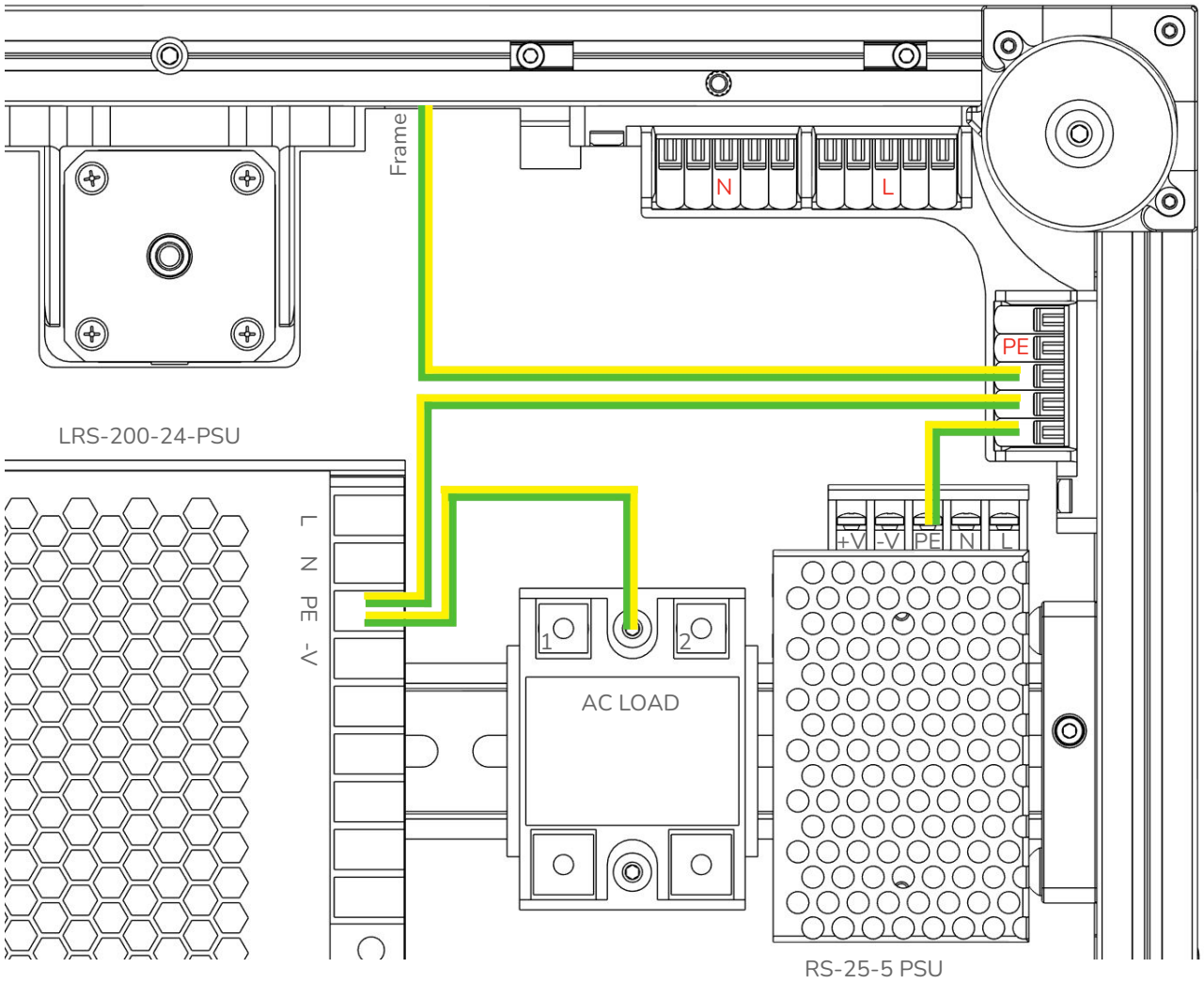
Leave the cables to the 5v PSU out.

MAINS WIRING CONTINUED

Secure the wires with cable clips / cable tie anchors.

The bed heater is powered by AC voltage and receives its PE in a later step.

Observe your local regulations in regards to the Protective Earth connections for the frame/other components.



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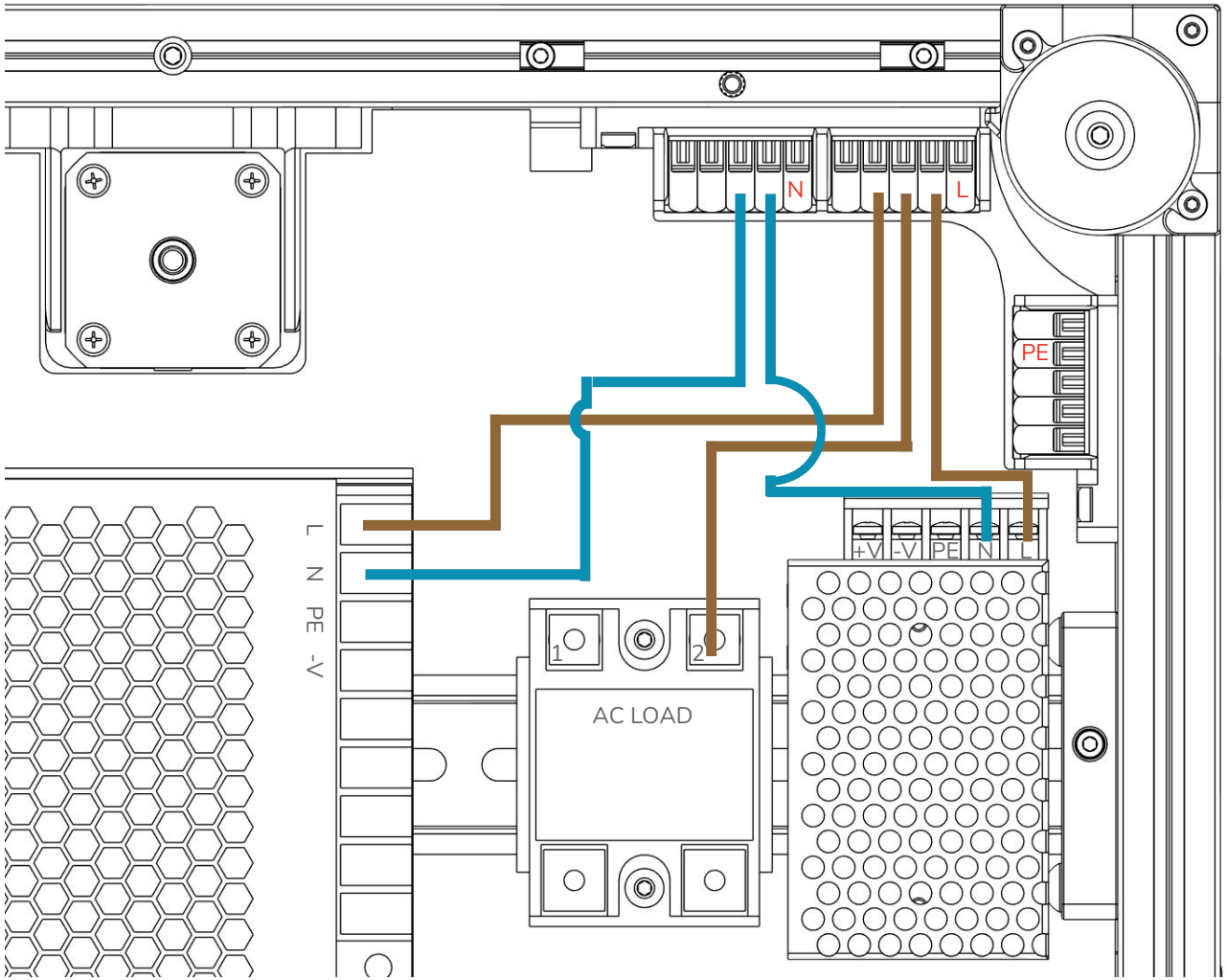
Leave the cables to the 5v PSU out.



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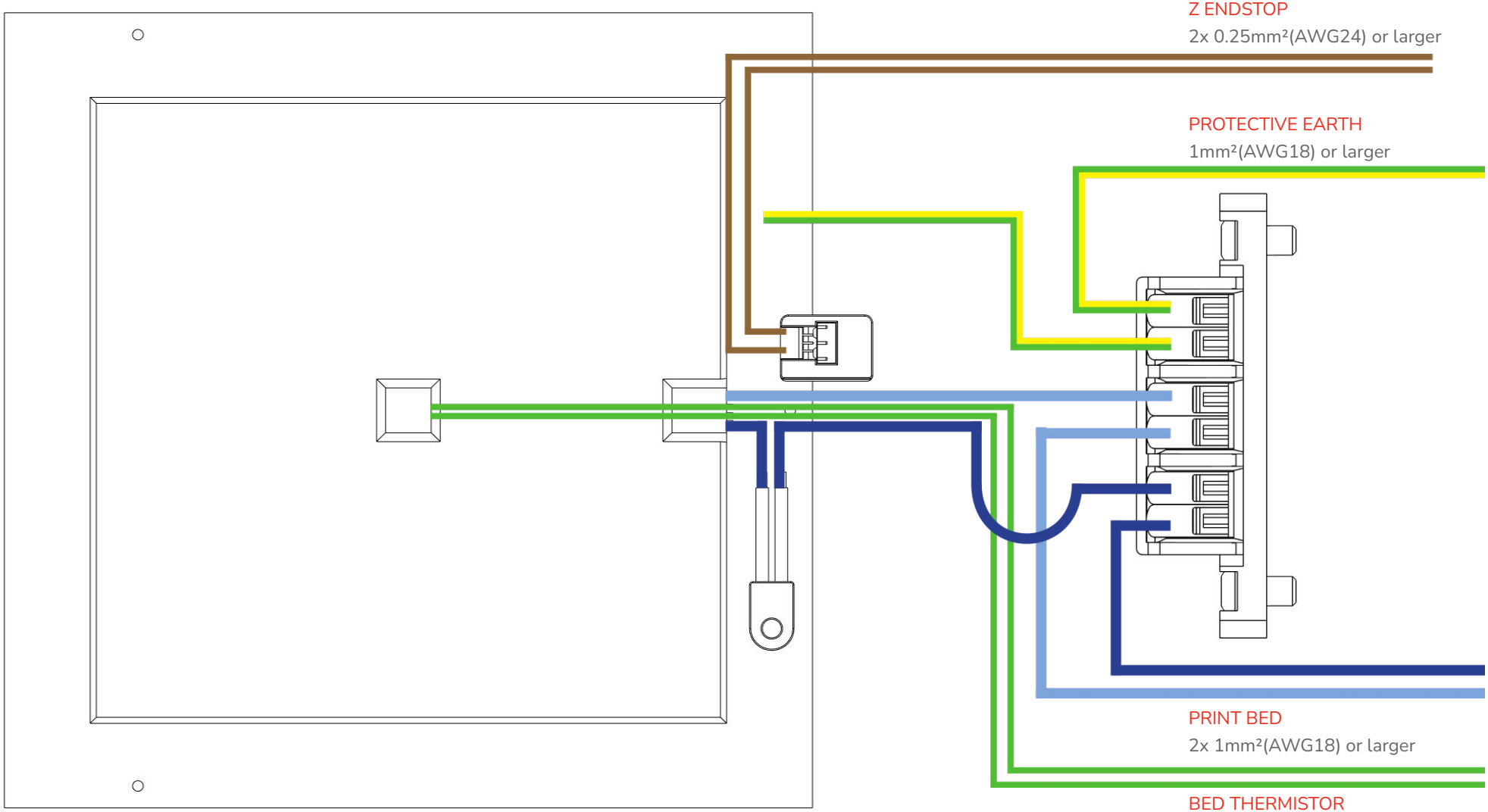
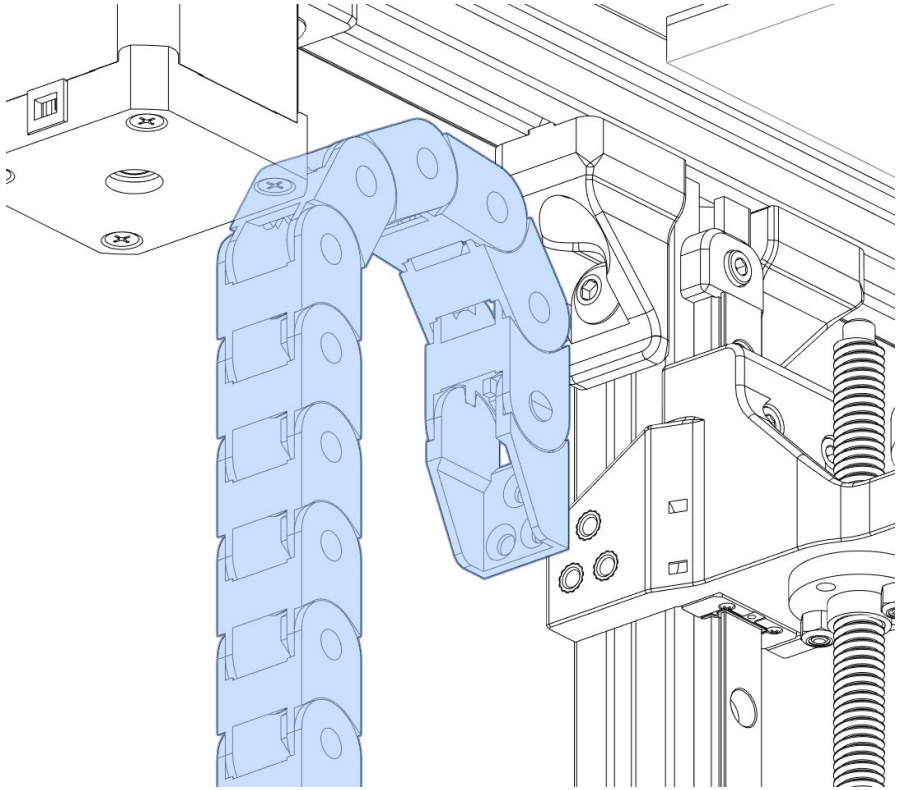
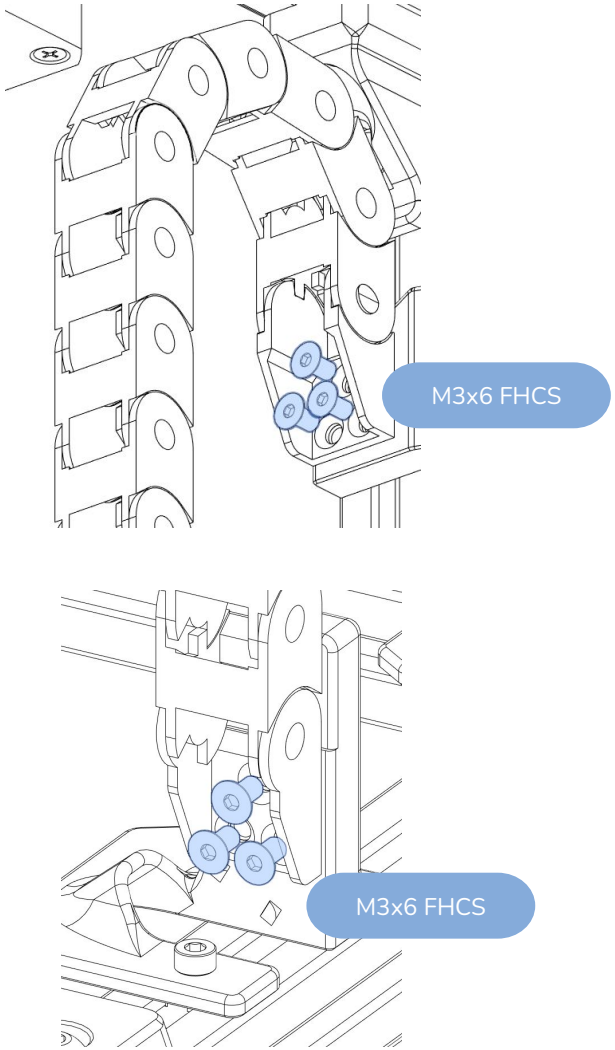
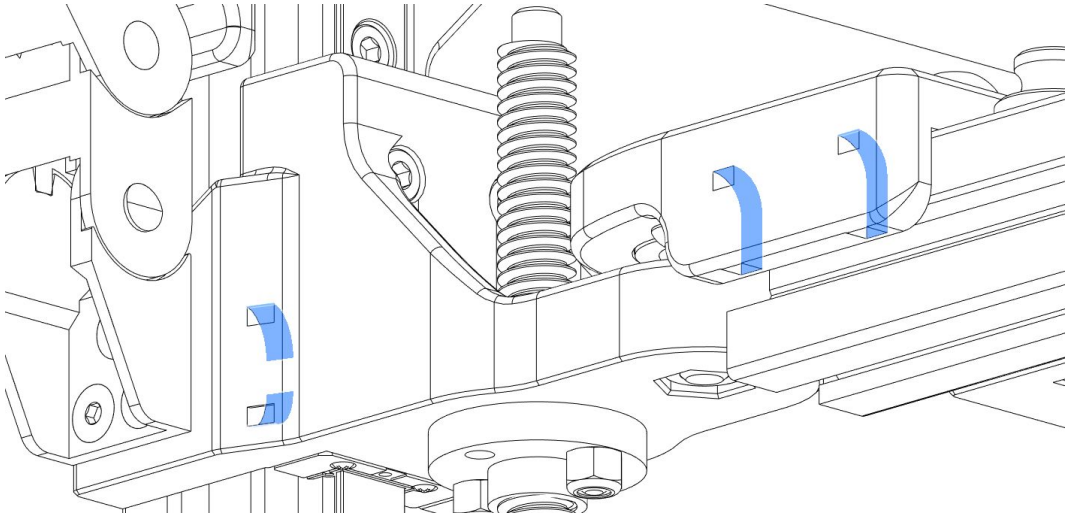

Z ENDSTOP
2x 0.25mm²(AWG24) or larger
PROTECTIVE EARTH
1mm²(AWG18) or larger
PRINT BED
2x 1mm²(AWG18) or larger
BED THERMISTOR
2x 0.25mm²(AWG24) or larger

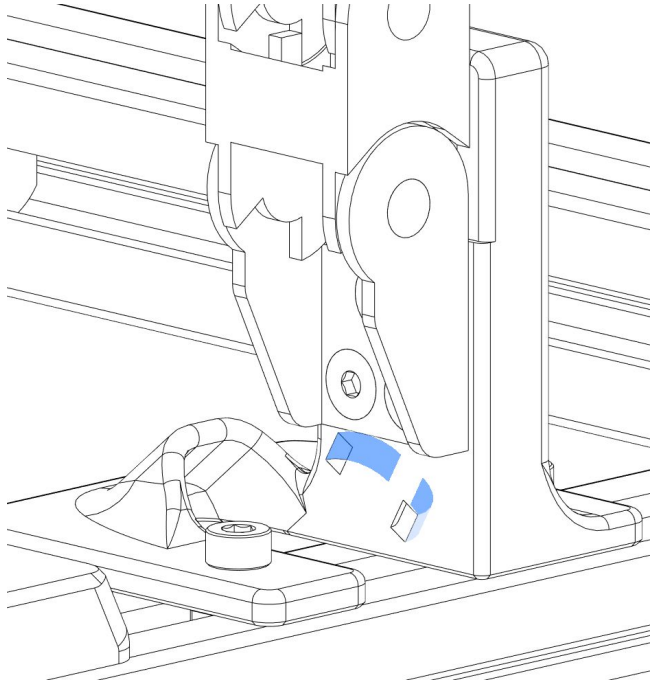


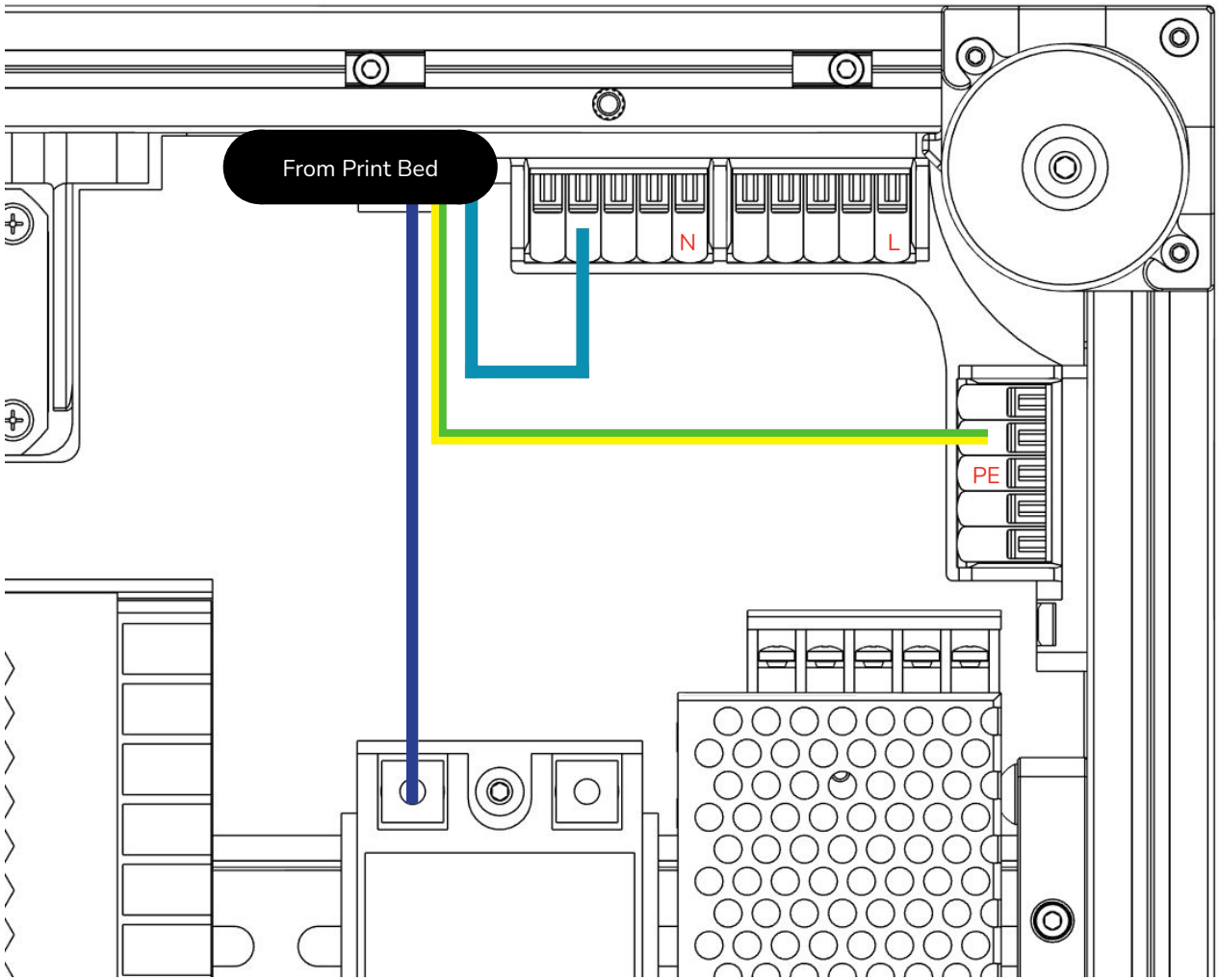

M3x6 FHCS
M3x6 FHCS



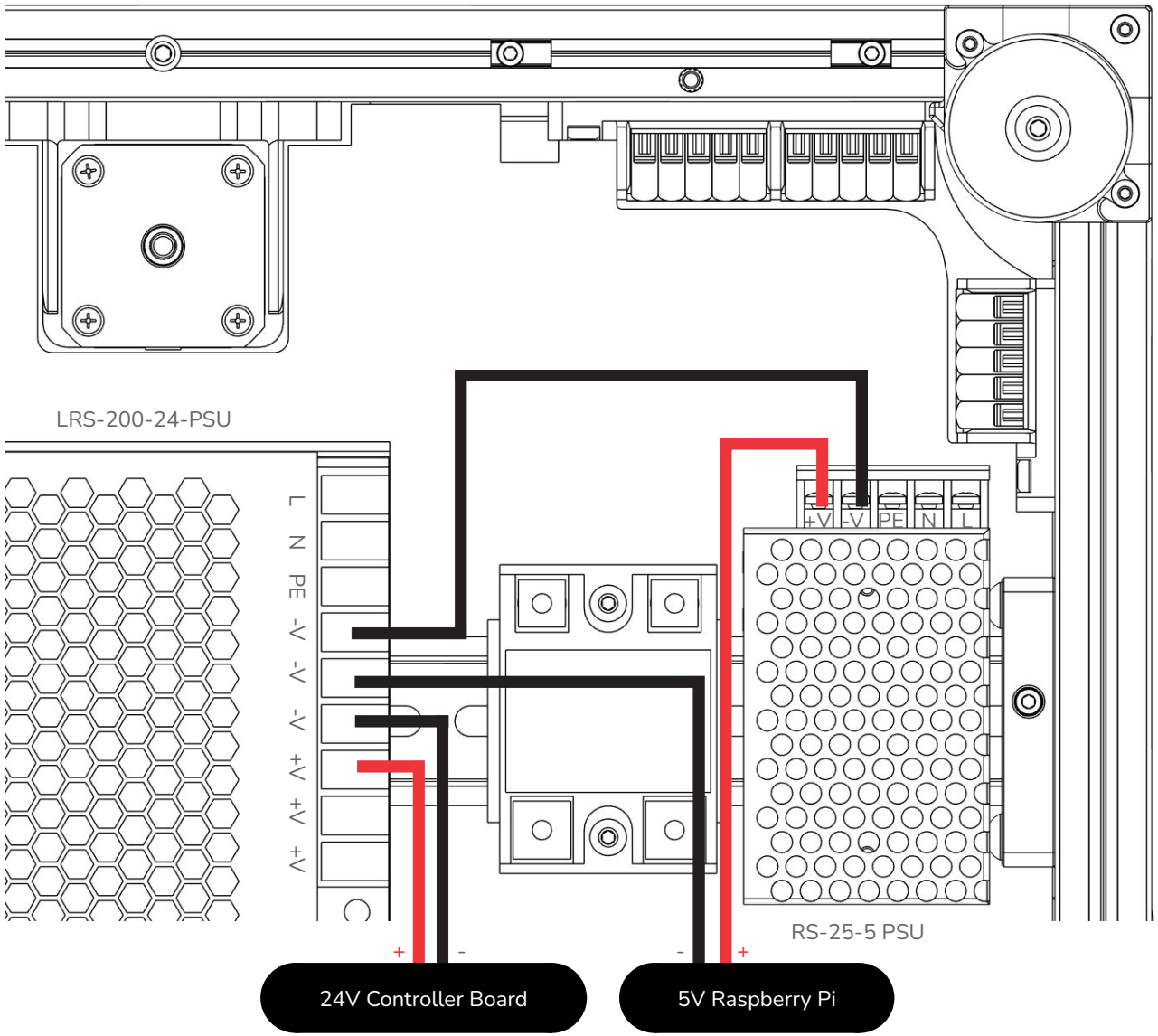
ZIP TIE LOOPS

Secure the wire bundle to the strain relief using small zip ties.




From Print Bed
N
L
PE

Leave the cables to the 5v PSU out.



CABLE CROSS SECTION

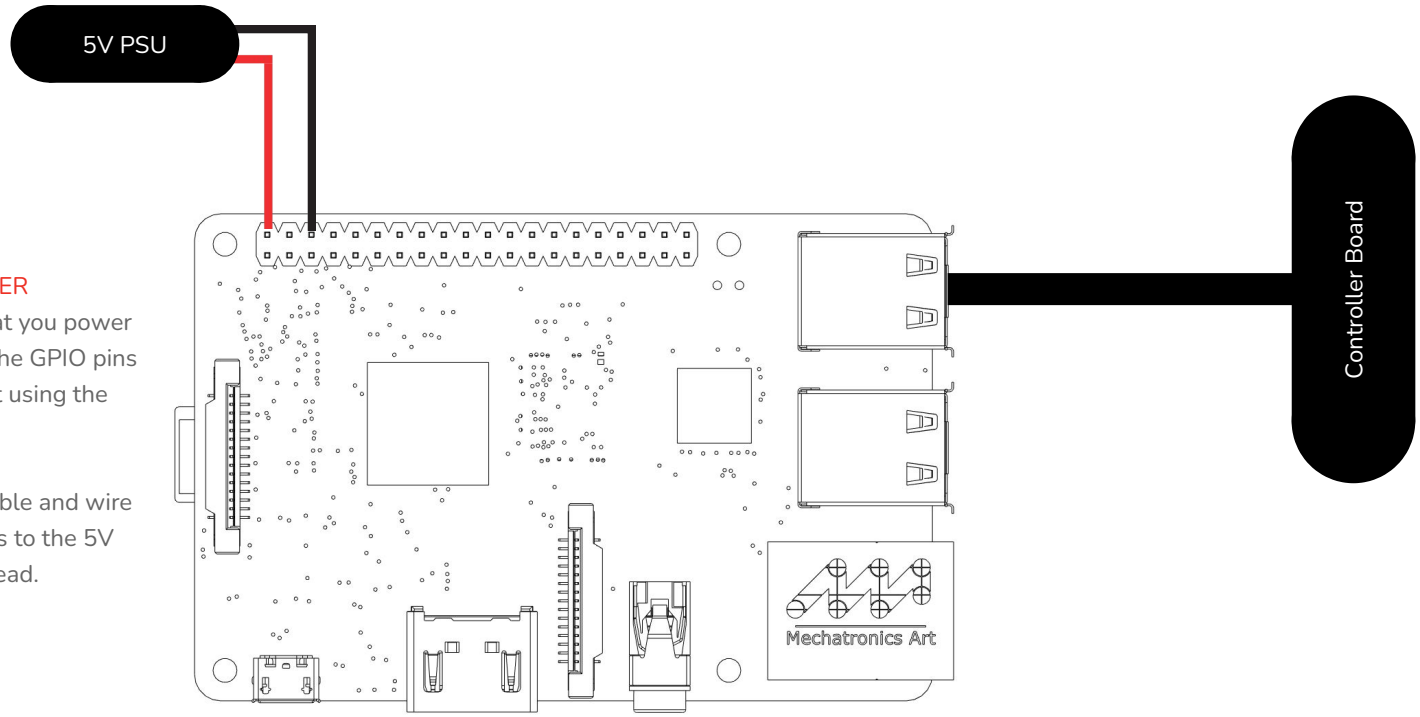
Cables to the controller board should be 1mm² (AWG18) or larger.

0.5mm² (AWG20) is sufficient for the connection to the Raspberry Pi.

TERMINAL COVER

After installing all cables install the Meanwell TBC-09 Terminal Cover included in the BOM on the PSU. It clips onto the the terminal block.

Skip, you dont use a standart Raspberry.



RASPBERRY PI POWER

While we suggest that you power the Raspberry Pi via the GPIO pins you may also power it using the “Power-In” USB port.

Cut a suitable USB cable and wire the + and ground lines to the 5V DC/DC converter instead.

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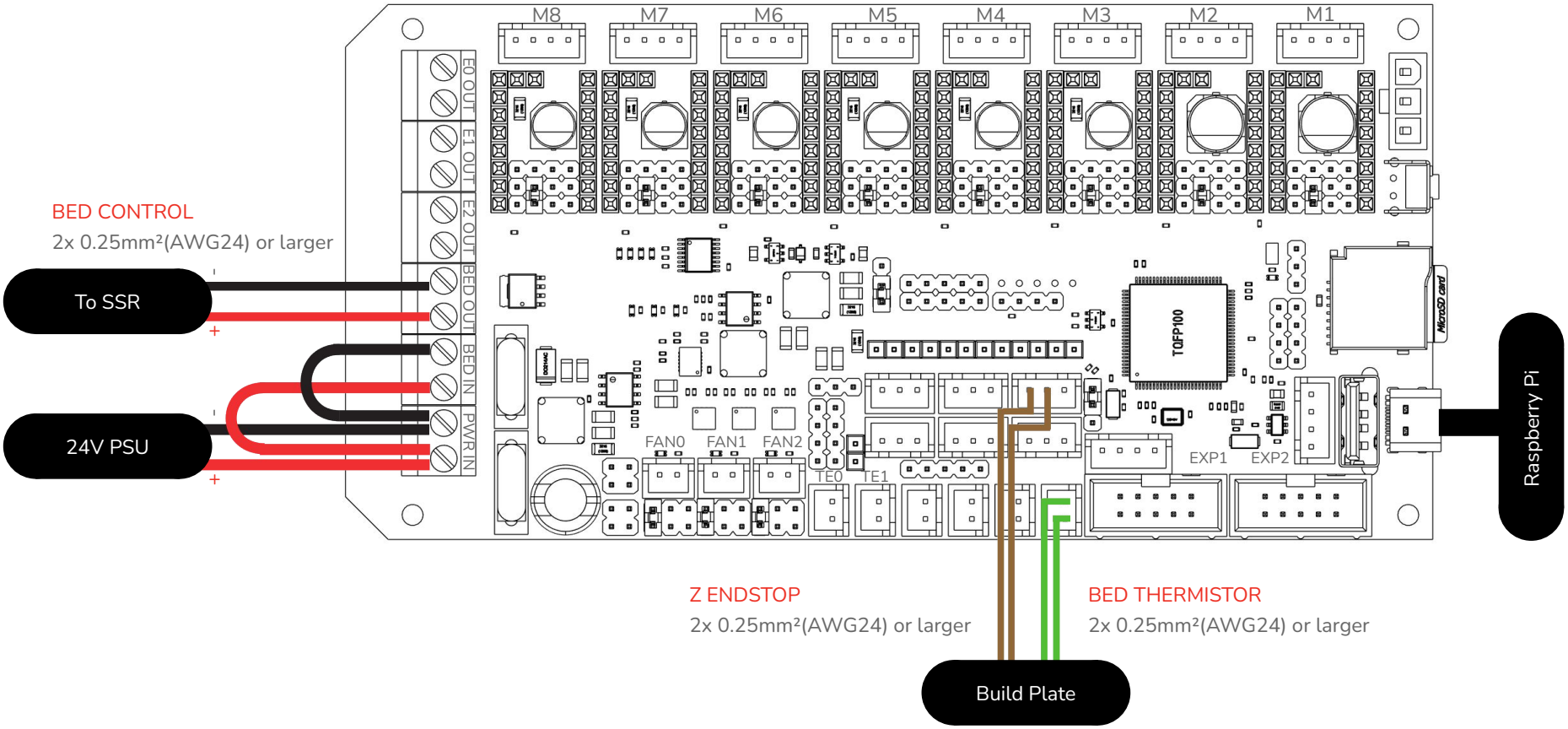
For a better Overview of the wiring, use the [wiring diagram] (https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/Wiring%20Diagram%20of%20Trident%20R1%20Pro.pdf).

The Z-Endstop cables are not needed.

CONTROLLER BOARD

The assembly manual will outline the wiring for a Fysetc Spider 2.2. You can find additional documentation and alternative configurations on docs.vorondesign.com.



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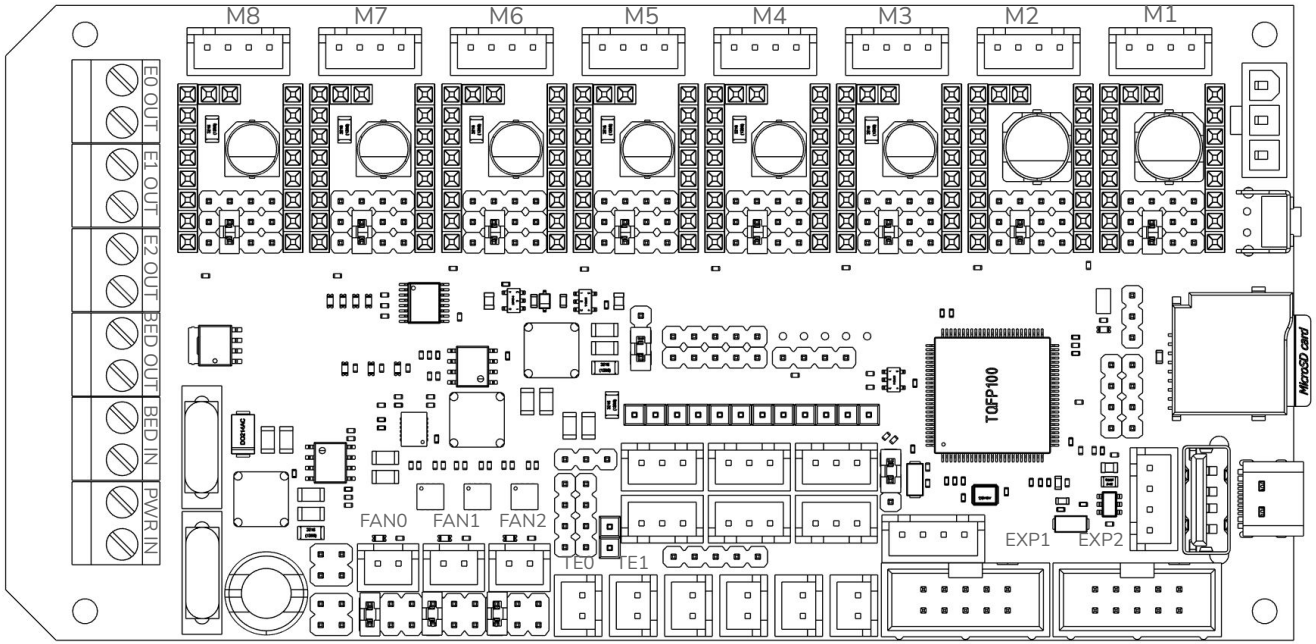
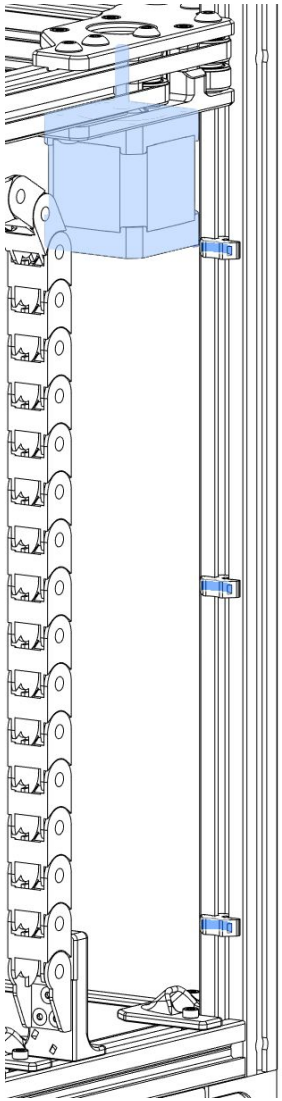
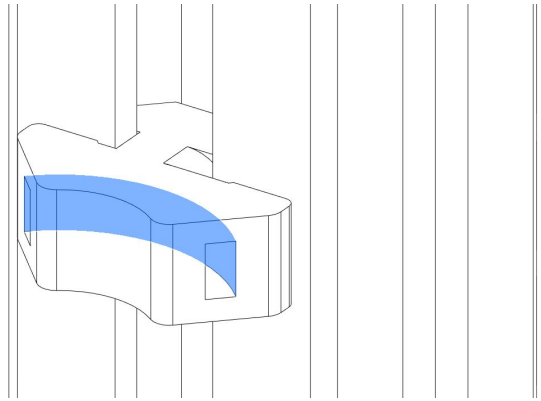


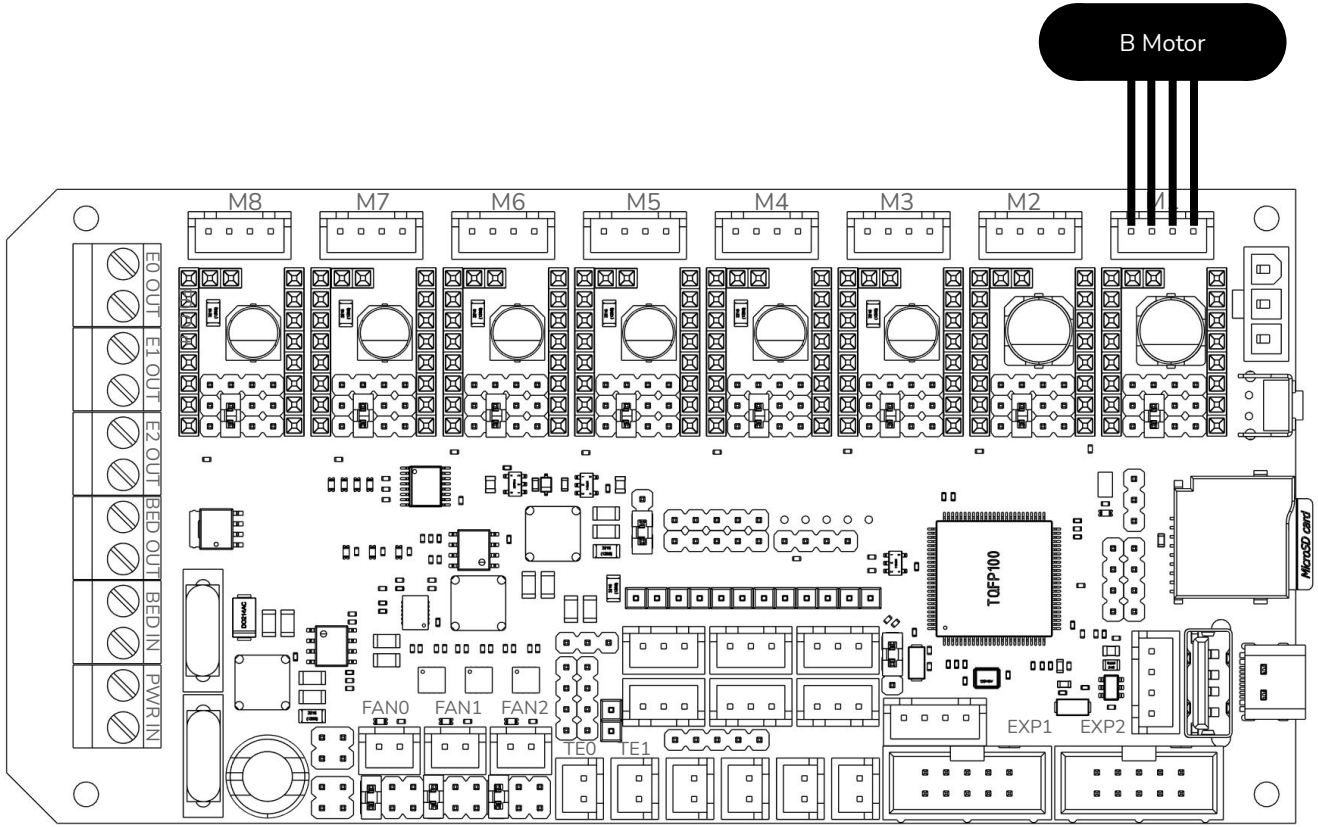




ZIP TIE LOOPS

Secure the wire bundle to the strain relief using small zip ties.

For a better Overview of the wiring, use the [wiring diagram] (https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/Wiring%20Diagram%20of%20Trident%20R1%20Pro.pdf).



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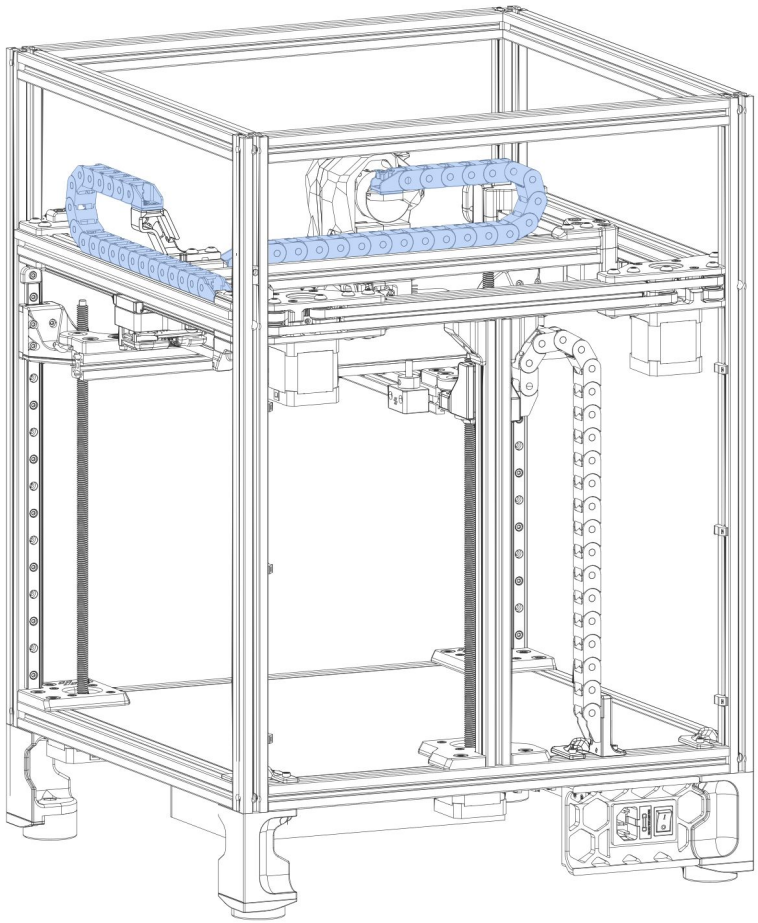
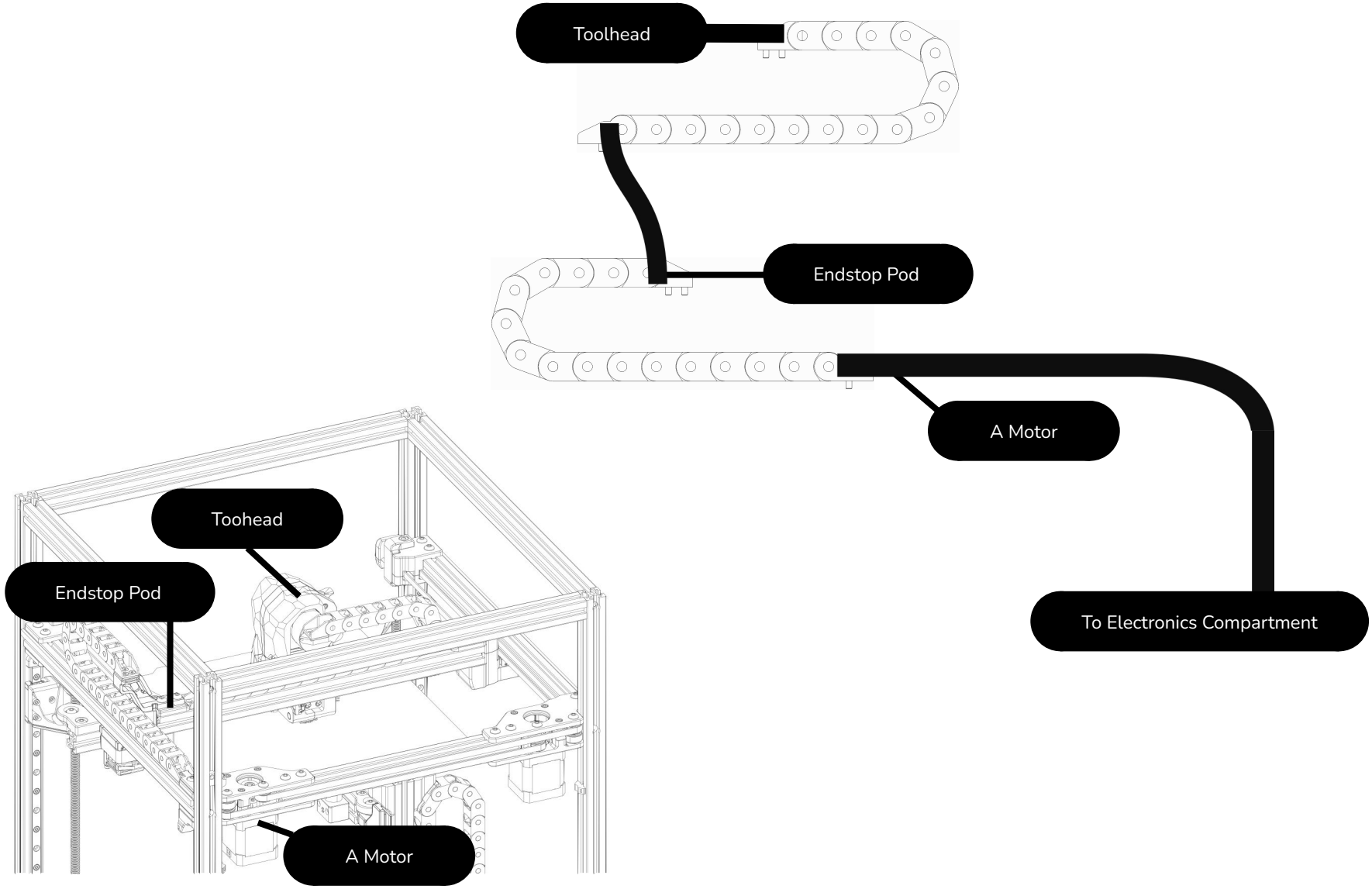


CABLE CHAINS INSTALL

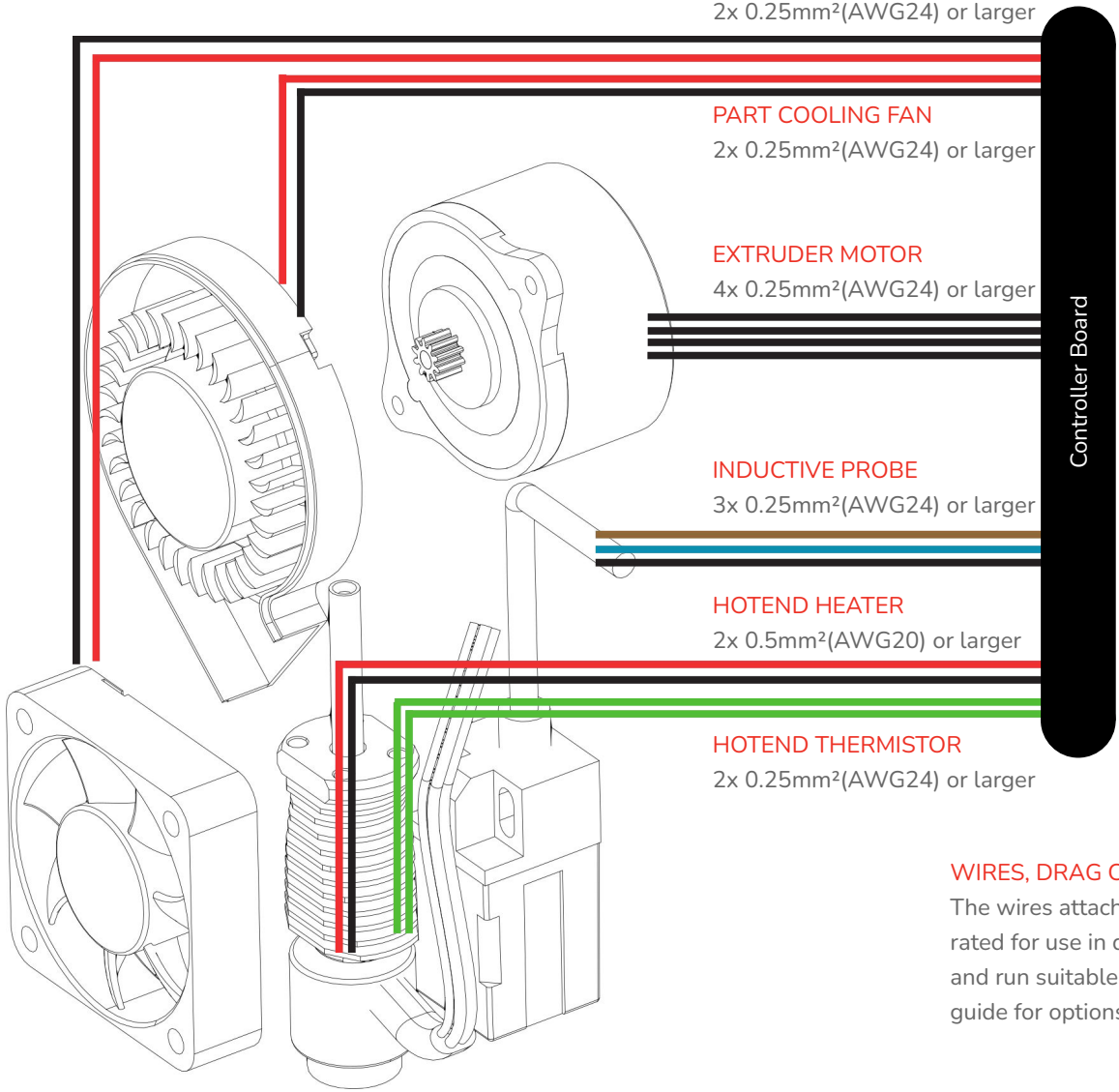
You can opt to install the chains now and fish the wires through the chains or build the complete harness outside of the printer and install it in one go. Either approach does work.

If you sourced a pre-built wire harness completing the harness outside of the printer is recommended.

For a better Overview of the wiring, use the [wiring diagram] (https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/Wiring%20Diagram%20of%20Trident%20R1%20Pro.pdf).



For a better Overview of the wiring, use the [wiring diagram] (https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/Wiring%20Diagram%20of%20Trident%20R1%20Pro.pdf).



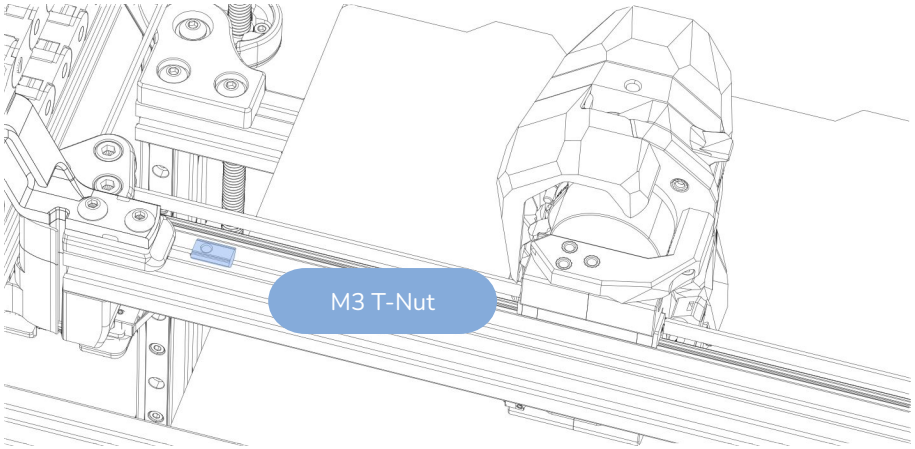
OPTION: TOOLHEAD PCB

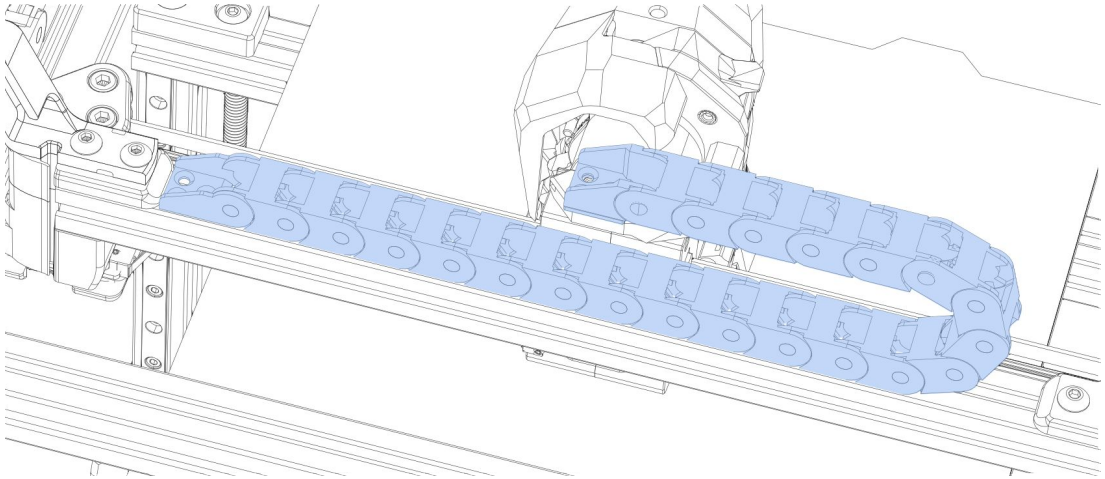
If you are planning to use a toolhead PCB consult the Board manufacturer for wiring instructions.

WIRES, DRAG CHAINS AND CRIMPS

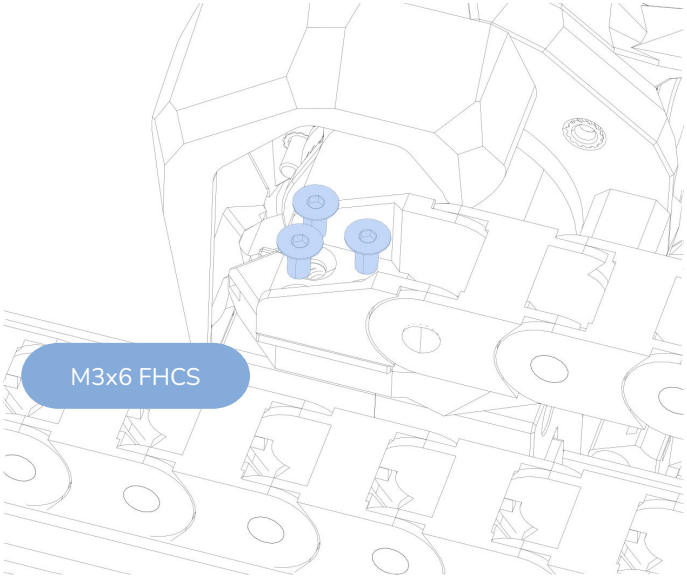
The wires attached to the probe, fans, heater, etc. are usually not rated for use in drag chains. Add crimp connectors at the toolhead and run suitable wire down the drag chains. Refer to the sourcing guide for options.

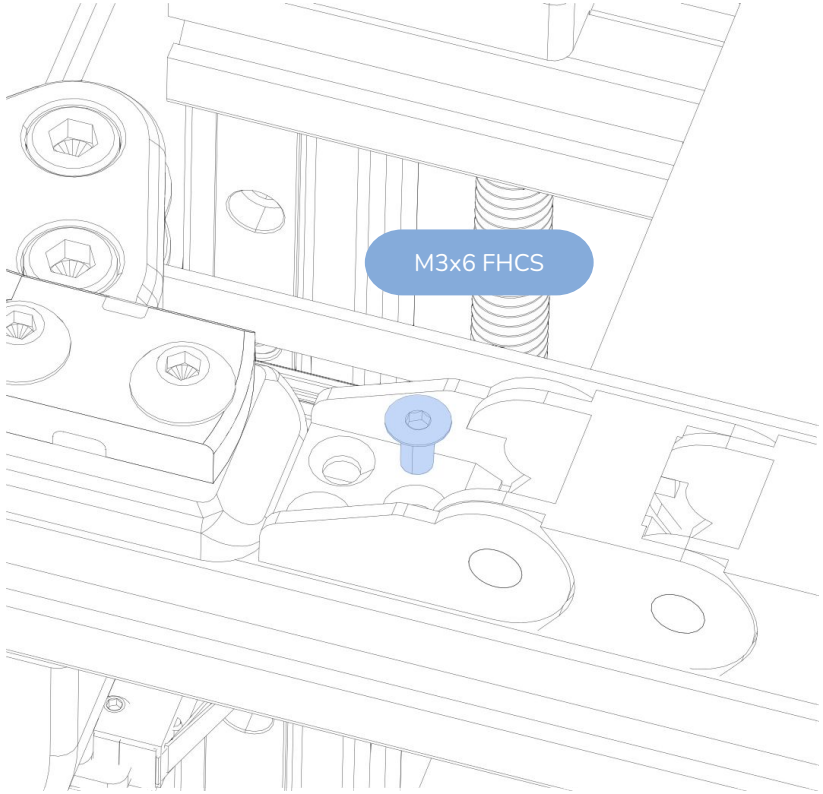
For a better Overview of the wiring, use the [wiring diagram] (https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/Wiring%20Diagram%20of%20Trident%20R1%20Pro.pdf).



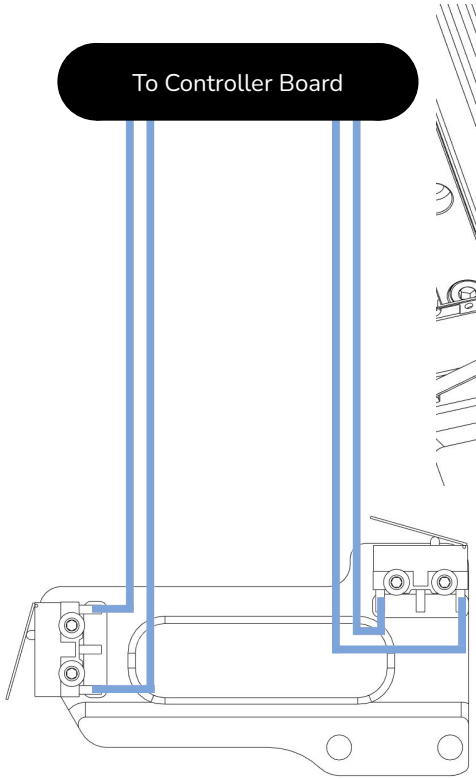


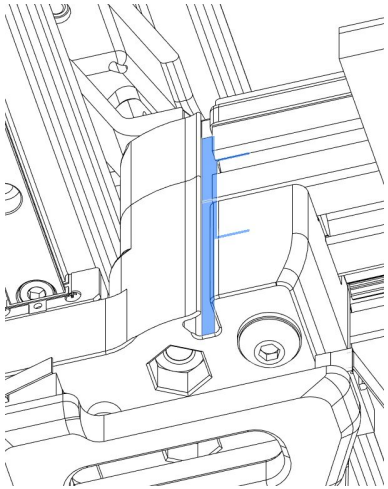
For a better Overview of the wiring, use the [wiring diagram] (https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/Wiring%20Diagram%20of%20Trident%20R1%20Pro.pdf).





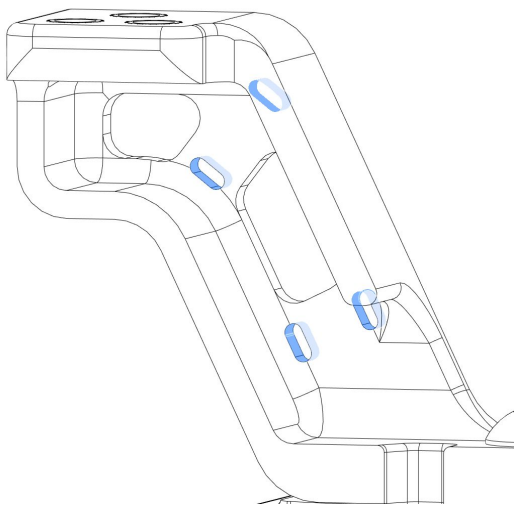
For a better Overview of the wiring, use the [wiring diagram] (https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/Wiring%20Diagram%20of%20Trident%20R1%20Pro.pdf).

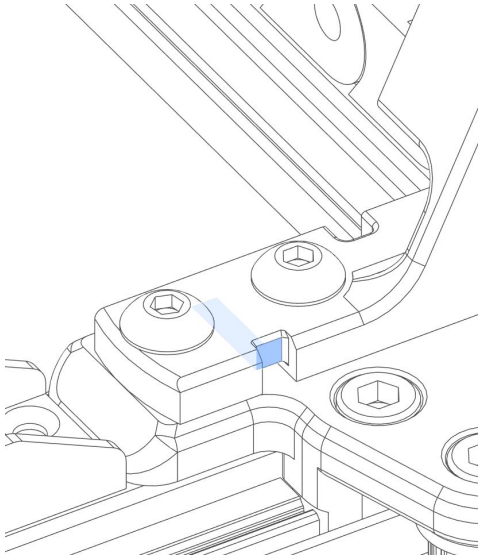




Y ENDSTOP
2x 0.25mm²(AWG24) or larger

X ENDSTOP
2x 0.25mm²(AWG24) or larger



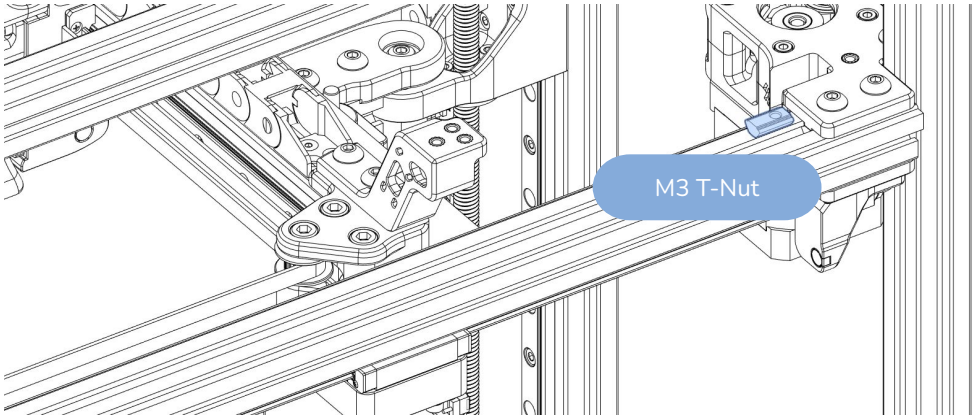


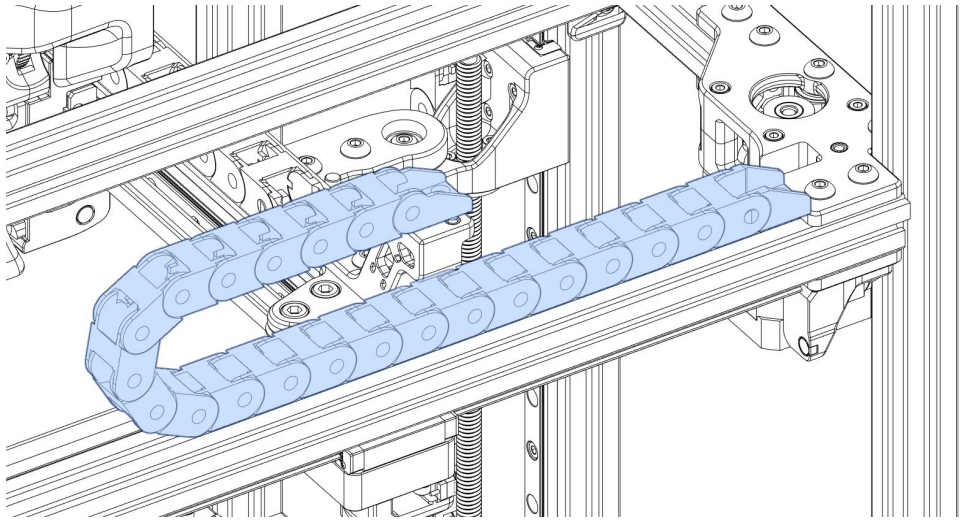
ZIP TIE LOOPS
Secure the wire bundle to the strain relief using small zip ties.

IGUS cable chains have a build-in strain relief. We omitted the loop on the part.

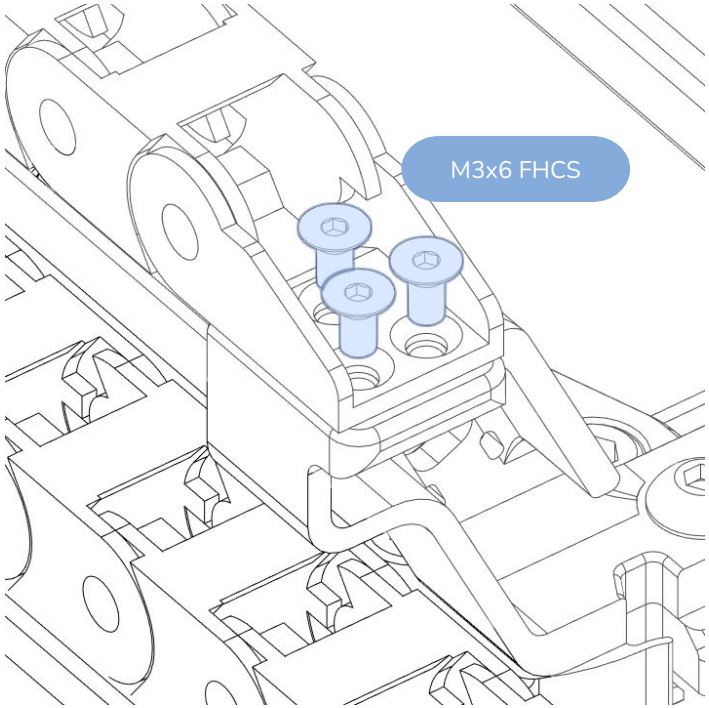
OPTION: ENDSTOP BOARD/HALL EFFECT BOARD
Those boards utilize a 4 pin connector instead. Please refer to https://voron.link/djhyygu and https://voron.link/d6qb7o6 for details.

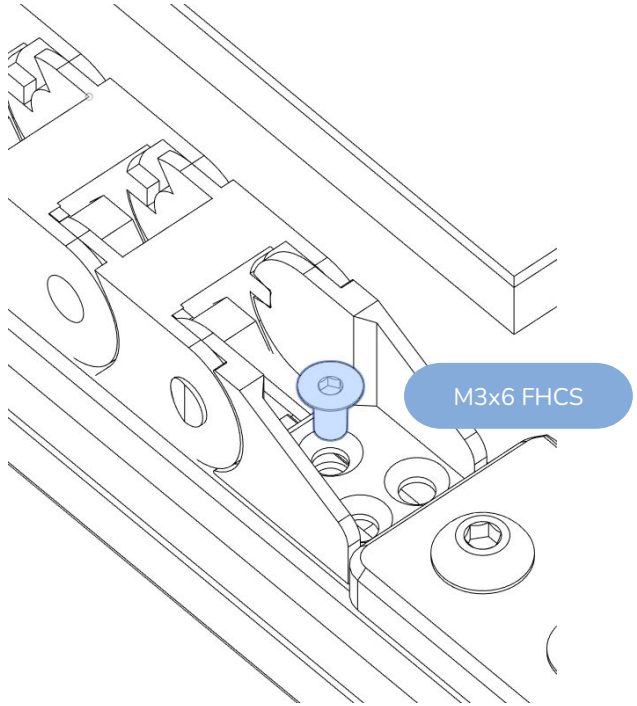
For a better Overview of the wiring, use the [wiring diagram] (https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/Wiring%20Diagram%20of%20Trident%20R1%20Pro.pdf).





For a better Overview of the wiring, use the [wiring diagram] (https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/Wiring%20Diagram%20of%20Trident%20R1%20Pro.pdf).

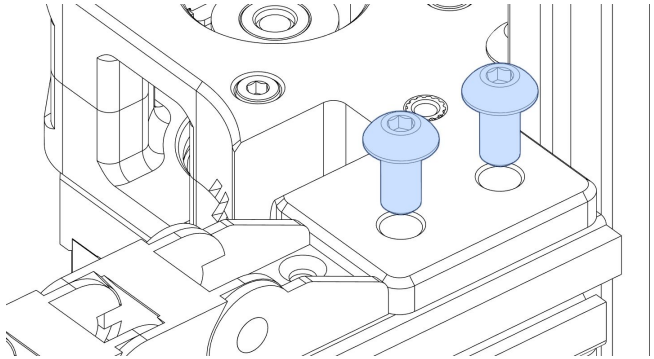


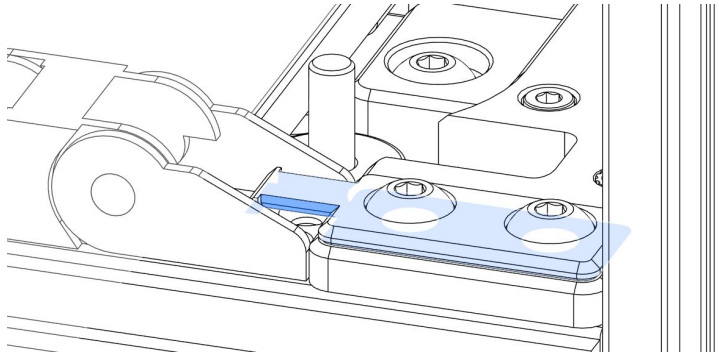


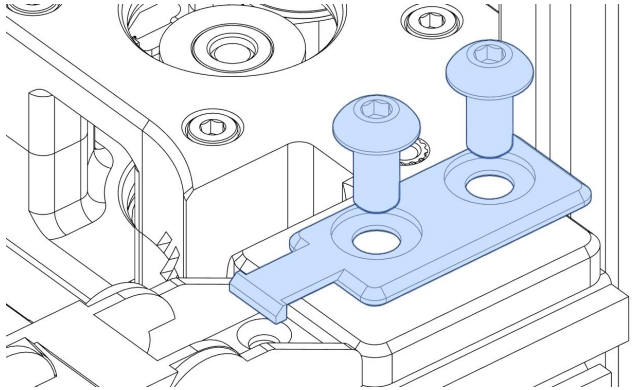
For a better Overview of the wiring, use the [wiring diagram] (https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/Wiring%20Diagram%20of%20Trident%20R1%20Pro.pdf).

GENERIC CABLE CHAINS

The chains generally don't include a strain relief. Undo the screws on the A drive and add the printed strain relief.







ZIP TIE LOOPS

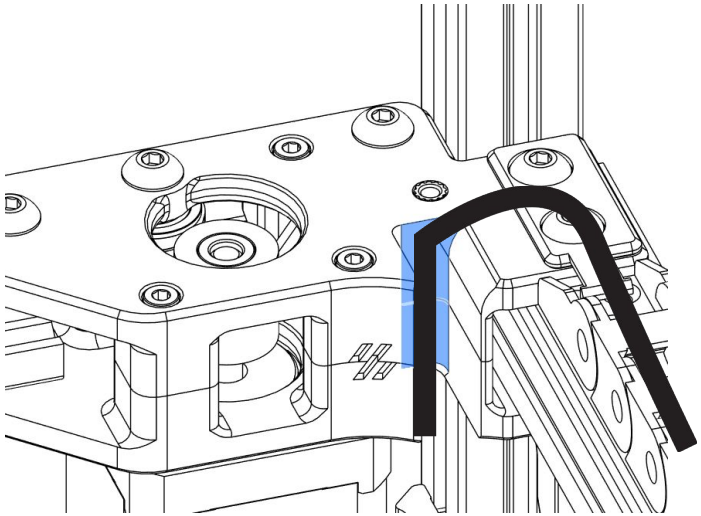
Secure the wire bundle to the strain relief using small zip ties.

IGUS cable chains have a build-in strain relief. Don't install the additional part.

For a better Overview of the wiring, use the [wiring diagram] (https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/Wiring%20Diagram%20of%20Trident%20R1%20Pro.pdf).

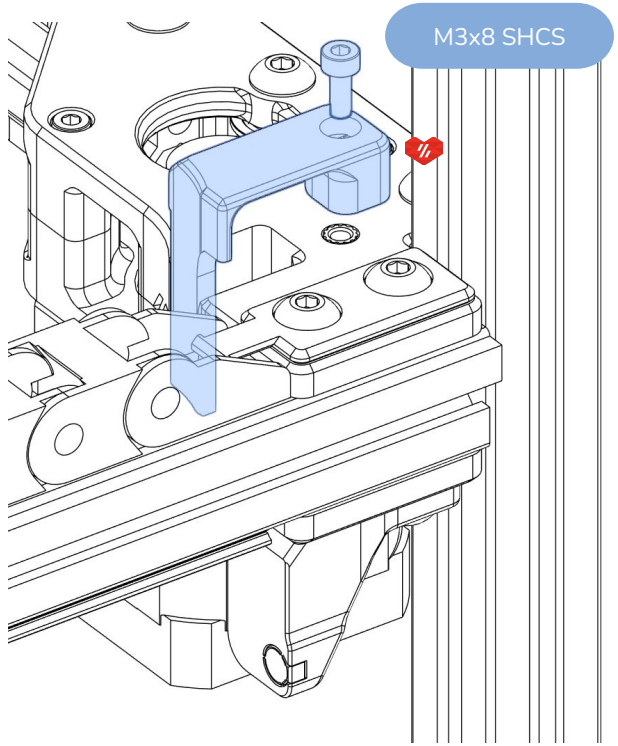
IGUS CABLE CHAINS

IGUS chains have 2 mounting holes.

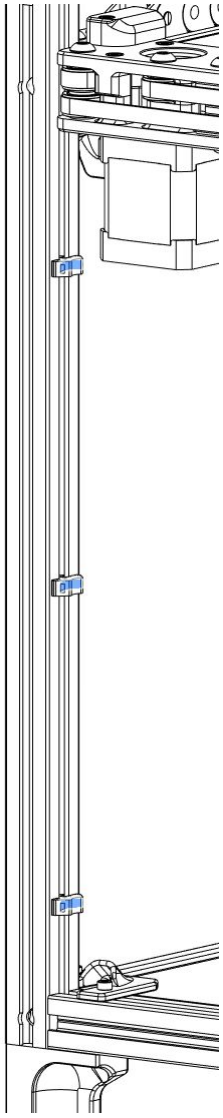


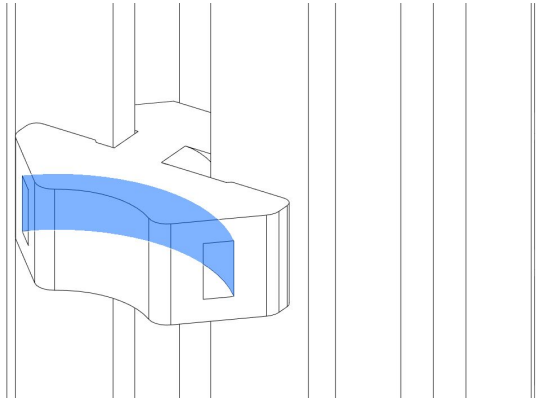
WIRING PATH

Guide the wires in the Highlighted path.



Route the Y Endstop cable and the A Motor cables down like the B Motor cable (but on the right side of the printer).

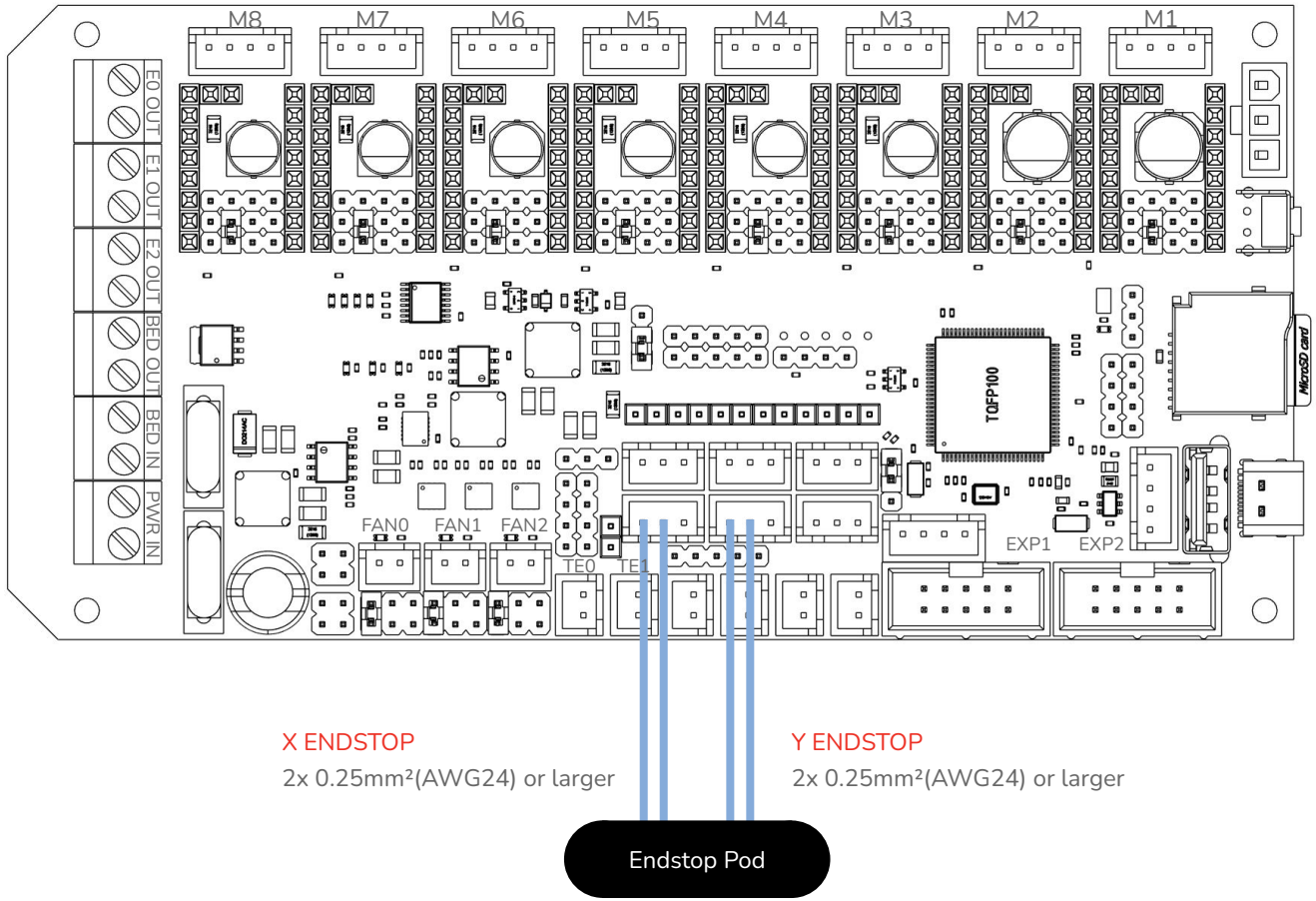




ZIP TIE LOOPS

Secure the wire bundle to the strain relief using small zip ties.

For a better Overview of the wiring, use the [wiring diagram] (https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/Wiring%20Diagram%20of%20Trident%20R1%20Pro.pdf).



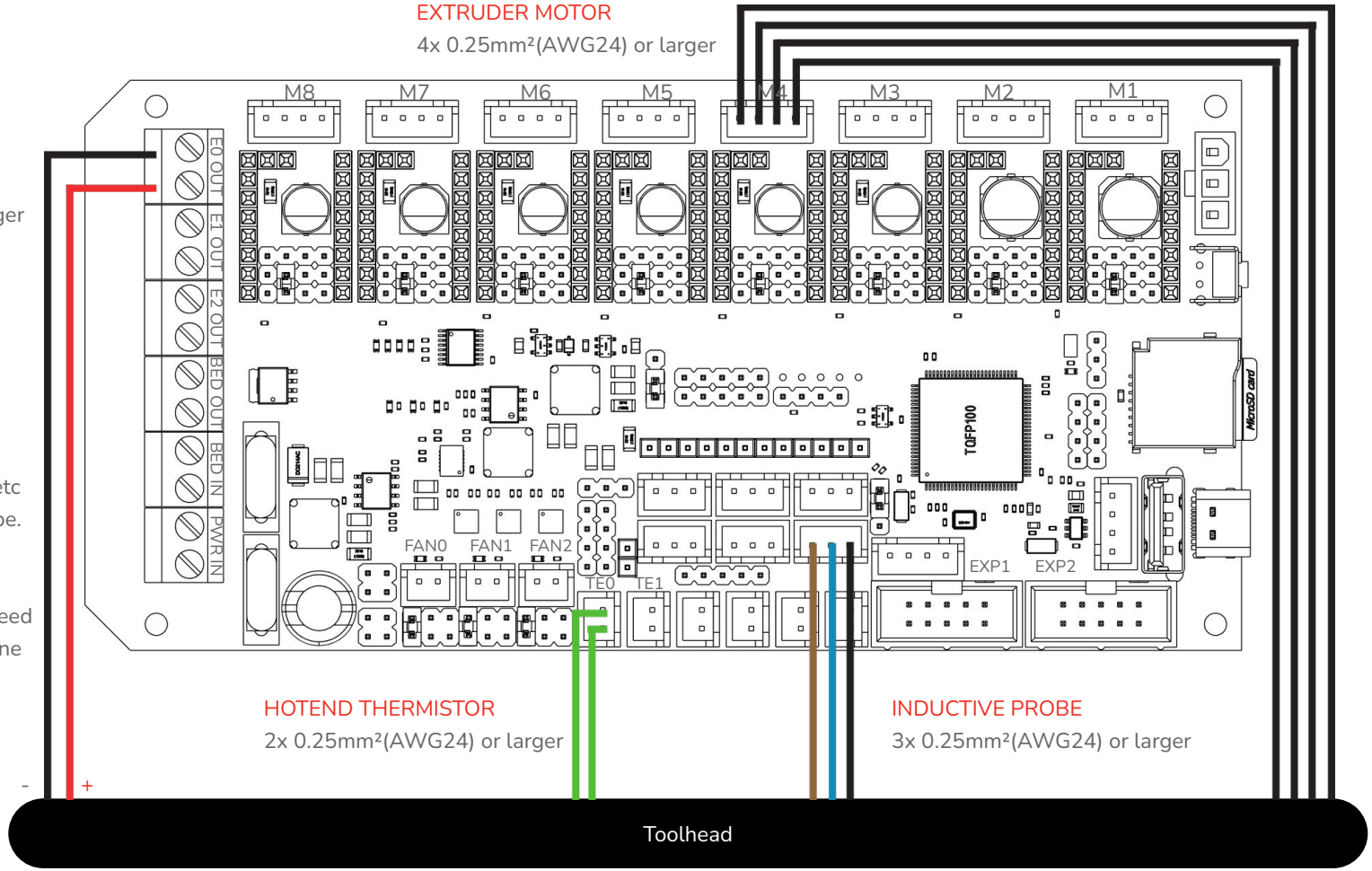
For a better Overview of the wiring, use the [wiring diagram] (https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/Wiring%20Diagram%20of%20Trident%20R1%20Pro.pdf).

EXTRUDER MOTOR
4x 0.25mm²(AWG24) or larger

HOTEND HEATER
2x 0.5mm²(AWG20) or larger

PROBE HOOKUP
The probe input of the Fysetc Spider supports a 24V probe.

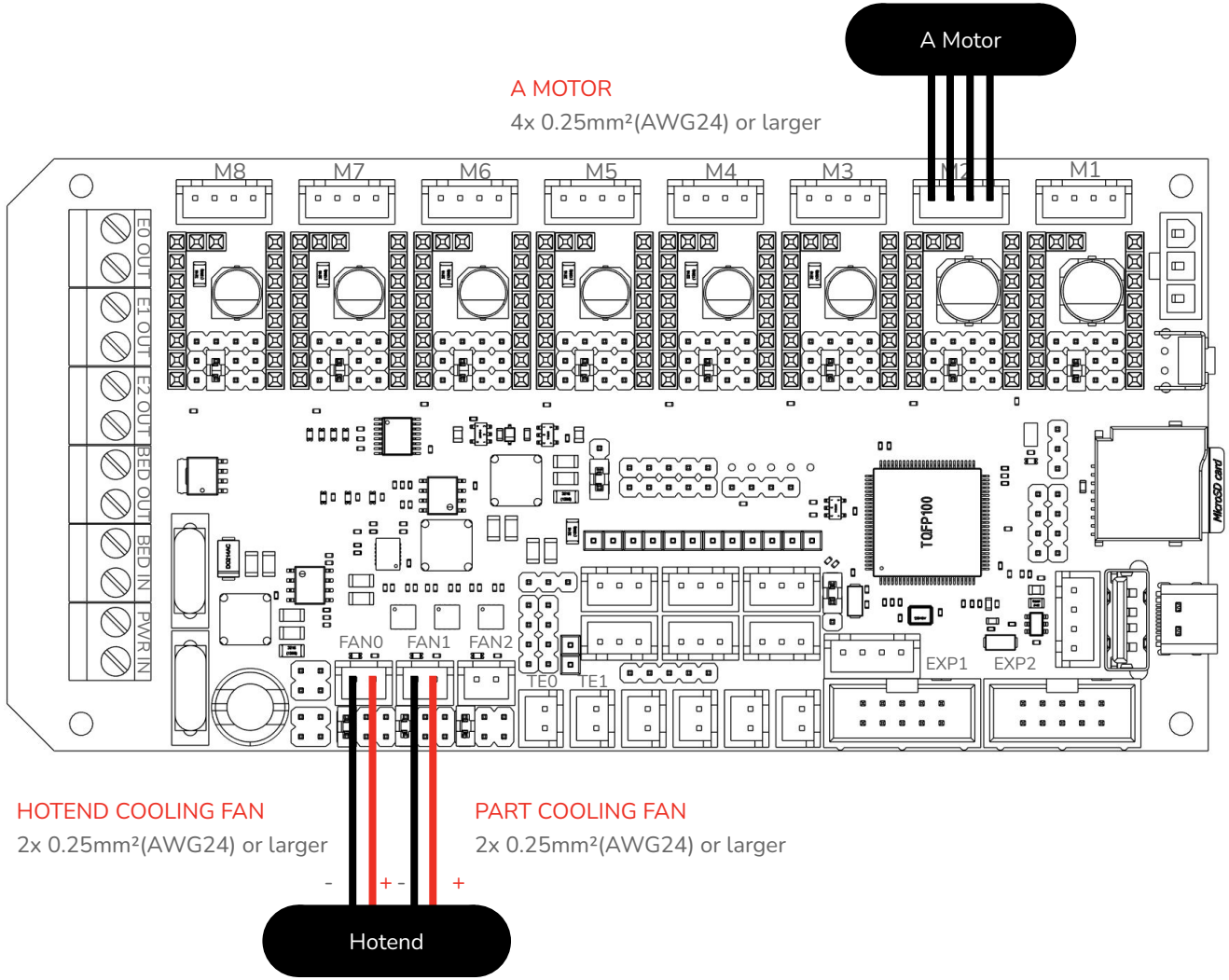
If you are using a different controller board you may need to wire the probe's signal line to an endstop input using a BAT85 diode.



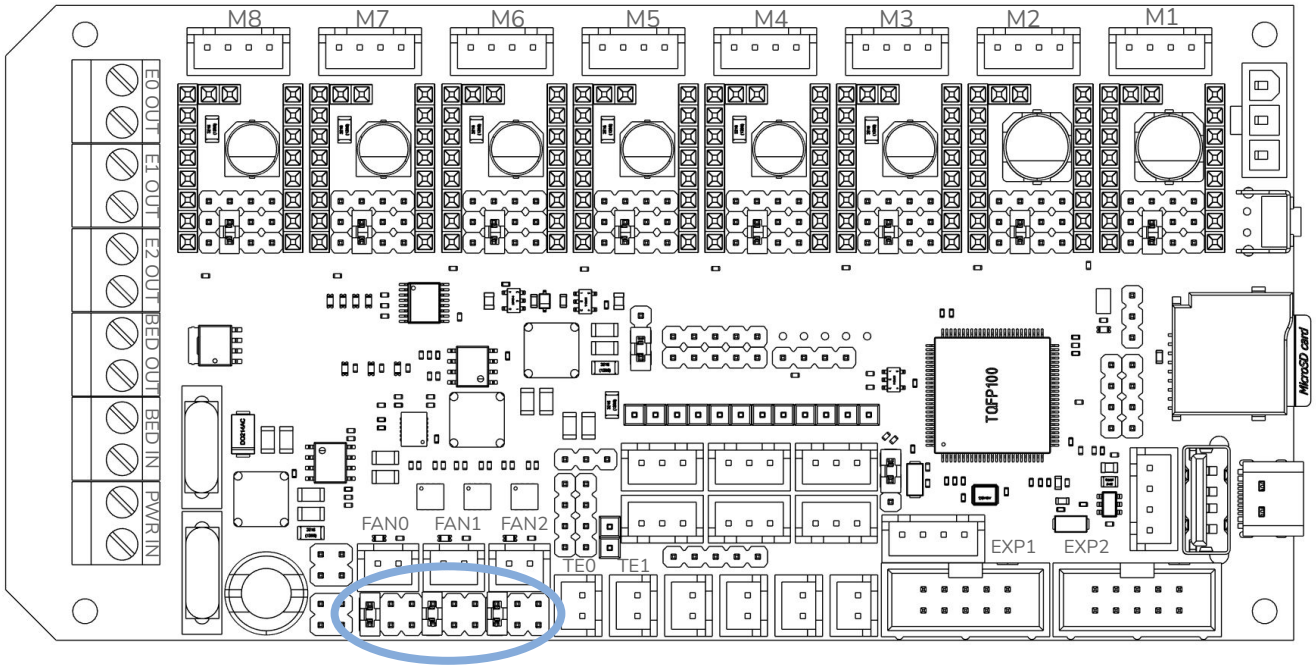
HOTEND THERMISTOR
2x 0.25mm²(AWG24) or larger

INDUCTIVE PROBE
3x 0.25mm²(AWG24) or larger

For a better Overview of the wiring, use the [wiring diagram] (https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/Wiring%20Diagram%20of%20Trident%20R1%20Pro.pdf).



For a better Overview of the wiring, use the [wiring diagram] (https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/Wiring%20Diagram%20of%20Trident%20R1%20Pro.pdf).

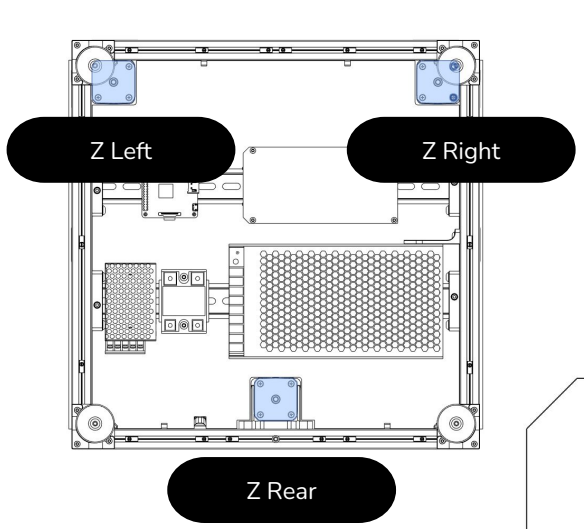


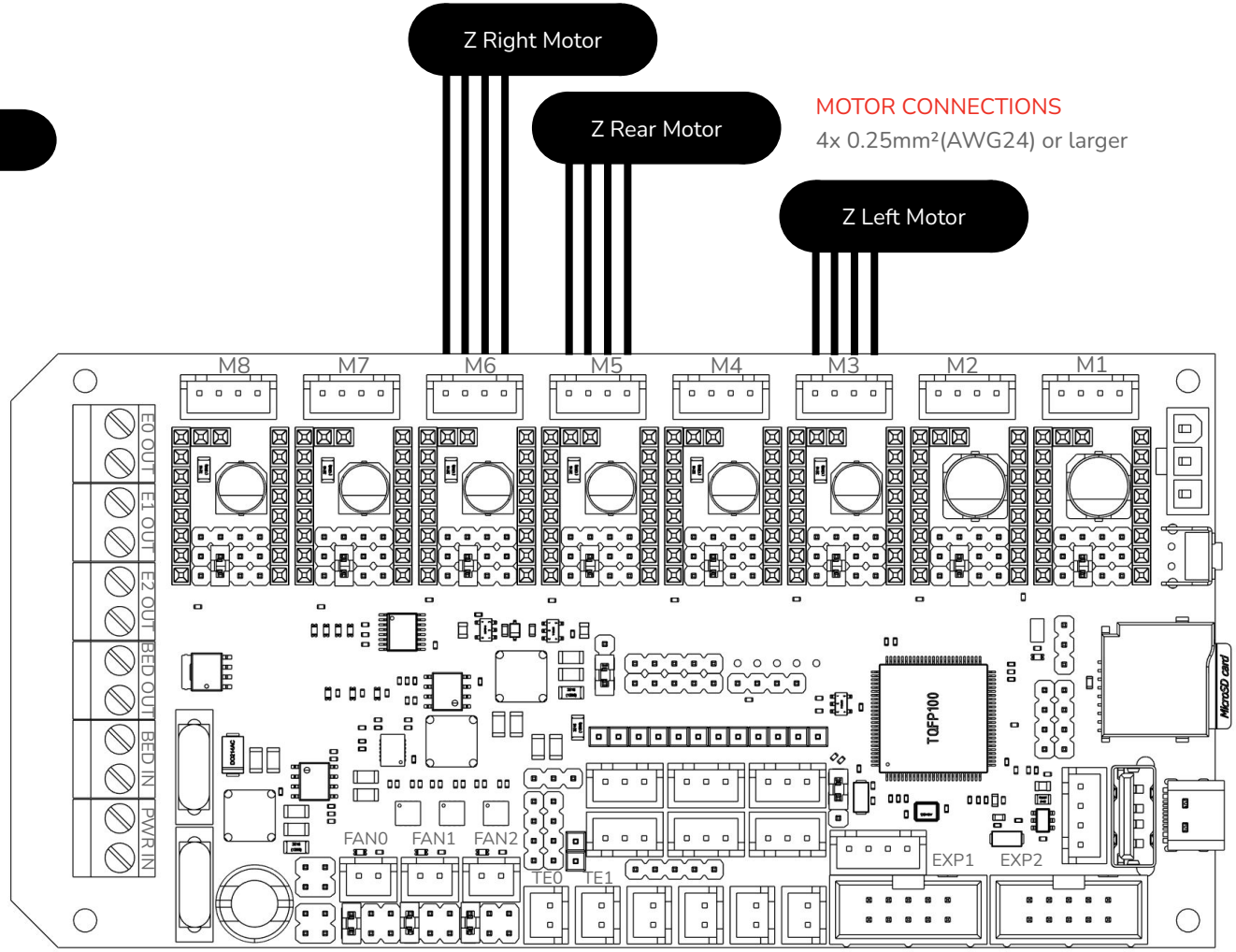
FAN VOLTAGE

The fans recommended in the sourcing guide are 24V fans.

Please check your hotend cooling (40x40x10 axial), part cooling (50x50x15 blower) and exhaust/electronics (60x60x20 axial) fans for their voltage rating and jumper the voltage selection accordingly. Refer to the Fysetc Spider manual for possible settings.

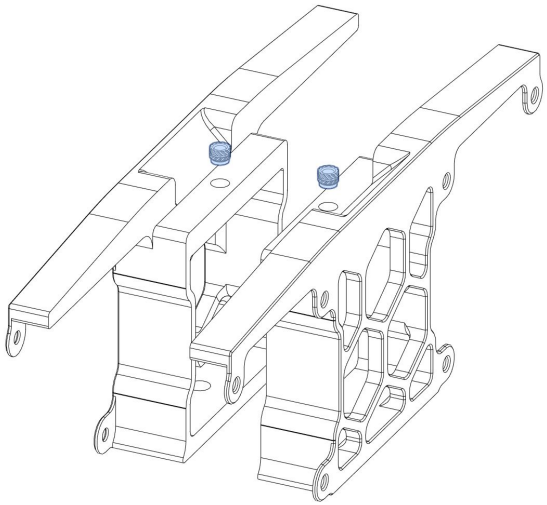
For a better Overview of the wiring, use the [wiring diagram] (https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/Wiring%20Diagram%20of%20Trident%20R1%20Pro.pdf).

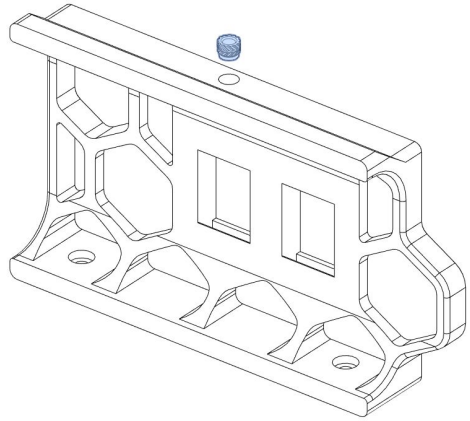




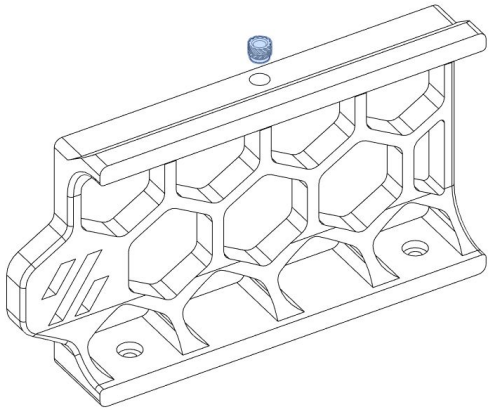
MOTOR CONNECTIONS

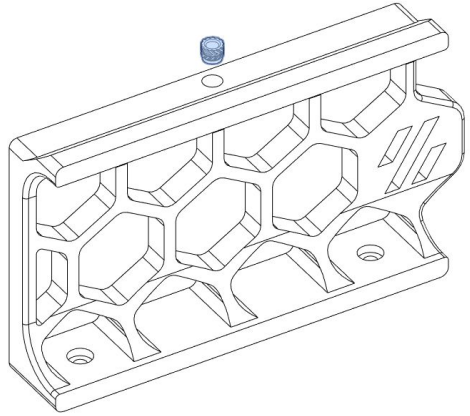
4x 0.25mm²(AWG24) or larger

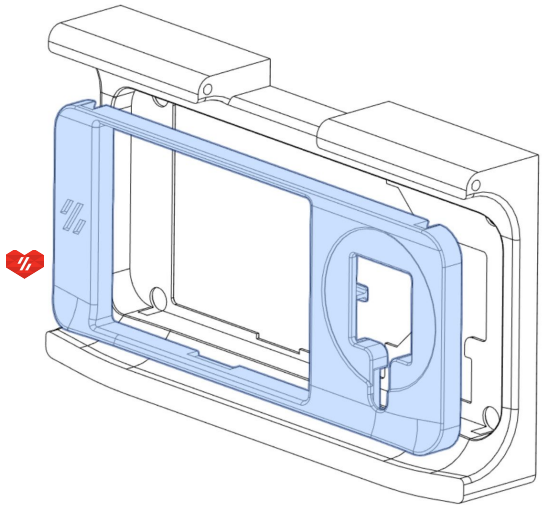


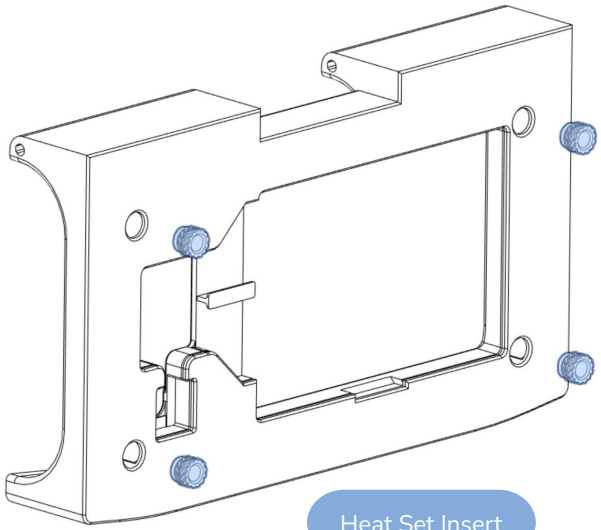


Heat Set Insert



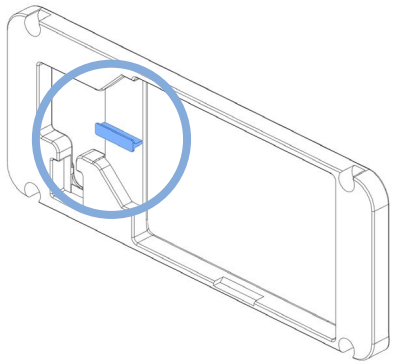






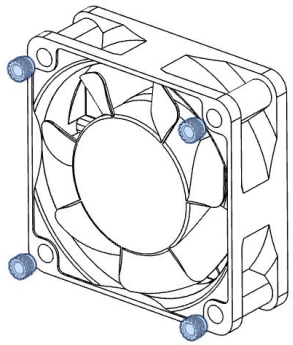
FRONT COVER

The front cover is held in place by the heat set inserts. Hold the front face firmly in place while inserting the heat set inserts.

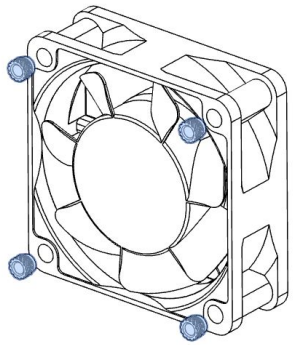


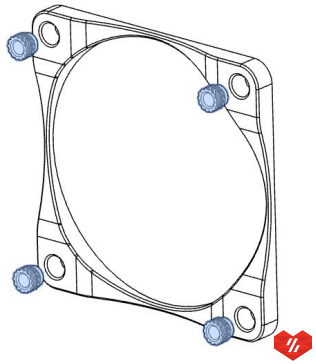
BUILT-IN SUPPORT

Remove the highlighted section. It's a built-in support for printability.

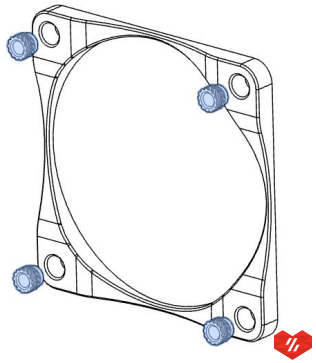


60x20 Fan

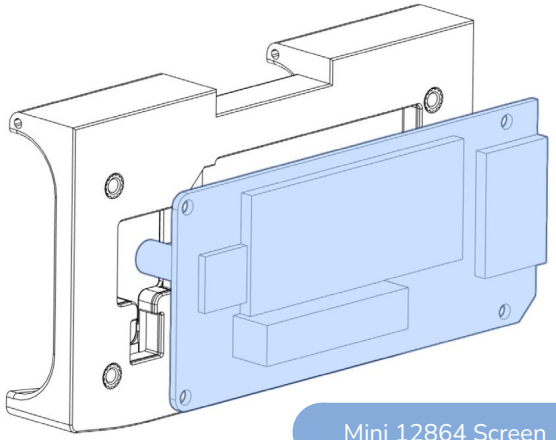




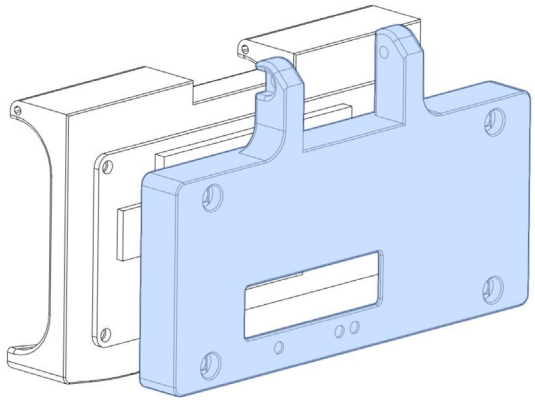
Heat Set Insert

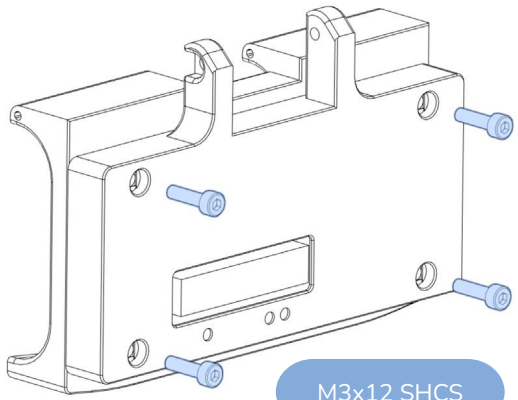


Skip these steps and instead mount the klipperscreen.



Mini 12864 Screen

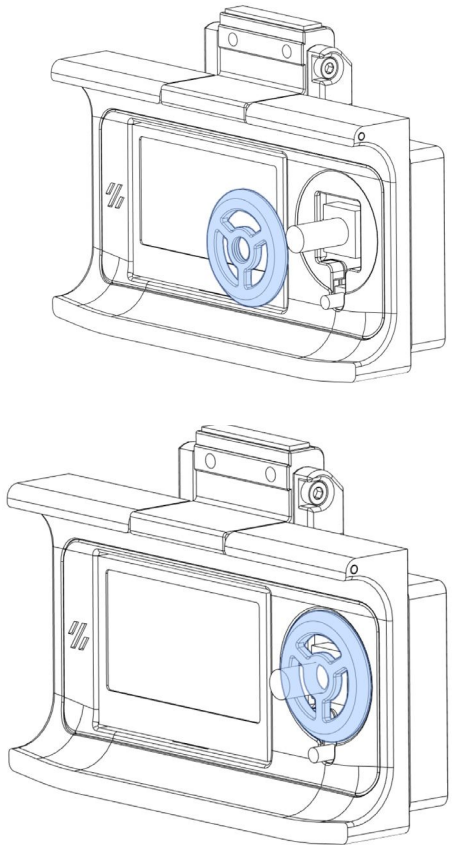


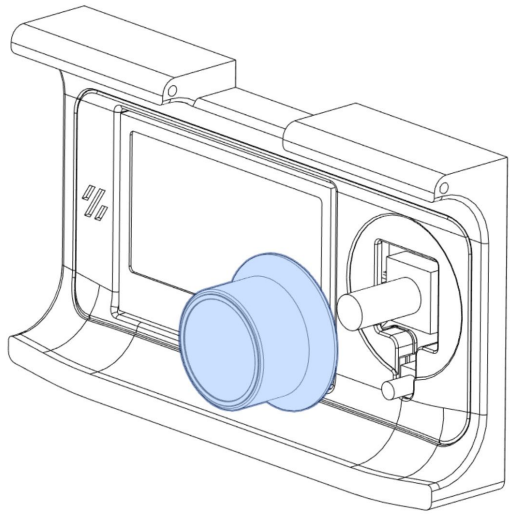


M3x12 SHCS

242

Skip these steps and instead mount the klipperscreen.



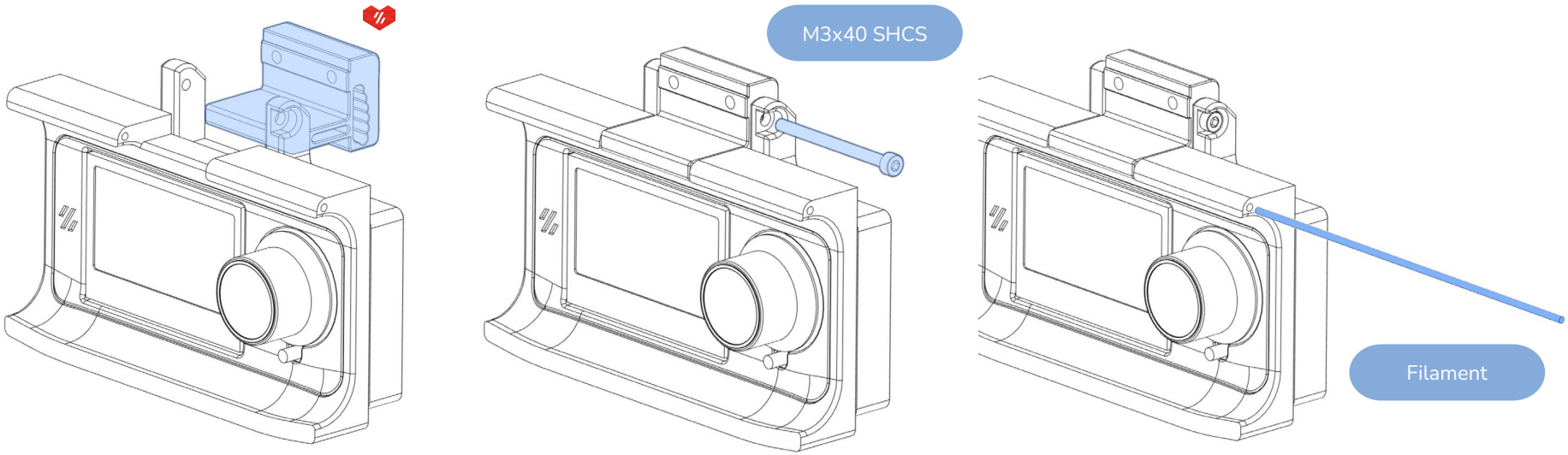


OPTION: LIGHT BLOCKER

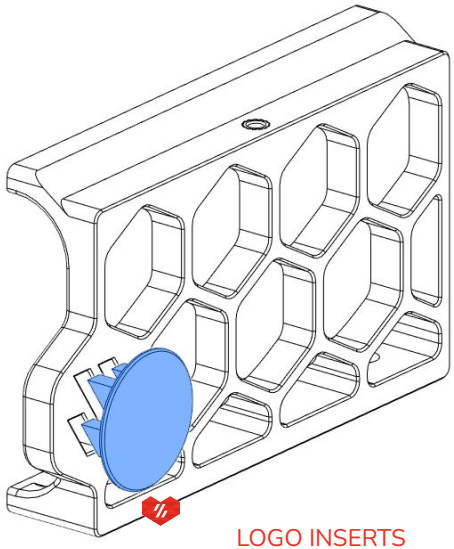
Some LCDs come with a smaller encoder knob. This extra piece prevents excess light bleed. Threads onto the encoder before the knob is pressed on.

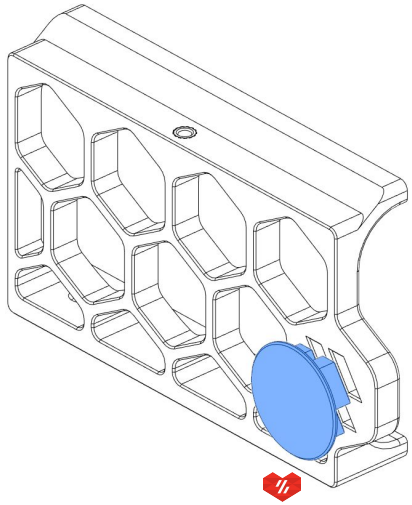
243

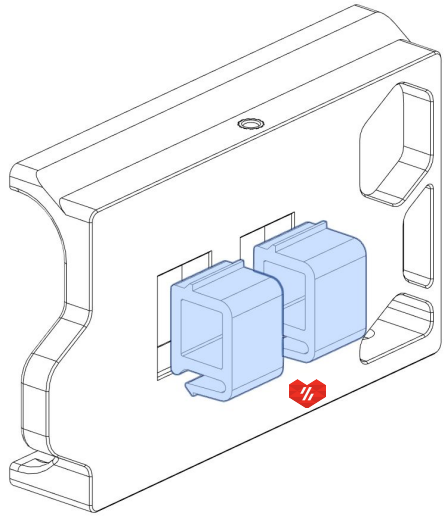
Skip these steps and instead mount the klipperscreen.



244

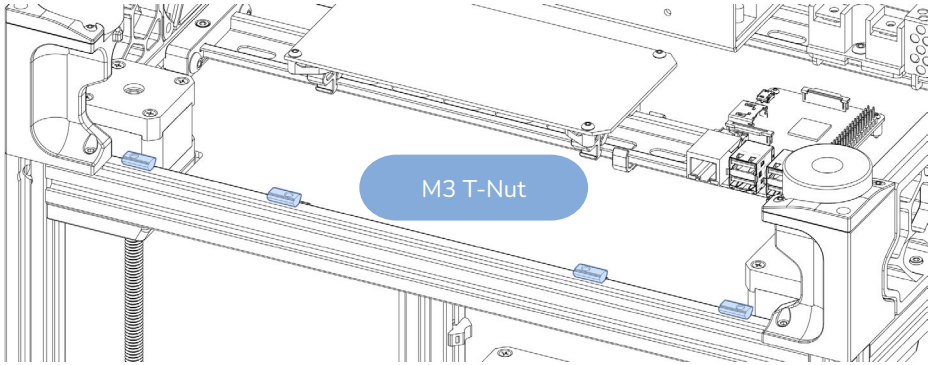


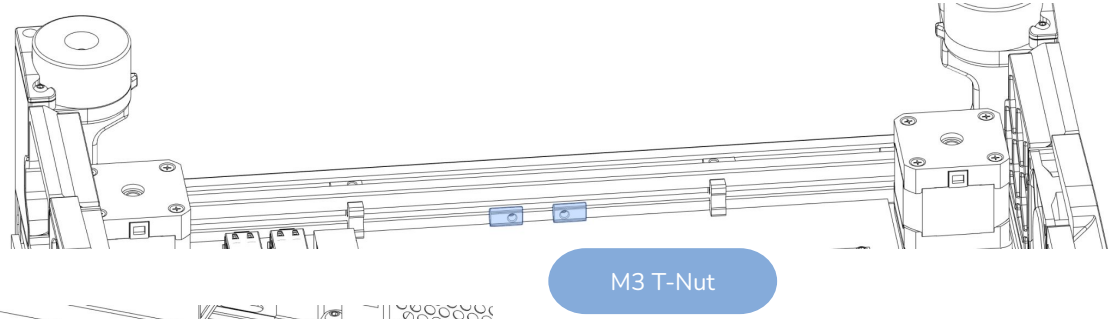


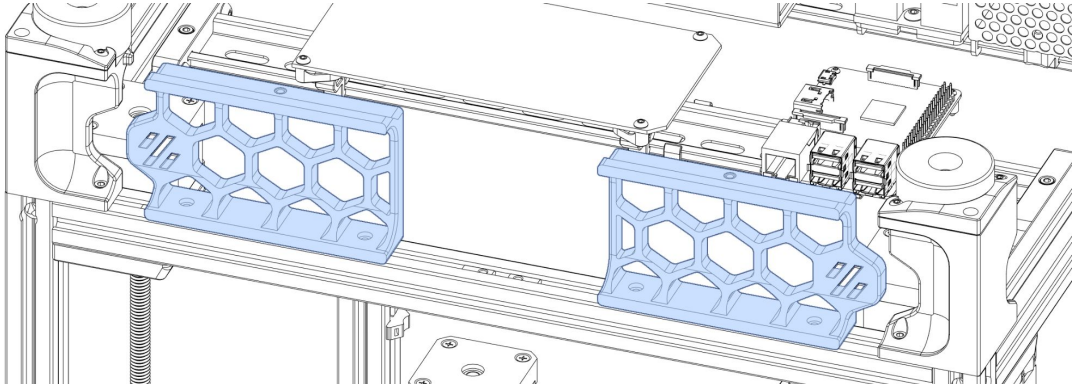


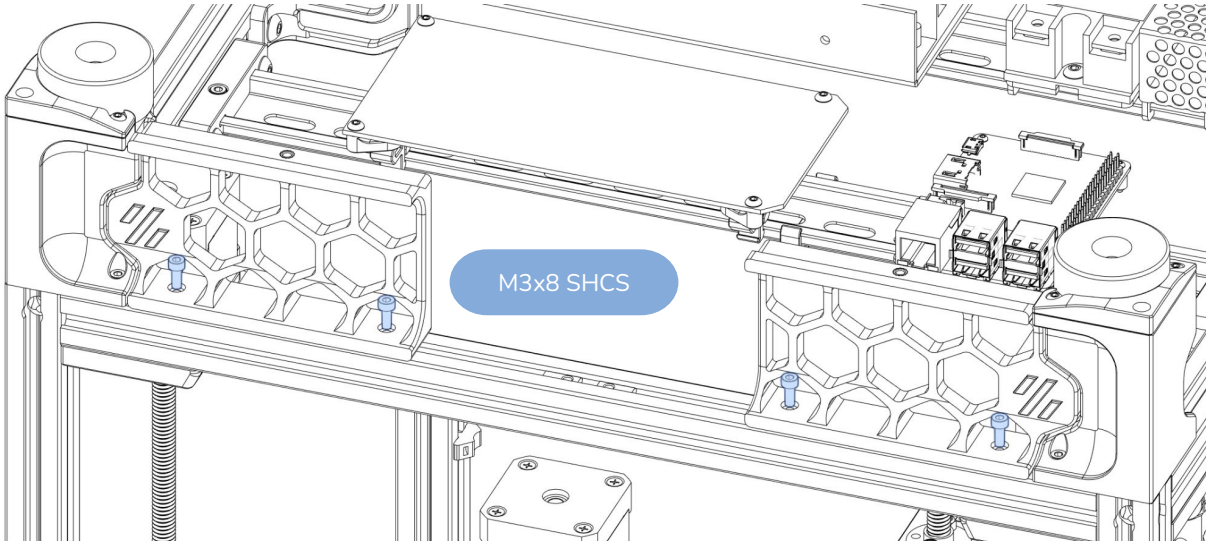
LOGO INSERTS

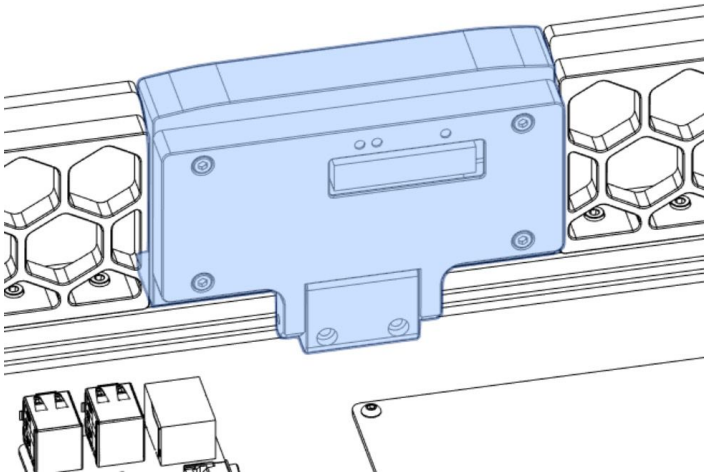
These will stay in place without any adhesive.
Alternatively they can be glued in place.

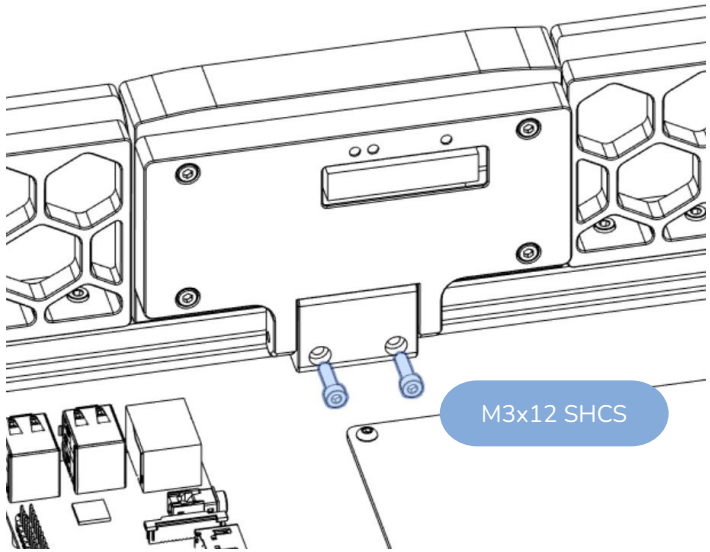

M3 T-Nut


M3 T-Nut




M3x8 SHCS

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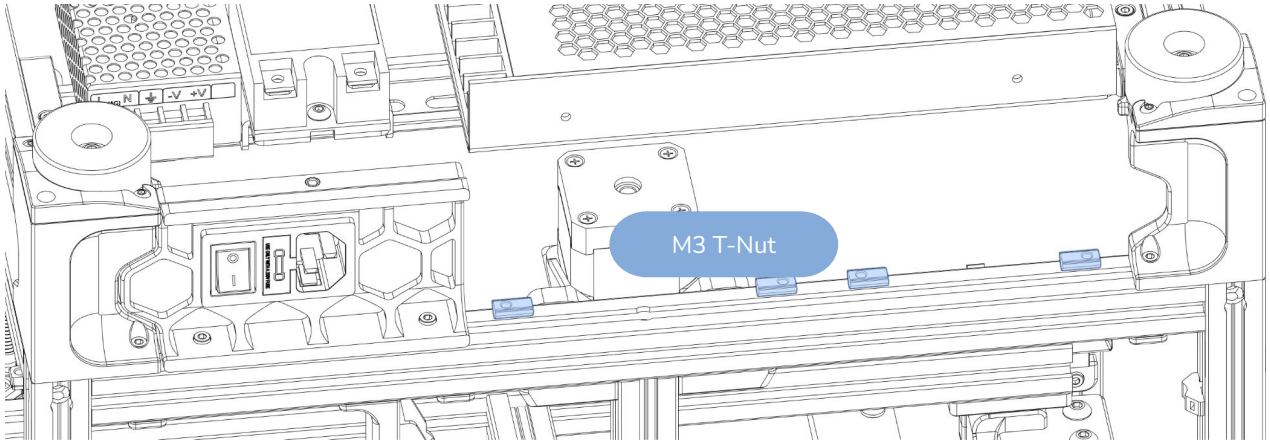
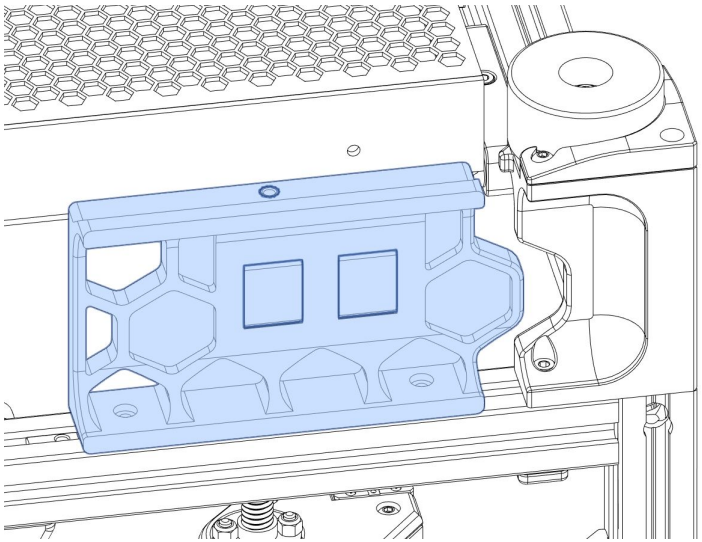
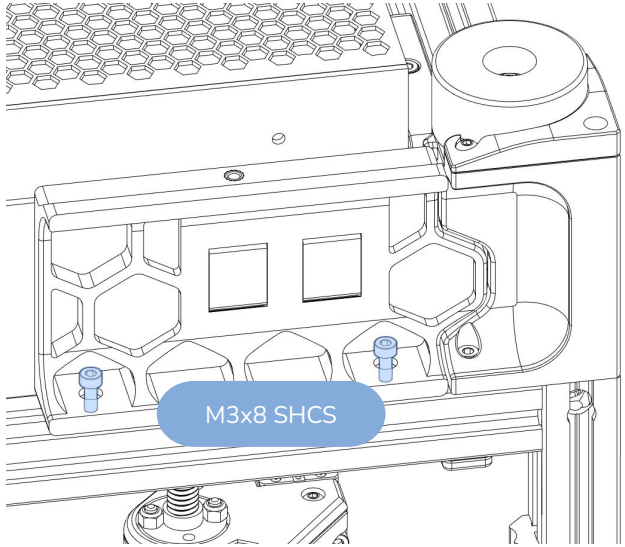

M3 T-Nut

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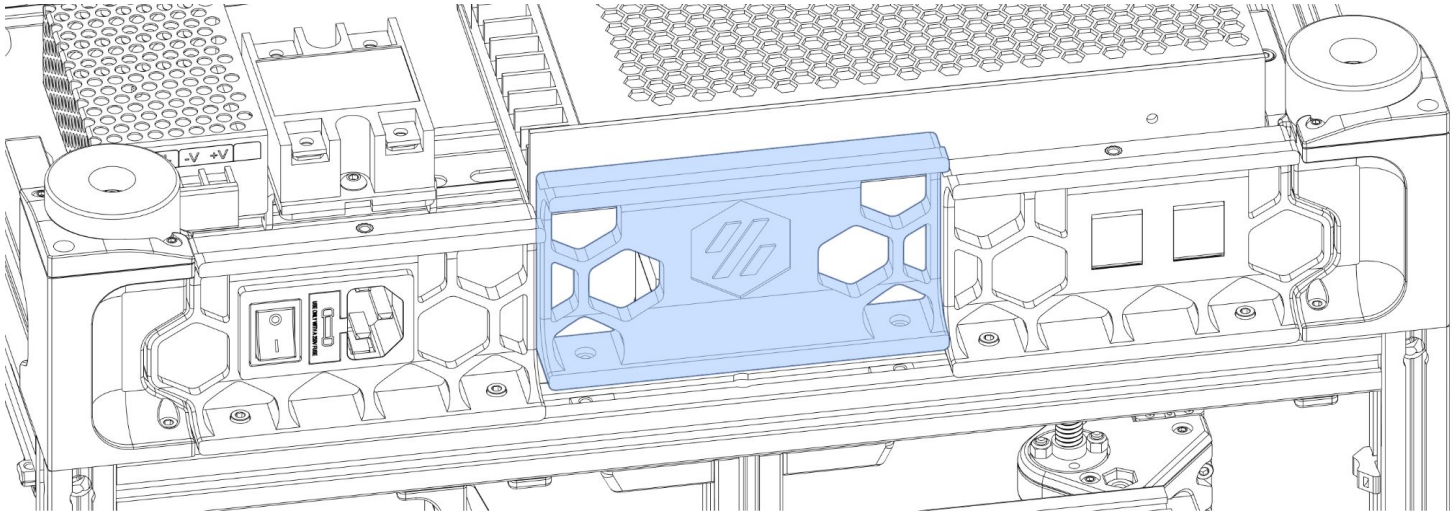
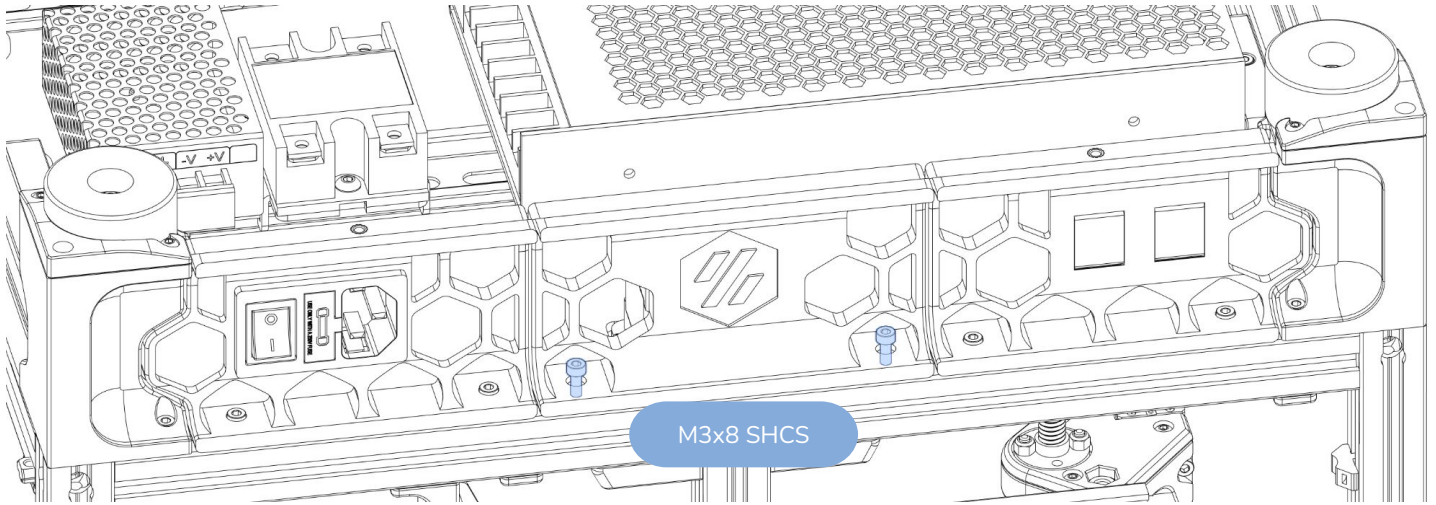
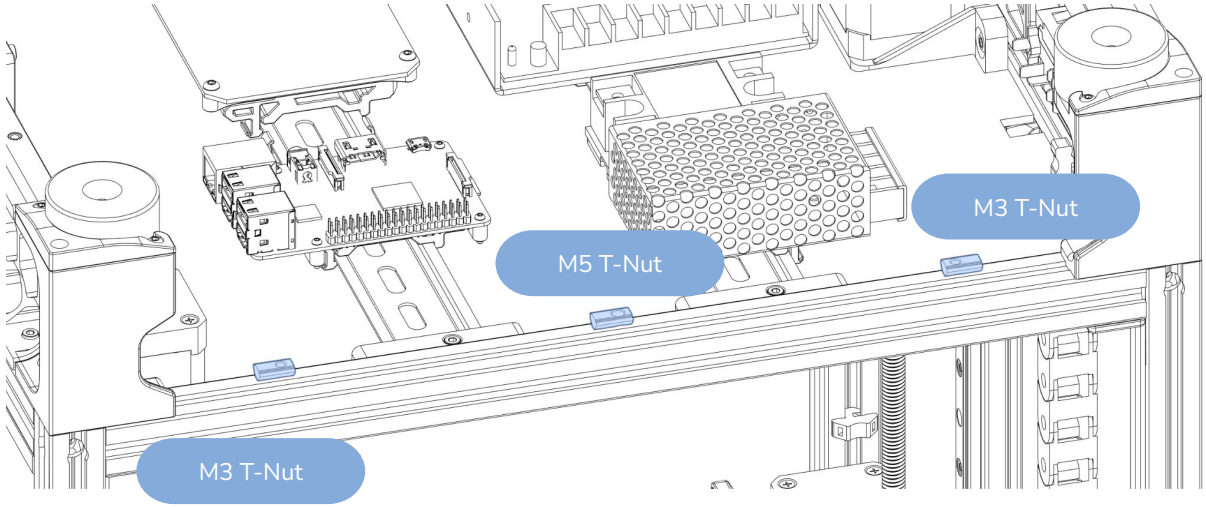
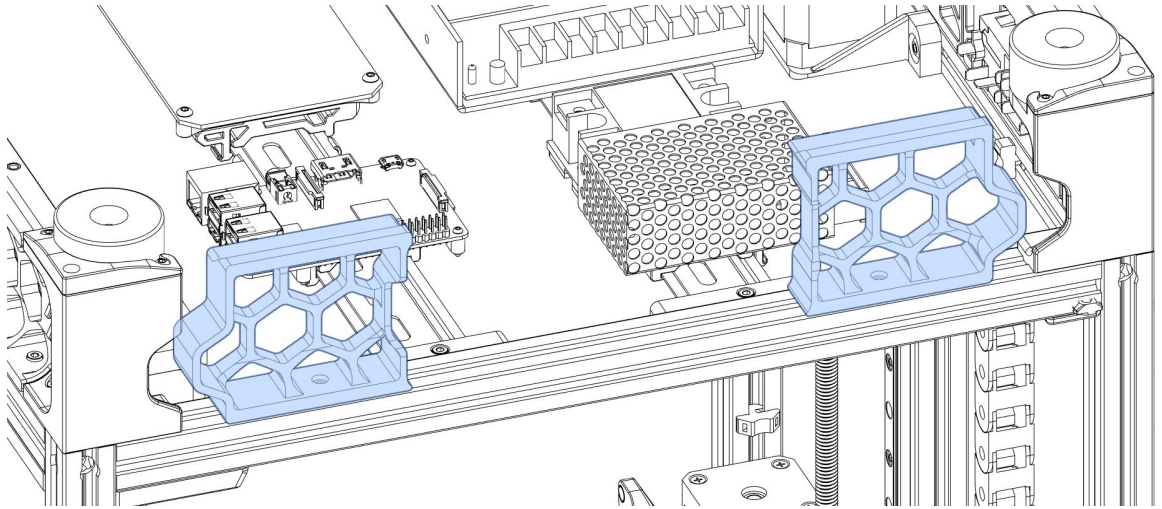
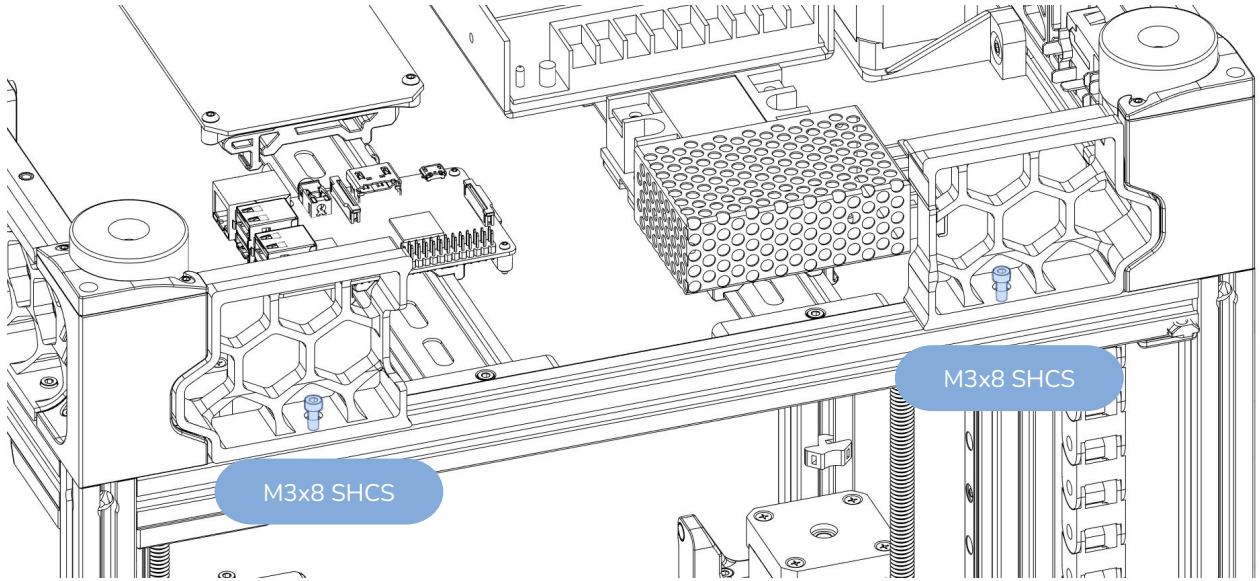
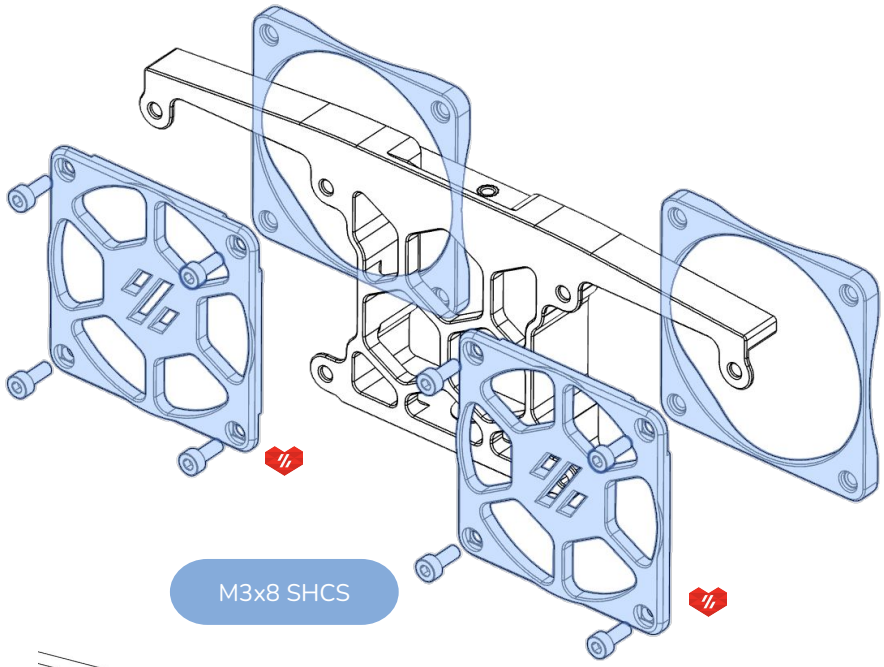


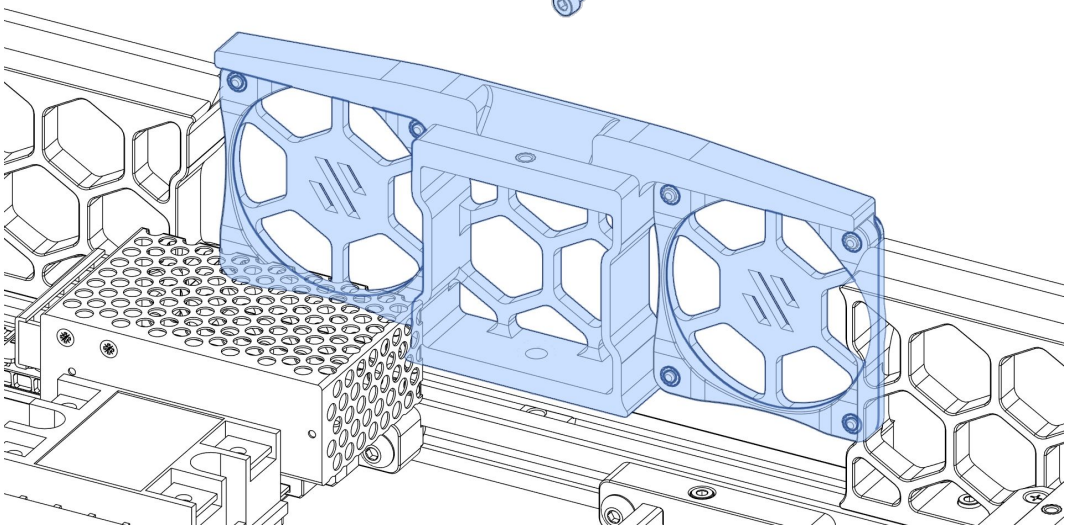

M3x8 SHCS

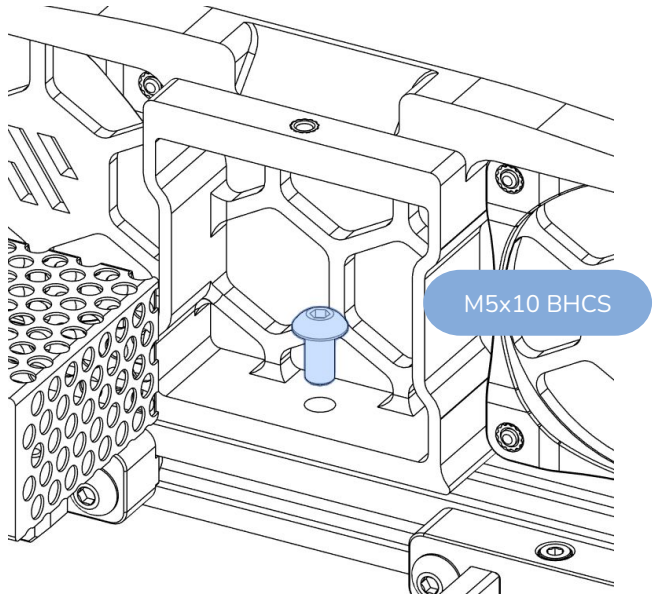

M3 T-Nut
M5 T-Nut
M3 T-Nut

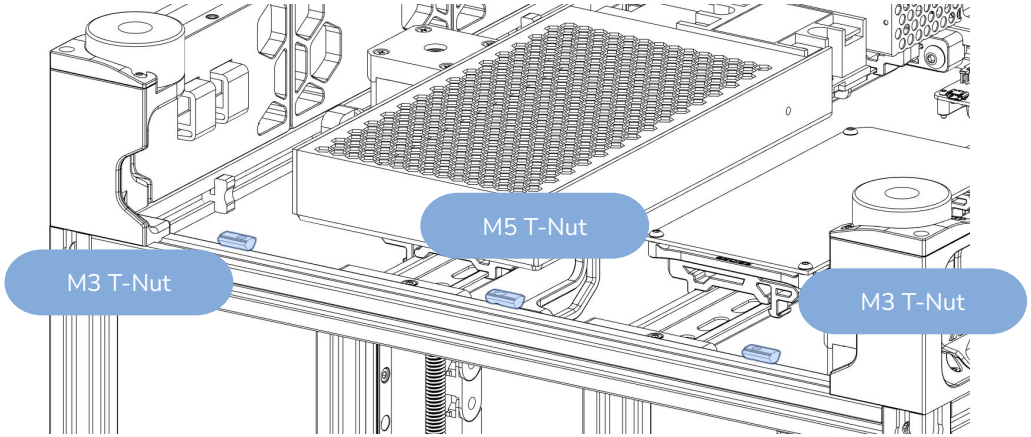


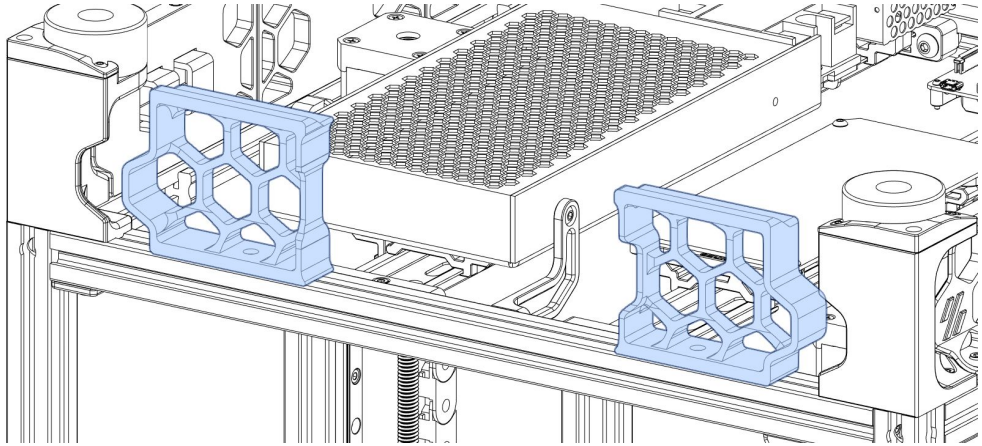

M3x8 SHCS
M3x8 SHCS

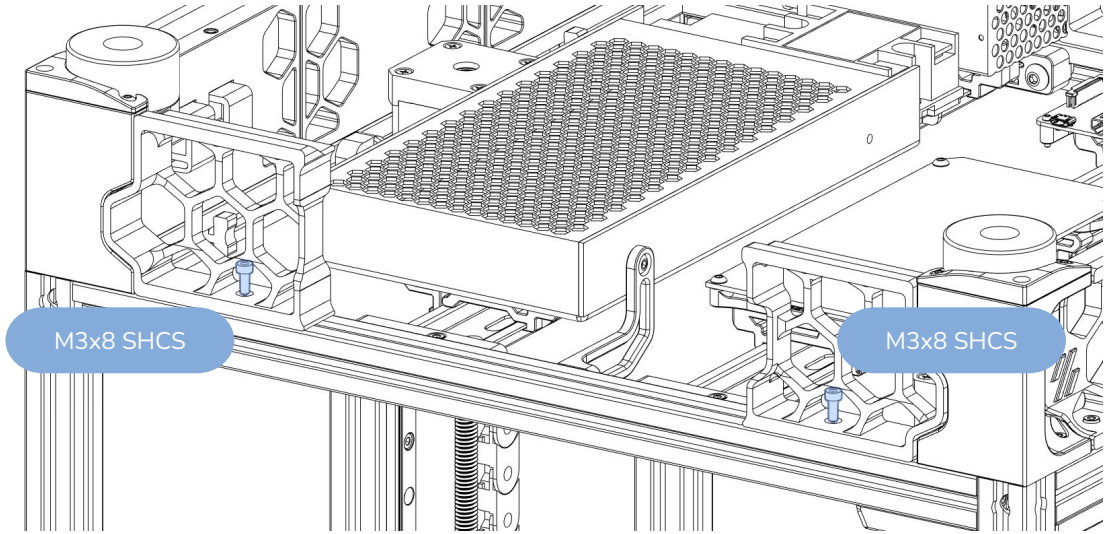

M3x8 SHCS

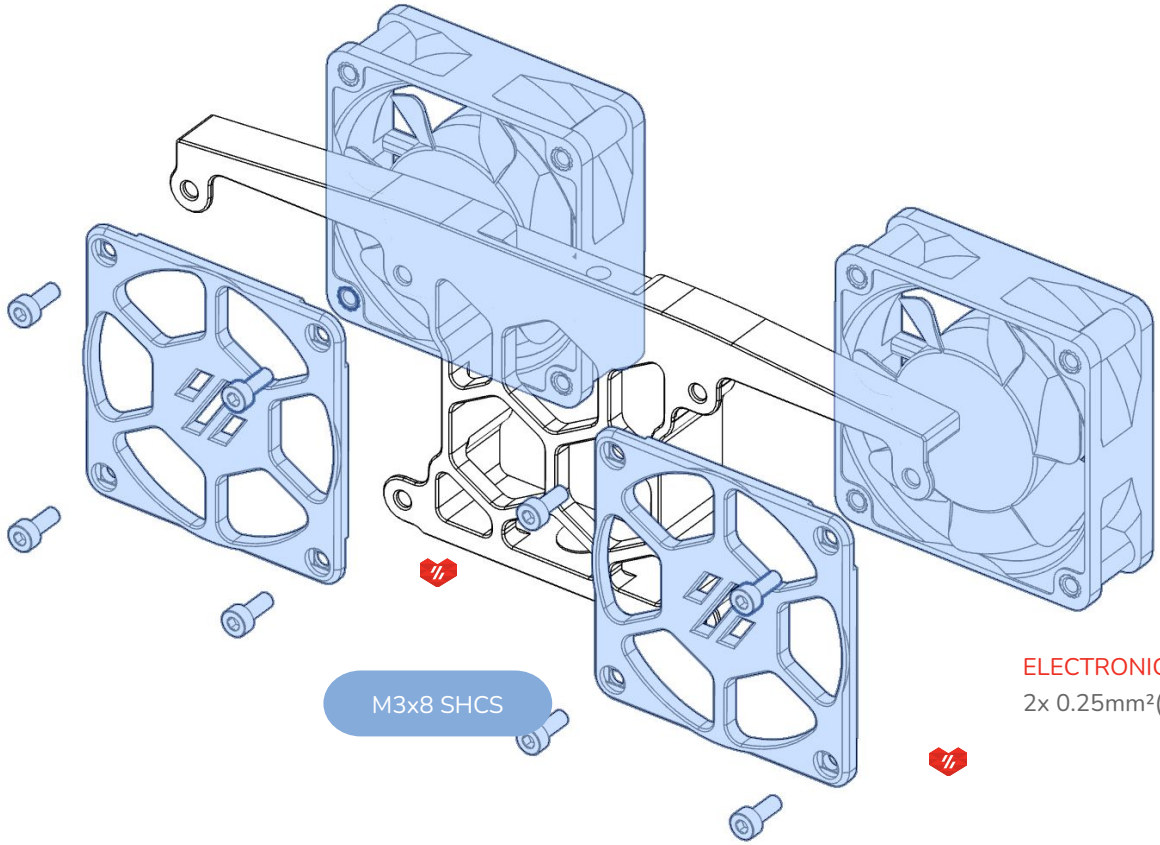


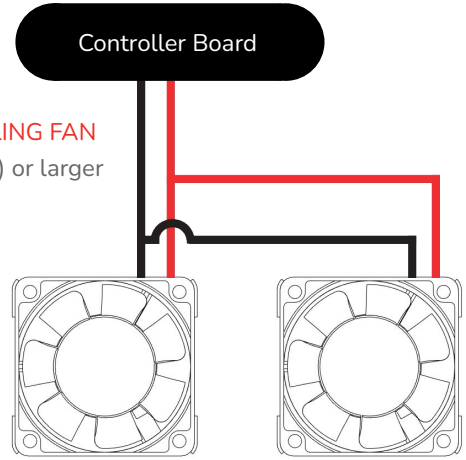

M5x10 BHCS


M5 T-Nut
M3 T-Nut
M3 T-Nut



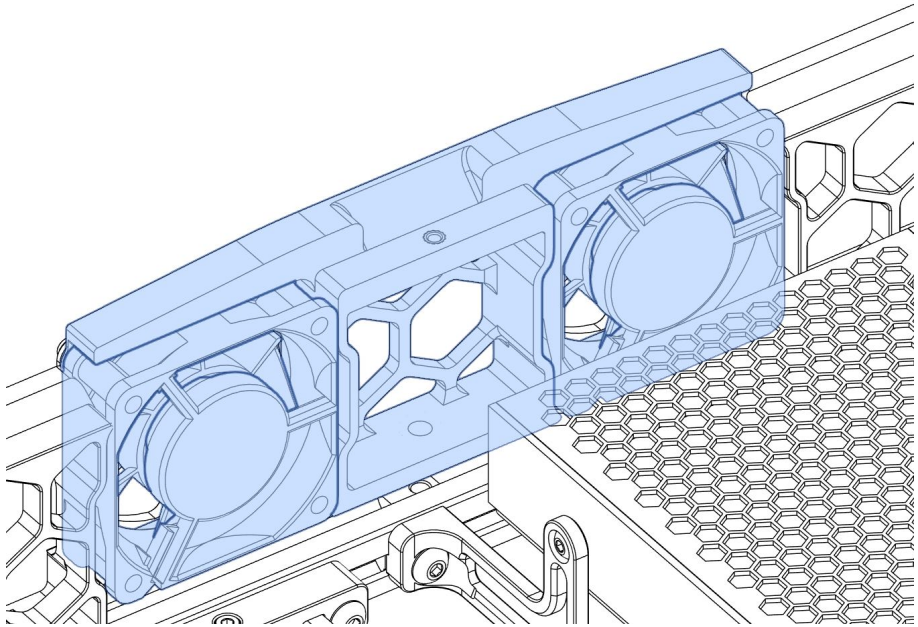

M3x8 SHCS
M3x8 SHCS

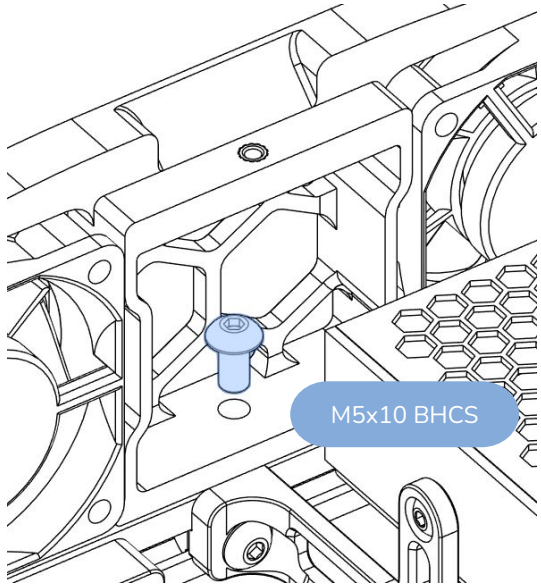




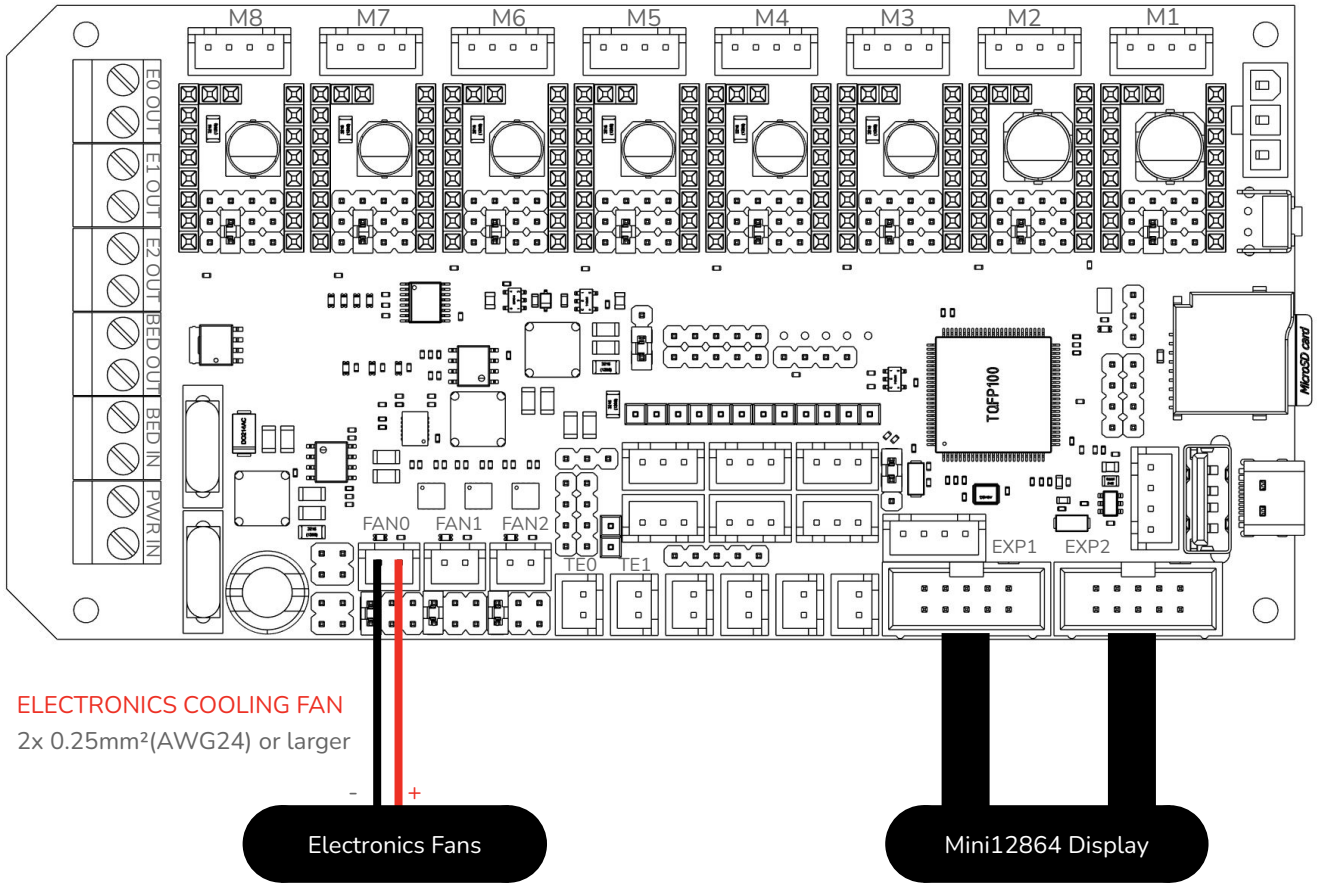
ELECTRONICS COOLING FAN

2x 0.25mm²(AWG24) or larger



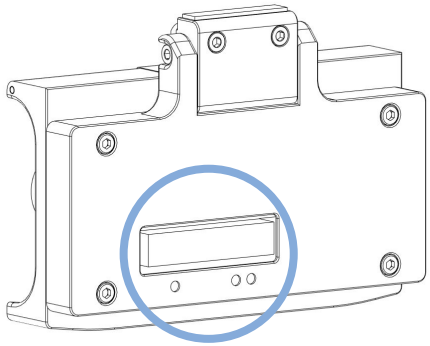

M5x10 BHCS

Skip these steps and instead mount the klipperscreen.



ELECTRONICS COOLING FAN

2x 0.25mm²(AWG24) or larger



WHICH IS WHICH?

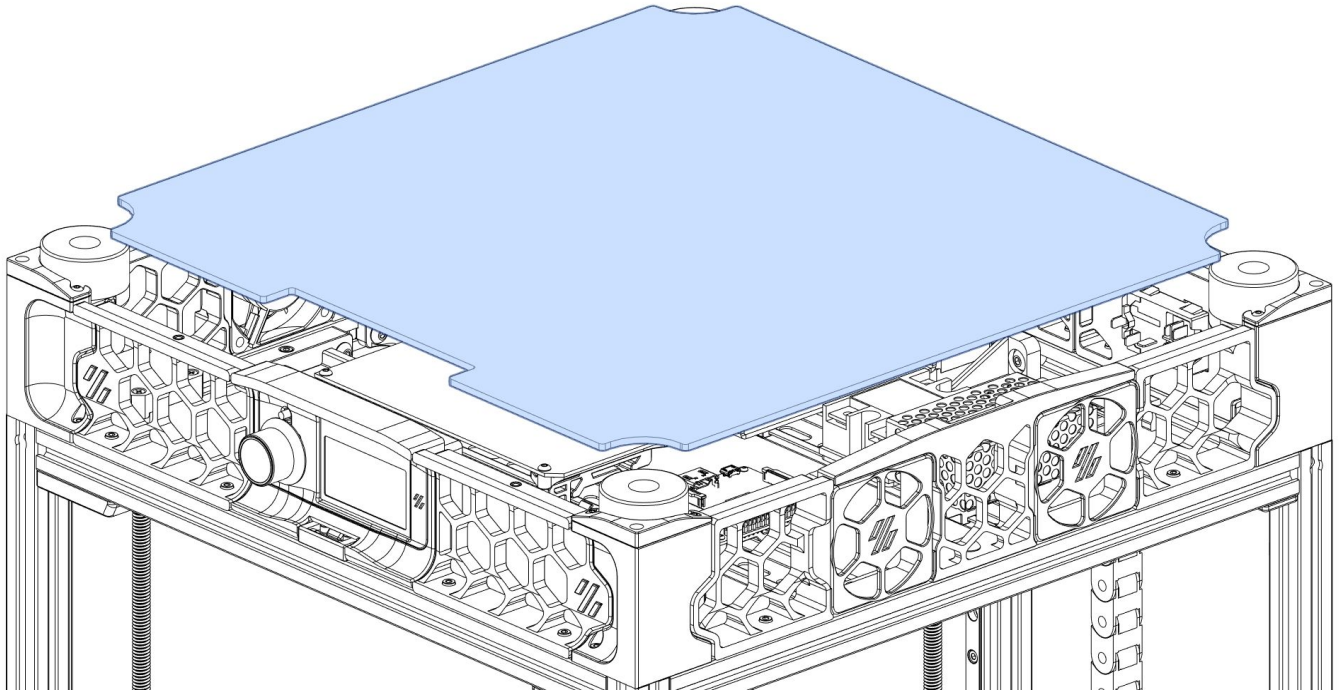
The socket with 1 dot below it is EXP1 and the socket with with 2 is EXP2.

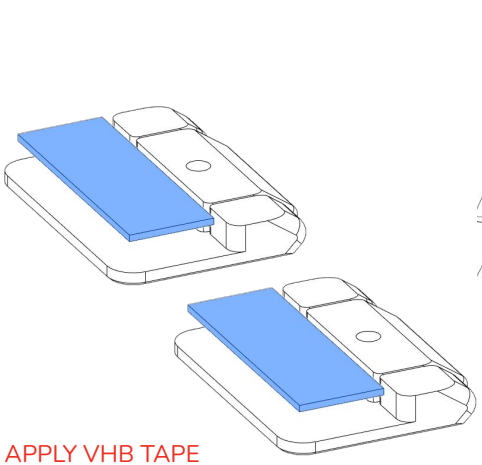
WARNING: DISPLAY HOOKUP

Before connecting the display make sure to review the documentation on the VORON documentation site (https://voron.link/ypdmcb2).

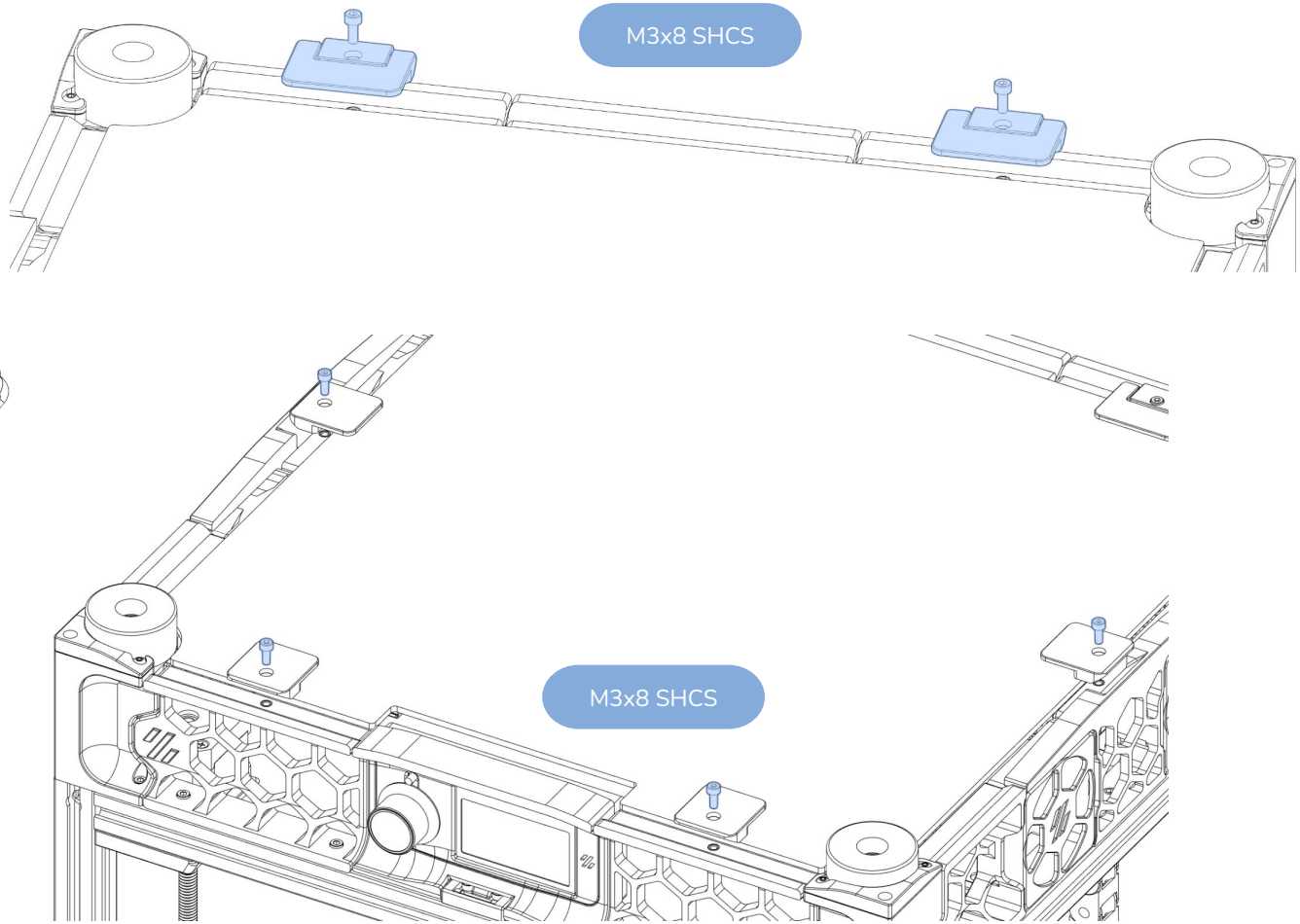
If you are using a “FYSETC mini12864” please review section 3.3 of the FYSETC Spider documentation (https://voron.link/m6etwnl).

Don't close the electronics compartment yet, you will need to acces this while flashing the boards.





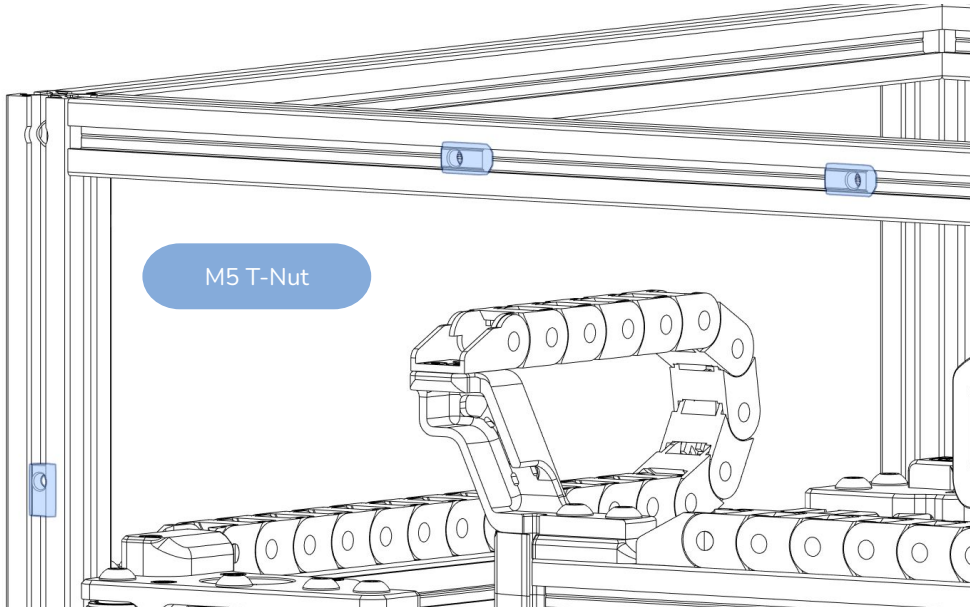
APPLY VHB TAPE

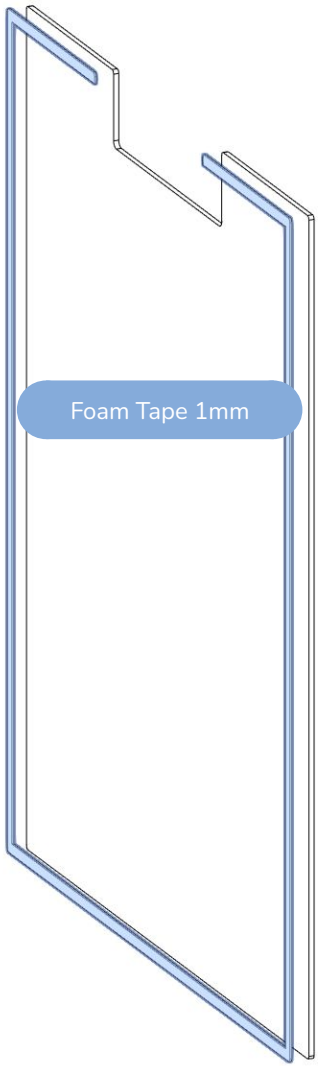


WHAT ARE THESE FOR?

These roll-in nuts will be used during the installation of the exhaust and spool holder.

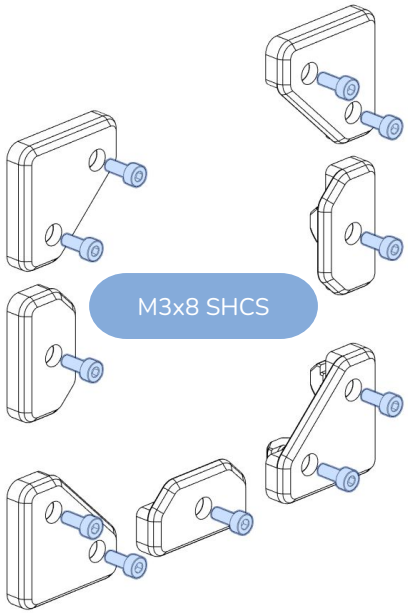
It's easier to install them now without the panel interfering.





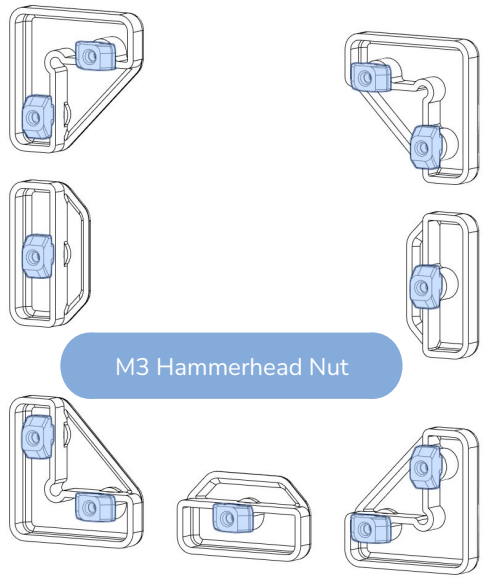
APPLY FOAM TAPE

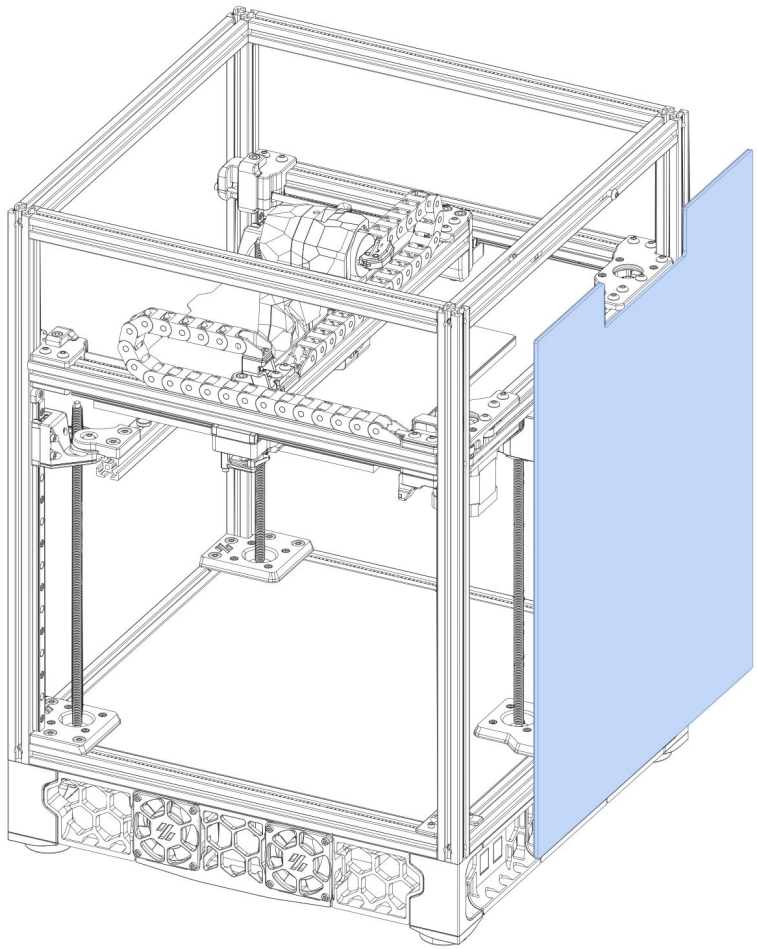
Use foam tape on the contact areas between the panels and the frame to mitigate noise from vibrations.

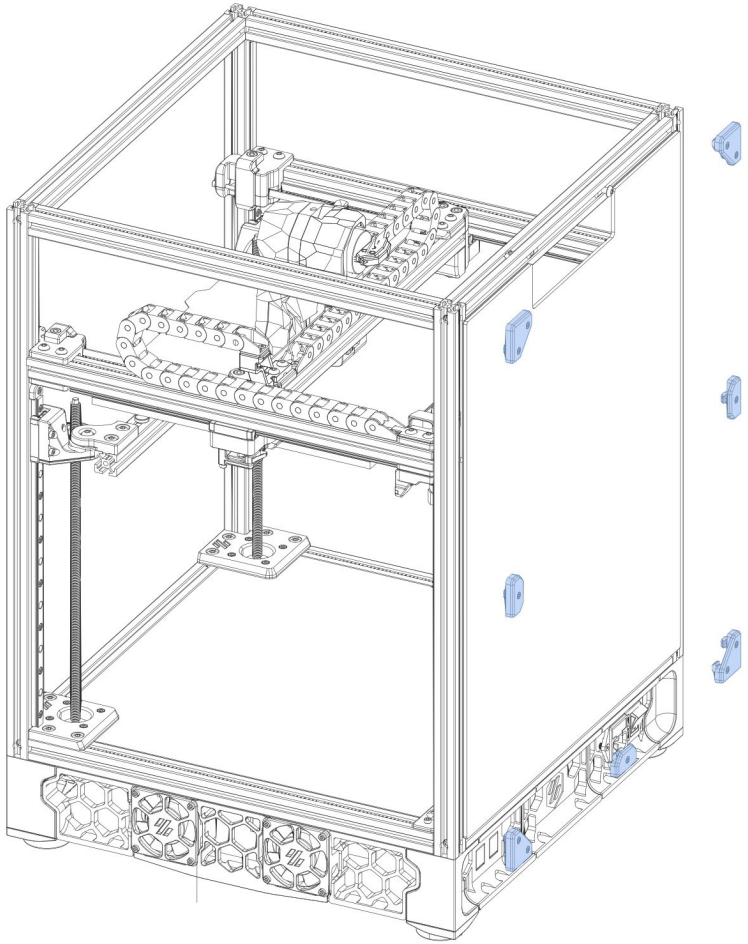


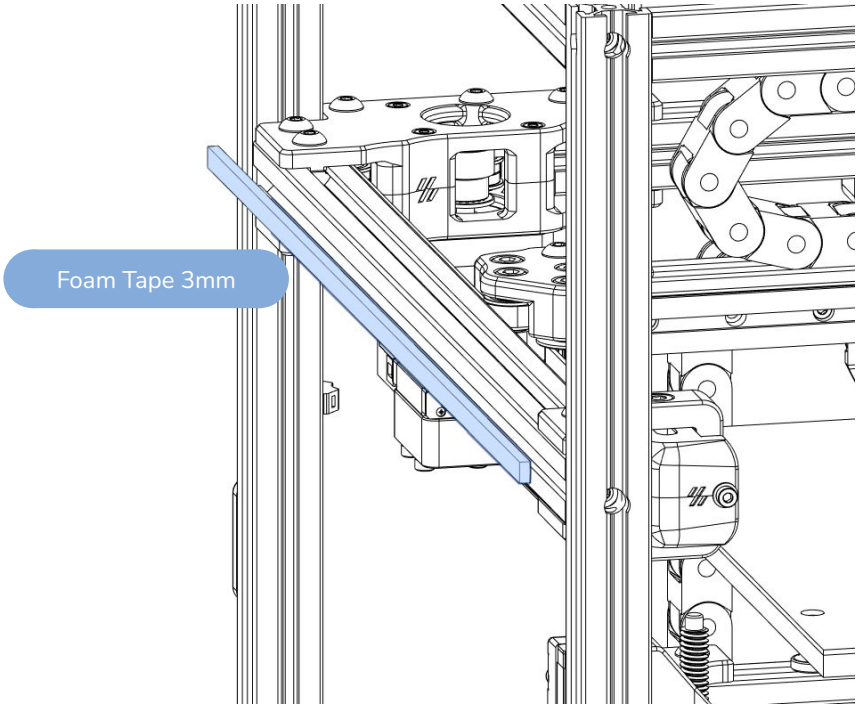
HAMMERHEAD NUTS?

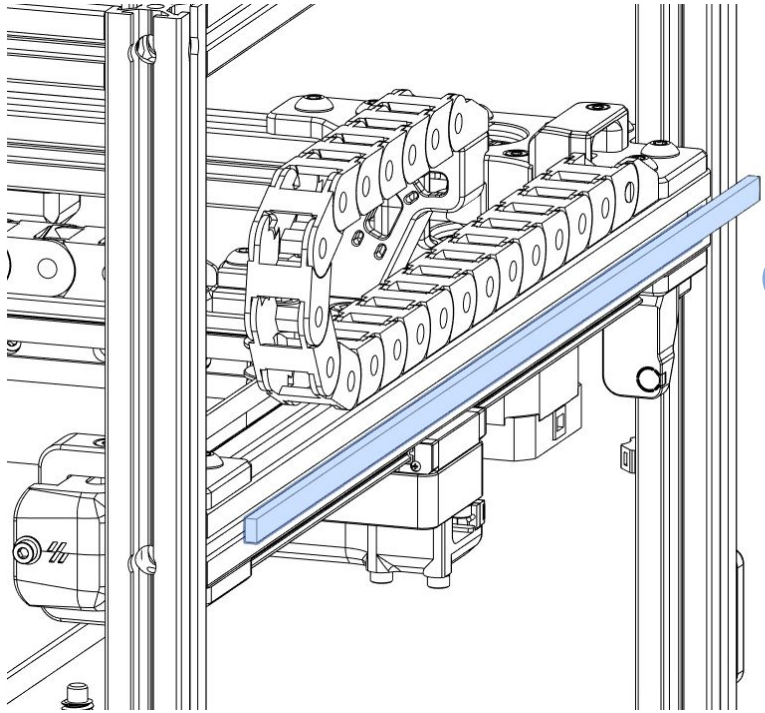
A drop of thread locker will turn the hammerhead nuts into a 1/4 turn quick release for the panels. Best done once the assembly is finished.





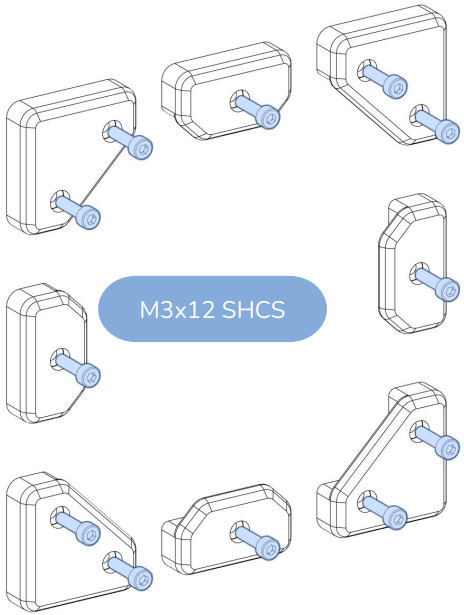


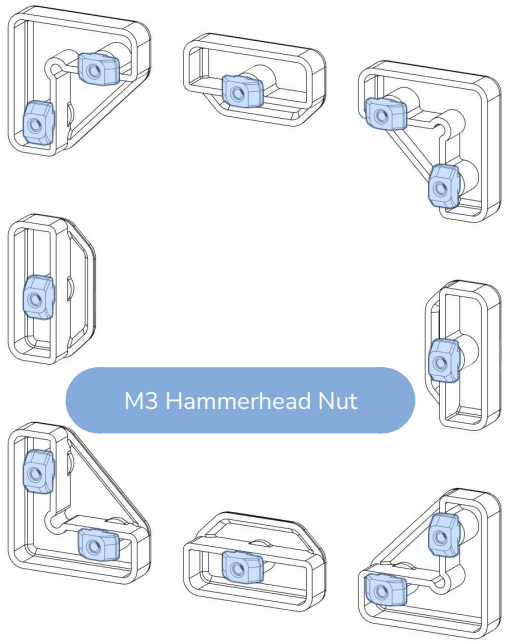




APPLY FOAM TAPE

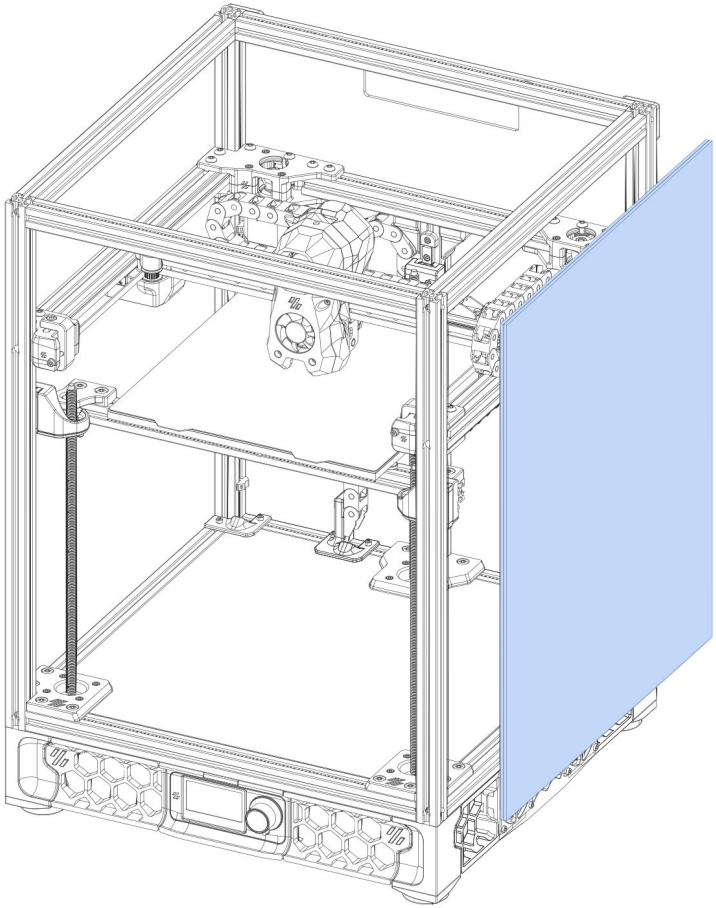
Use 3mm foam tape on the contact areas between the panels and the frame to mitigate noise from vibrations.

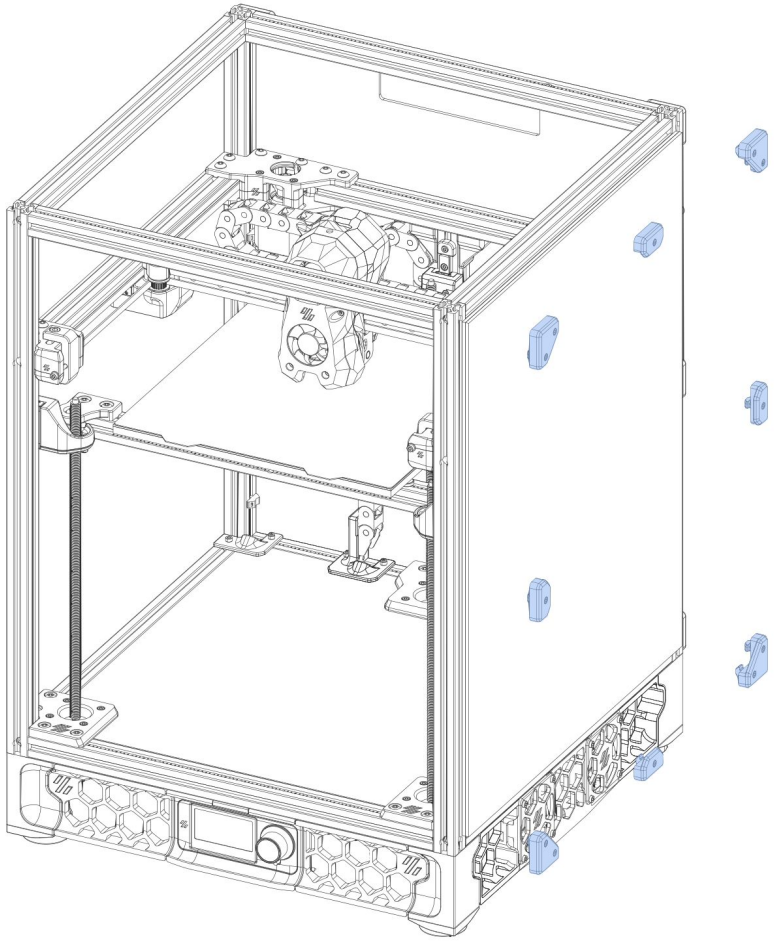


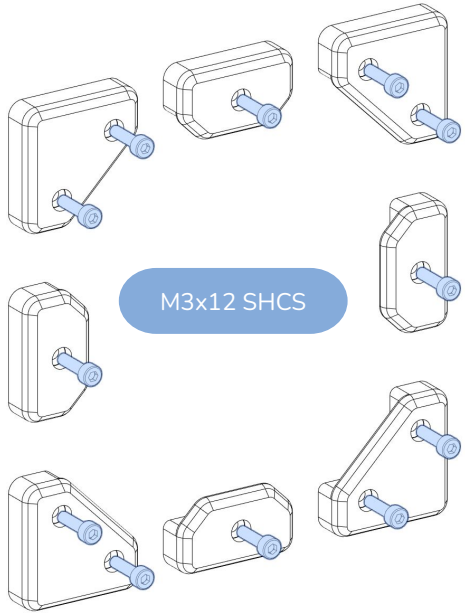


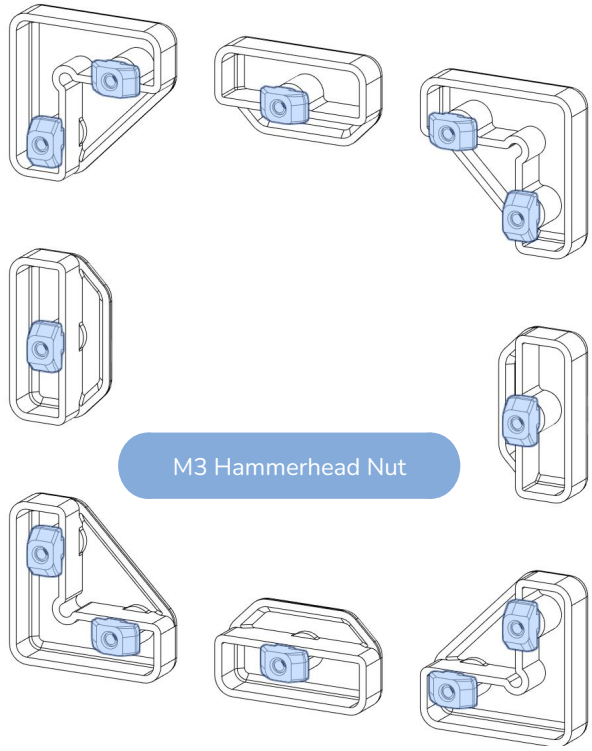
APPLY FOAM TAPE

Use 3mm foam tape on the contact areas between the panels and the frame to mitigate noise from vibrations.



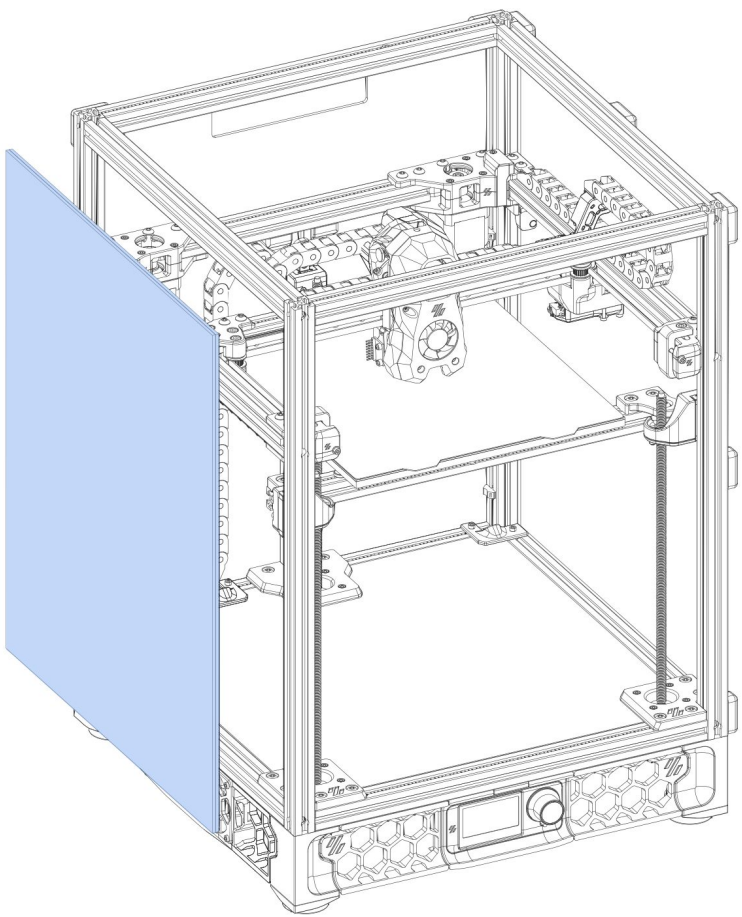


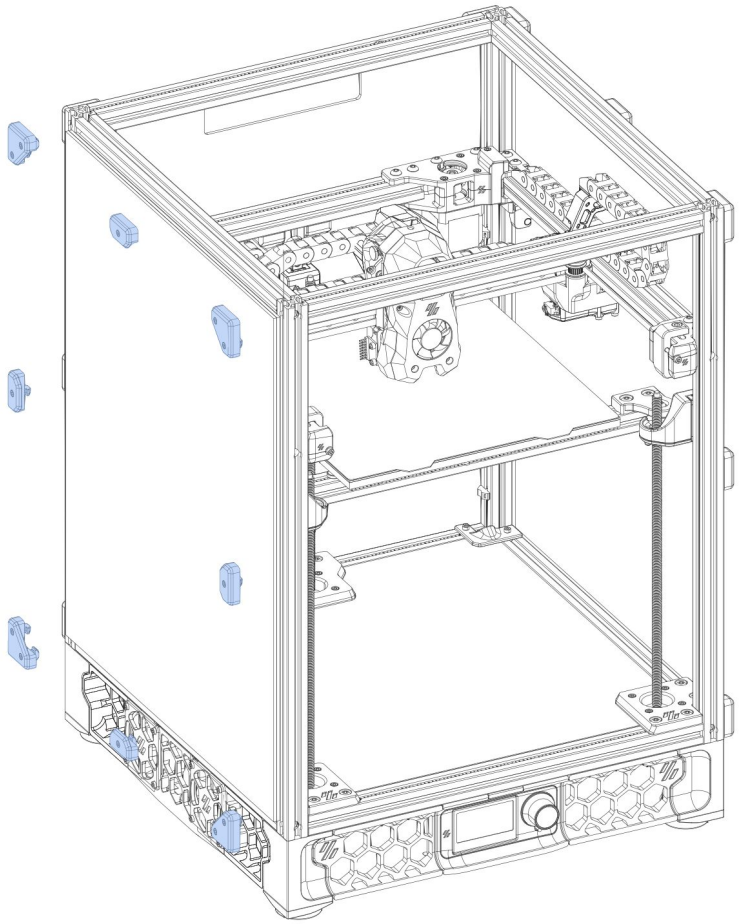


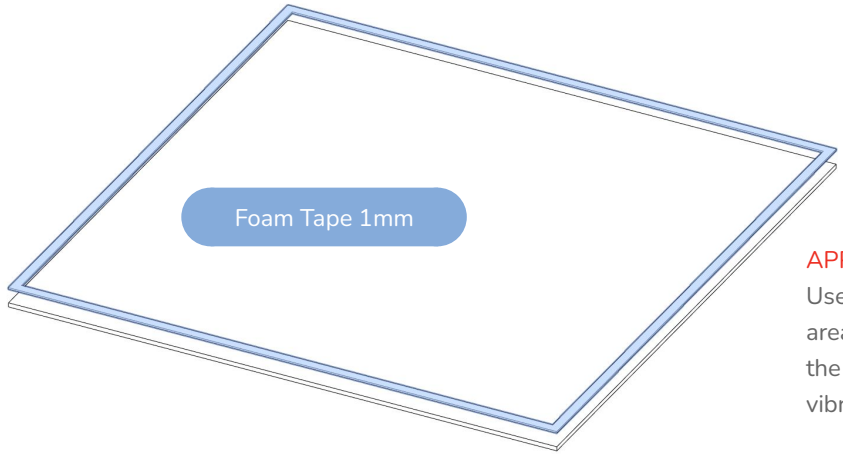


APPLY FOAM TAPE

Use 3mm foam tape on the contact areas between the panels and the frame to mitigate noise from vibrations.

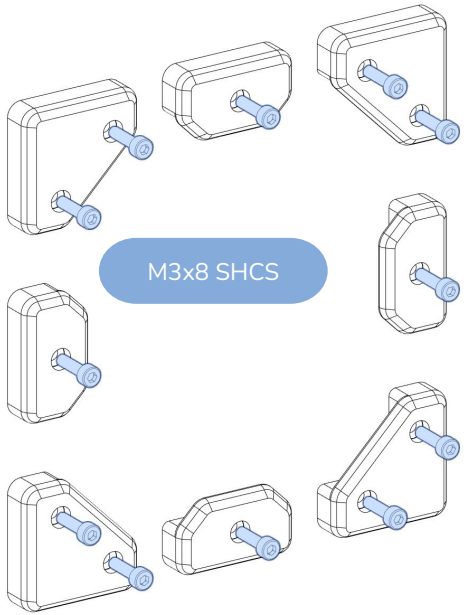


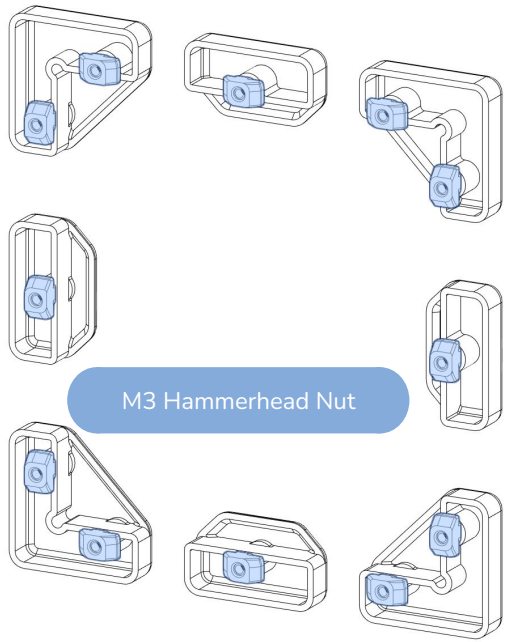


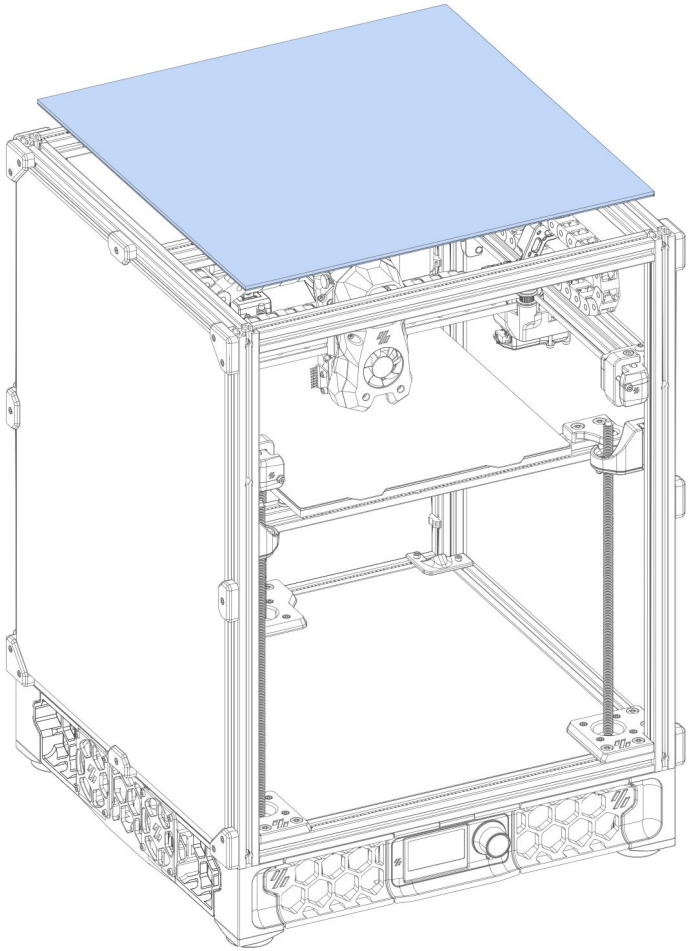


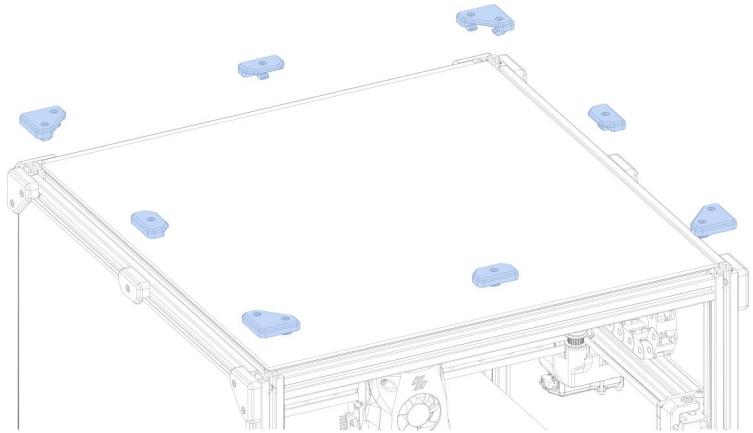
APPLY FOAM TAPE

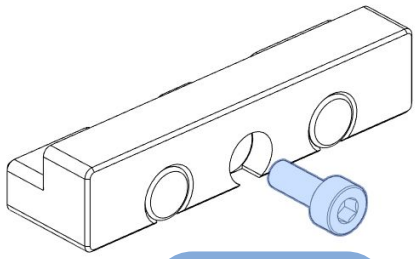
Use 1mm foam tape on the contact areas between the panels and the frame to mitigate noise from vibrations.



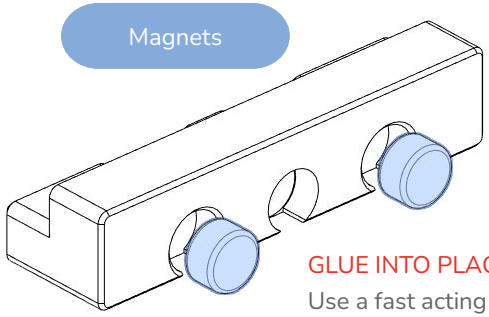




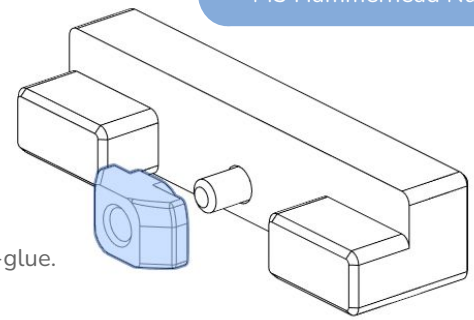




M3x8 SHCS



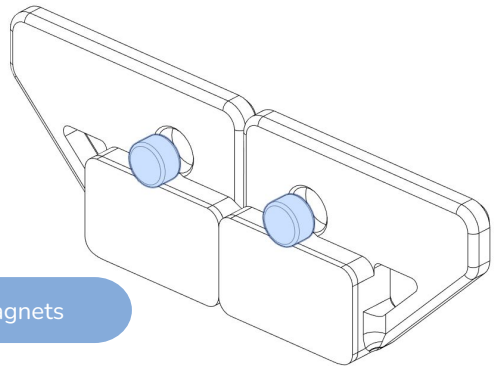
Magnets



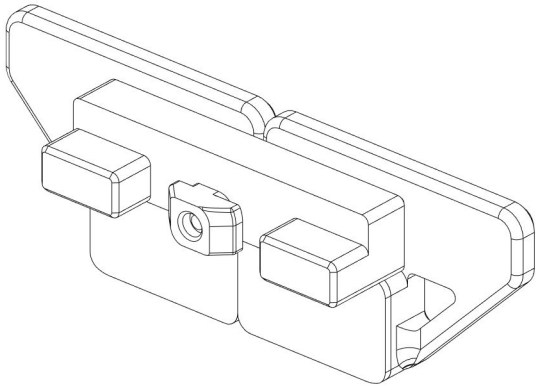
M3 Hammerhead Nut

GLUE INTO PLACE

Use a fast acting glue like super-glue.

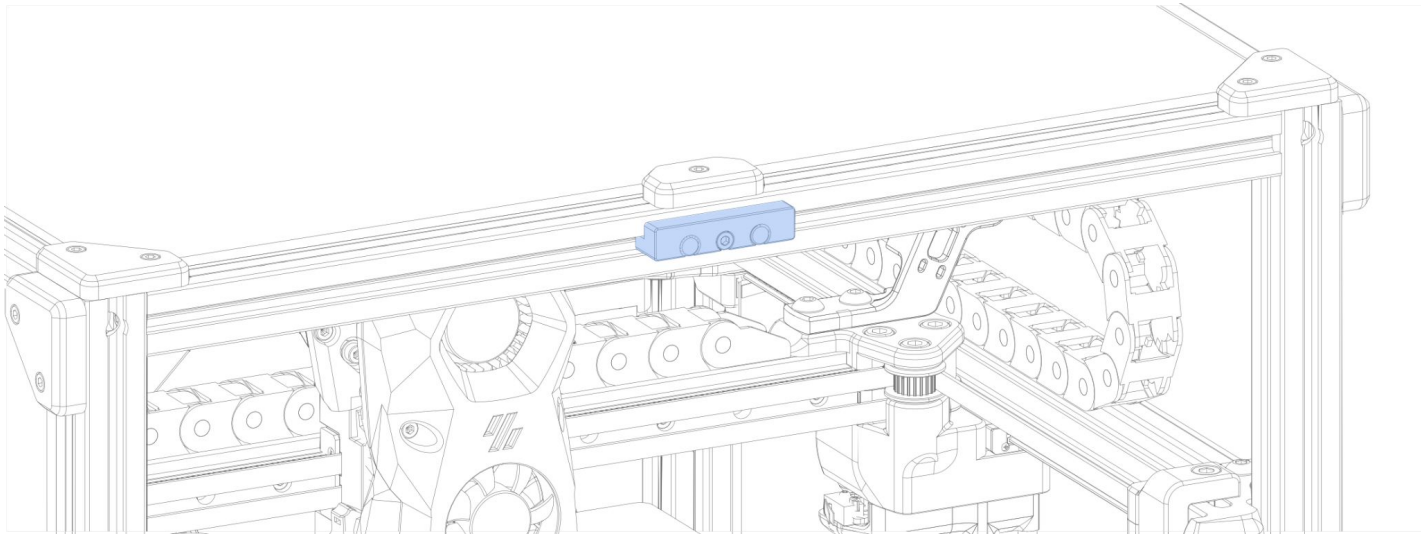


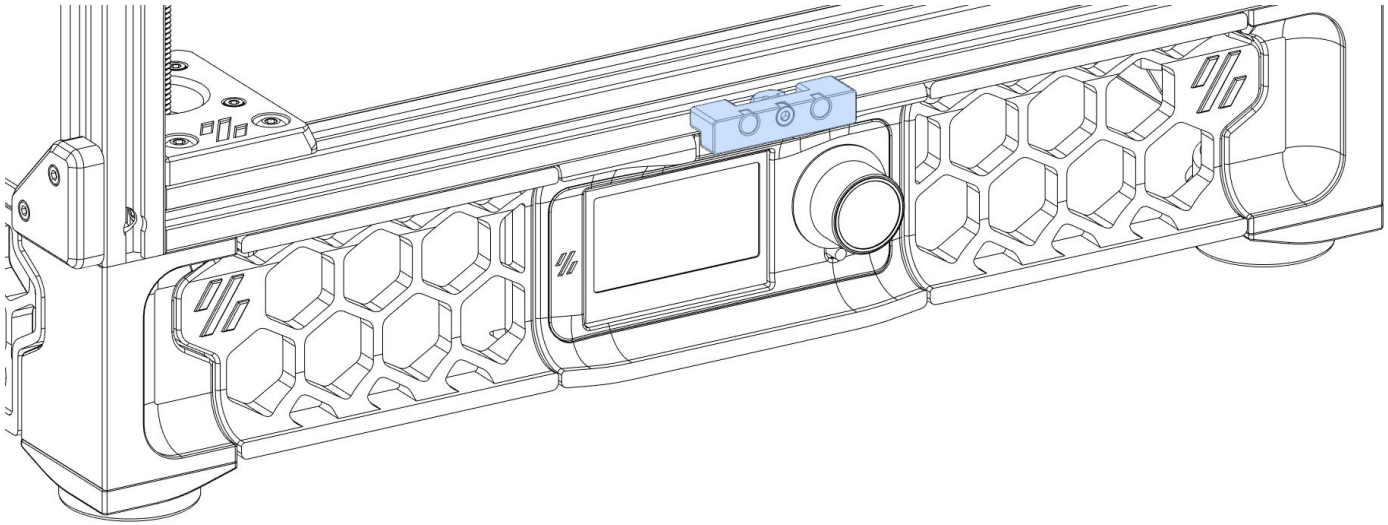
Magnets

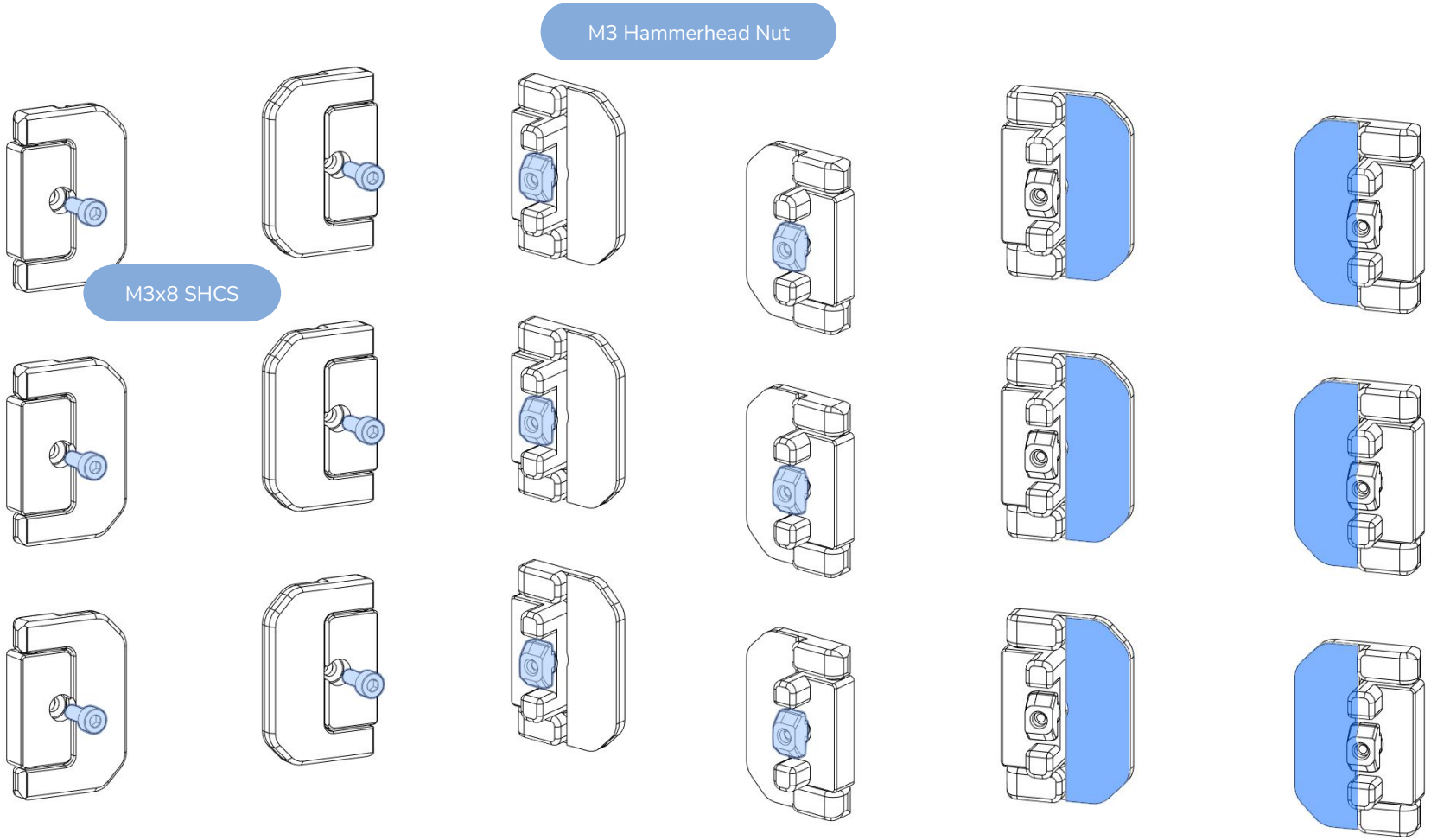


MIND THE MAGNET POLARITY

Ensure that the magnets are facing in the right direction prior to glueing them into place.

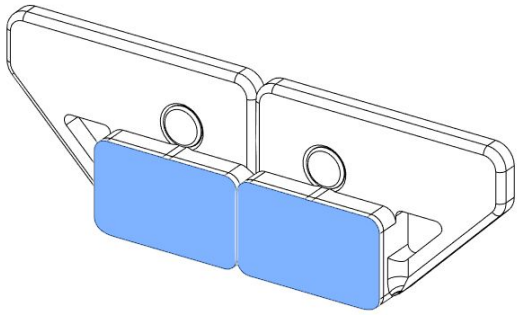






APPLY VHB TAPE

VHB Tape is a double sided adhesive tape.



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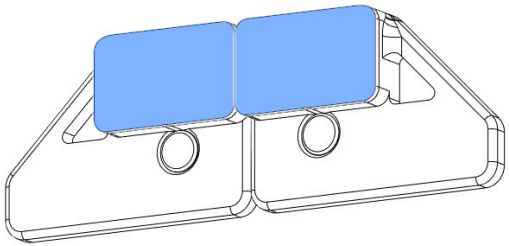
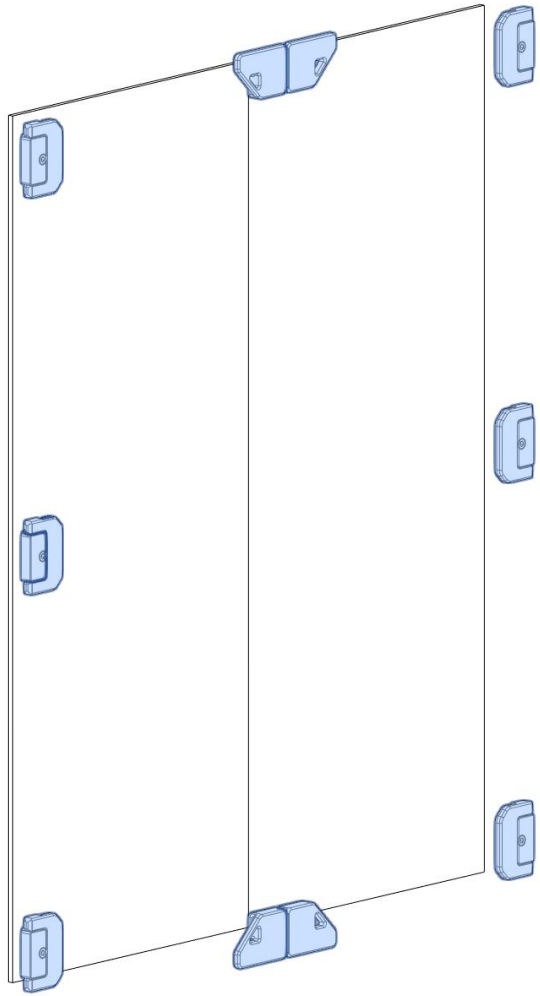
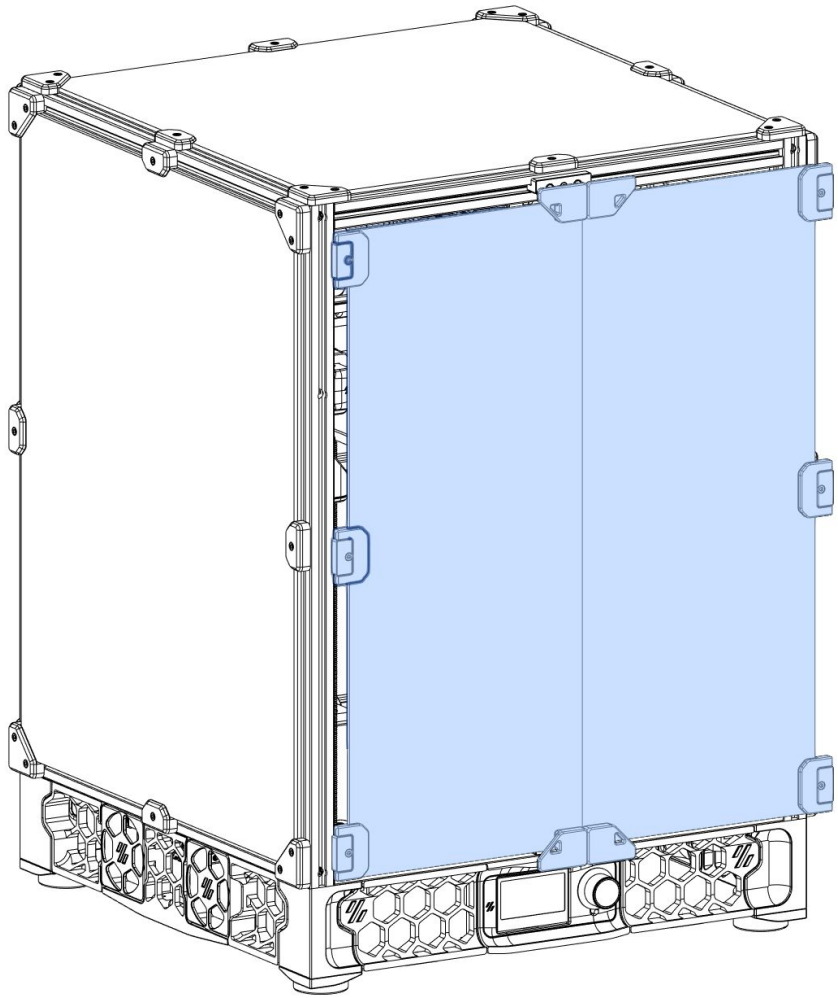
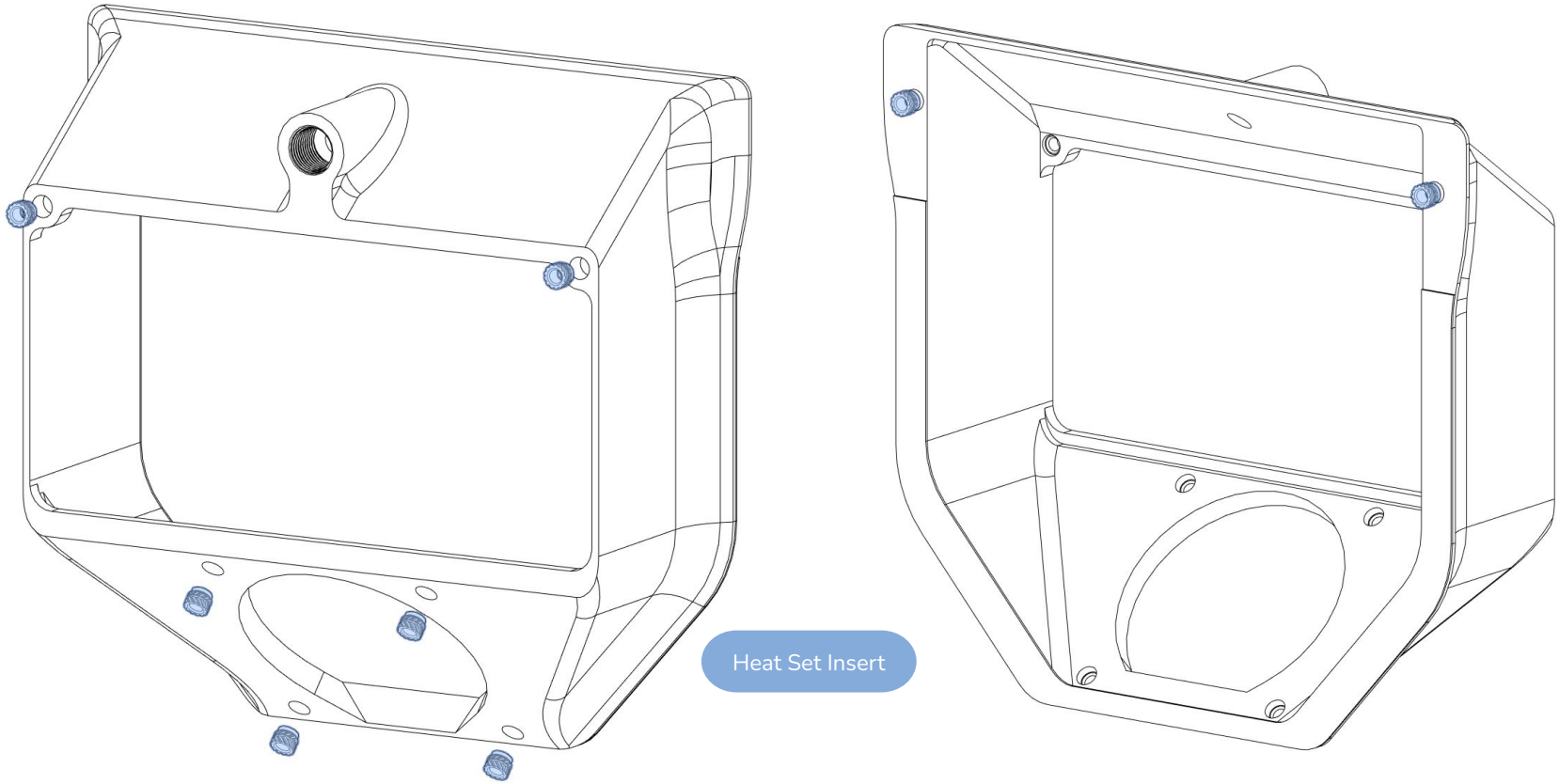
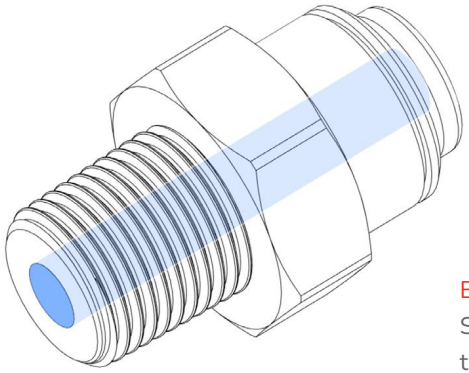
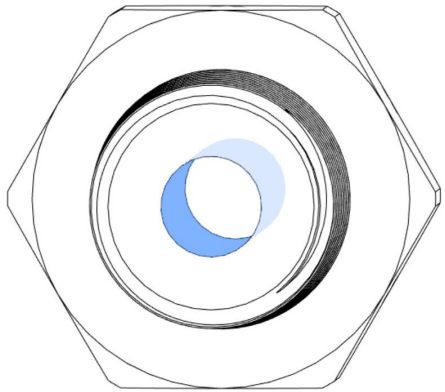







Heat Set Insert

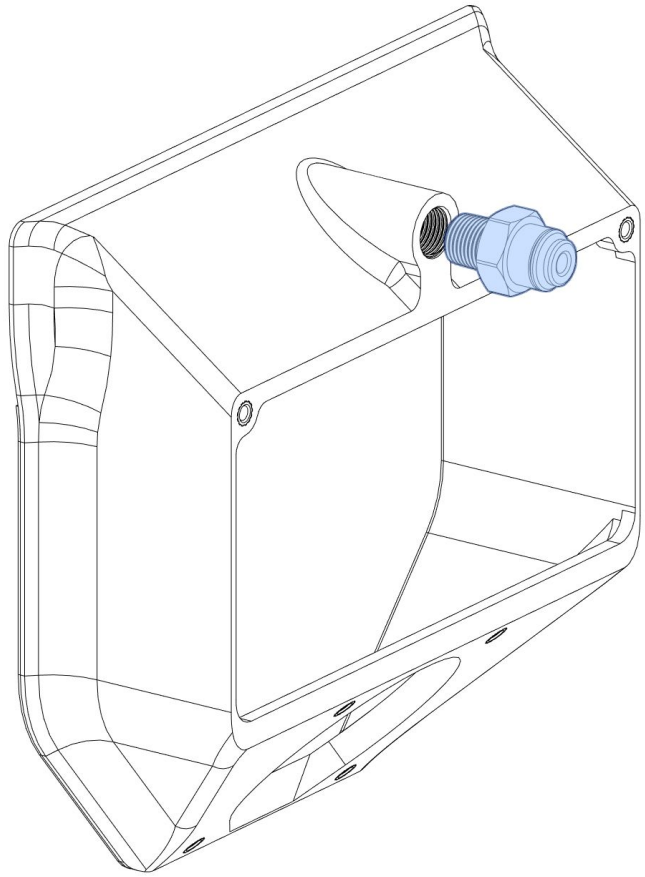


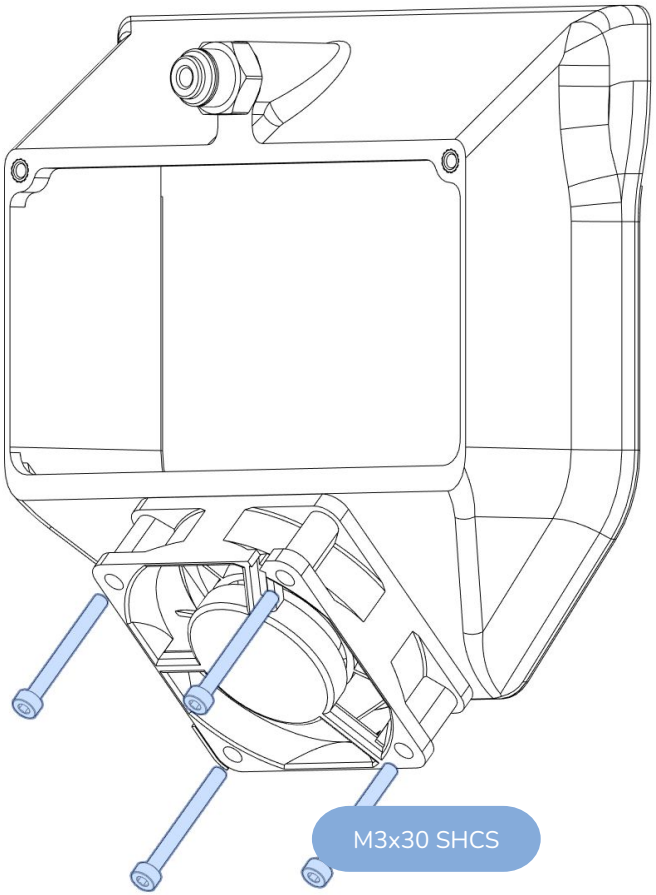


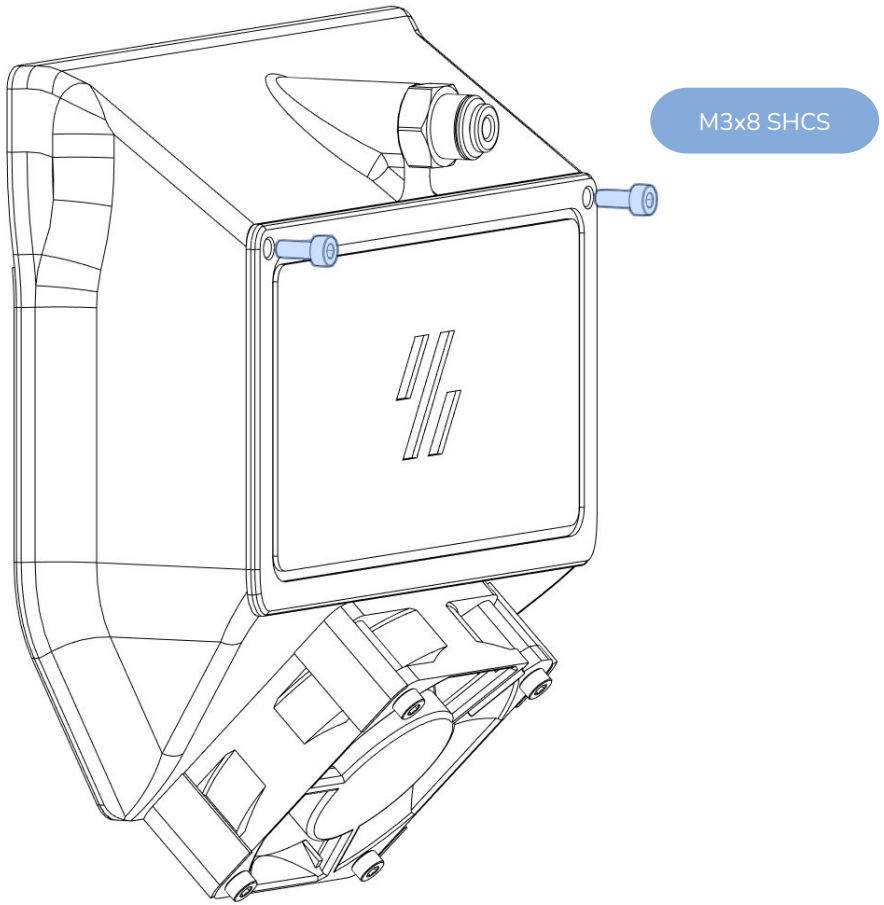
BSPP ADAPTER

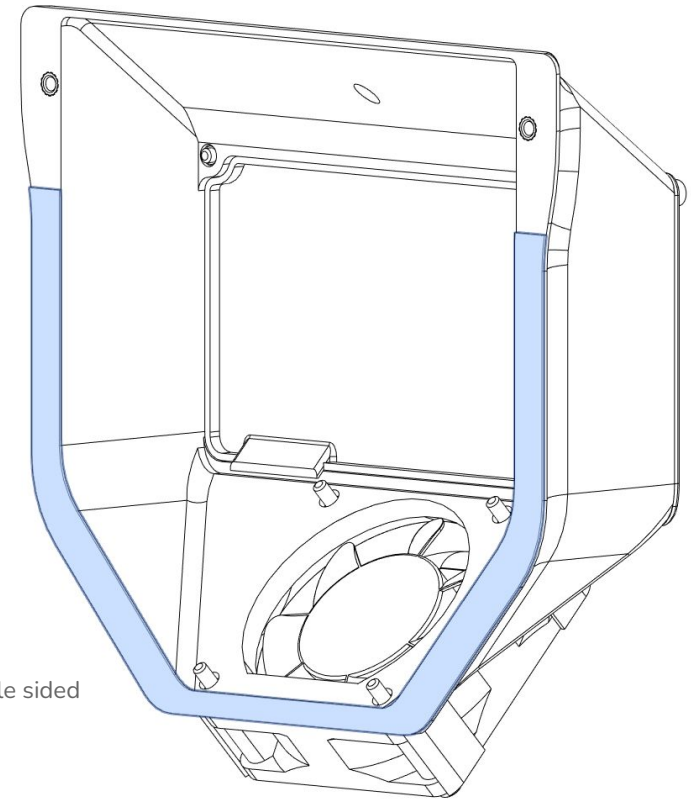
Some adapters have a small lip that prevents the PTFE tube from passing through.

Inspect the adapter and if necessary use a drill to carefully remove the lip.




M3x30 SHCS


M3x8 SHCS



APPLY VHB TAPE

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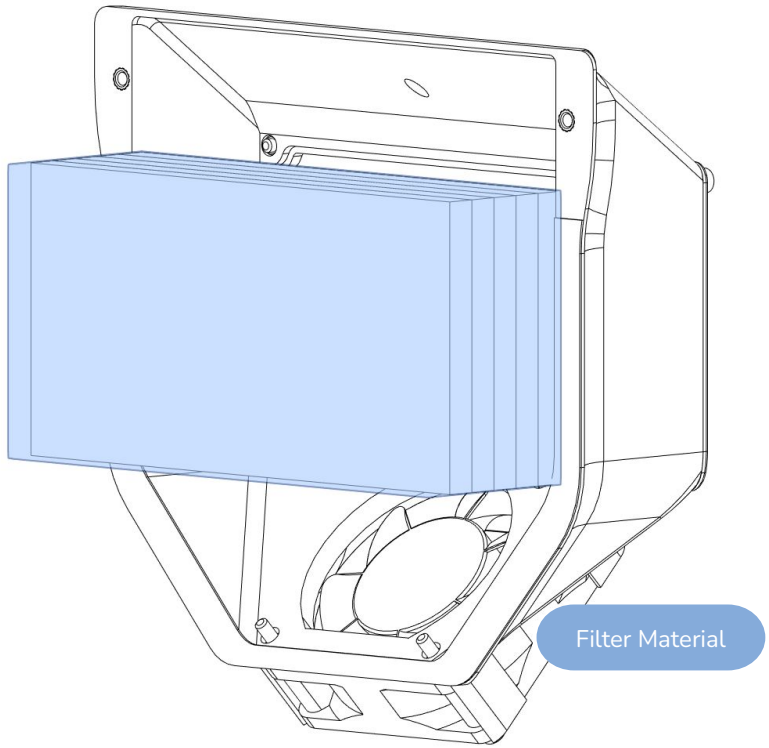
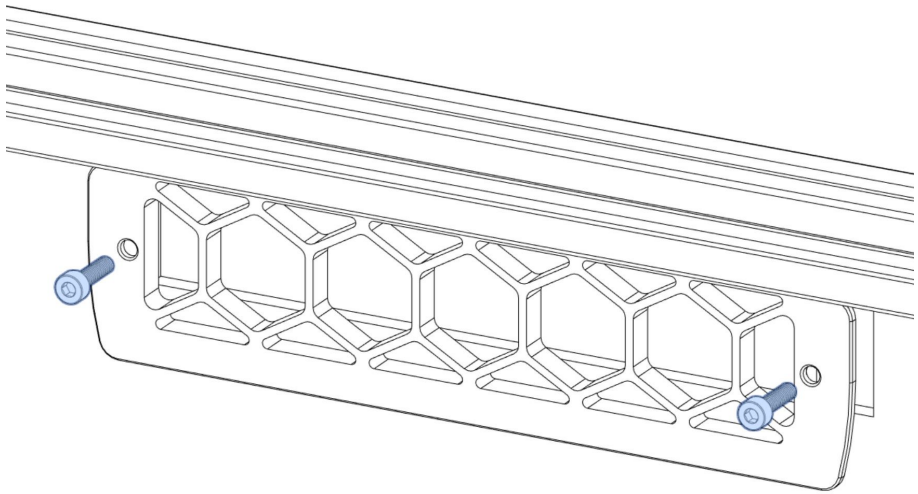
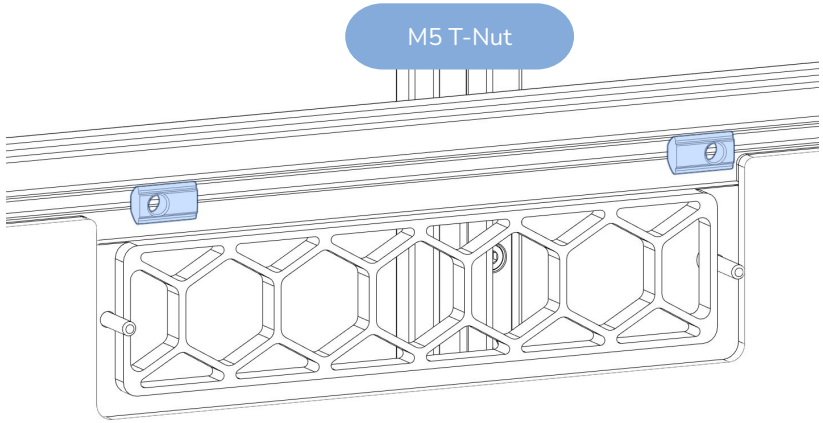
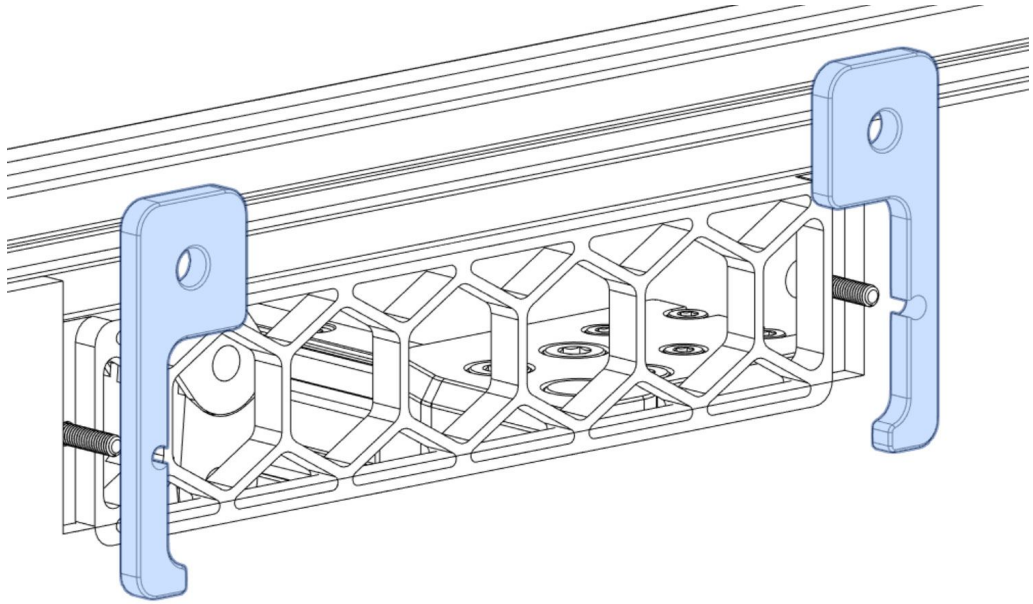




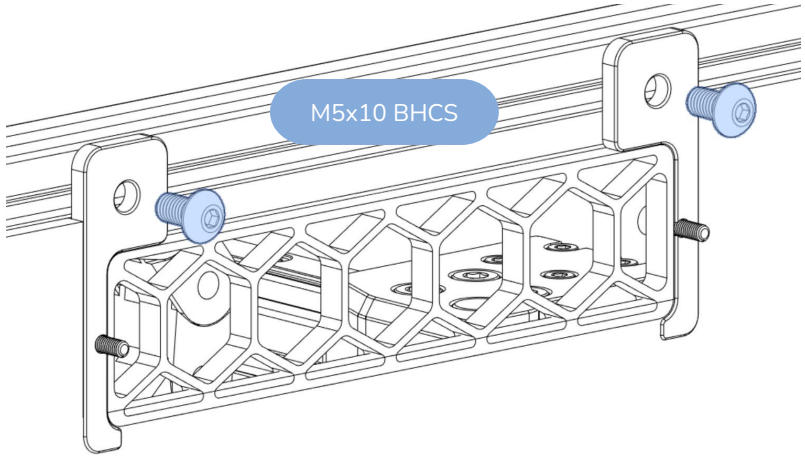
M3x12 SHCS



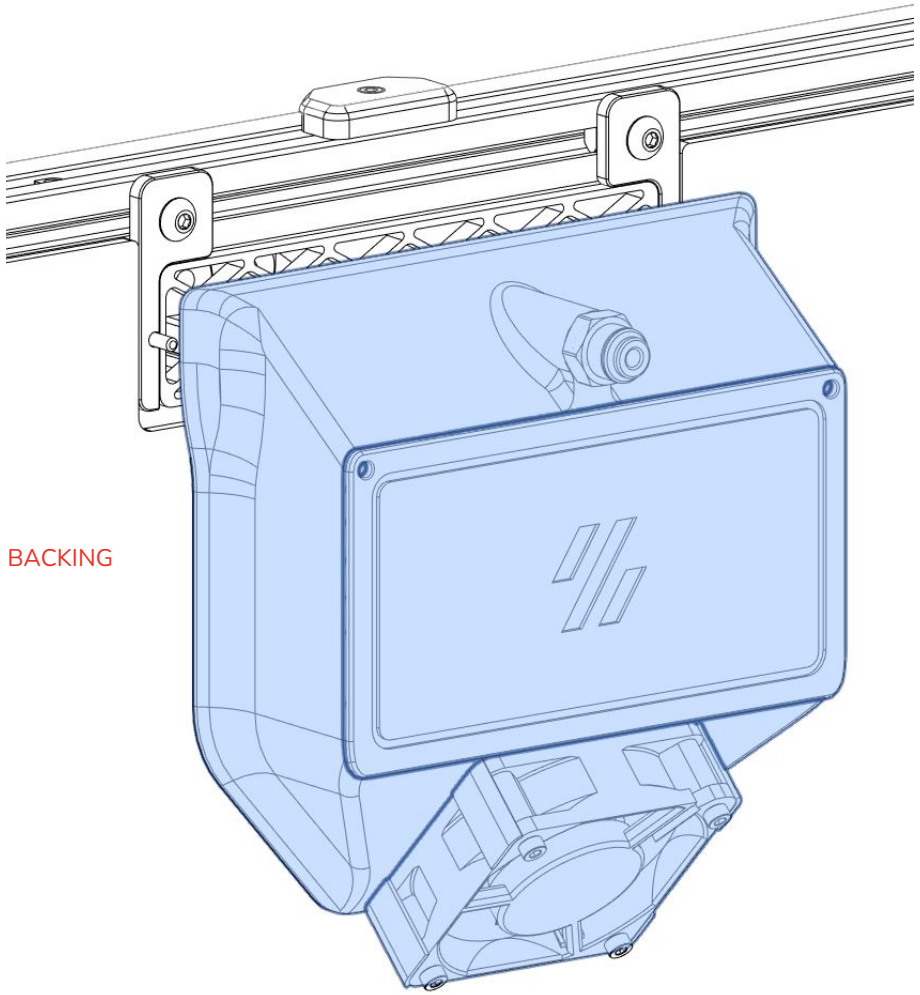
M5 T-Nut

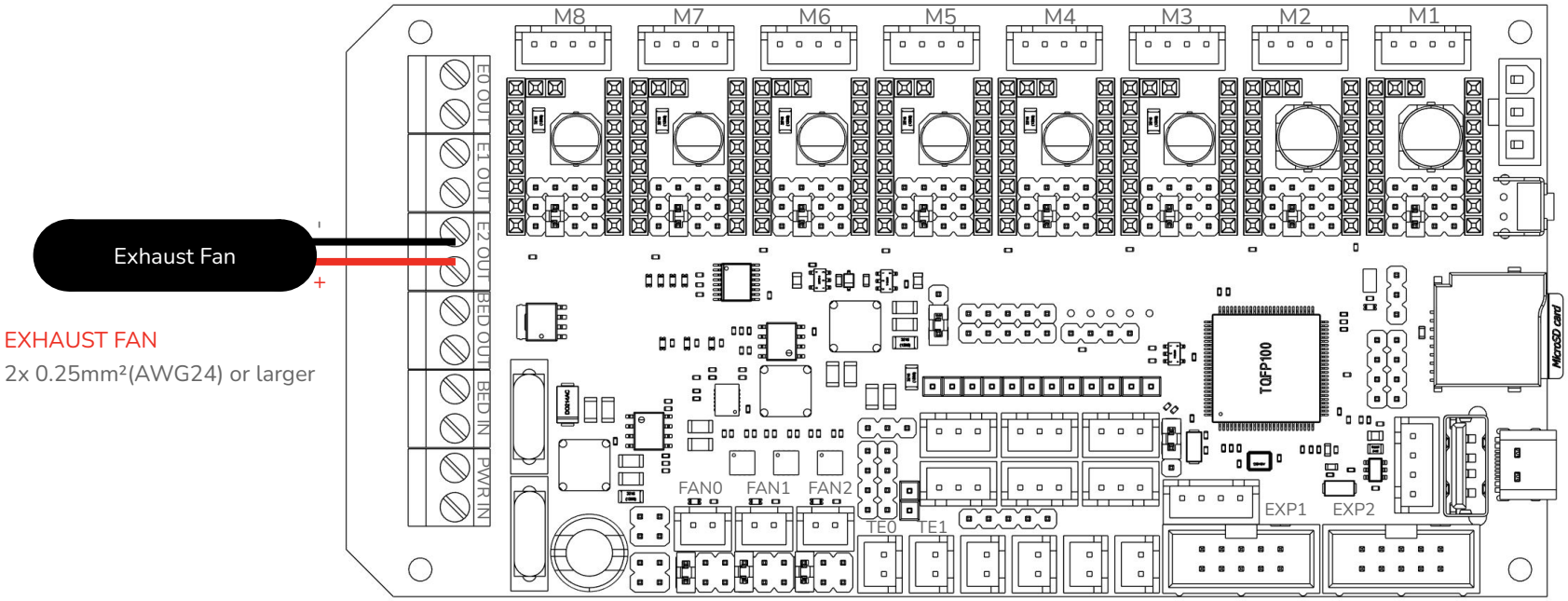


Use the big cable sleeve around the exhaust fan cable (if you use the stock exhaust).
 # Don't power on the printer if you are not 100% sure you wired everything correct!
And have a look into https://github.com/FORMBOT/Voron-Trident/blob/main/Diagrams/Instruction%20of%20Trident%20R1%20Pro.pdf



REMOVE TAPE BACKING

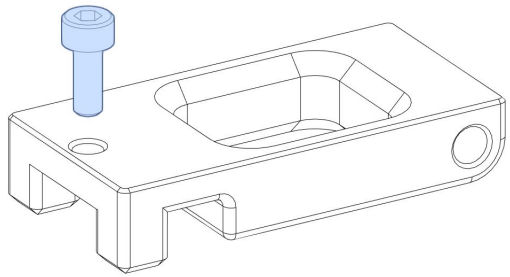


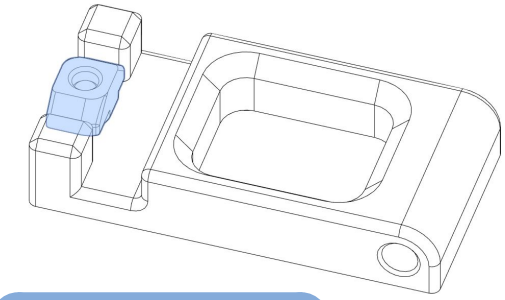


EXHAUST FAN

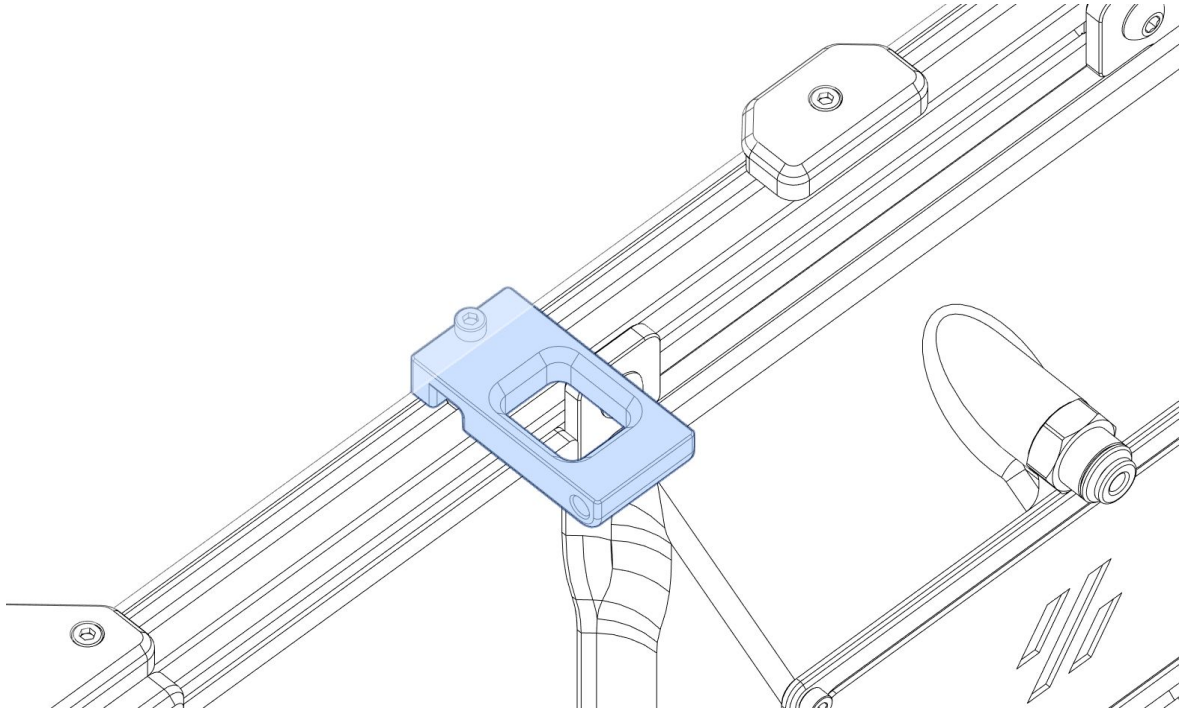
2x 0.25mm²(AWG24) or larger

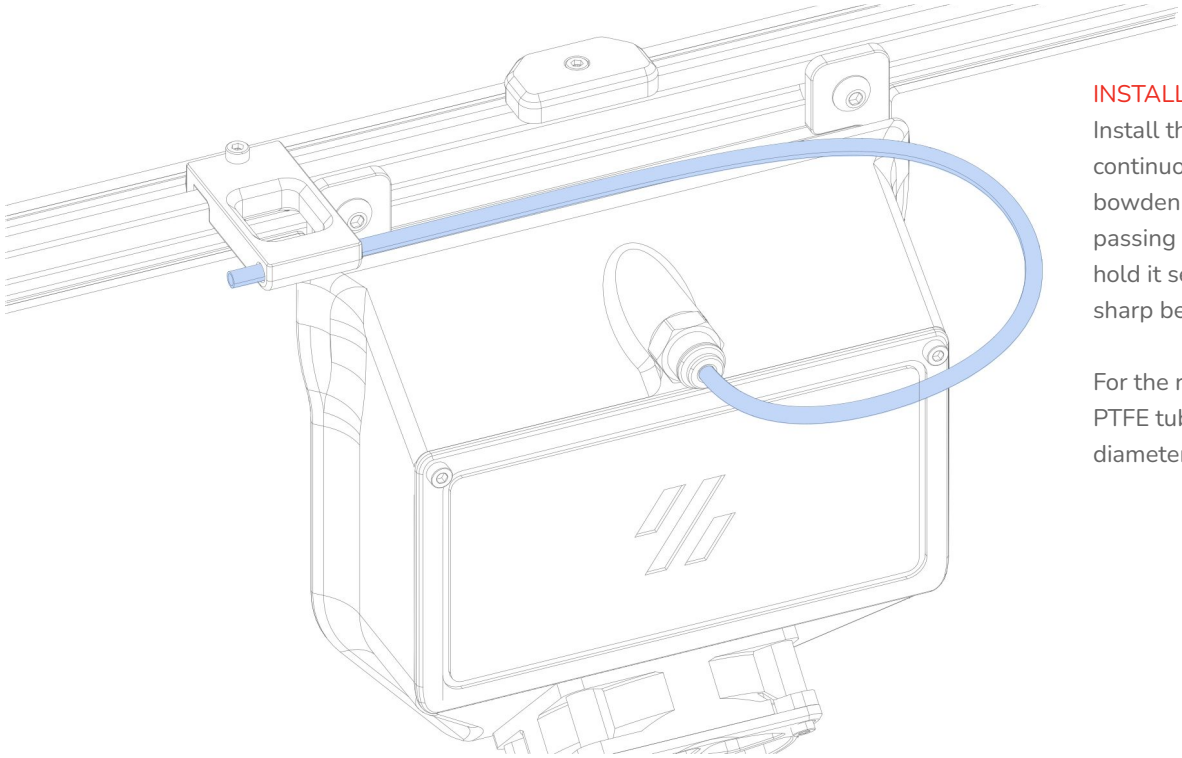
M3x8 SHCS





M3 Hammerhead Nut

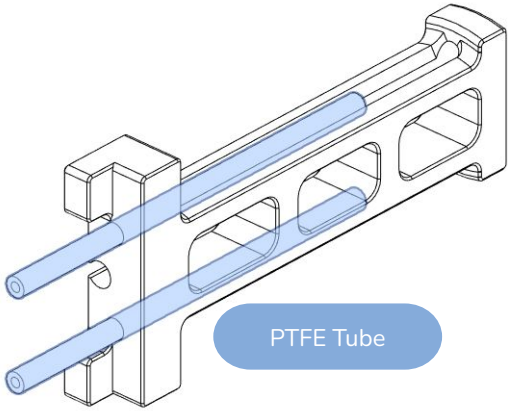




INSTALL PTFE TUBE

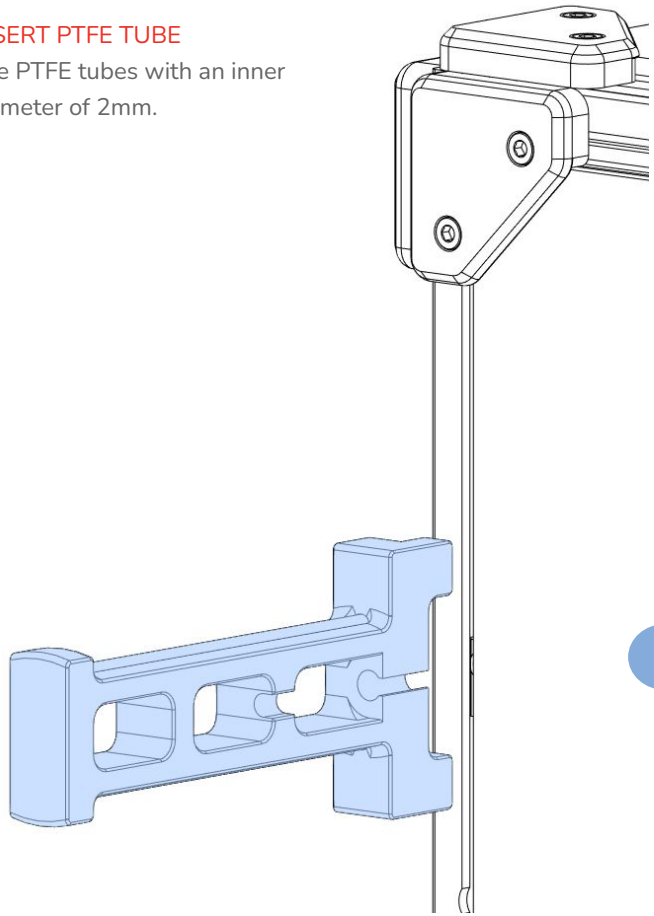
Install the PTFE tube as a continuous piece from the bowden retainer to the toolhead, passing it through the coupler to hold it securely in place. Avoid sharp bends in the filament path.

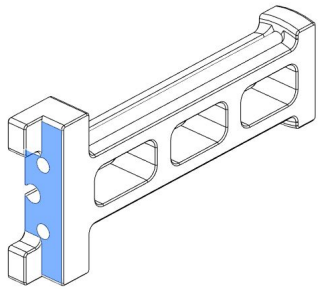
For the reverse bowden, use a PTFE tube with a 3 mm inner diameter.

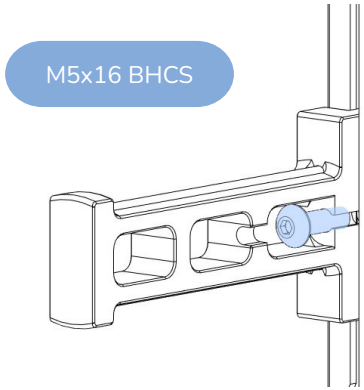


INSERT PTFE TUBE

Use PTFE tubes with an inner diameter of 2mm.







ASSEMBLY COMPLETED! ... NEXT STEP: SETUP & CALIBRATION

This manual is designed to be a reference manual for the build process of a VORON Trident printer. Additional details about the build and background on advanced topics can be found on our documentation page linked below.

The software setup and other initial setup steps with your new printer can also be found on our documentation page. We recommend starting here.

https://docs.vorondesign.com/

https://github.com/VoronDesign/Voron-Trident

HOW TO GET HELP

If you need assistance with your build, we're here to help. Head on over to our Discord group and post your questions. This is our primary medium to help VORON Users and we have a great community that can help you out if you get stuck. Alternatively, you can use our subreddit.

https://discord.gg/voron

https://www.reddit.com/r/VORONDesign

REPORTING ISSUES

Should you find an issue in this document or have a suggestion for an improvement please consider opening an issue on GitHub (https://github.com/VoronDesign/Voron-Trident/issues).

When raising an issue please include the relevant page numbers and a short description; annotated screenshots are also very welcome.

We periodically update the manual based on the feedback we get.

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There are some easter eggs hiding in this document. You might not spot them on a cursory glance.

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Website	GitHub	Docs	Discord
vorondesign.com	github.com/vorondesign	docs.vorondesign.com	discord.gg/voron